UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the Fiscal Year Ended December 31, 2023

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the Transition Period From to

Commission file number 1-8400

American Airlines Group Inc.

(Exact name of registrant as specified in its charter)

Delaware	**75-1825172**
(State or other jurisdiction of incorporation or organization)	*(I.R.S. Employer Identification No.)*
1 Skyview Drive, Fort Worth, Texas 76155	**(682) 278-9000**
(Address of principal executive offices, including zip code)	*Registrant's telephone number, including area code*

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock, $0.01 par value per share	**AAL**	**The Nasdaq Global Select Market**
Preferred Stock Purchase Rights	—	**(1)**

(1) Attached to the Common Stock

Securities registered pursuant to Section 12(g) of the Act: None

Commission file number 1-2691

American Airlines, Inc.

(Exact name of registrant as specified in its charter)

Delaware	**13-1502798**
(State or other jurisdiction of incorporation or organization)	*(I.R.S. Employer Identification No.)*
1 Skyview Drive, Fort Worth, Texas 76155	**(682) 278-9000**
(Address of principal executive offices, including zip code)	*Registrant's telephone number, including area code*

Securities registered pursuant to Section 12(b) of the Act: None

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act.

American Airlines Group Inc.	Yes	☒	No	☐
American Airlines, Inc.	Yes	☒	No	☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act.

American Airlines Group Inc.	Yes	☐	No	☒
American Airlines, Inc.	Yes	☐	No	☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days.

American Airlines Group Inc.	Yes	☒	No	☐
American Airlines, Inc.	Yes	☒	No	☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files).

American Airlines Group Inc.	Yes	☒	No	☐
American Airlines, Inc.	Yes	☒	No	☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of “large accelerated filer,” “accelerated filer,” “smaller reporting company,” and “emerging growth company” in Rule 12b-2 of the Exchange Act.

American Airlines Group Inc.	☒	Large accelerated filer	☐	Accelerated filer	☐	Non-accelerated filer	☐	Smaller reporting company	☐ Emerging growth company
American Airlines, Inc.	☐	Large accelerated filer	☐	Accelerated filer	☒	Non-accelerated filer	☐	Smaller reporting company	☐ Emerging growth company

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act.

American Airlines Group Inc.	☐
American Airlines, Inc.	☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management’s assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report.

American Airlines Group Inc.	Yes	☒	No	☐
American Airlines, Inc.	Yes	☒	No	☐

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements.

American Airlines Group Inc.	☐
American Airlines, Inc.	☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant’s executive officers during the relevant recovery period pursuant to §240.10D-1(b).

American Airlines Group Inc.	☐
American Airlines, Inc.	☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act).

American Airlines Group Inc.	Yes	☐	No	☒
American Airlines, Inc.	Yes	☐	No	☒

The aggregate market value of the voting stock held by non-affiliates of American Airlines Group Inc. as of June 30, 2023, was approximately $ 11.7 billion. As of February 16, 2024, there were 654,756,816 shares of American Airlines Group Inc. common stock outstanding.

As of February 16, 2024, there were 1,000 shares of American Airlines, Inc. common stock outstanding, all of which were held by American Airlines Group Inc.

OMISSION OF CERTAIN INFORMATION

American Airlines, Inc. meets the conditions set forth in General Instruction I(1)(a) and (b) of Form 10-K and has therefore omitted the information otherwise called for by Items 10-13 of Form 10-K as allowed under General Instruction I(2)(c).

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the proxy statement related to American Airlines Group Inc.'s 2024 Annual Meeting of Stockholders, which proxy statement will be filed under the Securities Exchange Act of 1934 within 120 days of the end of American Airlines Group Inc.'s fiscal year ended December 31, 2023, are incorporated by reference into Part III of this Annual Report on Form 10-K.

American Airlines Group Inc.
American Airlines, Inc.
Form 10-K
Year Ended December 31, 2023
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General

This report is filed by American Airlines Group Inc. (AAG) and its wholly-owned subsidiary American Airlines, Inc. (American). References in this Annual Report on Form 10-K to "we," "us," "our," the "Company" and similar terms refer to AAG and its consolidated subsidiaries. References in this report to "mainline" refer to the operations of American only and exclude regional operations.

Note Concerning Forward-Looking Statements

Certain of the statements contained in this report should be considered forward-looking statements within the meaning of the Securities Act of 1933, as amended (the Securities Act), the Securities Exchange Act of 1934, as amended (the Exchange Act), and the Private Securities Litigation Reform Act of 1995. These forward-looking statements may be identified by words such as "may," "will," "expect," "intend," "anticipate," "believe," "estimate," "plan," "project," "could," "should," "would," "continue," "seek," "target," "guidance," "outlook," "if current trends continue," "optimistic," "forecast" and other similar words. Such statements include, but are not limited to, statements about our plans, objectives, expectations, intentions, estimates and strategies for the future, and other statements that are not historical facts. These forward-looking statements are based on our current objectives, beliefs and expectations, and they are subject to significant risks and uncertainties that may cause actual results and financial position and timing of certain events to differ materially from the information in the forward-looking statements. These risks and uncertainties include, but are not limited to, those described below under Part I, Item 1A. Risk Factors, Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations and other risks and uncertainties listed from time to time in our filings with the Securities and Exchange Commission (the SEC).

All of the forward-looking statements are qualified in their entirety by reference to the factors discussed in Part I, Item 1A. Risk Factors and elsewhere in this report. There may be other factors of which we are not currently aware that may affect matters discussed in the forward-looking statements and may also cause actual results to differ materially from those discussed. We do not assume any obligation to publicly update or supplement any forward-looking statement to reflect actual results, changes in assumptions or changes in other factors affecting such statements other than as required by law. Any forward-looking statements speak only as of the date of this report or as of the dates indicated in the statements.

Summary of Risk Factors

Our business is subject to a number of risks and uncertainties that may affect our business, results of operations and financial condition, or the trading price of our common stock or other securities. We caution the reader that these risk factors may not be exhaustive. We operate in a continually changing business environment, and new risks and uncertainties emerge from time to time. Management cannot predict such new risks and uncertainties, nor can it assess the extent to which any of the risk factors below or any such new risks and uncertainties, or any combination thereof, may impact our business. These risks are more fully described in Part I, Item 1A. Risk Factors. These risks include, among others, the following:

Risks Related to our Business and Industry

- Downturns in economic conditions could adversely affect our business.
- We will need to obtain sufficient financing or other capital to operate successfully.
- Our high level of debt and other obligations may limit our ability to fund general corporate requirements and obtain additional financing, may limit our flexibility in responding to competitive developments and may cause our business to be vulnerable to adverse economic and industry conditions.
- We have significant pension and other postretirement benefit funding obligations, which may adversely affect our liquidity, results of operations and financial condition.
- If our financial condition worsens, provisions in our credit card processing and other commercial agreements may adversely affect our liquidity.
- The loss of key personnel upon whom we depend to operate our business or the inability to attract, develop and retain additional qualified personnel could adversely affect our business.
- Our business has been and will continue to be materially affected by many changing economic, geopolitical, commercial, regulatory and other conditions beyond our control, including global events that affect travel behavior, and our results of operations could be volatile and fluctuate materially due to changes in such conditions.
- The airline industry is intensely competitive and dynamic.
- Union disputes, employee strikes and other labor-related disruptions may adversely affect our operations and financial performance.
- If we encounter problems with any of our third-party regional operators or third-party service providers, our operations could be adversely affected by a resulting decline in revenue or negative public perception about our services.
- Any damage to our reputation or brand image could adversely affect our business or financial results.
- Changes to our business model that are designed to increase revenues may not be successful and may cause operational difficulties or decreased demand.
- Our intellectual property rights, particularly our branding rights, are valuable, and any inability to protect them may adversely affect our business and financial results.
- We may be a party to litigation in the normal course of business or otherwise, which could affect our financial position and liquidity.
- Our ability to utilize our NOLs and other carryforwards may be limited.

- We have a significant amount of goodwill, which is assessed for impairment at least annually. In addition, we may never realize the full value of our intangible assets or long-lived assets, causing us to record material impairment charges.
- The commercial relationships that we have with other companies, including any related equity investments, may not produce the returns or results we expect.
- Our business is very dependent on the price and availability of aircraft fuel. Continued periods of high volatility in fuel costs, increased fuel prices or significant disruptions in the supply of aircraft fuel could have a significant negative impact on consumer demand, our operating results and liquidity.
- Our business is subject to extensive government regulation, which may result in increases in our costs, disruptions to our operations, limits on our operating flexibility, reductions in the demand for air travel, and competitive disadvantages.
- We operate a global business with international operations that are subject to economic and political instability and have been, and in the future may continue to be, adversely affected by numerous events, circumstances or government actions beyond our control.
- We may be adversely affected by conflicts overseas, terrorist attacks or other acts of violence, domestically or abroad; the travel industry continues to face ongoing security concerns.
- We are subject to risks associated with climate change, including increased regulation of our greenhouse gas (GHG) emissions, changing consumer preferences and the potential for increased impacts of severe weather events on our operations and infrastructure.
- A shortage of pilots or other personnel has in the past and could continue to materially adversely affect our business.
- We depend on a limited number of suppliers for aircraft, aircraft engines and parts. Delays in scheduled aircraft deliveries, unexpected grounding of aircraft or aircraft engines whether by regulators or by us, or other loss of anticipated fleet capacity, and failure of new aircraft to receive regulatory approval, be produced or otherwise perform as and when expected, may adversely impact our business, results of operations and financial condition.
- We rely heavily on technology and automated systems to operate our business, and any failure of these technologies or systems could harm our business, results of operations and financial condition.
- Evolving data privacy requirements (in particular, compliance with applicable federal, state and foreign laws relating to handling of personal information about individuals) could increase our costs, and any significant data privacy incident could disrupt our operations, harm our reputation, expose us to legal risks and otherwise materially adversely affect our business, results of operations and financial condition.
- We are exposed to risks from cyberattacks, and any cybersecurity incidents involving us, our third-party service providers, or one of our AAdvantage partners or other business partners, could materially adversely affect our business, results of operations and financial condition.
- We rely on third-party distribution channels and must effectively manage the costs, rights and functionality of these channels.
- If we are unable to obtain and maintain adequate facilities and infrastructure throughout our system and, at some airports, adequate slots, we may be unable to operate our existing flight schedule and to expand or change our route network in the future, which may have a material adverse impact on our operations.

PART I

ITEM 1. BUSINESS

Overview

American Airlines Group Inc. (AAG), a Delaware corporation, is a holding company and its principal, wholly-owned subsidiaries are American Airlines, Inc. (American), Envoy Aviation Group Inc., PSA Airlines, Inc. (PSA) and Piedmont Airlines, Inc. (Piedmont). AAG was formed in 1982, under the name AMR Corporation (AMR), as the parent company of American, which was founded in 1934.

AAG's and American's principal executive offices are located at 1 Skyview Drive, Fort Worth, Texas 76155 and their telephone number is 682-278-9000.

Airline Operations

Together with our wholly-owned regional airline subsidiaries and third-party regional carriers operating as American Eagle, our primary business activity is the operation of a major network air carrier, providing scheduled air transportation for passengers and cargo through our hubs in Charlotte, Chicago, Dallas/Fort Worth, Los Angeles, Miami, New York, Philadelphia, Phoenix and Washington, D.C. and partner gateways, including in London, Doha, Madrid, Seattle/Tacoma, Sydney and Tokyo (among others). In 2023, approximately 211 million passengers boarded our flights. During 2023, we launched more than 50 new routes, providing service to close to 350 destinations around the world, and we announced several new destinations for customers to explore in 2024: Copenhagen, Denmark; Naples, Italy; Nice, France; Governor's Harbour, Bahamas; Tijuana, Mexico; Tulum, Mexico; Ocho Rios, Jamaica; Pasco, Washington and Hyannis, Massachusetts. In 2024, we announced new service to Brisbane, Australia and Veracruz, Mexico, as well as additional nonstop service between New York and Tokyo, Japan.

As of December 31, 2023, we operated 965 mainline aircraft supported by our regional airline subsidiaries and third-party regional carriers, which together operated an additional 556 regional aircraft. See Part I, Item 2. Properties for further discussion of our mainline and regional aircraft and *"Regional"* below for further discussion of our regional operations.

American is a founding member of the **one**world® Alliance, which brings together a global network of 13 world-class member airlines and their affiliates, working together to provide a superior and seamless travel experience. See *"Distribution and Marketing Agreements"* below for further discussion on the **one**world Alliance and other agreements with domestic and international airlines.

See Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations – *"2023 Financial Overview," "AAG's Results of Operations"* and *"American's Results of Operations"* for further discussion of AAG's and American's operating results and operating performance. Also, see Note 1(m) to each of AAG's and American's Consolidated Financial Statements in Part II, Items 8A and 8B, respectively, for passenger revenue by geographic region and Note 13 to AAG's Consolidated Financial Statements in Part II, Item 8A and Note 12 to American's Consolidated Financial Statements in Part II, Item 8B for information regarding operating segments.

Regional

Our regional carriers provide scheduled air transportation under the brand name "American Eagle." The American Eagle carriers include our wholly-owned regional carriers Envoy Air Inc. (Envoy), PSA and Piedmont, as well as third-party regional carriers including Republic Airways Inc. (Republic), SkyWest Airlines, Inc. (SkyWest) and Air Wisconsin Airlines LLC (Air Wisconsin). Our regional carriers are an integral component of our operating network. We rely heavily on regional carriers to serve small markets and also to drive connecting traffic to our hubs from markets that are not economical for us to serve with larger, mainline aircraft. In addition, regional carriers offer complementary service in many of our mainline markets. All American Eagle carriers use logos, service marks, aircraft paint schemes and uniforms similar to those of our mainline operations. In 2023, 46 million passengers boarded our regional flights, approximately 45% of whom connected to or from our mainline flights.

Our regional carrier arrangements are in the form of capacity purchase agreements with our third-party regional partners and similar arrangements with our wholly-owned affiliates which provide that all revenues, including passenger, in-flight, ancillary, mail and freight revenues, go to us. We control marketing, scheduling, ticketing, pricing and seat inventories. In return, we agree to pay predetermined fees to these airlines for operating an agreed-upon number of aircraft, without regard to the number of passengers on board. In addition, these agreements provide that we either reimburse or pay 100% of certain variable costs, such as airport landing fees, fuel and passenger liability insurance. In 2023, Air Wisconsin began operating scheduled flights under the American Eagle name.

Cargo

Our cargo division provides a wide range of freight and mail services, with facilities and interline connections available across the globe. In 2023, we served more than 21,000 unique origin and destination pairs, transporting over 900 million pounds of time-sensitive freight and mail across our network.

Distribution and Marketing Agreements

Passengers can purchase tickets for travel on American through several distribution channels, including our website (*www.aa.com*), our mobile app, our reservations centers and third-party distribution channels, including conventional travel agents, travel management companies and online travel agents (e.g., Expedia, including its booking sites Orbitz and Travelocity, and Booking Holdings, including its booking sites Kayak and Priceline). Over the last decade, American has been a leader in deploying new distribution technologies such as IATA New Distribution Capability (NDC) technology, which is now the primary means by which we distribute our content to third parties through aggregators (e.g., Amadeus, Sabre, Travelport and Travelfusion) or through direct connections. NDC technology provides customers access to enhanced content and functionality, providing a simplified booking experience, and enabling us to provide more relevant, tailored offers to customers.

To remain competitive, we will need to successfully manage our distribution costs and rights, increase our distribution flexibility and improve the functionality of our distribution channels, while maintaining an industry-competitive cost structure. For more discussion, see Part I, Item 1A. Risk Factors – "*We rely on third-party distribution channels and must effectively manage the costs, rights and functionality of these channels* ."

Member of oneworld Alliance

American is a founding member of the **one**world Alliance, which currently includes Alaska Airlines, British Airways, Cathay Pacific, Finnair, Iberia, Japan Airlines, Malaysia Airlines, Qantas Airways (Qantas), Qatar Airways, Royal Air Maroc, Royal Jordanian Airlines and SriLankan Airlines. Oman Air is expected to join the **one**world Alliance in 2024, and Fiji Airways is a **one**world connect partner offering select alliance benefits to **one**world frequent flyers. The **one**world Alliance links the networks of member carriers and their respective affiliates to enhance customer service and provide smooth connections to the destinations served by the alliance, including linking member carriers' loyalty programs and providing reciprocal access to the carriers' airport lounge facilities.

Joint Business Agreements and Other Cooperation Agreements

American has established a transatlantic joint business with British Airways, Aer Lingus, Iberia and Finnair, a transpacific joint business with Japan Airlines and a joint business covering Australia and New Zealand with Qantas. Joint business agreements enable the carriers involved to cooperate on flights between particular destinations and allow pooling and sharing of certain revenues and costs, enhanced loyalty program reciprocity and cooperation in other areas. Joint business agreements have become a common approach among major carriers to address key regulatory restrictions typically applicable to international airline service, including limitations on the foreign ownership of airlines and national laws prohibiting foreign airlines from carrying passengers beyond specific gateway cities.

We also have established a strategic alliance with Alaska Airlines covering certain routes on the West Coast of the United States and a strategic alliance with Qatar Airways covering the Middle East in order to provide customers with improved schedules and network connection opportunities, enhanced loyalty program reciprocity and cooperation in other areas.

In July 2010, in connection with a regulatory review related to our transatlantic joint business, we provided certain commitments to the European Commission (EC) regarding, among other things, the availability of take-off and landing slots at London Heathrow (LHR) or London Gatwick (LGW) airports. The commitments accepted by the EC were binding for 10 years. In anticipation of both the exit of the United Kingdom from the European Union (EU), commonly referred to as Brexit, and the expiry of the EC commitments in July 2020, the United Kingdom Competition and Markets Authority (CMA), in October 2018, opened an investigation into the transatlantic joint business. In September 2020 and April 2022, the CMA adopted interim measures that effectively extend the EC commitments until March 2026 in light of the uncertainty and other impacts resulting from the COVID-19 pandemic. The CMA restarted its investigation in September 2023 after a pause related to the COVID-19 pandemic and plans to complete the investigation before the scheduled expiration of the interim measures in March 2026. We continue to cooperate fully with the CMA.

Marketing Relationships

To improve access to each other's markets, various U.S. and foreign air carriers, including American, have established marketing agreements with other airlines. These marketing agreements vary in scope and are intended to provide enhanced customer choice by means of an expanded network with reciprocal loyalty program participation, but do not involve the same level of cooperation as our joint businesses or strategic alliances. As of December 31, 2023, in addition to the relationships described above, American had codeshare, marketing and/or loyalty program relationships with Air Tahiti Nui, Cape Air, Cathay Pacific, China Southern Airlines Company Limited (China Southern Airlines), EL AL Israel Airlines, Etihad Airways, Fiji Airways, GOL Linhas Aéreas Inteligentes S.A. (GOL), Gulf Air, Hawaiian Airlines, IndiGo, JetSMART, Jetstar, Jetstar Japan, Malaysia Airlines, Philippine Airlines, Royal Air Maroc, Royal Jordanian Airlines, Silver Airways, SriLankan Airlines and Vueling Airlines.

In 2023, we completed codeshare agreements with JetSMART, enabling American's customers to book travel on JetSMART's network beyond Santiago, Chile and Lima, Peru, and which will allow for further extension of our network to other markets in South America, such as Argentina, on JetSMART operated flights, subject to all necessary regulatory approvals.

Also in 2023, we launched a codeshare partnership with Philippine Airlines. This partnership introduced the first marketed flights by a Philippine carrier to several U.S. destinations and allows American's customers to travel to Manila and Cebu, Philippines.

We had a marketing relationship, the Northeast Alliance arrangement (NEA), with JetBlue Airways Corporation (JetBlue) that included an alliance agreement with reciprocal codesharing on certain domestic and international routes from New York (John F. Kennedy International Airport (JFK), LaGuardia Airport (LGA) and Newark Liberty International Airport) and Boston Logan International Airport. On May 19, 2023, the U.S. District Court for the District of Massachusetts issued an order permanently enjoining American and JetBlue from continuing and further implementing the NEA. In June 2023, JetBlue delivered a notice of termination of the NEA, effective July 29, 2023, and the carriers have commenced wind-down activities to accommodate mutual customers.

AAdvantage® Program

Our AAdvantage program was established to develop passenger loyalty by offering benefits and rewards to travelers for their continued patronage with American and our partners. AAdvantage members enjoy exclusive benefits and earn mileage credits for flying on eligible tickets on American, any **one**world Alliance airline or other partner airlines. For every dollar spent by flying on an eligible American ticket, members earn mileage credits, and AAdvantage Gold®, AAdvantage Platinum®, AAdvantage Platinum Pro® and AAdvantage Executive Platinum® status holders earn additional bonus mileage credits of 40%, 60%, 80% and 120%, respectively. Members also earn mileage credits by using the services of more than 1,000 non-flight partners, such as our co-branded credit cards, certain hotel and car rental companies and shopping and dining partners. The AAdvantage program in general, and our co-branded credit card programs in particular, are material assets of our business and have become increasingly important to our company over time. During 2023 and 2022, cash payments from co-branded credit card and other partners were $5.2 billion and $4.5 billion, respectively.

Mileage credits can be redeemed for travel and upgraded experiences on American and participating airlines, membership to our Admirals Club®, or for other non-flight awards, such as car rentals and hotels, from our program partners. Travel awards are available on all flights operated by American and, subject to capacity-controlled seating, on flights operated by our partners. A member's mileage credits generally do not expire if that member has any type of qualifying activity at least once every 24 months or if the AAdvantage member is the primary holder of a co-branded credit card. AAdvantage members qualify for status over a 12-month period beginning on March 1 of each year by earning

Loyalty Points, which can be earned through a variety of qualifying travel and non-travel activities, including use of our co-branded credit cards. Status members can enjoy additional travel benefits of the AAdvantage program, including complimentary upgrades, checked bags, and Preferred and Main Cabin Extra seats, as well as priority check-in, security, boarding and baggage delivery when traveling on American, any **one**world Alliance airline or select partner airlines. In addition, AAdvantage members can unlock benefits, rewards and choices before, between and beyond the traditional status tiers with Loyalty Point Rewards. In 2023, we introduced a new business loyalty program, AAdvantage Business, which rewards both eligible companies with AAdvantage miles and their travelers with additional Loyalty Points for booking business travel through our website or mobile app.

In 2023, the editorial staff of the digital news outlet, *The Points Guy*, selected AAdvantage as the Best U.S. Airline Loyalty Program. In addition, AAdvantage was recognized for the Best Elite Program in the Americas at the 2023 Freddie Awards, which is based entirely on votes from travelers around the world.

Under our agreements with AAdvantage members and program partners, we reserve the right to change the terms of the AAdvantage program at any time and without notice. Program rules, partners, special offers, awards and requisite mileage levels for awards are subject to change.

During 2023, our members redeemed approximately 13 million awards, including travel redemptions for flights and upgrades on American and other air carriers, as well as redemption of car and hotel awards, club memberships and merchandise. Approximately 8% of our 2023 total revenue passenger miles flown were from award travel.

See Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations – " *Critical Accounting Policies and Estimate*s" for more information on our loyalty program.

Industry Competition

Domestic

The markets in which we operate are highly competitive. On most of our domestic nonstop routes, we face competing service from other domestic airlines, including major network airlines, low-cost carriers and ultra-low-cost carriers such as Alaska Airlines, Allegiant Air, Delta Air Lines, Frontier Airlines, Hawaiian Airlines, JetBlue, Southwest Airlines, Spirit Airlines and United Airlines. Between cities that require a connection, where the major airlines compete via their respective hubs, competition is significant. In addition, we face competition on some of our connecting routes from airlines operating point-to-point service on such routes. We also compete with all-cargo and charter airlines and, particularly on shorter segments, ground and rail transportation.

In general, beyond nonstop city pairs, carriers that have the greatest ability to seamlessly connect passengers to and from markets have a competitive advantage. In some cases, however, foreign governments limit U.S. air carriers' rights to transport passengers beyond designated gateway cities in foreign countries. In order to improve access to domestic and foreign markets, we have arrangements with other airlines including through the **one**world Alliance, other cooperation agreements, joint business agreements and marketing relationships, as further discussed herein.

On all of our routes, pricing decisions are affected, in large part, by the need to meet competition from other airlines. Price competition occurs on a market-by-market basis through price discounts, changes in pricing structures, fare matching, targeted promotions and loyalty program initiatives. Airlines typically use discounted fares and other promotions to stimulate traffic during normally weak travel periods, when they begin service to new cities, when they have excess capacity, to generate cash flow, to maximize revenue per available seat mile or to establish, increase or preserve market share. Most airlines will quickly match price reductions in a particular market, and we have often elected to match discounted or promotional fares initiated by other air carriers in certain markets in order to compete in those markets. In addition, we face pricing pressures from so-called ultra-low-cost carriers, such as Allegiant Air, Frontier Airlines and Spirit Airlines, which compete in many of the markets in which we operate, with competition from these carriers increasing and new entrants regularly announcing their intention to start up new ultra-low-cost carriers.

In addition to price competition, airlines compete for market share by increasing the size of their route system and the number of markets they serve. The American Eagle regional carriers increase the number of markets we serve by flying to smaller markets and providing connections at our hubs. Many of our competitors also own or have agreements with regional airlines that provide similar services at their hubs and other locations. We also compete on the basis of scheduling (frequency and flight times), availability of nonstop flights, on-time performance, type of equipment, cabin configuration, amenities provided to passengers, loyalty programs, the automation of travel agent reservation systems, onboard products, health and safety, sustainability initiatives and other services.

International

In addition to our extensive domestic service, we provide international service to Canada, Mexico, the Caribbean, Central and South America, Europe, Qatar, China, Japan, Korea, India, Australia and New Zealand. In providing international air transportation, we compete with other U.S. airlines, foreign investor-owned airlines and foreign state-owned or state-affiliated airlines. Competition has also been increasing from low-cost airlines executing international long-haul expansion strategies, a trend we expect to continue, in particular with the planned introduction of long-range narrowbody aircraft in the coming years.

In order to increase our ability to compete in the market for international air transportation service, which is subject to extensive government regulation, U.S. and foreign carriers have entered into bilateral and multilateral marketing relationships, alliances, cooperation agreements and joint business agreements to exchange traffic among each other's flights and route networks. See "*Distribution and Marketing Agreements*" above for further discussion.

Sustainability

Operating a sustainable business that has the ability to serve our stakeholders over the long-term is an important part of our strategy. We have increased our focus over time on a number of elements that we view as important to build a more sustainable company, including those described below.

We have received recognition for our progress toward our sustainability goals. American was named the 2023 Air Transport World Eco-Airline of the Year, and in 2023 we were named to the Dow Jones Sustainability World Index for the first time, one of only two passenger airlines included in the index. We also returned to the Dow Jones Sustainability North America Index in 2023 for the third year in a row.

Climate

We recognize the challenge of climate change and have set ambitious goals to transition to operating a low-carbon airline over time. Our aim is to achieve net zero GHG emissions by 2050, and we have set an intermediate target to drive progress toward that goal. We have received validation from the Science Based Targets initiative (SBTi) that our 2035 GHG reduction target complies with the criteria in the SBTi's first aviation pathway.

The vast majority of our direct GHG emissions comes from the use of jet fuel in our operations. Our current strategy for reaching net zero GHG emissions by 2050 is focused on running a more fuel-efficient operation, with more fuel-efficient aircraft, powered by low-carbon fuel. To do so, we are working to drive progress across several key levers, including:

- Continuing to replace older, less fuel-efficient aircraft with new, more efficient aircraft over time;
- Helping scale the production of sustainable aviation fuel (SAF) with the aim of transitioning to lower-carbon fuels. Currently, SAF is not available at the cost or scale necessary to meet our industry's needs. We continue to enter into agreements to purchase SAF as part of our goal to replace 10% of our conventional jet fuel with SAF in 2030 and to encourage investment in SAF; and
- Evaluating and investing in innovations that may enable commercial aircraft to be powered by low- and no-carbon fuel sources over the long term. For example, we have made direct investments in companies working to develop hydrogen-electric propulsion technology and green hydrogen distribution. We are also an anchor partner of Breakthrough Energy Catalyst, which aims to make investments to accelerate the development of new clean energy technologies, including SAF.

Achieving our ambitious goals will require significant action and investments by governments, manufacturers and other stakeholders. We are committed to engaging with our stakeholders to seek to advance these initiatives, and we have dedicated resources to advance our own progress. Our Board and Corporate Governance and Public Responsibility Committee receive updates on our climate strategy, progress and key risks regularly. Our Chief Executive Officer is responsible for oversight of our climate change strategy.

Safety

The safety of our customers and team members is a top priority. Our approach to safety is guided by our FAA-approved safety management systems (SMS), an organization-wide approach to identifying and managing risk. Each SMS is comprised of four components: Safety Policy, Safety Assurance, Safety Risk Management and Safety Promotion. Our Safety Policy sets safety objectives while striving to comply with applicable regulatory requirements and laws in the countries where we operate and establishing standards for acceptable operational behaviors.

The Safety Assurance component of our SMS specifies how we use data and conduct quality assurance and internal oversight to validate the effectiveness of risk controls and the performance of the SMS. The Safety Risk Management (SRM) element of our SMS provides a decision-making process for identifying hazards and mitigating risk based on a thorough understanding of our systems and their operating environment. We employ SRM whenever there is a significant change to our operations, such as the delivery of new aircraft. Lastly, the Safety Promotion component includes training and raising awareness among team members so that they can spot potential safety events.

Customers

We fly to close to 350 destinations in the United States and internationally, and we are committed to providing our customers with a world-class travel experience. We continued to rigorously measure and track customer satisfaction through passenger surveys in 2023, efforts that led to further improvements in our operations and the services we provide. In 2023, we achieved our best-ever full year completion factor, with the lowest number of cancellations annually since the 2013 merger with US Airways Group, Inc., which led to a record Likelihood to Recommend score for the full year. Additionally in 2023, we were recognized for the sixth consecutive year with the prestigious Five Star rating in The APEX Official Airline Ratings – Global Airline category. This rating is based on verified customer feedback on the overall travel experience.

Our People

The airline business is labor intensive, and our team members are critical to delivering for our customers. The operational complexity of our business requires a diverse team of personnel trained and experienced in a variety of technical areas such as flight operations, ground operations, safety and maintenance, customer service and airline scheduling and planning. Fostering a culture where our team members feel supported to take care of our customers is critical to our success. To do this, we must continue to build a diverse and inclusive environment, helping all team members reach their full potential and providing them with the right resources and support.

In 2023, mainline and regional salaries, wages and benefits were our largest expense and represented 34% of our total operating expenses. As of December 31, 2023, we had approximately 132,100 active full-time equivalent employees, approximately 87% of whom were represented by various labor unions responsible for negotiating the collective bargaining agreements (CBAs) governing their compensation and job duties, among other things.

Talent Development

We focus on providing our team members the tools, training and resources they need to do their best work. We maintain a suite of programs aimed at helping our people develop the skills and experience they need to succeed in their roles and build rewarding, long-term careers within our company. Additionally, we have partnered with leading online learning platforms to make professional development available on-demand to all of our team members.

Diversity, Equity and Inclusion

Cultivating an environment that celebrates diversity, equity and inclusion (DEI) is a priority for us, and we seek to create a workplace where diverse perspectives and experiences are welcomed and encouraged, where team members feel comfortable to be their authentic selves and where we are always learning from one another. Our goal is to make culture a competitive advantage so people will want to work with us, fly with us and invest in us. We are implementing a multiyear strategy focused on embedding DEI throughout our company by:

- Hiring, engaging and retaining talent for growth;
- Delivering excellence in our operations to serve and expand our global markets;
- Striving to have our teams effectively serve the communities we represent; and
- Driving innovation to build competitive advantages.

In 2023, we received a perfect score on the Disability Equality Index for the eighth consecutive year and were named one of the best places to work for disability inclusion. We also received a top score of 100 on the Human Rights Campaign Foundation's 2023-2024 Corporate Equality Index, an assessment of LGBTQ+ workplace equality.

Competitive Pay and Comprehensive Benefits

We seek to offer competitive pay, comprehensive benefits and a wide variety of resources designed to support the physical, behavioral and financial well-being of our team members and their families, including medical coverage that is intended to be affordable and flexible along with healthcare navigation and support tools.

Our internal recognition programs give team members and customers the opportunity to show their appreciation for a job well done, including through our Nonstop Thanks program whereby team members can award each other points for exceptional service or as an expression of gratitude. Recognition points earned through the recognition program can be redeemed for items in an online catalog. In 2023, our team members were recognized by customers, peers and company leaders approximately three million times and more than 1,600 peer nominations were submitted for the annual Circle of Excellence, the highest honor that we bestow upon our team members for their career achievements.

Our future success depends in large part on our ability to attract, develop and retain highly qualified management, technical and other personnel. Retaining and recruiting people with the appropriate skills became particularly challenging as the economy in general, and the airline industry in particular, recovered from the COVID-19 pandemic, and there remains intense competition for the human resources necessary to operate our business successfully. Like many other airlines, we have experienced and continue to experience periodic shortages of frontline team members as a result. For more discussion, see Part I, Item 1A. Risk Factors – *"The loss of key personnel upon whom we depend to operate our business or the inability to attract, develop and retain additional qualified personnel could adversely affect our business."*

Labor Relations

Labor relations in the air transportation industry are regulated under the Railway Labor Act (RLA), which vests in the National Mediation Board (NMB) certain functions with respect to disputes between airlines and labor unions relating to union representation and CBAs.

The following table shows our domestic airline employee groups that are represented by unions:

Union	Class or Craft	Employees [1]	Contract Amendable Date
Mainline:			
Allied Pilots Association (APA)	Pilots	14,500	2027
Association of Professional Flight Attendants (APFA)	Flight Attendants	24,950	2019
Airline Customer Service Employee Association – Communications Workers of America and International Brotherhood of Teamsters (CWA-IBT)	Passenger Service	14,650	2029
Transport Workers Union and International Association of Machinists & Aerospace Workers (TWU-IAM Association)	Mechanics and Related	12,350	2025
TWU-IAM Association	Fleet Service	19,100	2025
TWU-IAM Association	Stock Clerks	2,000	2025
TWU-IAM Association	Flight Simulator Engineers	150	2025
TWU-IAM Association	Maintenance Control Technicians	190	2025
TWU-IAM Association	Maintenance Training Instructors	100	2025
Professional Airline Flight Control Association (PAFCA)	Dispatchers	570	2025
Transport Workers Union (TWU)	Flight Crew Training Instructors	390	2025

Union	Class or Craft	Employees [1]	Contract Amendable Date
Envoy:			
Air Line Pilots Associations (ALPA)	Pilots	2,070	2029
Association of Flight Attendants-CWA (AFA)	Flight Attendants	1,850	2026
TWU	Ground School Instructors	10	2027
TWU	Mechanics and Related	1,200	2027
TWU	Stock Clerks	130	2027
TWU	Simulator Instructors	20	2026
TWU	Fleet Service	4,020	2026
TWU	Dispatchers	70	2025
Communications Workers of America (CWA)	Passenger Service	7,000	2026
Piedmont:			
ALPA	Pilots	640	2029
AFA	Flight Attendants	310	2026
International Brotherhood of Teamsters (IBT)	Mechanics and Related	470	2026
IBT	Stock Clerks	60	2026
CWA	Fleet and Passenger Service	6,650	2023
IBT	Dispatchers	40	2025
ALPA	Flight Crew Training Instructors	70	2029
PSA:			
ALPA	Pilots	1,500	2028
AFA	Flight Attendants	1,190	2023
International Association of Machinists & Aerospace Workers (IAM)	Mechanics and Related	680	2027
TWU	Dispatchers	40	2024
ALPA	Flight Crew Training Instructors	80	2028

(1) Represents approximate number of active employees as of December 31, 2023.

In 2023, a new four-year CBA was ratified by the APA, the union representing our mainline pilots. Additionally, in January 2024, a new five-year CBA was ratified by the CWA-IBT, which is amendable in 2029. The CBA covering our mainline flight attendants is now amendable and negotiations continue. Among our wholly-owned regional subsidiaries, Piedmont fleet and passenger service and PSA flight attendants have agreements that are now amendable and are engaged in negotiations.

For more discussion, see Part I, Item 1A. Risk Factors – "*Union disputes, employee strikes and other labor-related disruptions may adversely affect our operations and financial performance*."

Aircraft Fuel

Our operations and financial results are materially affected by the availability and price of aircraft fuel, which represents one of the largest single cost items in our business. Based on our 2024 forecasted mainline and regional fuel consumption, we estimate that a one cent per gallon increase in the price of aircraft fuel would increase our 2024 annual fuel expense by approximately $45 million.

The following table shows annual aircraft fuel consumption and costs, including taxes, for our mainline and regional operations for 2023 and 2022 (gallons and aircraft fuel expense in millions).

Year	Gallons	Average Price per Gallon	Aircraft Fuel Expense	Percent of Total Operating Expenses
2023	4,140	$2.96	$12,257	25%
2022	3,901	$3.54	$13,791	29%

As of December 31, 2023, we did not have any fuel hedging contracts outstanding to hedge our fuel consumption. Our current policy is not to enter into transactions to hedge our fuel consumption, although we review this policy from time to time based on market conditions and other factors. As such, and assuming we do not enter into any future transactions to hedge our fuel consumption, we will continue to be fully exposed to fluctuations in aircraft fuel prices.

Aircraft fuel prices have in the past, and may in the future, experience substantial volatility. We cannot predict the future availability, price volatility or cost of aircraft fuel. For more discussion, see Part I, Item 1A. Risk Factors – "*Our business is very dependent on the price and availability of aircraft fuel. Continued periods of high volatility in fuel costs, increased fuel prices or significant disruptions in the supply of aircraft fuel could have a significant negative impact on consumer demand, our operating results and liquidity*."

Seasonality and Other Factors

Due to the greater demand for air travel during the summer months, revenues in the airline industry exhibit seasonal patterns based on the peak travel periods. General economic conditions, fears of terrorism or war, fare initiatives, fluctuations in fuel prices, labor actions, weather, natural disasters, outbreaks of disease, geopolitical factors and other factors could impact this seasonal pattern. Therefore, our quarterly results of operations are not necessarily indicative of operating results for the entire year, and historical operating results in a quarterly or annual period are not necessarily indicative of future operating results.

Domestic and Global Regulatory Landscape

General

Airlines are subject to extensive domestic and international regulatory requirements. Domestically, the DOT and the Federal Aviation Administration (FAA) exercise significant regulatory authority over air carriers.

The DOT, among other things, oversees and regulates domestic and international codeshare agreements, international route authorities, competition and consumer protection matters including accessibility, the display and sharing of ancillary fee information and refund practices. The Antitrust Division of the Department of Justice, along with the DOT in certain instances, have jurisdiction over airline antitrust matters.

The FAA similarly exercises safety oversight and regulates most operational matters of our business, including how we operate and maintain our aircraft. FAA requirements cover, among other things, required technology and necessary onboard equipment; systems, procedures and training necessary to ensure the continuous airworthiness of our fleet of aircraft; safety measures and equipment; crew scheduling limitations and experience requirements; and many other technical aspects of airline operations. Additionally, our pilots and other employees are subject to rigorous certification standards, and our pilots and other crew members must adhere to flight time and rest requirements.

The FAA also controls the national airspace system, including operational rules and fees for air traffic control (ATC) services. The efficiency, reliability and capacity of the ATC network has a significant impact on our costs and on the timeliness of our operations.

The U.S. Postal Service has jurisdiction over certain aspects of the transportation of mail and related services.

Airport Access and Operations

Domestically, any U.S. airline authorized by the DOT is generally free to operate scheduled passenger service between any two points within the U.S. and its territories, with the exception of certain airports that require landing and take-off rights and authorizations (slots) and other facilities, and certain airports that impose geographic limitations on operations or curtail operations based on the time of day. Operations at three major domestic airports we serve (JFK and LGA in New York City, and Ronald Reagan Washington National Airport (DCA) near Washington, D.C.) and many foreign airports we serve (including LHR) are regulated by governmental entities through allocations of slots or similar regulatory mechanisms

that limit the rights of carriers to conduct operations at those airports. Each slot represents the authorization to land at and take off from the particular airport during a specified time period. In addition to slot restrictions, operations at DCA and LGA are also limited based on a so-called "perimeter rule" which generally limits the stage length of the flights that can be operated from those airports to 1,250 and 1,500 miles, respectively. Generally, our ability to retain slots is conditioned on the continued use of such slots, and in the absence of use, the slots are subject to forfeiture. In certain circumstances, such as during the COVID-19 pandemic, regulators may issue slot waivers which temporarily suspend or amend slot usage requirements, and we have used slot waivers at times to reduce flying levels during periods of reduced demand for travel. Moreover, on multiple occasions in 2023, the FAA issued slot waivers for New York City area airports as a result of operational challenges arising from air traffic control staffing shortages; those waivers expire in October 2024, and we cannot guarantee that such waivers will be made available to us, or that upon expiration or cancellation of such waivers it will be economical for us to resume prior levels of flying to destinations where we have operated a reduced service. If we are forced to surrender slots or other rights, we may be unable to provide our desired level of service to or from certain destinations in the future. For more discussion, see Part I, Item 1A. Risk Factors – "*If we are unable to obtain and maintain adequate facilities and infrastructure throughout our system and, at some airports, adequate slots, we may be unable to operate our existing flight schedule and to expand or change our route network in the future, which may have a material adverse impact on our operations."*

Our ability to provide service can also be impaired at airports where the airport gates and other facilities are currently inadequate to accommodate all of the service that we would like to provide, or where we have no access to gates at all.

Existing law also permits domestic local airport authorities to implement procedures and impose restrictions designed to abate noise, provided such procedures and restrictions do not unreasonably interfere with interstate or foreign commerce or the national transportation system. In some instances, these restrictions have caused curtailments in service or increases in operating costs.

Airline Fares, Taxes and User Fees

Airlines are permitted to establish their own domestic fares without governmental regulation. The DOT maintains authority over certain international fares, rates and charges, but only applies this authority on a limited basis. In addition, international fares and rates are sometimes subject to the jurisdiction of the governments of the foreign countries which we serve.

Airlines are obligated to collect a federal excise tax, commonly referred to as the "ticket tax," on domestic and international air transportation, and to collect other taxes and charge other fees, such as foreign taxes, security fees and passenger facility charges. Although these taxes and fees are not our operating expenses, they represent an additional cost to our customers. These taxes and fees are subject to increase from time to time.

DOT Passenger Protection Rules

The DOT regulates airline interactions with passengers through the ticketing process, at the airport and onboard the aircraft. Among other things, these regulations govern how our fares are displayed online, required customer disclosures, access by disabled passengers, handling of long onboard flight delays and reporting of mishandled bags. In 2023, the DOT finalized rules for accessible lavatories on single-aisle aircraft and has continued to work through proposals for a number of disability regulations that will impact us, including penalties for wheelchair loss or damage and prompt wheelchair assistance. The DOT has also proposed rules requiring refunds for cancellations and significant delays and rules mandating the display of ancillary fees during the initial itinerary search.

International

International air transportation is subject to extensive government regulation, including aviation agreements between the U.S. and other countries or governmental authorities, such as the EU. Moreover, our alliances with international carriers may be subject to the jurisdiction and regulations of various foreign agencies. The U.S. government has negotiated "open skies" agreements with more than 130 trading partners, which allow unrestricted route authority access between the U.S. and the foreign markets.

In addition, foreign countries impose passenger protection rules, which are analogous to, and often meet or exceed the requirements of, the DOT passenger protection rules discussed above. In cases where these foreign requirements exceed the DOT rules, we may bear additional burdens and liabilities. Further, various foreign airport authorities impose noise and curfew restrictions at their local airports.

Security

All aspects of civil aviation and border security in the U.S. affecting U.S. carriers are controlled or regulated by the federal government through the Transportation Security Administration (TSA) and the U.S. Customs and Border Protection (CBP). The TSA is responsible for the security of the nation's transportation systems. The TSA's requirements for aviation security include, among other things, screening of passengers, baggage, cargo, mail, employees and vendors; carriage of federal air marshals at no charge; and continuous background checks of all employees and vendor employees with access to secure areas of airports. Funding for the TSA is provided by a combination of air carrier fees, passenger fees and taxpayer funds. The CBP is responsible for securing the nation's borders by combining customs, immigration and agricultural protection. The CBP regulatory requirements include the transmission of advanced passport data to facilitate the U.S. entry process. Funding for a portion of CBP operations is provided by a combination of fees collected by airlines. Our international service further requires us to comply with host government civil aviation security regimes and foreign border control authorities.

Environmental Matters

Environmental Regulation

The airline industry is subject to various laws and government regulations concerning environmental matters in the U.S. and other countries. U.S. federal laws that have a particular impact on our operations include the Airport Noise and Capacity Act of 1990, the Clean Air Act, the Resource Conservation and Recovery Act, the Clean Water Act, the Safe Drinking Water Act and the Comprehensive Environmental Response, Compensation and Liability Act. The U.S. Environmental Protection Agency (EPA) and other federal agencies may promulgate regulations that have an impact on our operations. In addition to these federal activities, various states have been delegated certain authorities under the aforementioned federal statutes. Many state and local governments have adopted environmental laws and regulations that are similar to or stricter than federal requirements.

Revised underground storage tank regulations issued by the EPA in 2015 have affected certain airport fuel hydrant systems, with modifications of such systems needed in order to comply with applicable portions of the revised regulations. In addition, related to the EPA and state regulations pertaining to storm water management, several U.S. airport authorities are actively engaged in efforts to limit discharges of deicing fluid into the environment, often by requiring airlines to participate in the building or reconfiguring of airport deicing facilities. Additionally, compliance with updated federal and state regulations governing fire extinguishing foams are expected to require modification to fire suppression systems that we operate, as well as those maintained by airports. On November 23, 2022, the EPA also published the final rule for particulate matter emission standards and test procedures for civil aircraft engines, which took effect on December 23, 2022. These or similar regulations could directly or indirectly result in increased compliance costs, but at this time we do not expect these costs to be material.

The environmental laws include those related to responsibility for potential soil and groundwater contamination. We are conducting investigation and remediation activities to address soil and groundwater conditions at several sites, including airports and maintenance bases. We presently anticipate that the ongoing costs of such activities will not have a material impact on our operations.

We employ an environmental management system that provides a systematic approach for compliance with environmental regulations and management of a broad range of environmental issues, including but not limited to air emissions, hazardous waste, underground tanks, and aircraft water quality.

Global and Domestic Regulation Related to Climate Change

Climate change-related regulatory activity and developments may adversely affect our business and financial results by requiring us to adapt to rapidly evolving domestic and international regulation and to achieve emission reductions before cost-effective technologies are available, for example, through requirements to make capital investments to purchase specific types of equipment or technologies, purchase carbon offset credits or otherwise incur additional costs related to our emissions. Such trends may also impact us indirectly by increasing our operating costs, including fuel costs.

The Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA)

We are subject to the requirements of the CORSIA, an international, market-based emissions reduction program adopted by the International Civil Aviation Organization (ICAO) in 2016. CORSIA is intended to achieve carbon-neutral growth in the international aviation sector from 2021 until 2035 through the purchase of certain types of carbon offset credits or the use of eligible renewable fuels.

For each year from 2021 through 2032, CORSIA requires airlines to compensate for the rate of growth of GHG emissions of the aviation sector as a whole, relative to a predetermined baseline as determined by ICAO. ICAO originally defined the baseline as the average emissions from covered flights in 2019 and 2020. However, due to the impact of the COVID-19 pandemic on air travel, in June 2020, ICAO removed 2020 from the baseline calculation for the CORSIA pilot phase (2021-2023). In October 2022, ICAO member countries agreed that 85% of 2019 emissions would be used as the baseline for the remainder of CORSIA's term (2024-2035).

The CORSIA program is being implemented in three phases: a pilot phase that ran from 2021 through 2023, followed by a first phase of the program beginning in 2024 through 2026 and a second phase beginning in 2027 through 2035. ICAO member countries are expected to enact legislation to implement CORSIA. We expect to be required to purchase carbon offset credits to comply with CORSIA's first phase, however, the U.S. government has not yet enacted implementation legislation.

Our future costs of CORSIA compliance are uncertain due to the uncertainty with respect to the future growth of covered GHG emissions, the supply and price of CORSIA-eligible carbon offset credits and development of the market for eligible renewable fuels.

European GHG Emissions Regulations

On May 16, 2023, revisions to the EU Emissions Trading System (EU ETS) were published in the Official Journal of the EU. Pursuant to these revisions, the allocation of emissions allowances currently granted for free to aircraft operators under the EU ETS will be phased out by 2026, and CORSIA will apply to flights to and from EU countries that are ICAO member countries. The EC will also be required to undertake a review in 2026 to determine whether CORSIA is sufficiently delivering on the goals of the Paris Agreement and, to the extent it is determined not to be, would extend the scope of the EU ETS to include all departing flights from the European Economic Area (EEA) (and not just flights within the EEA and flights departing the EEA to the United Kingdom and Switzerland).

In 2023, the European Parliament and the European Council formally adopted the EU's ReFuelEU Aviation initiative to create a SAF blending mandate for aviation fuel suppliers. The agreed text requires fuel suppliers to ensure that minimum shares of SAF are made available to aircraft operators at EU airports starting January 1, 2025. Such minimum requirements are 2% in 2025, 6% in 2030, 20% in 2035, 34% in 2040, 42% in 2045 and 70% in 2050. In addition, a specific proportion of the fuel mix (1.2% in 2030, 2% in 2032, 5% in 2035 and progressively reaching 35% in 2050) must comprise synthetic fuels such as e-kerosene, and as of 2025, there will be an EU label for the environmental performance of flights, such that airlines may market their flights indicating the expected carbon footprint per passenger. The potential effects on our business of such requirements are uncertain at this time. The UK and other countries have adopted or are considering adoption of a SAF blending mandate similar to that of the EU.

U.S. Emissions Standards for Aircraft Engines

In January 2021, the EPA adopted GHG emission standards for new aircraft engines, which are aligned with the 2017 ICAO aircraft engine GHG emission standards. Like the ICAO standards, the final EPA standards for new aircraft engines would not apply retroactively to engines on in-service aircraft. On November 15, 2021, the EPA announced that it would not rewrite the existing aircraft engine GHG emissions standards but would seek more ambitious new aircraft GHG emission standards within the ICAO process. Since then, the EPA and ICAO's Committee on Aviation Environmental Protection have had several meetings on this issue, but no further progress has been made. In addition, several states and environmental groups have challenged the EPA's standards and on June 30, 2023, the U.S. Court of Appeals for the D.C. Circuit denied such petitions and upheld the EPA's GHG emissions standards.

For more information on our approach to climate change, see our 2022 Sustainability Report on our website *www.aa.com* available under "Environmental, Social and Governance." None of the information or contents under our "Environmental, Social and Governance" page, 2022 Sustainability Report, or our website are incorporated into this Annual Report on Form 10-K.

Impact of Regulatory Requirements on Our Business

Regulatory requirements, including but not limited to those discussed above, affect operations and increase operating costs for the airline industry, including our airline subsidiaries, and future regulatory developments may continue to do the same. For additional information, see Part I, Item 1A. Risk Factors – "*Evolving cybersecurity and data privacy requirements (in particular, compliance with applicable federal, state and foreign laws relating to handling of personal information about individuals) could increase our costs, and any significant cybersecurity or data privacy incident could disrupt our operations, harm our reputation, expose us to legal risks and otherwise materially adversely affect our business, results of operations and financial condition," "If we are unable to obtain and maintain adequate facilities and infrastructure throughout our system and, at some airports, adequate slots, we may be unable to operate our existing flight schedule and to expand or change our route network in the future, which may have a material adverse impact on our operations," "Our business is subject to extensive government regulation, which may result in increases in our costs, disruptions to our operations, limits on our operating flexibility, reductions in the demand for air travel, and competitive disadvantages," "The airline industry is heavily taxed," "We are subject to many forms of environmental and noise regulation and may incur substantial costs as a result,"* and *"We are subject to risks associated with climate change, including increased regulation of our GHG emissions, changing consumer preferences and the potential for increased impacts of severe weather events on our operations and infrastructure."*

Available Information

Use of Websites to Disclose Information

Our website is located at *www.aa.com*. We have made, and expect in the future to make, public disclosures to investors and the general public of information regarding AAG and its subsidiaries by means of the investor relations section of our website as well as through the use of our social media sites, including Facebook and X. In order to receive notifications regarding new postings to our website, investors are encouraged to enroll on our website to receive automatic email alerts (see *https://americanairlines.gcs-web.com/email-alerts*), "follow" American (@AmericanAir) on X and "like" American on our Facebook page (*www.facebook.com/AmericanAirlines*). None of the information or contents of our website or social media postings is incorporated into this Annual Report on Form 10-K.

Availability of SEC Reports

A copy of this Annual Report on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K and amendments to those reports are available free of charge on our website as soon as reasonably practicable after we electronically file such material with, or furnish it to, the SEC. The SEC also maintains a website that contains reports, proxy and information statements, and other information regarding issuers that file electronically with the SEC at *www.sec.gov*.

ITEM 1A. RISK FACTORS

Below are certain risk factors that may affect our business, results of operations and financial condition, or the trading price of our common stock or other securities. We caution the reader that these risk factors may not be exhaustive. We operate in a continually changing business environment, and new risks and uncertainties emerge from time to time. Management cannot predict such new risks and uncertainties, nor can it assess the extent to which any of the risk factors below or any such new risks and uncertainties, or any combination thereof, may impact our business.

Risks Related to our Business and Industry

Downturns in economic conditions could adversely affect our business.

Due to the discretionary nature of business and leisure travel spending and the highly competitive nature of the airline industry, our revenues are heavily influenced by the condition of the U.S. economy and economies in other regions of the world. Unfavorable conditions in these broader economies have resulted, and may result in the future, in decreased passenger demand for air travel, changes in booking practices and related reactions by our competitors, all of which in turn have had, and may have in the future, a strong negative effect on our business. For example, the COVID-19 pandemic and associated decline in economic activity and increase in unemployment levels had a severe and prolonged effect on the global economy generally and, in turn, resulted in a prolonged period of depressed demand for air travel. In addition, a rapid economic expansion following the height of the COVID-19 pandemic resulted in significant inflationary pressures and volatility in certain currencies, which have increased our costs for aircraft fuel, wages and benefits and other goods and services we require to operate our business, as well as increasing the interest expense on our variable-rate indebtedness.

We will need to obtain sufficient financing or other capital to operate successfully.

Our business plan contemplates continued significant investments related to our fleet, improving the experience of our customers and updating our facilities. Significant capital resources will be required to execute this plan. We estimate that, based on our commitments as of December 31, 2023, our planned aggregate expenditures for aircraft purchase commitments and certain engines for calendar years 2024 through 2028 would be approximately $11.7 billion. We may also require financing to refinance maturing obligations and to provide liquidity to fund other corporate requirements. Accordingly, we will need substantial liquidity, financing or other capital resources to finance such aircraft and engines and meet such other liquidity needs. If needed, it may be difficult for us to raise additional capital on acceptable terms, or at all, due to, among other factors: our substantial level of existing indebtedness, particularly following transactions we completed in response to the impact of the COVID-19 pandemic; our non-investment grade credit rating; volatile or otherwise unfavorable market conditions; and the availability of assets to use as collateral for loans or other indebtedness, which has been reduced significantly as a result of certain financing transactions we have undertaken since the beginning of 2020 and may be further reduced. If we are unable to arrange any such required financing at customary advance rates and on terms and conditions acceptable to us, we may need to use cash from operations or cash on hand to purchase aircraft and engines or fund our other corporate requirements, or may seek to negotiate deferrals for such aircraft and engines with the applicable manufacturers or otherwise defer corporate obligations. Depending on numerous factors applicable at the time we seek capital, many of which are out of our control, such as the state of the domestic and global economies, the capital and credit markets' view of our prospects and the airline industry in general, and the general availability of debt and equity capital, the financing or other capital resources that we will need may not be available to us, or may be available only on onerous terms and conditions. Furthermore, we hold significant balances of cash and short-term investments, including as necessary to conduct our day-to-day operations, some of which are held in deposit accounts at commercial banks in excess of the government-provided deposit insurance. There can be no assurance that we will be successful in obtaining financing or other needed sources of capital to operate successfully or to fund our committed expenditures. An inability to obtain necessary financing on acceptable terms would limit our ability to execute necessary capital projects and would have a material adverse impact on our business, results of operations and financial condition.

Our high level of debt and other obligations may limit our ability to fund general corporate requirements and obtain additional financing, may limit our flexibility in responding to competitive developments and may cause our business to be vulnerable to adverse economic and industry conditions.

We have significant amounts of indebtedness and other financial obligations, including pension obligations, obligations to make future payments on flight equipment and property leases related to airport and other facilities, and substantial non-cancelable obligations under aircraft and related spare engine purchase agreements. Moreover, currently a very significant portion of our assets are pledged to secure our indebtedness. Our substantial indebtedness and other

obligations, which are generally greater than the indebtedness and other obligations of our competitors, could have important consequences. For example, they may:

- make it more difficult for us to satisfy our obligations under our indebtedness;
- limit our ability to obtain additional funding for working capital, capital expenditures, acquisitions, investments and general corporate purposes, and adversely affect the terms on which such funding can be obtained;
- require us to dedicate a substantial portion of our liquidity or cash flow from operations to payments on our indebtedness and other obligations, thereby reducing the funds available for other purposes;
- make us more vulnerable to economic downturns, industry conditions and catastrophic external events, particularly relative to competitors with lower relative levels of financial leverage;
- significantly constrain our ability to respond, or respond quickly, to unexpected disruptions in our own operations, the U.S. or global economies, or the businesses in which we operate, or to take advantage of opportunities that would improve our business, operations, or competitive position versus other airlines;
- limit our ability to withstand competitive pressures and reduce our flexibility in responding to changing business and economic conditions;
- bear interest at floating rates, subjecting us to volatility in interest expenses as interest rates fluctuate;
- contain covenants requiring us to maintain an aggregate of at least $2.0 billion of unrestricted cash and cash equivalents and amounts available to be drawn under revolving credit facilities and collateral coverage ratios and peak debt service coverage ratios;
- impact availability of borrowings under revolving lines of credit; and
- contain restrictive covenants that could, among other things:
    - limit our ability to merge, consolidate, sell assets, incur additional indebtedness, issue preferred stock, make investments and pay dividends; and
    - if breached, result in an event of default under our other indebtedness.

In addition, during the COVID-19 pandemic we were required to obtain a significant amount of additional financing from a variety of sources and we cannot guarantee that we will not need to obtain additional financing in the future. Such financing may include the issuance of additional unsecured or secured debt securities, equity securities and equity-linked securities as well as additional bilateral and syndicated secured and/or unsecured credit facilities, among other items. There can be no assurance as to the timing of any such financing transactions, which may be in the near term, or that we will be able to obtain such additional financing on favorable terms, or at all. Any such actions may be material in nature, could result in the incurrence and issuance of significant additional indebtedness or equity and could impose significant covenants and restrictions to which we are not currently subject. Moreover, as a result of the financing activities we undertook in response to the COVID-19 pandemic, the number of financings with respect to which such covenants and provisions apply has increased, thereby subjecting us to more substantial risk of cross-default and cross-acceleration in the event of breach, and additional covenants and provisions could become binding on us should we seek additional liquidity in the future.

The obligations discussed above, including those imposed as a result of any additional financings we may undertake, could also impact our ability to obtain additional financing, if needed, and our flexibility in the conduct of our business, and could materially adversely affect our liquidity, results of operations and financial condition.

Further, a substantial amount of our long-term indebtedness bears interest at floating interest rates, which tend to fluctuate based on general short-term interest rates, rates set by the U.S. Federal Reserve and other central banks, the supply of and demand for credit in treasury repurchase or other markets and general economic conditions. We have not hedged our interest rate exposure with respect to our floating rate debt. Accordingly, our interest expense for any particular period will fluctuate based on the relevant benchmark rate and other variable interest rates. In 2022 and 2023, in response to rising inflation which coincided with a rapid rebound of economic activity as governments lifted restrictions and economies reopened following the COVID-19 pandemic, central banks around the world—including the U.S. Federal

Reserve, the European Central Bank and the Bank of England—undertook a cycle of raising interest rates, which has consequently increased the interest we pay on our floating-rate indebtedness. To the extent the interest rates applicable to our floating rate debt remain elevated or continue to increase, our interest expense will increase, in which event we may have difficulties making interest payments and funding our other fixed costs, and our available cash flow for general corporate requirements may be adversely affected.

In connection with the phase-out of the London Interbank Offered Rate (LIBOR) as a reference rate in June 2023, the U.S. Federal Reserve, in conjunction with the Alternative Reference Rates Committee, chose the Secured Overnight Financing Rate (SOFR), and specifically Term SOFR, as the recommended risk-free reference rate for the U.S. (calculated based on repurchase agreements backed by treasury securities). Prior to the discontinuation of LIBOR, we amended substantially all of our LIBOR-based financing arrangements to transition them to successor rates, primarily Term SOFR. We cannot predict the extent to which Term SOFR will gain widespread acceptance as a replacement for LIBOR, the consequences of the replacement of LIBOR on financial markets generally or on our business, financial condition or results of operations specifically, and our transition to successor rates could cause the amount of interest payable on our long-term debt to be different or higher than expected.

We have significant pension and other postretirement benefit funding obligations, which may adversely affect our liquidity, results of operations and financial condition.

Our pension funding obligations are significant. The amount of our pension funding obligations will depend on the performance of investments held in trust by the pension plans, interest rates for determining liabilities and actuarial experience. We also have significant obligations for retiree medical and other postretirement benefits.

Additionally, we participate in the IAM National Pension Fund (the IAM Pension Fund). The funding status of the IAM Pension Fund is subject to the risk that other employers may not meet their obligations, which under certain circumstances could cause our obligations to increase. On March 29, 2019, the actuary for the IAM Pension Fund certified that the fund was in “endangered” status despite reporting a funded status of over 80%. Additionally, the IAM Pension Fund’s Board voluntarily elected to enter into “critical” status on April 17, 2019. Upon entry into critical status, the IAM Pension Fund was required by law to adopt a rehabilitation plan aimed at restoring the financial health of the pension plan and did so on April 17, 2019 (the Rehabilitation Plan). Under the Rehabilitation Plan, American was subject to an immaterial contribution surcharge, which ceased to apply on June 14, 2019 upon American’s mandatory adoption of a contribution schedule under the Rehabilitation Plan. The contribution schedule requires 2.5% annual increases to its contribution rate. This contribution schedule will remain in effect through the earlier of December 31, 2031 or the date the IAM Pension Fund emerges from critical status. Furthermore, if we were to withdraw from the IAM Pension Fund, if the IAM Pension fund were to terminate, or if the IAM Pension Fund were to undergo a mass withdrawal, we could be subject to liability as imposed by law.

If our financial condition worsens, provisions in our credit card processing and other commercial agreements may adversely affect our liquidity.

We have agreements with companies that process customer credit card transactions for the sale of air travel and other services. These agreements allow these credit card processing companies, under certain conditions (including, with respect to certain agreements, our failure to maintain certain levels of liquidity), to hold an amount of our cash (referred to as a holdback) equal to some or all of the advance ticket sales that have been processed by that credit card processor, but for which we have not yet provided the air transportation. Additionally, such credit card processing companies may require cash or other collateral reserves to be established. These credit card processing companies are not currently entitled to maintain any holdbacks pursuant to these requirements. These holdback requirements can be implemented at the discretion of the credit card processing companies upon the occurrence of specific events, including material adverse changes in our financial condition or the triggering of a liquidity covenant. The imposition of holdback requirements, up to and including 100% of relevant advanced ticket sales, would materially reduce our liquidity. Likewise, other of our commercial agreements contain provisions that allow counterparties to impose less-favorable terms, including the acceleration of amounts due, in the event of material adverse changes in our financial condition. For example, we maintain certain letters of credit as well as insurance- and surety-related agreements under which counterparties may require collateral, including cash collateral.

The loss of key personnel upon whom we depend to operate our business or the inability to attract, develop and retain additional qualified personnel could adversely affect our business.

We believe that our future success will depend in large part on our ability to attract, develop and retain highly qualified management, technical and other personnel. Retaining and recruiting people with the appropriate skills is particularly challenging as the economy in general, and the airline industry in particular, continue to recover from the COVID-19 pandemic, resulting in competition for the human resources necessary to operate our business successfully. We may not be successful in attracting, developing or retaining key personnel or other highly qualified personnel. In addition, competition for skilled personnel has intensified and may continue to intensify if overall industry capacity continues to increase and/or we were to incur attrition at levels higher than we have historically. Any inability to attract, develop and retain significant numbers of qualified management and other personnel would have a material adverse effect on our business, results of operations and financial condition.

Our business has been and will continue to be materially affected by many changing economic, geopolitical, commercial, regulatory and other conditions beyond our control, including global events that affect travel behavior, and our results of operations could be volatile and fluctuate materially due to changes in such conditions.

Our business, results of operations and financial condition have been and will continue to be affected by many changing economic, geopolitical, commercial, regulatory and other conditions beyond our control, including, among others:

- actual or potential changes in international, national, regional and local economic, business and financial conditions, including recession, inflation and higher interest rates;
- the occurrence of wars, conflicts, terrorist attacks and geopolitical instability;
- changes in consumer preferences, perceptions, spending patterns and demographic trends;
- changes in the competitive environment due to industry consolidation, changes in airline alliance affiliations and other factors;
- delays in scheduled aircraft deliveries, unexpected grounding of aircraft or aircraft engines whether by regulators or by us, or other loss of anticipated fleet capacity, and failure of new aircraft to receive regulatory approval, be produced or otherwise perform as and when expected;
- actual or potential disruptions to the U.S. National Airspace System (the ATC system);
- increases in costs of safety, security and environmental measures;
- increases in costs related to meeting our climate goals or obligations, including in respect of the costs to be incurred to migrate to increased use of SAF in lieu of conventional aviation fuel;
- outbreaks of diseases or other public health or safety concerns that affect travel behavior, such as occurred during the COVID-19 pandemic; and
- weather and natural disasters, including increases in frequency, severity or duration of such disasters, and related costs caused by more severe weather due to climate change.

The COVID-19 pandemic, along with the measures governments and private organizations worldwide implemented in an attempt to contain its spread, resulted in significant volatility in demand for air travel, which adversely affected our business, operations and financial condition to an unprecedented extent and for a prolonged period. Measures implemented during the COVID-19 pandemic—such as travel restrictions, including testing regimes, “stay at home” and quarantine orders, limitations on public gatherings, cancellation of public events and many others—initially resulted in a precipitous decline in demand for both domestic and international business and leisure travel. In response to this material deterioration in demand, we took a number of aggressive actions to ameliorate the impacts to our business, operations and financial condition. While governments have loosened or lifted COVID-19-related travel restrictions, the potential for a resurgence of COVID-19, including the emergence and spread of any new variants, and its after effects remain uncertain, and there can be no assurance that any mitigating actions we take in response will be sufficient to avert a deterioration in our business, financial condition and results of operations. Additionally, the COVID-19 pandemic necessitated changes in business practices which may persist. For example, businesses and other travelers may continue to forego air travel in

favor of remote or flexible working policies and communication alternatives such as videoconferencing. In addition, businesses may seek to reduce travel costs by requiring the purchase of less expensive tickets, thereby potentially impacting our average revenue per available seat mile.

In addition to the effects of the COVID-19 pandemic, an outbreak of another contagious disease—such as has occurred in the past with the Ebola virus, Middle East Respiratory Syndrome, Severe Acute Respiratory Syndrome, H1N1 influenza virus, avian flu, Zika virus or any other similar illness—if it were to become associated with air travel or persist for an extended period, could materially affect the airline industry and us by reducing revenues and adversely impacting our operations and passengers' travel behavior. As a result of these or other conditions beyond our control, our results of operations could be volatile and subject to rapid and unexpected change. In addition, due to generally weaker demand for air travel during the winter, our revenues in the first and fourth quarters of the year could be weaker than revenues in the second and third quarters of the year.

The airline industry is intensely competitive and dynamic.

Our competitors include other major domestic airlines and foreign, regional and new entrant airlines, as well as joint ventures formed by some of these airlines, many of which have greater financial or other resources and/or lower cost structures than ours, as well as other forms of transportation, such as rail and private automobiles or alternatives to commuting or business travel including remote or flexible working policies and communication alternatives such as videoconferencing. In many of our markets, we compete with at least one low-cost carrier (including so-called ultra-low-cost carriers). Our revenues are sensitive to the actions of other carriers in many areas, including pricing, scheduling, capacity, fees (including cancellation, change and baggage fees), amenities, loyalty benefits and promotions, which can have a substantial adverse impact not only on our revenues, but on overall industry revenues. These factors may become even more significant in periods when the industry experiences large losses (such as occurred during the COVID-19 pandemic), as airlines under financial stress, or in bankruptcy, may institute pricing or fee structures intended to attract more customers to achieve near-term survival at the expense of long-term viability.

Low-cost carriers (including so-called ultra-low-cost carriers) have a profound impact on industry revenues. Using the advantage of low unit costs, these carriers offer lower fares in order to shift demand from larger, more established airlines, and represent significant competitors, particularly for customers who fly infrequently or are price sensitive and therefore tend not to be loyal to any one particular carrier. Many of these carriers, including several that have recently commenced operations, have announced growth strategies including commitments to acquire significant numbers of new aircraft for delivery in the next few years. These low-cost carriers are attempting to continue to increase their market share through growth and consolidation, and are expected to continue to have an impact on our revenues and overall performance. We and several other large network carriers have implemented "Basic Economy" fares designed to more effectively compete against low-cost carriers, but we cannot predict whether these initiatives will be successful. While historically these carriers have provided competition in domestic markets, we have recently experienced new competition from low-cost carriers on international routes, including low-cost airlines executing international long-haul expansion strategies, a trend likely to continue, in particular with the planned introduction of long-range narrowbody aircraft in coming years. Additionally, other carriers focused on premium passenger travel are attempting to implement growth strategies. The actions of existing or future carriers, including those described above, could have a material adverse effect on our operations and financial performance.

In certain instances, other air carriers are attempting to operate scheduled service with a business model that relies on FAA Part 135, a regulatory environment that is generally less stringent than the rules applicable to our airline and similar airlines that operate under FAA Part 121 and which provides those airlines certain competitive advantages that Part 121 airlines cannot replicate. We have objected to the DOT and TSA that the less stringent Part 135 rules were never intended as a basis for scheduled passenger service and that business model should not be permissible, and the agencies' review is ongoing. A DOT or TSA decision to allow scheduled passenger service under Part 135 and the actions of existing or future carriers using that business model, including those described above, could adversely impact our business, financial condition and results of operations.

We provide air travel internationally, directly as well as through joint businesses, strategic alliances, codeshare and similar arrangements to which we are a party. While our network is comprehensive, compared to some of our key global competitors, we generally have somewhat greater relative exposure to certain regions (for example, Latin America) and somewhat lower relative exposure to others (for example, Asia). Our financial performance relative to our key competitors will therefore be influenced significantly by macro-economic conditions in particular regions around the world and the relative exposure of our network to the markets in those regions, including the duration of any declines in demand for

travel to specific regions as a result of health emergencies (such as during the COVID-19 pandemic), geopolitical instability or other factors, and the speed with which demand for travel to these regions returns.

Our international service exposes us to foreign economies and the potential for reduced demand when any foreign country we serve suffers adverse local economic conditions or if governments restrict commercial air service to or from any of these markets. For example, the COVID-19 pandemic resulted in a precipitous and prolonged decline in demand for air travel, in particular international travel, in part as a result of the imposition by the U.S. and foreign governments of restrictions on travel from certain regions. In addition, open skies agreements, which are now in place with a substantial number of countries around the world, provide international airlines with open access to U.S. markets, potentially subjecting us to increased competition on our international routes. See also *"Our business is subject to extensive government regulation, which may result in increases in our costs, disruptions to our operations, limits on our operating flexibility, reductions in the demand for air travel, and competitive disadvantages."*

To the extent alliances formed by our competitors can undertake activities that are not available to us, including as to regulatory approvals, access slots, gates and routes and other matters, our ability to effectively compete may be hindered. Our ability to attract and retain customers is dependent upon, among other things, our ability to offer our customers convenient access to desired markets. Our business could be adversely affected if we are unable to maintain or obtain alliance and marketing relationships with other air carriers in desired markets.

American has established a transatlantic joint business with British Airways, Aer Lingus, Iberia and Finnair, a transpacific joint business with Japan Airlines and a joint business relating to Australia and New Zealand with Qantas. We have also established a strategic alliance with Alaska Airlines relating to certain routes on the West Coast of the United States and a strategic alliance relating to the Middle East with Qatar Airways. In July 2010, in connection with a regulatory review related to our transatlantic joint business, we provided certain commitments to the EC regarding, among other things, the availability of take-off and landing slots at LHR or LGW airports. The commitments accepted by the EC were binding for 10 years. In anticipation of both the exit of the United Kingdom from the EU, commonly referred to as Brexit, and the expiry of the EC commitments in July 2020, the CMA, in October 2018, opened an investigation into the transatlantic joint business. In September 2020 and April 2022, the CMA adopted interim measures that effectively extend the EC commitments until March 2026 in light of the uncertainty and other impacts resulting from the COVID-19 pandemic. The CMA restarted its investigation in September 2023 after a pause related to the COVID-19 pandemic and plans to complete the investigation before the scheduled expiration of the interim measures in March 2026. We continue to cooperate fully with the CMA. The foregoing arrangements are important aspects of our international network and we are dependent on the performance and continued cooperation of the other airlines party to those arrangements.

On May 19, 2023, the U.S. District Court for the District of Massachusetts issued an order permanently enjoining American and JetBlue from continuing and further implementing the NEA. In June 2023, JetBlue delivered a notice of termination of the NEA, effective July 29, 2023, and the carriers have commenced wind-down activities to accommodate mutual customers. American has appealed the District Court's decision to the Court of Appeals for the First Circuit; American's opening brief was filed on December 6, 2023. Separately, in December 2022, two putative class action lawsuits were filed in the U.S. District Court for the Eastern District of New York alleging that American and JetBlue violated U.S. antitrust law in connection with the previously disclosed NEA. In February 2023, private party plaintiffs filed two additional putative class action antitrust complaints against American and JetBlue in the U.S. District Court for the District of Massachusetts and the U.S. District Court for the Eastern District of New York, respectively. All cases have since been consolidated in the U.S. District Court for the Eastern District of New York. American, together with JetBlue, filed a motion to dismiss on September 21, 2023, which remains pending. The motions to dismiss argue, among other things, that the plaintiffs each waived their right to bring class action claims. We believe these complaints are without merit and are defending against them vigorously.

No assurances can be given as to any benefits that we may derive from any of the foregoing arrangements or any other arrangements that may ultimately be implemented, or whether regulators will, or if granted continue to, approve or impose material conditions on our business activities.

Other mergers and other forms of airline partnerships, including regulatory approvals such as antitrust immunity grants, may take place and may not involve us as a participant, or could result in unforeseen impacts on the industry generally and our company in particular. Depending on which carriers combine or integrate and which assets, if any, are sold or otherwise transferred to other carriers in connection with any such transactions, our competitive position relative to the post-transaction carriers or other carriers that acquire such assets could be harmed. In addition, as carriers combine through traditional mergers or integrate their operations through other arrangements, their route networks will grow, and that growth will result in greater overlap with our network, which in turn could decrease our overall market share and

revenues. Such combination or collaboration is not limited to the U.S., but could include further transactions among international carriers in Europe and elsewhere that result in broader networks offered by rival airlines.

Additionally, our AAdvantage program, which is an important element of our sales and marketing programs, faces significant and increasing competition from the loyalty programs offered by other travel companies, as well as from similar loyalty benefits offered by banks and other financial services companies. Competition among loyalty programs is intense regarding the rewards, fees, required usage, and other terms and conditions of these programs. In addition, we have used certain assets from our AAdvantage program as collateral for the AAdvantage Financing, which contains covenants that impose restrictions on certain amendments or changes to certain of our AAdvantage program agreements provided as collateral under the AAdvantage Financing and other aspects of the AAdvantage program. These competitive factors and covenants (to the extent applicable) may affect our ability to attract and retain customers, increase usage of our loyalty program and maximize the revenue generated by our loyalty program.

We may also be impacted by competition regulations affecting certain of our major commercial partners, including our co-branded credit card partners. For example, there has been bipartisan legislation proposed in Congress called the Credit Card Competition Act designed to increase credit card transaction routing options for merchants which, if enacted, could result in a reduction of the fees levied on credit card transactions. If this legislation or any similar legislation or regulation were enacted, it could fundamentally alter the profitability of our agreements with co-branded credit card partners and the benefits we provide to our consumers through the co-branded credit cards issued by these partners.

Union disputes, employee strikes and other labor-related disruptions may adversely affect our operations and financial performance.

Relations between air carriers and labor unions in the U.S. are governed by the RLA. Under the RLA, CBAs generally contain “amendable dates” rather than expiration dates, and the RLA requires that a carrier maintain the existing terms and conditions of employment following the amendable date through a multi-stage and usually lengthy series of bargaining processes overseen by the NMB. As of December 31, 2023, approximately 87% of our employees were represented for collective bargaining purposes by labor unions, and 34% were covered by CBAs that are currently amendable or that will become amendable within one year. For the dates that the CBAs with our major work groups become amendable under the RLA, see *“Labor Relations”* under Part I, Item 1. Business – *“Sustainability – Our People*.”

In the case of a CBA that is amendable under the RLA, if no agreement is reached during direct negotiations between the parties, either party may request that the NMB appoint a federal mediator. The RLA prescribes no timetable for the direct negotiation and mediation processes, and it is not unusual for those processes to last for many months or even several years. If no agreement is reached in mediation, the NMB in its discretion may declare that an impasse exists and proffer binding arbitration to the parties. Either party may decline to submit to arbitration, and if arbitration is rejected by either party, a 30-day “cooling off” period commences. During or after that period, a Presidential Emergency Board (PEB) may be established, which examines the parties’ positions and recommends a solution. The PEB process lasts for 30 days and is followed by another 30-day “cooling off” period. At the end of this “cooling off” period, unless an agreement is reached or action is taken by Congress, the labor organization may exercise “self-help,” such as a strike, which could materially adversely affect our business, results of operations and financial condition.

None of the unions representing our employees presently may lawfully engage in concerted slowdowns or refusals to work, such as strikes, sick-outs or other similar activity, against us. Nonetheless, there is a risk that employees, either with or without union involvement, could engage in one or more concerted refusals to work that could individually or collectively harm the operation of our airline and impair our financial performance. Additionally, some of our unions have brought and may continue to bring grievances to binding arbitration, including those related to wages. If successful, there is a risk these arbitral avenues could result in material additional costs that we did not anticipate.

Currently, we believe our labor costs are generally competitive relative to the other large network carriers. However, personnel shortages, in particular for pilots, and general wage inflation stand to impact our labor costs moving forward. In July 2023, we reached a tentative agreement with the union representing our mainline pilots, which was subsequently ratified by the pilots in August 2023. The new agreement, which became effective in the third quarter of 2023, includes significant increases in pilot pay and benefits, in line with agreements recently concluded by our large network competitors with their pilots’ unions. We remain in negotiations for other new labor agreements and anticipate that any new contracts we agree to with our labor groups will include material increases in salaries and other benefits, which will significantly increase our labor expense.

If we encounter problems with any of our third-party regional operators or third-party service providers, our operations could be adversely affected by a resulting decline in revenue or negative public perception about our services.

A significant portion of our regional operations are conducted by third-party operators on our behalf and are provided for under capacity purchase agreements. Due to our reliance on third parties to provide these essential services, we are subject to the risk of disruptions to their operations, which has in the past and may in the future result from many of the same risk factors disclosed in this report, such as the impact of adverse economic conditions, the inability of third parties to hire or retain skilled personnel, including in particular pilots and mechanics, and other risk factors, such as an out-of-court or bankruptcy restructuring of any of our regional operators. Several of these third-party regional operators provide significant regional capacity that we would be unable to replace in a short period of time should that operator fail to perform its obligations to us. Disruptions to capital markets, shortages of pilots, mechanics and other skilled personnel and adverse economic conditions in general have subjected certain of these third-party regional operators to significant financial pressures, which have in the past and may in the future lead to bankruptcies among these operators. In particular, the severe decline in demand for air travel resulting from the COVID-19 pandemic and related governmental restrictions on travel materially impacted demand for services provided by our regional carriers and, as a result, we temporarily significantly reduced our regional capacity. Further, as airlines attempt to restore capacity in line with increased demand for air travel following the height of the COVID-19 pandemic, these third-party operators have experienced difficulties in recruiting and retaining sufficient personnel to operate significantly increased schedules, and have in some instances been required to offer significant increases in pay and other benefits to recruit and retain pilots and other personnel. Periods of volatility in travel demand have the potential to adversely affect our regional operators, some of whom may experience significant financial stress, declare bankruptcy or otherwise cease to operate. We may also experience disruption to our regional operations or incur financial damages if we terminate the capacity purchase agreement with one or more of our current operators or transition the services to another provider. Any significant disruption to our regional operations would have a material adverse effect on our business, results of operations and financial condition.

In addition, our reliance upon others to provide essential services on our behalf in our operations may result in our relative inability to control the efficiency and timeliness of contract services. We have entered into agreements with contractors to provide various facilities and services required for our operations, including distribution and sale of airline seat inventory, reservations, provision of information technology and services, regional operations, aircraft maintenance, fueling, catering, ground services and facilities and baggage handling. Similar agreements may be entered into in any new markets we decide to serve. These agreements are generally subject to termination after notice by the third-party service provider. We are also at risk should one of these service providers cease operations, and there is no guarantee that we could replace these providers on a timely basis with comparably priced providers, or at all. These third parties are also facing challenges retaining and recruiting people with the appropriate skills to meet our requirements as the economy in general, and the airline industry in particular, continue to recover from the COVID-19 pandemic. The COVID-19 pandemic also caused significant disruption in global supply chains and staffing shortages, which have affected and may continue to affect the availability and timely delivery and fulfillment of many goods, including certain of those that we purchase directly or which are required by third parties to perform contracted services for us. We rely on the operation of complex supply chains and a large number of third parties for the procurement and fulfillment of parts, components, consumable or disposable goods and other products and services essential to our business. Following a faster than expected return of demand for air travel as COVID-19 cases declined worldwide and governments lifted travel restrictions, suppliers and many of the airports we serve experienced acute shortages of personnel, resulting in increased delays, cancellations and, in certain cases, restrictions on passenger numbers or the number of flights to or from certain airports. We cannot guarantee that, as a result of ongoing or future supply chain disruptions or staffing shortages, we, our third-party partners, or the airports we serve will be able to timely source all of the products and services we require in the course of our business, or that we will be successful in procuring suitable alternatives. Any material problems with the adequacy, efficiency and timeliness of contract services, resulting from financial hardships, personnel shortages or otherwise, could have a material adverse effect on our business, results of operations and financial condition.

Any damage to our reputation or brand image could adversely affect our business or financial results.

Maintaining a good reputation globally is critical to our business. Our reputation or brand image could be adversely impacted by, among other things, any failure to maintain high ethical, social and environmental sustainability practices for all of our operations and activities, our impact on the environment, public pressure from investors or policy groups to change our policies, such as movements to institute a “living wage,” customer perceptions of our advertising campaigns, sponsorship arrangements or marketing programs, customer perceptions of our use of social media, including greenwashing concerns regarding our advertising campaigns and marketing programs related to our sustainability

initiatives, or customer perceptions of statements made by us, our employees and executives, agents or other third parties. In addition, we operate in a highly visible industry that has significant exposure to social media. Negative publicity, including as a result of misconduct by our customers, vendors or employees, can spread rapidly through social media. Should we not respond in a timely and appropriate manner to address negative publicity, our brand and reputation may be significantly harmed. Damage to our reputation or brand image or loss of customer confidence in our services could adversely affect our business and financial results, as well as require additional resources to rebuild our reputation.

Moreover, an outbreak and spread of an infectious disease could adversely impact consumer perceptions of the health and safety of travel, and in particular airline travel, such as occurred during the COVID-19 pandemic. Actual or perceived risk of infection on our flights could have a material adverse effect on the public's perception of us and may harm our reputation and business. We have in the past, and may in the future be required to take extensive measures to reassure our team members and the traveling public of the safety of air travel, and we could incur significant costs implementing safety, hygiene-related or other actions to limit the actual or perceived threat of infection among our employees and passengers. However, we cannot assure that any actions we might take in response to an infectious disease outbreak will be sufficient to restore the confidence of consumers in the safety of air travel. In addition, as a result of mask mandates and other mitigating measures that airports and carriers were required by law to implement to limit the spread of COVID-19, we experienced an increase in the incidence of aggressive customer behavior and physical confrontation on our flights, certain of which resulted in injuries to our personnel. While the rate of these incidents has declined following the lifting of mask mandates and other COVID-19 measures, if our employees feel unsafe or believe that we are not doing enough to prevent and prosecute such incidents, we could experience higher rates of employee absence or attrition and we may suffer reputational harm which could make it more difficult to attract and retain employees, and which could in turn negatively affect our business, financial condition and results of operations.

We are at risk of losses and adverse publicity stemming from any public incident involving our company, our people or our brand, including any accident or other public incident involving our personnel or aircraft, or the personnel or aircraft of our regional, codeshare or joint business operators.

We are at risk of adverse publicity stemming from any public incident involving our company, our people or our brand, particularly given the ease with which individuals can now capture and rapidly disseminate information via social media. Such an incident could involve the actual or alleged behavior of any of our employees, contractors or passengers. Further, if our personnel, one of our aircraft, a type of aircraft in our fleet, or personnel of, or an aircraft that is operated under our brand by, one of our regional operators or an airline with which we have a marketing alliance, joint business or codeshare relationship, were to be involved in a public incident, accident, catastrophe or regulatory enforcement action, we could be exposed to significant reputational harm and potential legal liability. The insurance we carry may be inapplicable or inadequate to cover any such incident, accident, catastrophe or action. In the event that our insurance is inapplicable or inadequate, we may be forced to bear substantial losses from an incident or accident. In addition, any such incident, accident, catastrophe or action involving our personnel, one of our aircraft (or personnel and aircraft of our regional operators and our codeshare partners), or a type of aircraft in our fleet could create an adverse public perception, which could harm our reputation, result in air travelers being reluctant to fly on our aircraft or those of our regional operators or codeshare partners, and adversely impact our business, results of operations and financial condition.

Changes to our business model that are designed to increase revenues may not be successful and may cause operational difficulties or decreased demand.

We have in the past instituted, and intend to institute in the future, changes to our business model designed to increase revenues and offset costs. These measures include further segmentation of the classes of service we offer, such as Premium Economy service and Basic Economy service, enhancements to our AAdvantage program, charging separately for services that had previously been included within the price of a ticket, changes to our practices and contracts with providers of distribution systems to provide additional content flexibility, commercial practices related to ticket distribution channels, including efforts by us to migrate an increasing portion of our customers to our modern, direct distribution channels in lieu of third party channels, changing (whether it be increasing, decreasing or eliminating) other pre-existing fees, reconfiguration of our aircraft cabins, and efforts to optimize our network including by focusing growth on a limited number of large hubs and entering into agreements with other airlines. For example, in 2020, we eliminated change fees for most domestic and international tickets, which has reduced our change fee revenue, a trend which is expected to continue assuming this policy remains in place. We may introduce additional initiatives in the future; however, as time goes on, we expect that it will be more difficult to identify and implement additional initiatives. We cannot assure that these measures or any future initiatives will be successful in increasing our revenues or offsetting our costs. Additionally, the implementation of these initiatives may create logistical challenges that could harm the operational performance of our airline or result in decreased demand. Also, our implementation of any new or increased fees might result in adverse

brand perceptions, reputational harm or regulatory scrutiny, and could reduce the demand for air travel on our airline or across the industry in general, particularly if weakened economic conditions make our customers more sensitive to increased travel costs or provide a significant competitive advantage to other carriers that determine not to institute similar charges.

Our intellectual property rights, particularly our branding rights, are valuable, and any inability to protect them may adversely affect our business and financial results.

We consider our intellectual property rights, particularly our branding rights such as our trademarks applicable to our airline and AAdvantage program, to be a significant and valuable aspect of our business. We protect our intellectual property rights through a combination of trademark, copyright and other forms of legal protection, contractual agreements and policing of third-party misuses of our intellectual property. Our failure to obtain or adequately protect our intellectual property or any change in law that lessens or removes the current legal protections of our intellectual property may diminish our competitiveness and adversely affect our business and financial results. Any litigation or disputes regarding intellectual property may be costly and time-consuming and may divert the attention of our management and key personnel from our business operations, either of which may adversely affect our business and financial results.

In addition, we have used certain of our branding and AAdvantage program intellectual property as collateral for various financings (including the AAdvantage Financing, defined in the accompanying notes to the consolidated financial statements to this Annual Report on Form 10-K), which contain covenants that impose restrictions on the use of such intellectual property and, in the case of the AAdvantage Financing, on certain amendments or changes to our AAdvantage program. These covenants may have an adverse effect on our ability to use such intellectual property.

We may be a party to litigation in the normal course of business or otherwise, which could affect our financial position and liquidity.

From time to time, we are a party to or otherwise involved in legal proceedings, claims and government inspections or investigations and other legal matters, both inside and outside the United States, arising in the ordinary course of our business or otherwise. We are currently involved in various legal proceedings and claims that have not yet been fully resolved, and additional claims may arise in the future. Legal proceedings can be complex and take many months, or even years, to reach resolution, with the final outcome depending on a number of variables, some of which are not within our control. Litigation is subject to significant uncertainty and may be expensive, time-consuming, and disruptive to our operations. Although we will vigorously defend ourselves in such legal proceedings, their ultimate resolution and potential financial and other impacts on us are uncertain. For these and other reasons, we may choose to settle legal proceedings and claims, regardless of their actual merit. If a legal proceeding is resolved against us, it could result in significant compensatory damages, and in certain circumstances punitive or trebled damages, disgorgement of revenue or profits, remedial corporate measures or injunctive relief imposed on us. If our existing insurance does not cover the amount or types of damages awarded, or if other resolution or actions taken as a result of the legal proceeding were to restrain our ability to operate or market our services, our consolidated financial position, results of operations or cash flows could be materially adversely affected. In addition, legal proceedings, and any adverse resolution thereof, can result in adverse publicity and damage to our reputation, which could adversely impact our business. Additional information regarding certain legal matters in which we are involved can be found in Note 11(e) to AAG's Consolidated Financial Statements in Part II, Item 8A and Note 10(e) to American's Consolidated Financial Statements in Part II, Item 8B.

Our ability to utilize our NOLs and other carryforwards may be limited.

Under the Internal Revenue Code of 1986, as amended (the Code), a corporation is generally allowed a deduction for net operating losses (NOLs) carried over from prior taxable years. At December 31, 2023, we had approximately $13.7 billion of gross federal NOLs and $4.7 billion of other carryforwards available to reduce future federal taxable income, of which $3.4 billion will expire beginning in 2029 if unused and $15.0 billion can be carried forward indefinitely. We also had approximately $5.5 billion of NOL carryforwards to reduce future state taxable income at December 31, 2023, which will expire in taxable years 2023 through 2043 if unused. Our NOL carryforwards are subject to adjustment on audit by the Internal Revenue Service and the respective state taxing authorities. Additionally, due to the impact of the COVID-19 pandemic and other economic factors, certain of the NOL carryforwards may expire before we can generate sufficient taxable income to use them.

Our ability to use our NOLs and other carryforwards depends on the amount of taxable income generated in future periods. There can be no assurance that an additional valuation allowance on our net deferred tax assets will not be required should our financial performance be negatively impacted in the future. Such valuation allowance could be material.

A corporation's ability to deduct its federal NOL carryforwards and to utilize certain other available tax attributes can be substantially constrained under the general annual limitation rules of Section 382 of the Code (Section 382) if it undergoes an "ownership change" as defined in Section 382 (generally where cumulative stock ownership changes among material stockholders exceed 50% during a rolling three-year period). In 2013, we experienced an ownership change in connection with our emergence from bankruptcy and US Airways Group, Inc. (US Airways Group) experienced an ownership change in connection with the merger of US Airways Group and AMR Corporation (the Merger). The general limitation rules for a debtor in a bankruptcy case are liberalized where the ownership change occurs upon emergence from bankruptcy. We elected to be covered by certain special rules for federal income tax purposes that permitted approximately $9.0 billion (with $3.0 billion of unlimited NOLs still remaining at December 31, 2023) of our federal NOL carryforwards to be utilized without regard to the annual limitation generally imposed by Section 382. If the special rules are determined not to apply, our ability to utilize such federal NOL carryforwards may be subject to limitation. Potential future transactions involving warrants, stock options, common or preferred stock or other equity, may increase the possibility that the Company will experience a future "ownership change" under Section 382. Substantially all of our remaining federal NOL carryforwards attributable to US Airways Group and its subsidiaries are subject to limitation under Section 382 as a result of the Merger; however, our ability to utilize such NOL carryforwards is not anticipated to be effectively constrained as a result of such limitation. Similar limitations may apply for state income tax purposes.

Notwithstanding the foregoing, an ownership change may severely limit or effectively eliminate our ability to utilize our NOL carryforwards and other tax attributes. In connection with the expiration in December 2021 of certain transfer restrictions applicable to substantial shareholders contained in our Certificate of Incorporation, the Board of Directors of AAG adopted a tax benefits preservation plan (the Tax Benefit Preservation Plan) in order to preserve our ability to use our NOLs and certain other tax attributes to reduce potential future income tax obligations. The Tax Benefit Preservation Plan was subsequently ratified by our stockholders at the 2022 Annual Meeting of Stockholders of AAG. The Tax Benefit Preservation Plan is designed to reduce the likelihood that we experience an ownership change by deterring certain acquisitions of AAG common stock. There is no assurance, however, that the deterrent mechanism will be effective, and such acquisitions may still occur. In addition, the Tax Benefit Preservation Plan may adversely affect the marketability of AAG common stock by discouraging existing or potential investors from acquiring AAG common stock or additional shares of AAG common stock, because any non-exempt third party that acquires 4.9% or more of the then-outstanding shares of AAG common stock would suffer substantial dilution of its ownership interest in AAG.

New U.S. tax legislation may adversely affect our financial condition, results of operations and cash flows.

We are subject to taxation at the federal, state and local levels in the United States. The U.S. government may enact significant changes to the taxation of business entities. For example, on August 16, 2022, the Inflation Reduction Act was signed into law, introducing, among other changes, a corporate minimum tax on certain corporations and an excise tax on certain stock repurchases by certain corporations. While certain other draft legislation has been proposed, the likelihood of any proposed changes to the tax law being enacted or implemented is unclear, and we are currently unable to predict whether such changes will occur. If any such changes are implemented, we are currently unable to predict the ultimate impact on our business and therefore there can be no assurance our business will not be adversely affected.

We have a significant amount of goodwill, which is assessed for impairment at least annually. In addition, we may never realize the full value of our intangible assets or long-lived assets, causing us to record material impairment charges.

Goodwill and indefinite-lived intangible assets are not amortized, but are assessed for impairment at least annually, or more frequently if conditions indicate that an impairment may have occurred. In accordance with applicable accounting standards, we first assess qualitative factors to determine whether it is necessary to perform a quantitative impairment test. In addition, we are required to assess certain of our other long-lived assets for impairment if conditions indicate that an impairment may have occurred.

Future impairment of goodwill, intangible assets or other long-lived assets could be recorded in results of operations as a result of changes in assumptions, estimates, or circumstances, some of which are beyond our control. There can be no assurance that a material impairment charge of goodwill or tangible or intangible assets will be avoided. The value of our aircraft could be impacted in future periods by changes in supply and demand for these aircraft. Such changes in supply

and demand for certain aircraft types could result from grounding of aircraft by us or other airlines, including as a result of significant or prolonged declines in demand for air travel and corresponding reductions to capacity. We can provide no assurance that a material impairment loss of tangible or intangible assets will not occur in a future period; we have previously incurred significant impairment charges associated with our decision to retire certain aircraft as a result of the severe decline in demand for air travel due to the COVID-19 pandemic, and the risk of future material impairments remains uncertain. Such impairment charges could have a material adverse effect on our business, results of operations and financial condition.

The commercial relationships that we have with other companies, including any related equity investments, may not produce the returns or results we expect.

An important part of our strategy to expand our network has been to initiate or expand our commercial relationships with other airlines, such as by entering into global alliance, joint business and codeshare relationships, and, in certain instances, including China Southern Airlines, GOL and JetSMART, by making an equity investment in another airline in connection with initiating or expanding such a commercial relationship. We may explore additional investments in, and joint ventures and strategic alliances with, other carriers as part of our global business strategy. We face competition in forming and maintaining these commercial relationships since there are a limited number of potential arrangements and other airlines are looking to enter into similar relationships, and our inability to form or maintain these relationships, or inability to form as many of these relationships as our competitors, may have an adverse effect on our business. Any such existing or future investment could involve significant challenges and risks, including that we may not realize a satisfactory return on our investment, if any, or that they may not generate the expected revenue synergies, and they may distract management focus from our operations or other strategic options. We may also be subject to consequences from any illegal conduct of joint business partners as well as to any political or regulatory change that negatively impacts or prohibits our arrangements with any such business partners. In addition, as a result of the COVID-19 pandemic and subsequent economic recovery, the industry experienced significant volatility in demand for air travel both internationally and domestically, which is expected to continue into the foreseeable future and could materially disrupt our partners' abilities to provide air service, the timely execution of our strategic operating plans, including the finalization, approval and implementation of new strategic relationships or the maintenance or expansion of existing relationships. If any carriers with which we partner or in which we hold an equity stake were to cease trading or be declared insolvent, we could lose the value of any such investment or experience significant operational disruption, which is a risk that we are subject to with respect to our investment in and commercial arrangements with GOL in light of its commencement in January 2024 of bankruptcy proceedings in the U.S. Federal Bankruptcy Court for the Southern District of New York. These events could have a material adverse effect on our business, results of operations and financial condition.

We may also from time to time pursue commercial relationships with companies outside the airline industry, which relationships may include equity investments or other financial commitments. Any such relationship or related investment could involve unique risks, particularly where these relationships involve new industry participants, emerging technologies or industries with which we are unfamiliar.

Our business is very dependent on the price and availability of aircraft fuel. Continued periods of high volatility in fuel costs, increased fuel prices or significant disruptions in the supply of aircraft fuel could have a significant negative impact on consumer demand, our operating results and liquidity.

Our operating results are materially impacted by changes in the availability, price volatility and cost of aircraft fuel, which represents one of the largest single cost items in our business and thus is a significant factor in the price of airline tickets. Market prices for aircraft fuel have fluctuated substantially over the past several years and prices continue to be highly volatile, with market spot prices ranging from a low of approximately $1.32 per gallon to a high of approximately $4.40 per gallon during the period from January 1, 2021 to December 31, 2023. Aircraft fuel prices reflect not only the price of underlying crude oil, but also the price charged to refine crude oil into aircraft fuel (often referred to as the “crack spread”), transportation costs, handling costs and taxes, and increases in any of these underlying components would increase the price we ultimately pay for aircraft fuel.

Because of the amount of fuel needed to operate our business, even a relatively small increase or decrease in the price of fuel can have a material effect on our operating results and liquidity. Due to the competitive nature of the airline industry and unpredictability of the market for air travel, we can offer no assurance that we may be able to increase our fares, impose fuel surcharges or otherwise increase revenues or decrease other operating costs sufficiently to offset fuel price increases. Similarly, we cannot predict actions that may be taken by our competitors in response to changes in fuel prices.

We cannot predict the future availability, price volatility or cost of aircraft fuel. Natural disasters (including hurricanes or similar events in the U.S. Southeast and on the Gulf Coast where a significant portion of domestic refining capacity is located), political disruptions or armed conflicts involving oil-producing countries or impacting global trade routes, changes in production levels of individual nations or associations of oil-producing states, economic sanctions imposed against oil-producing countries or specific industry participants, changes in fuel-related governmental policy, the strength of the U.S. dollar against foreign currencies, changes in the cost to transport or store petroleum products and any related staffing or transportation equipment shortages, changes in access to petroleum product pipelines and terminals, speculation in the energy futures markets, changes in aircraft fuel production capacity, environmental concerns and other unpredictable events, may result in fuel supply shortages, variations in the applicable crack spread, distribution challenges, additional fuel price volatility and cost increases in the future. Any of these factors or events could cause a disruption in or increased demands on oil production, refinery operations, pipeline capacity or terminal access and possibly result in significant increases in the price of aircraft fuel and diminished availability of aircraft fuel supply.

Our aviation fuel purchase contracts generally do not provide meaningful price protection against increases in fuel costs. Our current policy is not to enter into transactions to hedge our fuel consumption, although we review this policy from time to time based on market conditions and other factors. Accordingly, as of December 31, 2023, we did not have any fuel hedging contracts outstanding to hedge our fuel consumption. As such, and assuming we do not enter into any future transactions to hedge our fuel consumption, we will continue to be fully exposed to fluctuations in fuel prices. See also the discussion in Part II, Item 7A. Quantitative and Qualitative Disclosures About Market Risk – *"Aircraft Fuel."*

In addition, as part of our emissions reduction targets, we and other airlines have committed to increasing the use of SAF in our fleet. Currently, industrial production of SAF is small in scale and inadequate to meet growing industry demand, and while additional production capacity is expected to become operational in the coming years, we anticipate that competition for SAF among industry participants will remain intense. As a result, SAF may be significantly more costly than conventional jet fuel. To secure future SAF supply, we have entered into multiple agreements for the purchase of future SAF production, and we continue to engage with producers regarding potential future SAF purchases, which may include investments and other commitments to support these producers. Certain existing or potential future agreements pertain to SAF production from facilities that are planned but not yet financed, and which may utilize technology that has not been proven at commercial scale. There is no assurance that these facilities will be built or that they will meet contracted production timelines and volumes. In the event that the SAF is not delivered on schedule or in sufficient volumes, there can be no assurance that we will be able to source a supply of SAF sufficient to meet our stated goals, or that we will be able to do so on favorable economic terms.

Our business is subject to extensive government regulation, which may result in increases in our costs, disruptions to our operations, limits on our operating flexibility, reductions in the demand for air travel, and competitive disadvantages.

Airlines are subject to extensive domestic and international regulatory requirements. In the last several years, Congress and state and local governments have passed laws and regulatory initiatives, and the DOT, the FAA, the TSA and several of their respective international counterparts have issued regulations and a number of other directives that affect the airline industry. These requirements impose substantial costs on us and restrict the ways we may conduct our business.

For example, the FAA from time to time issues directives and other regulations relating to the maintenance and operation of aircraft that require significant expenditures or operational restrictions. These requirements can be issued with little or no notice, or can otherwise impact our ability to efficiently or fully utilize our aircraft, and in some instances have resulted in the temporary and prolonged grounding of aircraft or engine types altogether including, for example, the March 2019 grounding of all Boeing 737 MAX Family aircraft, which was not lifted in the United States until November 2020, the January 2024 grounding of 737-9 MAX aircraft (a model that we do not operate), and the significant limitations imposed on the use of Pratt & Whitney GTF aircraft engines on certain Airbus aircraft (an engine that we do not use in our fleet), or otherwise caused substantial disruption and resulted in material costs to us and lost revenues. The recent telecom industry roll-out of 5G technology, and concerns regarding its possible interference with aircraft navigation systems, also resulted in regulatory uncertainty and the potential for operational impacts, including possible suspension of service to certain airports or the operation of certain aircraft, though the issue has since been resolved. See "*We rely heavily on technology and automated systems to operate our business, and any failure of these technologies or systems could harm our business, results of operations and financial condition."* The FAA also exercises comprehensive regulatory authority over nearly all technical aspects of our operations. Our failure to comply with such requirements has in the past and may in the future result in fines and other enforcement actions by the FAA or other regulators. In the future, any new

regulatory requirements, particularly requirements that limit our ability to operate or price our products, could have a material adverse effect on us and the industry.

In 2018, Congress passed a five-year funding authorization for the FAA which was scheduled to expire on September 30, 2023, but was recently extended to March 8, 2024. The legislative process to renew this authorization (the FAA Authorization Renewal) could impact us, and commercial aviation more generally, in numerous ways. As part of the FAA Authorization Renewal, Congress could seek to impose new rules or regulations concerning, among other things, customer service, aviation safety, labor requirements, investments in FAA staffing and resources, improvements to the ATC system and managing new entrants in the U.S. national airspace system, as well as new or increased fees or taxes intended to fund these policies. Any new or enhanced requirements resulting from the FAA Authorization Renewal have the potential to increase our costs or impact our operation. Congressional action on the FAA Authorization Renewal has already begun and Congress has indicated that their goal is to pass the bill in advance of the newly set March 8, 2024 expiration. If Congress fails to pass the FAA Authorization Renewal, we expect passage of an additional extension of the current law to prevent a lapse in authorities.

DOT consumer rules, and rules promulgated by certain analogous agencies in other countries we serve, dictate procedures for many aspects of our customer's journey, including at the time of ticket purchase, at the airport and onboard the aircraft. DOT requires multiple disclosures of airline fares, taxes and baggage fees and is further changing these requirements to increase the number of disclosures and the time at which they must be disclosed. DOT also recently issued a proposed rule mandating refunds in certain circumstances, such as a global pandemic. DOT has also proposed rules requiring disclosure of certain ancillary fees by air carriers and travel agents. Finally, the DOT finalized rules in 2023 for accessible lavatories on single-aisle aircraft and has continued to work through proposals for a number of disability regulations that will impact us, including penalties for wheelchair loss or damage and prompt wheelchair assistance.

The Aviation and Transportation Security Act mandates the federalization of certain airport security procedures and imposes additional security requirements on airports and airlines, most of which are funded by a per-ticket tax on passengers and a tax on airlines. Present and potential future security requirements can have the effect of imposing costs and inconvenience on travelers, potentially reducing the demand for air travel.

Similarly, there are a number of legislative and regulatory initiatives and reforms at the state and local levels in the U.S. These initiatives include increasingly stringent laws to protect the environment, wage/hour requirements, mandatory paid sick or family leave and healthcare mandates. These laws could affect our relationship with our workforce and the vendors that serve our airline and cause our expenses to increase without an ability to pass through these costs. In recent years, the airline industry has experienced an increase in litigation over the application of state and local employment laws, particularly in California. Application of these laws may result in operational disruption, increased litigation risk and impact our negotiated labor agreements. For example, we are currently involved in legal proceedings in California concerning alleged violations of the state's labor code including, among other things, violations of certain meal and rest break laws, and an adverse determination in any of these cases could adversely impact our operational flexibility and result in the imposition of damages and fines, which could potentially be significant. We have reached an agreement to settle a class litigation brought by flight attendants in California and anticipate final approval by the court in the first quarter of 2024. In addition, legislation passed by the California legislature in March 2023 should effectively foreclose future meal and rest break claims from flight attendants in California. However, there is still risk of future litigation from flight attendants and other work groups involving other types of wage and hour laws in California and other jurisdictions which could seek to implement similar laws.

The results of our operations, demand for air travel and the manner in which we conduct business each may be affected by changes in law and future actions taken by governmental agencies, including:

- changes in law that affect the services that can be offered by airlines in particular markets and at particular airports, or the types of fares offered or fees that can be charged to passengers;
- the granting and timing of certain governmental approvals (including antitrust or foreign government approvals) needed for codesharing alliances, joint businesses and other arrangements with other airlines, and the imposition of regulatory investigations or commencement of litigation related to any of the foregoing;
- restrictions on competitive practices (for example, court orders, or agency regulations or orders, that would curtail an airline's ability to respond to a competitor);
- the adoption of new passenger security standards or regulations that impact customer service standards;

- restrictions on airport operations, such as restrictions on the use of slots at airports or the auction or reallocation of slot rights currently held by us;
- the adoption of more restrictive locally-imposed noise restrictions; and
- restrictions on travel or special guidelines regarding aircraft occupancy or hygiene in response to outbreaks of illness, such as occurred during the COVID-19 pandemic, including the imposition of preflight testing regimes or vaccination confirmation requirements which have in the past and may in the future have the effect of reducing demand for air travel in the markets where such requirements are imposed.

Each additional regulation or other form of regulatory oversight increases costs and adds greater complexity to airline operations and, in some cases, may reduce the demand for air travel. There can be no assurance that the increased costs or greater complexity associated with our compliance with new rules, anticipated rules or other forms of regulatory oversight will not have a material adverse effect on us.

Any significant reduction in air traffic capacity at and in the airspace serving key airports in the U.S. or overseas could have a material adverse effect on our business, results of operations and financial condition. In addition, the ATC system is not successfully modernizing to meet the growing demand for U.S. air travel. Air traffic controllers rely on outdated procedures and technologies that routinely compel airlines, including ourselves, to fly inefficient routes or take significant delays on the ground. The ATC system's inability to manage existing travel demand, including due to significant staffing shortages, has led government agencies to implement short-term capacity constraints during peak travel periods or adverse weather conditions in certain markets, resulting in delays and disruptions of air traffic. The outdated technologies also cause the ATC system to be less resilient in the event of a failure, and past system disruptions have resulted in large-scale flight cancellations and delays. We experienced this challenge in January 2023 when an outage in the ATC Notice to Air Missions system led to a nationwide ground-stop for nearly two hours, resulting in significant operational disruption throughout the day.

In the early 2000s, the FAA embarked on a path to modernize the national airspace system, including migration from the current radar-based ATC system to a GPS-based system. This modernization of the ATC system, generally referred to as "NextGen," has been plagued by delays and cost overruns, and it remains uncertain when the full array of benefits expected from this modernization will be available to the public and the airlines, including ourselves. Failure to update the ATC system and the substantial costs that may be imposed on airlines, including ourselves, to fund a modernized ATC system may have a material adverse effect on our business.

Further, our business has been adversely impacted when government agencies have ceased to operate as expected, including due to partial shutdowns, sequestrations or similar events and the COVID-19 pandemic. These events have resulted in, among other things, reduced demand for air travel, an actual or perceived reduction in air traffic control and security screening resources and related travel delays, as well as disruption in the ability of the FAA to grant required regulatory approvals, such as those that are involved when a new aircraft is first placed into service.

Our operating authority in international markets is subject to aviation agreements between the U.S. and the respective countries or governmental authorities, such as the EU, and in some cases, fares and schedules require the approval of the DOT and/or the relevant foreign governments. Moreover, alliances with international carriers may be subject to the jurisdiction and regulations of various foreign agencies. The U.S. government has negotiated "open skies" agreements with more than 130 trading partners, which agreements allow unrestricted route authority access between the U.S. and the foreign markets. While the U.S. has worked to increase the number of countries with which open skies agreements are in effect, a number of markets important to us, including China, do not have open skies agreements. For example, the open skies air services agreement between the U.S. and the EU, which took effect in March 2008, provides airlines from the U.S. and EU member states open access to each other's markets, with freedom of pricing and unlimited rights to fly from the U.S. to any airport in the EU. As a result of the agreement and a subsequent open skies agreement involving the U.S. and the United Kingdom, which was agreed in anticipation of Brexit, we face increased competition in these markets, including LHR. Bilateral and multilateral agreements among the U.S. and various foreign governments of countries we serve but which are not covered by an open skies treaty are subject to periodic renegotiation. We currently operate a number of international routes under government arrangements that limit the number of airlines permitted to operate on the route, the capacity of the airlines providing services on the route, or the number of airlines allowed access to particular airports. If an open skies policy were to be adopted for any of these markets, it could adversely impact us and could result in impairments of our related tangible and intangible assets. In addition, competition from foreign airlines, revenue-sharing

joint ventures, joint business agreements, and other alliance arrangements by and among other airlines could impair the value of our business and assets on the open skies routes.

On May 1, 2021 the EU and United Kingdom entered into a new trade and cooperation agreement (the EU-UK Trade and Cooperation Agreement) to govern certain aspects of their relationship following Brexit. We face risks associated with Brexit, notably given the extent of our passenger and cargo traffic and that of our joint business partners that flows through LHR in the United Kingdom. The EU-UK Trade and Cooperation Agreement includes provisions in relation to commercial air service that we expect to be sufficient to sustain our current services under the transatlantic joint business. However, the scope of traffic rights under the EU-UK Trade and Cooperation Agreement is less extensive than before Brexit and therefore the full impact of the EU-UK Trade and Cooperation Agreement is uncertain. For example, on December 4, 2023, the United Kingdom government launched a consultation on the reform of the rules applicable to airport slots in the United Kingdom. At this stage, the impact of this consultation and any consequent changes to the United Kingdom slot rules on our operations or those of our joint business partners at LHR is uncertain, but could be material. As a result, the continuation of our current services, and those of our partners could be disrupted. This could materially adversely affect our business, results of operations and financial condition. More generally, changes in U.S. or foreign government aviation policies could result in the alteration or termination of such agreements, diminish the value of route authorities, slots or other assets located abroad, or otherwise adversely affect our international operations.

We operate a global business with international operations that are subject to economic and political instability and have been, and in the future may continue to be, adversely affected by numerous events, circumstances or government actions beyond our control.

We operate a global business with significant operations outside of the U.S. Our current international activities and prospects have been, and in the future could be, adversely affected by government policies, reversals or delays in the opening of foreign markets, increased competition in international markets, the performance of our alliance, joint business and codeshare partners in a given market, exchange controls or other restrictions on repatriation of funds, currency and political risks (including changes in exchange rates and currency devaluations), environmental regulation, increases in taxes and fees and changes in international governmental regulation of our operations, including the inability to obtain or retain needed route authorities and/or slots, and new or evolved policies related to consumer protections. In particular, the COVID-19 pandemic severely impacted the demand for international travel for a prolonged period, and resulted in the imposition of significant governmental restrictions on commercial air service to or from certain regions. We responded by temporarily suspending a significant portion of our long-haul international flights and delaying the introduction of certain new long-haul international routes. While many countries have largely eliminated their pandemic restrictions, we can provide no assurance as to when demand for international travel will return to pre-COVID-19 pandemic levels in certain markets, if at all, or whether certain international destinations we previously served will be economical in the future.

We are subject to varying registration requirements and ongoing reporting obligations in the countries where we operate. Our permission to continue doing business in these countries may depend on our ability to timely fulfil or remedy any noncompliance with these and other governmental requirements. We may also be subject to the risk that relevant government agencies will be delayed in granting or renewing required approvals, including as a result of shutdowns (such as occurred in certain jurisdictions during the COVID-19 pandemic), cybersecurity incidents or other events. Any lapse, revocation, suspension or delay in approval of our authority to do business in a given jurisdiction may prevent us from serving certain destinations and could adversely impact our business, financial condition and results of operations.

More generally, our industry may be affected by any deterioration in global trade relations, including shifts in the trade policies of individual nations. For example, much of the demand for international air travel is the result of business travel in support of global trade. Should protectionist governmental policies, such as increased tariff or other trade barriers, travel limitations and other regulatory actions, have the effect of reducing global commercial activity, the result could be a material decrease in the demand for international air travel. Additionally, certain of the products and services that we purchase, including certain of our aircraft and related parts, are sourced from suppliers located outside the U.S., and the imposition of new tariffs, or any increase in existing tariffs, by the U.S. government in respect of the importation of such products could materially increase the amounts we pay for them.

We face risks associated with Brexit, notably given the extent of our passenger and cargo traffic and that of our joint business partners that flows through LHR in the United Kingdom. The EU-UK Trade and Cooperation Agreement includes provisions in relation to commercial air service that we expect to be sufficient to sustain our current services under the transatlantic joint business. However, the scope of traffic rights under the EU-UK Trade and Cooperation Agreement is less extensive than before Brexit and therefore the full impact of the EU-UK Trade and Cooperation Agreement is uncertain. As a result, the continuation of our current services, and those of our partners could be disrupted. Moreover,

Brexit has created uncertainty as to the future trade relationship between the EU and the United Kingdom, including air traffic services. LHR is presently a very important element of our international network, however it may become less desirable as a destination or as a hub location after Brexit when compared to other airports in Europe, where we do not have as strong a presence. This could materially adversely affect our business, results of operations and financial condition.

Brexit has also led to legal and regulatory uncertainty such as new regulatory action and/or potentially divergent treaties, laws and regulations as the United Kingdom determines which EU treaties, laws and regulations to replace or replicate, including those governing aviation, labor, environmental, data protection/privacy, competition and other matters applicable to the provision of air transportation services by us or our alliance, joint business or codeshare partners. The impact on our business of any treaties, laws and regulations that replace the existing EU counterparts, or other governmental or regulatory actions taken by the United Kingdom or the EU in connection with or subsequent to Brexit, cannot be predicted, including whether or not regulators will continue to approve or impose material conditions on our business activities such as the transatlantic joint business. See also *"The airline industry is intensely competitive and dynamic."* Any of these effects, and others we cannot anticipate, could materially adversely affect our business, results of operations and financial condition.

Additionally, fluctuations in foreign currencies, including devaluations, exchange controls and other restrictions on the repatriation of funds, have significantly affected and may continue to significantly affect our operating performance, liquidity and the value of any cash held outside the U.S. in local currency. Such fluctuations in foreign currencies, including devaluations, cannot be predicted by us and can significantly affect the value of our assets located outside the United States. These conditions, as well as any further delays, devaluations or imposition of more stringent repatriation restrictions, may materially adversely affect our business, results of operations and financial condition.

We may be adversely affected by conflicts overseas, terrorist attacks or other acts of violence, domestically or abroad; the travel industry continues to face ongoing security concerns.

Acts of terrorism and other violence, domestically or abroad, or fear of such attacks, including elevated national threat warnings, wars or other military conflicts, may depress air travel, particularly on international routes, and cause declines in revenues and increases in costs. The attacks of September 11, 2001 and continuing terrorist threats, attacks and attempted attacks materially impacted and continue to impact air travel. Increased security procedures introduced at airports since the attacks of September 11, 2001 and any other such measures that may be introduced in the future generate higher operating costs for airlines. The Aviation and Transportation Security Act mandated improved flight deck security, deployment of federal air marshals on-board flights, improved airport perimeter access security, airline crew security training, enhanced security screening of passengers, baggage, cargo, mail, employees and vendors, enhanced training and qualifications of security screening personnel, additional provision of passenger data to the U.S. Customs and Border Protection Agency and enhanced background checks. A concurrent increase in airport security charges and procedures, such as restrictions on carry-on baggage, has also had and may continue to have a disproportionate impact on short-haul travel, which constitutes a significant portion of our flying and revenue. Implementation of and compliance with increasingly complex security and customs requirements will continue to result in increased costs for us and our passengers, and have caused and likely will continue to cause periodic service disruptions and delays. We have at times found it necessary or desirable to make significant expenditures to comply with security-related requirements while seeking to reduce their impact on our customers, such as expenditures for automated security screening lines at airports. As a result of competitive pressure, and the need to improve security screening throughput to support the pace of our operations, it is unlikely that we will be able to capture all security-related costs through increased fares. We cannot forecast what new security requirements may be imposed in the future, or their impact on our business. In addition, avoiding areas of armed conflict or locations inaccessible to us due to geopolitical factors can impact our operations and financial results. For instance, airspace closures or restrictions may require us to alter flight paths, thereby increasing the distance, duration and amount of fuel required to operate certain international flights, in particular relative to competitors not subject to these airspace restrictions. Armed conflicts in or affecting international markets we serve could also adversely impact our business by, among other things, depressing demand for travel to certain regions or requiring us to suspend air service to certain destinations. For example, in October 2023, we suspended our service to Tel Aviv, Israel, and cannot predict when, or if, we will be in a position to restore such service. The outbreak or spread of armed conflict could force us to make additional reductions or changes to our service and could result in volatility in oil markets and disruptions to global trade, which could materially increase our costs or impact our supply chains.

We are subject to risks associated with climate change, including increased regulation of our GHG emissions, changing consumer preferences and the potential for increased impacts of severe weather events on our operations and infrastructure.

Efforts to combat climate change have increased the focus by regulators worldwide on the need to reduce GHG emissions, including those from the airline industry. Concerns over GHG emissions are likely to result in continued attempts to adopt requirements or change business environments related to aviation that, if successful, may result in increased costs to the airline industry and us. In addition, several countries and U.S. states have adopted or are considering adopting programs, including potentially new taxes, to regulate GHG emissions. In addition, certain airports have proposed, and could in the future adopt, GHG emission or climate-related goals or measures that could impact our operations or require us to make changes or investments in our infrastructure. In particular, ICAO has adopted rules, including those pertaining to CORSIA, which will require us to mitigate the growth of GHG emissions associated with a significant majority of our international flights.

At this time, the costs of complying with our future obligations under CORSIA are uncertain, primarily due to significant uncertainty with respect to the future growth of covered GHG emissions, the supply and price of eligible carbon credits and the future development of the market for eligible renewable fuels. Due to the competitive nature of the airline industry and unpredictability of the market for air travel, we can offer no assurance that we may be able to increase our fares, impose surcharges or otherwise increase revenues or decrease other operating costs sufficiently to offset the costs of meeting our obligations under CORSIA.

Due to the uncertainty surrounding the applicability of CORSIA to our operations in the long-term, along with the recent implementation of and potential for other new regulatory initiatives to reduce airline GHG emissions, we and other airlines are increasingly subject to an unpredictable and inconsistent array of national or regional emissions restrictions, creating a patchwork of complex regulatory requirements that could lead to increased expenses related to the emissions of our flights. For more information on these regulatory developments, see "*Aircraft Emissions and Climate Change Requirements"* under Part I, Item 1. Business – *"Domestic and Global Regulatory Landscape – Environmental Matters."*

In addition, as part of our emissions reduction targets, we and other airlines have committed to increasing the use of SAF in our fleet. Currently, industrial production of SAF is small in scale and inadequate to meet growing industry demand, and while additional production capacity is expected to become operational in the coming years, we anticipate that competition for SAF among industry participants will remain intense. As a result, SAF may be significantly more costly than conventional jet fuel. To secure future SAF supply, we have entered into multiple agreements for the purchase of future SAF production, and we continue to engage with producers regarding potential future SAF purchases, which may include investments and other commitments to support these producers. Certain existing or potential future agreements pertain to SAF production from facilities that are planned but not yet financed, and which may utilize technology that has not been proven at commercial scale. There is no assurance that these facilities will be built or that they will meet contracted production timelines and volumes. In the event that the SAF is not delivered on schedule or in sufficient volumes, there can be no assurance that we will be able to source a supply of SAF sufficient to meet our stated goals, or that we will be able to do so on favorable economic terms.

Additionally, growing recognition among consumers of the dangers of climate change may mean some customers choose to fly less frequently or fly on an airline they perceive as operating in a manner that is more sustainable to the climate. Business customers may choose to use alternatives to travel, such as virtual meetings and workspaces. Greater development of high-speed rail in markets now served by short-haul flights could provide passengers with lower-carbon alternatives to flying with us. Customers may also elect to travel on flights that produce comparatively fewer GHG emissions, particularly after commencement of the EU environmental labelling scheme for flights in 2025. Our collateral to secure loans, in the form of aircraft, spare parts and airport slots, could lose value as customer demand shifts and economies move to low-carbon alternatives, which may increase our financing cost.

We have published a number of sustainability-related targets and goals, including with respect to reducing our GHG emissions. These goals are often long-term in nature, and in many cases rely on assumptions about the future availability and efficacy of technologies that do not yet exist or are not yet commercially viable. Our ability to meet our publicly stated targets is dependent on a number of factors outside our control, including the ability of third parties, such as engine and airframe manufacturers, SAF producers and other industry participants, to timely develop and commercialize these technological solutions. Additionally, we face risks associated with allegations or similar claims that our public statements concerning our sustainability efforts and achievements are exaggerated or unsubstantiated, sometimes referred to as "greenwashing," and could be subject to litigation or regulatory enforcement actions challenging the basis for such statements which could be costly and disruptive, whether or not meritorious.

Finally, the potential acute and chronic physical effects of climate change, such as increased frequency and severity of storms, floods, fires, sea-level rise, excessive heat, longer-term changes in weather patterns and other climate-related events, could affect our operations, infrastructure and financial results as well as the safety of our team members. Operational impacts, such as more frequent or widespread flight cancellations, could result in loss of revenue. We could incur significant costs to improve the climate resiliency of our infrastructure and otherwise prepare for, respond to, and mitigate such physical effects of climate change. We are not able to predict accurately the materiality of any potential losses or costs associated with the physical effects of climate change.

We are subject to many forms of environmental and noise regulation and may incur substantial costs as a result.

We are subject to a number of increasingly stringent federal, state, local and foreign laws, regulations and ordinances relating to the protection of human health and the environment and noise reduction, including those relating to emissions to the air, discharges to land and surface and subsurface waters, safe drinking water, and the management of hazardous substances, oils and waste materials. This universe of substances is evolving to encompass many substances not previously regulated. Compliance with environmental laws and regulations can require significant expenditures, and violations can lead to significant fines and penalties, as well as civil liability.

We are also subject to other environmental laws and regulations, including those that require us to investigate and remediate soil or groundwater to meet certain remediation standards. Under federal law, generators of waste materials, and current and former owners or operators of facilities, can be subject to liability for investigation and remediation costs at locations that have been identified as requiring response actions. Liability under these laws may be retroactive, strict, joint and several, meaning that we could be liable for the costs of cleaning up environmental contamination regardless of when it occurred, fault or the amount of waste directly attributable to us. We have liability for investigation and remediation costs at various sites, although such costs currently are not expected to have a material adverse effect on our business.

Governmental authorities in the U.S. and abroad are increasingly focused on potential contamination resulting from the use of certain chemicals, most notably per- and polyfluoroalkyl, substances (PFAS). Products containing PFAS have been used in manufacturing, industrial, and consumer applications over many decades, including those related to aviation. Among other things, recent changes to federal requirements for firefighting foams containing PFAS, as well as related state regulations affecting their use, will require operational changes. In August 2022, the EPA published for public comment a new rulemaking that would designate two PFAS substances (perfluorooctanoic acid and perfluorooctanesulfonic acid) as hazardous substances under the Comprehensive Environmental Response, Compensation, and Liability Act. This rulemaking, which is expected to be finalized in early 2024, would require entities to immediately report current and past releases that meet or exceed the reportable quantity for such substances to EPA's National Response Center. Depending on the final outcome of this rulemaking and the introduction of any additional state or federal regulations, we may incur costs in connection with reporting obligations and costs related to historic usage of PFAS-containing materials, transitioning away from the usage of PFAS-containing products, disposing of PFAS-containing waste or remediating any residual environmental impacts.

We have various leases and agreements with respect to real property, tanks and pipelines with airports and other operators. Under these leases and agreements, we have agreed to indemnify the lessor or operator against environmental liabilities associated with the real property or operations described under the agreement, even in certain cases where we are not the party responsible for the initial event that caused the environmental damage. We also participate in leases with other airlines in fuel consortiums and fuel committees at airports, and such indemnities are generally joint and several among the participating airlines.

Governmental authorities in several U.S. and foreign cities are also considering, or have already implemented, aircraft noise reduction programs, including the imposition of nighttime curfews and limitations on daytime take offs and landings as well as setting an annual flight cap from specific cities. We have been able to accommodate local noise restrictions imposed to date, but our operations could be adversely affected if locally-imposed regulations become more restrictive or widespread. The FAA is also currently evaluating possible changes to how aircraft noise is measured, and the resulting standards that are based on them. Ultimately, these changes could have an impact on, or limit, our operations, or make it more difficult for the FAA to modernize and increase the efficiency of the airspace and airports we utilize.

A high level of pilot retirements, more stringent duty time regulations, increased flight hour requirements for commercial airline pilots, reductions in the number of military pilots entering the commercial workforce, increased training requirements and other factors have caused a shortage of pilots that could materially adversely affect our business.

Large numbers of pilots in the industry accepted early retirement during the COVID-19 pandemic or are approaching the FAA's mandatory retirement age of 65. Our pilots and other employees are subject to rigorous certification standards, and our pilots and other crew members must adhere to flight time and rest requirements. Commencing in 2013, the minimum flight hour requirement to achieve a commercial pilot's license in the United States increased from 250 to 1,500 hours, thereby significantly increasing the time and cost commitment required to become licensed to fly commercial aircraft. Additionally, the number of military pilots being trained by the U.S. armed forces and available as commercial pilots upon their retirement from military service has been decreasing. Further, in the course of the domestic airline industry rapidly restoring capacity during the recovery from the COVID-19 pandemic, the significant training requirements to return large numbers of pilots to active flying have been time consuming and disruptive.

These and other factors have contributed to a shortage of qualified, entry-level pilots, shortages of experienced pilots trained and ready for duty, principally at our regional affiliates, and increased compensation costs materially for pilots throughout the industry. We believe that this industry-wide pilot shortage will remain a significant problem for regional airlines in the United States for the foreseeable future. We have recently implemented a number of recruitment initiatives intended to recruit qualified pilots to our regional airlines, including offering significant financial incentives, but we cannot guarantee that such efforts will be successful. Notwithstanding these efforts, our regional airline subsidiaries and other regional partners have recently been unable to hire adequate numbers of pilots to meet their needs, resulting in a reduction in the number of flights offered, operational disruptions, increased compensation expense and costs of operations, financial difficulties and other adverse effects, and these circumstances may become more severe in the future and thereby cause a material adverse effect on our business.

As part of the FAA Authorization Renewal process, Congress has proposed increasing the pilot retirement age from 65 to 67 to help address the pilot shortage. Raising the mandatory retirement age could help to mitigate the pilot shortage at regional airlines and other carriers operating domestically, but it could create potentially significant challenges to mainline carriers operating internationally, as the international standard for pilot retirement is currently 65.

We depend on a limited number of suppliers for aircraft, aircraft engines and parts. Delays in scheduled aircraft deliveries, unexpected grounding of aircraft or aircraft engines whether by regulators or by us, or other loss of anticipated fleet capacity, and failure of new aircraft to receive regulatory approval, be produced or otherwise perform as and when expected, may adversely impact our business, results of operations and financial condition.

We depend on a limited number of suppliers for aircraft, aircraft engines and many aircraft and engine parts. For example, all of our mainline aircraft were manufactured by either Airbus or Boeing and all of our regional aircraft were manufactured by either Bombardier or Embraer. Further, our supplier base continues to consolidate as evidenced by recent transactions involving Airbus and Bombardier and Mitsubishi and Bombardier, and the cessation of production of certain Bombardier regional aircraft that we and our regional partners currently operate in large numbers. Due to the limited number of suppliers, constraints on production capacity, large order books and long production lead times, manufacturers may face challenges in timely fulfilling our aircraft on order, and we may face competition from other carriers in securing an adequate supply of aircraft in the future. If new aircraft orders are not filled on a timely basis, we could face higher financing and operating costs than planned. The limited number of these suppliers may also result in reduced competition and potentially higher prices than if the supplier base was less concentrated. In addition, we are vulnerable to any problems associated with the performance of these suppliers' obligation to supply key aircraft, parts and engines, including design defects, mechanical problems, contractual performance by suppliers or adverse perception by the public that would result in customer avoidance of any of our aircraft. If the aircraft we receive do not meet expected performance or quality standards, including with respect to fuel efficiency, safety and reliability, we could also face higher financing and operating costs than planned and our business, results of operations and financial condition could be adversely impacted. We are also subject to the risk that action by the FAA or any other regulatory authority could result in an inability to certify or operate our aircraft, even temporarily. For instance, in March 2019, the FAA ordered the grounding of all Boeing 737 MAX Family aircraft, which remained in place for over a year and was not lifted in the United States until November 2020. An additional grounding of Boeing aircraft occurred in January 2024 involving the Boeing 737-9 MAX, a model that we do not operate. Further, significant limitations imposed on the use of Pratt & Whitney GTF aircraft engines (an engine that we do not use in our fleet) on certain Airbus aircraft have resulted in very significant numbers of the related aircraft being grounded while awaiting refurbished engines. Regulatory concerns raised by the FAA also previously forced

Boeing to suspend deliveries of certain 787 aircraft, temporarily resulting in significant reductions to our planned long-haul flying. More generally, we have recently experienced delivery delays across manufacturers due to regulatory matters such as those described above, regulatory restrictions on production rate increases (such as those that the FAA has announced it intends to impose on Boeing 737 production), supply chain limitations, development delays, and other factors, which have created significant challenges in planning our fleet, and those challenges are likely to continue. There is also the prospect that new aircraft models will continue to face certification delays further impeding the delivery of new aircraft to the airline industry and increasing competition for the production capacity that is available.

The success of our business depends on, among other things, effectively managing the number and types of aircraft we operate. If, for any reason, we are unable to accept or secure deliveries of new aircraft on contractually scheduled delivery timelines, our business, results of operations and financial condition could be negatively impacted. Our failure to integrate newly purchased aircraft into our fleet as planned might require us to seek extensions of the terms for some leased aircraft or otherwise delay the exit of certain aircraft from our fleet. Such unanticipated extensions or delays, which as noted above have recently been relatively commonplace among manufacturers of commercial aircraft, may require us to operate existing aircraft beyond the point at which it is economically optimal to retire them, resulting in increased maintenance costs, or reductions to our schedule, thereby reducing revenues. Repeated or prolonged delays in the production, delivery or induction of our new aircraft could also require us to scale back our growth plans, reduce frequencies or forgo service entirely to certain markets, which could adversely affect our business, financial condition and results of operations.

We rely heavily on technology and automated systems to operate our business, and any failure of these technologies or systems could harm our business, results of operations and financial condition.

We are highly dependent on existing and emerging technology and automated systems to operate our business. These technologies and systems include but may not be limited to our computerized airline reservation system, flight operations and crew scheduling systems, financial planning, management and accounting systems, telecommunications systems, website, maintenance systems and check-in kiosks. In order for our operations to work efficiently, our website and reservation system must be able to accommodate a high volume of traffic, maintain secure information and deliver flight information, as well as issue electronic tickets and process critical financial information in a timely manner. Substantially all of our tickets are issued to passengers as electronic tickets. We depend on our reservation system, which is hosted and maintained under a long-term contract by a third-party service provider, to be able to issue, track and accept these electronic tickets. If our technologies or automated systems are not functioning or if our third-party service providers were to fail to adequately provide technical support, system maintenance or timely software upgrades for any one of our key existing systems, we could experience service disruptions or delays, which could harm our business and result in the loss of important data, increase our expenses and decrease our revenues. Furthermore, certain critical aspects of our operation rely on legacy technological systems which may grow more difficult or expensive to support and maintain over time, and such systems may fail to perform as required or become more vulnerable to malfunction or failure over time. In the event that one or more of our primary technology or systems vendors goes into bankruptcy, ceases operations or fails to perform as promised, replacement services may not be readily available on a timely basis, at competitive rates or at all, and any transition time to a new system may be significant.

Our aircraft employ a number of sophisticated radio and satellite-based navigation and safety technologies, and we are subject to risks associated with the introduction or expansion of technologies that could interfere with the safe operation of these flight systems. For example, telecommunications companies are expanding and increasing the commercial and consumer applications of 5G cellular communication networks, and regulators, manufacturers and operators have expressed concerns that certain 5G applications could interfere with certain flight systems. On December 23, 2021, the FAA issued a special airworthiness information bulletin (SAIB), in which it indicated that further testing and assessment is needed regarding the effects of 5G on certain aircraft equipped with radar altimeters, which measure the aircraft's altitude and guide pilots during landings. If it were determined that 5G signals posed an interference risk to these altimeters or other systems, the FAA indicated in its SAIB that it could restrict flight operations in areas where such interference could occur. On June 17, 2022, the FAA and the telecommunications industry reached an agreement to delay the full implementation of 5G deployment near airports until July 1, 2023. The delayed implementation allowed the aviation industry time to retrofit the radio altimeters on aircraft to prevent potential interference from 5G signals. American has completed the retrofit of its impacted mainline and regional aircraft, and we now expect operational certainty as it pertains to 5G until 2028, when the current operating agreement between the FAA, Federal Communications Commission and the telecommunications industry expires.

Our technologies and automated systems are not completely protected against events that are beyond our control, including natural disasters, power failures, terrorist attacks, cyberattacks, data theft, defects, errors, equipment and

software failures, computer viruses or telecommunications failures. When service interruptions occur as a result of any of the aforementioned events, we address them in accordance with applicable laws, rules and regulations. However, substantial or sustained system failures could cause service delays or failures and result in our customers purchasing tickets from other airlines. We cannot assure that our security measures, change control procedures or disaster recovery plans are adequate to prevent disruptions or delays. Disruption in or changes to these technologies or systems could result in a disruption to our business and the loss of important data. Any of the foregoing could result in a material adverse effect on our business, results of operations and financial condition.

Evolving data privacy requirements (in particular, compliance with applicable federal, state and foreign laws relating to handling of personal information about individuals) could increase our costs, and any significant data privacy incident could disrupt our operations, harm our reputation, expose us to legal risks and otherwise materially adversely affect our business, results of operations and financial condition.

In the normal course of our business, we collect, process, use and disclose personal information about individuals and rely on third party service providers to host or otherwise process personal information. Many federal, state and foreign governmental bodies and agencies have adopted, or are considering adopting, laws and regulations that impose limits on the collection, processing, use, disclosure and security of personal information about individuals. In some cases, such laws and regulations can be enforced by private parties in addition to government entities. In addition, privacy advocacy and industry groups may propose new and different self-regulatory standards or guidance that may legally or contractually apply to us and our vendors. These non-uniform laws, regulations, standards and guidance are complex and currently evolving and can be subject to significant change and interpretation, and may be inconsistently applied and enforced from one jurisdiction to another.

Our business requires the secure processing and storage of personal information relating to our customers, employees, business partners and others, and other data such as confidential information. However, like any global enterprise operating in today's digital business environment, we and our third party service providers have experienced cybersecurity incidents and data breaches. For example, in July 2022, a minor phishing incident resulted in certain employee email accounts being accessed and acquired without authorization that contained personal information about a very limited number of individuals, including travelers (following which we notified the individuals). We react and respond to these cybersecurity incidents in accordance with the applicable legal requirements, our own cybersecurity protocols, as well as our commercial partners' standards (as appropriate), but we cannot ensure that our responses (or those of our partners and service providers) will be sufficient to prevent or mitigate the potential adverse impacts of these cybersecurity incidents, which may be material.

There has been heightened legislative and regulatory focus on data privacy and cybersecurity in the U.S., EU, U.K., China and elsewhere, particularly with respect to critical infrastructure providers, including those in the transportation sector. As a result, we must comply with a proliferating and fast-evolving set of legal requirements in this area, including substantive data privacy and cybersecurity standards as well as requirements for notifying regulators and affected individuals in the event of a cybersecurity incident. In addition, we are subject to an increasing number of reporting obligations in respect of material cybersecurity incidents. These reporting requirements have been proposed or implemented by a number of regulators in different jurisdictions, may vary in their scope and application, and could contain conflicting requirements. Certain of these rules and regulations may require us to report a cybersecurity incident before we have been able to fully assess its impact or remediate the underlying issue. Efforts to comply with such reporting requirements could divert management's attention from our cybersecurity incident response and could potentially reveal system vulnerabilities to threat actors. Failure to timely report cybersecurity incidents under these rules could also result in regulatory investigations, litigation, monetary fines, sanctions, or subject us to other forms of liability. Even though we believe we and our third party service providers are generally in compliance with applicable laws, rules and regulations relating to privacy and data security, the regulatory environment is increasingly challenging as data privacy and cybersecurity laws, rules, regulations, industry standards and other requirements are continually developing. These changing requirements, along with their evolving application, interpretation, and amendment, may present material obligations and risks to our business, including significantly expanded compliance burdens, costs and enforcement risks.

In addition, many of our commercial partners, including credit card companies, have imposed data security standards that we must meet. In particular, we are required by the Payment Card Industry Security Standards Council, founded by the credit card companies, to comply with their highest level of data security standards (the Payment Card Industry Data Security Standard (PCI DSS)). While we and our service providers continue our efforts to meet these standards, new and revised standards may be imposed that may be difficult for us to meet and could increase our costs, and if we are unable to comply with revised standards, we may be subject to fines, restrictions or other liability, which could materially and

adversely affect our business. Moreover, it is not guaranteed that PCI DSS compliance will prevent illegal or improper use of our payment systems or the theft, loss or misuse of payment card data or transaction information.

Litigation, claims and enforcement related to data privacy, biometrics and other provisions of state privacy laws may involve new interpretations of privacy laws. There has also been a noticeable uptick in class actions in the U.S. wherein plaintiffs have utilized a variety of laws, including state wiretapping laws, in relation to companies' use of tracking technologies, such as cookies and pixels. Compliance with these laws and regulations may be inconsistent from jurisdiction to jurisdiction, increasing the cost of compliance and our risk of liability from litigation. Any litigation, claims or enforcement actions to which we are or become a party could potentially result in substantial monetary damages or fines, and negative reputational impacts that cause us to lose existing or future customers, which could materially adversely affect our business, results of operations and financial condition.

We are exposed to risks from cyberattacks, and any cybersecurity incidents involving us, our third-party service providers, or one of our AAdvantage partners or other business partners, could materially adversely affect our business, results of operations and financial condition.

Significant cybersecurity incidents involving us, our third-party service providers, or one of our AAdvantage partners or other business partners, have in the past and may in the future result in a range of potentially material negative consequences for us, including unauthorized access to, disclosure, modification, misuse, loss or destruction of company systems or data; theft of sensitive, regulated or confidential data, such as personal information or our intellectual property; the loss of functionality of critical systems through ransomware, denial of service or other cyberattacks; a diminished ability to retain or attract new customers; a deterioration in our relationships with business partners and other third parties; interruptions or failures in our payment related systems; and business delays, service or system disruptions, damage to equipment and injury to persons or property. The methods used to obtain unauthorized access, disable or degrade service or sabotage systems are constantly evolving and may be difficult to anticipate or to detect for long periods of time. The constantly changing nature of the threats means that we cannot and have not been able to prevent all data security breaches or misuse of data, and there is a risk that our security measures will not be fully effective in the future. Similarly, we depend on the ability of our key commercial partners, including AAdvantage partners, other business partners, our regional carriers, distribution partners and technology vendors, to conduct their businesses in a manner that complies with applicable security standards and assures their ability to perform on a timely basis. A security failure, including a failure to meet PCI DSS requirements, breach or other significant cybersecurity incident affecting one of our partners, interruptions or failures in our payment related systems, could result in potentially material negative consequences for us, including loss of critical data, service interruptions, delays in operations, and the potential for fines, restrictions and expulsion from card acceptance programs. In addition, we use third party service providers to help us deliver services to customers. These service providers may store personal information, credit card information and/or other confidential information. Such information has been and will be the target of unauthorized access or subject to security breaches because of third-party action, employee error, malfeasance or otherwise. Any of these could (a) result in the loss of information, litigation, indemnity obligations, expensive and inconsistent cybersecurity incident and data breach notification requirements, damage to our reputation, regulatory scrutiny, and other liability, or (b) have a material adverse effect on our business, financial condition and results of operations.

The threat of cybersecurity incidents continues to increase as the frequency, intensity and sophistication of cyberattacks and intrusions increase around the world. Diverse threat actors, such as state-sponsored organizations, opportunistic hackers and hacktivists, as well as diverse attack vectors such as social engineering/phishing, malware (including ransomware), malfeasance by insiders, human or technological error, denial of service attacks or exploitation of vulnerabilities, threaten the confidentiality, integrity, and availability of our and our third party service providers' information systems, personal information and confidential information. Geopolitical issues also continue to increase our cybersecurity risk and potential for cybersecurity incidents, for example, the conflict involving Russia and Ukraine, which has resulted in a heightened risk of cyberattacks against companies like ours that have operations, vendors and/or supply chain providers located in or around the region of conflict or are otherwise related to the conflict. Despite ongoing efforts to maintain and improve the security of our information systems and digital information, individuals, including employees, contractors, and external threat actors, may be able to circumvent the security measures we put in place, and we may be unable to anticipate new techniques used for these attacks and intrusions and implement adequate preventative measures. We, our business partners and service providers have been the target of cybersecurity attacks in the past and expect that we, our business and service partners, will continue to experience cybersecurity incidents in the future.

The costs and operational consequences of defending against, preparing for, responding to and remediating a cybersecurity incident are substantial. As cybersecurity incidents become more frequent, intense and sophisticated, costs of proactive defense measures are increasing. Further, we could be exposed to litigation, regulatory enforcement or other

legal action as a result of an incident, carrying the potential for damages, fines, sanctions or other penalties, as well as injunctive relief and enforcement actions requiring costly compliance measures. A significant number of recent data privacy and cybersecurity incidents, including those involving other large airlines, have resulted in very substantial adverse financial consequences to those companies. A cybersecurity incident could also impact our brand, including that of the AAdvantage program, harm our reputation and adversely impact our relationship with our customers, employees and stockholders. The increased regulatory focus on data privacy practices apart from how personal information is secured, such as how personal information is collected, used for marketing purposes, and shared with third parties, also may require changes to our processes and increase compliance costs. There is also an increased risk to our business in the event of a significant cybersecurity or data privacy violation, including additional compliance costs, reputational harm, disruption to the manner in which we provide our services, including the geographies we service, and being subject to complaints and/or regulatory investigations, significant monetary liability, fines, penalties, regulatory enforcement, individual or class action lawsuits, public criticism, loss of customers, loss of goodwill or other additional liabilities, such as claims by industry groups or other third parties. Accordingly, failure to appropriately address data privacy and cybersecurity issues could result in material financial and other liabilities and cause significant reputational harm to our company.

We rely on third-party distribution channels and must effectively manage the costs, rights and functionality of these channels.

While our priority is to migrate an increasing portion of our customers to our modern, direct distribution channels in lieu of third party channels, we continue to rely on third-party distribution channels, including those provided by or through global distribution systems (GDSs) (e.g., Amadeus, Sabre and Travelport), conventional travel agents, travel management companies and online travel agents (OTAs) (e.g., Expedia, including its booking sites Orbitz and Travelocity, and Booking Holdings, including its booking sites Kayak and Priceline), to distribute a significant portion of our airline tickets, and we expect in the future to continue to rely on these channels. We are also dependent upon the ability and willingness of these distribution channels to expand their ability to distribute and collect revenues for ancillary products (e.g., fees for selective seating). These distribution channels are more expensive and at present have less functionality in respect of ancillary product offerings than those we operate ourselves, such as our website at *www.aa.com*. Certain of these distribution channels also effectively restrict the manner in which we distribute our products generally.

To remain competitive, we will need to manage successfully our distribution costs and rights, increase our distribution flexibility, continue to migrate the distribution of tickets to our proprietary and other modern distribution channels, and improve the functionality of our distribution channels, while maintaining an industry-competitive cost structure and a high level of customer satisfaction. Further, as distribution technology changes we will need to continue to update our technology by acquiring new technology from third parties, building the functionality ourselves, or a combination, which in any event will likely entail significant technological and commercial risk and involve potentially material investments. These imperatives may affect our relationships with conventional travel agents, travel management companies, GDSs and OTAs, including if consolidation of conventional travel agents, travel management companies, GDSs or OTAs continues, or should any of these parties seek to acquire other technology providers thereby potentially limiting our technology alternatives. Any inability to manage our third-party distribution costs, rights and functionality at a competitive level or any material diminishment or disruption in the distribution of our tickets could have a material adverse effect on our business, results of operations and financial condition.

If we are unable to obtain and maintain adequate facilities and infrastructure throughout our system and, at some airports, adequate slots, we may be unable to operate our existing flight schedule and to expand or change our route network in the future, which may have a material adverse impact on our operations.

In order to operate our existing and proposed flight schedule and, where desirable, add service along new or existing routes, we must be able to maintain and/or obtain adequate gates, check-in counters, operations areas, operations control facilities and administrative support space. As airports around the world become more congested, it may not be possible for us to ensure that our plans for new service can be implemented in a commercially viable manner, given operating constraints at airports throughout our network, including those imposed by inadequate facilities at desirable airports.

In light of constraints on existing facilities, there is presently a significant amount of capital spending underway at major airports in the United States, including large projects underway at a number of airports where we have significant operations, such as O'Hare International Airport, Dallas/Fort Worth International Airport and Los Angeles International Airport. More generally, following long periods of underinvestment, there is a trend among airports in the United States to engage in significant, expensive expansion, remodeling and infrastructure improvement projects. This spending is expected to result in increased costs to airlines and the traveling public that use those facilities as the airports seek to

recover their investments through increased rental, landing and other facility costs. In some circumstances, such costs could be imposed by the relevant airport authority without our approval. Accordingly, our operating costs are expected to increase significantly at many airports at which we operate, including a number of our hubs and gateways, as a result of capital spending projects currently underway and additional projects that we expect to commence over the next several years.

In addition, operations at three major domestic airports, certain smaller domestic airports and many foreign airports we serve are regulated by governmental entities through allocations of slots or similar regulatory mechanisms that limit the rights of carriers to conduct operations at those airports. Each slot represents the authorization to land at or take off from the particular airport during a specified time period and may impose other operational restrictions as well. In the U.S., the DOT and the FAA currently regulate the allocation of slots or slot exemptions at DCA and two New York City airports: JFK and LGA. Our operations at these airports generally require the allocation of slots or similar regulatory authority. In addition to slot restrictions, operations at DCA and LGA are also limited based on a so-called "perimeter rule" which generally limits the stage length of the flights that can be operated from those airports to 1,250 and 1,500 miles, respectively. Similarly, our operations at LHR, international airports in Frankfurt, Paris, Tokyo and other airports outside the U.S. are regulated by local slot authorities pursuant to the International Airline Trade Association Worldwide Scheduling Guidelines and/or applicable local law. Termination of slot controls or other operational restrictions at some or all of the foregoing airports could affect our operational performance and competitive position. We currently have sufficient slots or analogous authorizations to operate our existing flights and we have generally, but not always, been able to obtain the rights to expand our operations and to change our schedules. However, there is no assurance that we will be able to obtain sufficient slots or analogous authorizations in the future or as to the cost of acquiring such rights because, among other reasons, such allocations are often sought after by other airlines and are subject to changes in governmental policies. During periods of reduced demand for air travel, such as during the COVID-19 pandemic, we may rely on exemptions granted by applicable authorities from the requirement that we continuously use certain slots, gates and routes or risk having such operating rights revoked, and depending on the applicable authority these exemptions can vary in the way they are structured and applied. We cannot predict whether such exemptions will be made available, whether they will be granted on the same or similar terms as in past instances, or whether we ultimately could be at risk of losing valuable operating rights. If we are forced to surrender slots or other rights, we may be unable to provide our desired level of service to or from certain destinations in the future. We cannot provide any assurance that regulatory changes resulting in changes in the application of slot controls or the allocation of or any reallocation of existing slots, the continued enforcement or termination of a perimeter rule or similar regulatory regime will not have a material adverse impact on our operations.

Our ability to provide service can also be impaired at airports where the airport gates and other facilities are currently inadequate to accommodate all of the service that we would like to provide, or where we have no access to gates at all.

Any limitation on our ability to acquire or maintain adequate gates, ticketing facilities, operations areas, operations control facilities, slots (where applicable), or office space could have a material adverse effect on our business, results of operations and financial condition.

Interruptions or disruptions in service at one of our key facilities could have a material adverse impact on our operations.

We operate principally through our hubs in Charlotte, Chicago, Dallas/Fort Worth, Los Angeles, Miami, New York, Philadelphia, Phoenix and Washington, D.C. and partner gateways including London Heathrow (among others). Substantially all of our flights either originate at or fly into one of these locations. A significant interruption or disruption in service at one of our hubs, gateways or other airports where we have a significant presence, resulting from air traffic control delays, weather conditions, natural disasters, growth constraints, performance by third-party service providers (such as electric utility or telecommunications providers), failure of computer systems, disruptions at airport facilities or other key facilities used by us to manage our operations (including as a result of social or environmental activism), labor relations, power supplies, fuel supplies, terrorist activities, or otherwise could result in the cancellation or delay of a significant portion of our flights and, as a result, could have a severe impact on our business, results of operations and financial condition. We have limited control, particularly in the short term, over the operation, quality or maintenance of many of the services on which our operations depend and over whether vendors of such services will improve or continue to provide services that are essential to our business.

Increases in insurance costs or reductions in insurance coverage may adversely impact our operations and financial results.

The terrorist attacks of September 11, 2001 led to a significant increase in insurance premiums and a decrease in the insurance coverage available to commercial air carriers. Accordingly, our insurance costs increased significantly, and our ability to continue to obtain insurance even at current prices remains uncertain. The occurrence or persistence of certain events, including armed conflicts, could also impact our ability to obtain commercial insurance coverage against certain risks, or to obtain such insurance on commercially acceptable terms. If we are unable to maintain adequate insurance coverage or to secure suitable alternatives outside the commercial insurance markets, our business could be materially and adversely affected. Additionally, severe disruptions in the domestic and global financial markets could adversely impact the claims paying ability of some insurers. Future downgrades in the ratings of enough insurers could adversely impact both the availability of appropriate insurance coverage and its cost. Because of competitive pressures in our industry, our ability to pass along additional insurance costs to passengers is limited. As a result, further increases in insurance costs or reductions in available insurance coverage could have an adverse impact on our financial results.

The airline industry is heavily taxed.

The airline industry is subject to extensive government fees and taxation that negatively impact our revenue and profitability. The U.S. airline industry is one of the most heavily taxed of all industries. These fees and taxes have grown significantly in the past decade for domestic flights, and various U.S. fees and taxes also are assessed on international flights. For example, as permitted by federal legislation, most major U.S. airports impose a per-passenger facility charge on us. In addition, the governments of foreign countries in which we operate impose on U.S. airlines, including us, various fees and taxes, and these assessments have been increasing in number and amount in recent years. Moreover, we are obligated to collect a federal excise tax, commonly referred to as the “ticket tax,” on domestic and international air transportation. We collect the excise tax, along with certain other U.S. and foreign taxes and user fees on air transportation (such as passenger security fees), and pass along the collected amounts to the appropriate governmental agencies. Although these taxes and fees are not our operating expenses, they represent an additional cost to our customers. There are continuing efforts in Congress and in other countries to raise different portions of the various taxes, fees, and charges imposed on airlines and their passengers, including the passenger facility charge, and we may not be able to recover all of these charges from our customers. Increases in such taxes, fees and charges could negatively impact our business, results of operations and financial condition.

Under DOT regulations, all governmental taxes and fees must be included in the prices we quote or advertise to our customers. Due to the competitive revenue environment, many increases in these fees and taxes have been absorbed by the airline industry rather than being passed on to the customer. Further increases in fees and taxes may reduce demand for air travel, and thus our revenues.

Risks Related to Ownership of AAG Common Stock and Convertible Notes

The price of AAG common stock has been and may in the future be volatile.

The market price of AAG common stock has fluctuated substantially in the past, and may fluctuate substantially in the future, due to a variety of factors, many of which are beyond our control, including:

- the effects of external events, such as the COVID-19 pandemic, on our business or the U.S. and global economies;
- macro-economic conditions, including the price of fuel;
- changes in market values of airline companies as well as general market conditions;
- our operating and financial results failing to meet the expectations of securities analysts or investors;
- changes in financial estimates or recommendations by securities analysts;
- changes in our level of outstanding indebtedness and other obligations;
- changes in our credit ratings;
- material announcements by us or our competitors;

- expectations regarding any future capital deployment program, including share repurchase programs and any future dividend payments that may be declared by our Board of Directors, or any subsequent determination to cease repurchasing stock or paying dividends;
- new regulatory pronouncements and changes in regulatory guidelines;
- general and industry-specific economic conditions;
- changes in our key personnel;
- inclusion of our common stock in broad market indexes favored by passive investors;
- investor preferences to invest in certain sectors, including large technology companies in lieu of industrial or transportation companies;
- public or private sales of a substantial number of shares of AAG common stock or issuances of AAG common stock upon the exercise or conversion of restricted stock unit awards, stock appreciation rights, or other securities that may be issued from time to time, including warrants we have issued in connection with our receipt of funds under the CARES Act, the PSP Extension Law and the ARP;
- increases or decreases in reported holdings by insiders or other significant stockholders;
- fluctuations in trading volume; and
- technical factors in the public trading market for our stock that may produce price movements that may or may not comport with macro, industry or company-specific fundamentals, including, without limitation, the sentiment of retail investors (including as may be expressed on financial trading and other social media sites), the amount and status of short interest in our securities, access to margin debt, trading in options and other derivatives on our common stock and any related hedging and other technical trading factors.

The closing price of our common stock on the Nasdaq Global Select Market varied from $10.92 to $18.80 during 2023 and $12.93 to $15.36 during 2024 year-to-date through February 16, 2024. At times, fluctuations in our stock price have been rapid, imposing risks on investors due to the possibility of significant, short-term price volatility. While we believe that in recent years this wide range of trading prices has largely reflected the changing prospects for a large airline facing the challenges imposed by the COVID-19 pandemic, we also believe, based in part on the commentary of market analysts, that the trading price of our common stock has at times been influenced by the technical trading factors discussed in the last bullet above. On some occasions, market analysts have explained fluctuations in our stock price by reference to purported “short squeeze” activity. A “short squeeze” is a technical market condition that occurs when the price of a stock increases substantially, forcing market participants who had taken a position that its price would fall (i.e., who had sold the stock “short”), to buy it, which in turn may create significant, short-term demand for the stock not for fundamental reasons, but rather due to the need for such market participants to acquire the stock in order to forestall the risk of even greater losses. A “short squeeze” condition in the market for a stock can lead to short-term conditions involving very high volatility and trading that may or may not track fundamental valuation models.

If we decide to make repurchases of or pay dividends on our common stock, we cannot guarantee that we will continue to do so or that such a capital deployment program will enhance long-term stockholder value.

If we determine to make any share repurchases in the future, such repurchases may be made through a variety of methods, which may include open market purchases, privately negotiated transactions, block trades or accelerated share repurchase transactions. Our future repurchases of AAG common stock, if any, may be limited, suspended or discontinued at any time at our discretion and without prior notice.

If we determine to make any dividends in the future, such dividends that may be declared and paid from time to time will be subject to market and economic conditions, applicable legal requirements and other relevant factors. The amount and timing of any future dividends, if any, may vary, and the payment of any dividend does not assure that we will pay dividends in the future.

In addition, any future repurchases of AAG common stock or payment of dividends, or any determination to cease repurchasing stock or paying dividends, could affect our stock price and increase its volatility. The existence of a future share repurchase program and any future dividends could cause our stock price to be higher than it would otherwise be and could potentially reduce the market liquidity for our stock. Additionally, any future repurchases of AAG common stock

or payment of dividends will diminish our cash reserves, which may impact our ability to finance future growth and to pursue possible future strategic opportunities and acquisitions. Further, our repurchase of AAG common stock may fluctuate such that our cash flow may be insufficient to fully cover our share repurchases. Under the recently enacted IRA, we may become subject to an excise tax on the fair market value of AAG common stock repurchased after December 31, 2022, which may adversely affect our financial condition. Although our share repurchase programs are intended to enhance long-term stockholder value, there is no assurance that they will do so.

AAG's Certificate of Incorporation, Bylaws and Tax Benefit Preservation Plan include provisions that limit voting and acquisition and disposition of our equity interests and specify an exclusive forum for certain stockholder disputes.

Our Certificate of Incorporation and Bylaws include significant provisions that limit voting and ownership and disposition of our equity interests as described in Part II, Item 5. Market for American Airlines Group's Common Stock, Related Stockholder Matters and Issuer Purchases of Equity Securities - "*Ownership Restrictions*" and AAG's Description of the Registrants' Securities Registered Pursuant to Section 12 of the Securities Exchange Act of 1934, which is filed as Exhibit 4.1 hereto. Further restrictions are set forth in our Tax Benefit Preservation Plan, which was filed as Exhibit 4.1 to AAG's Current Report on Form 8-K filed on December 22, 2021. These restrictions may adversely affect the ability of certain holders of AAG common stock and our other equity interests to vote such interests and adversely affect the ability of persons to acquire shares of AAG common stock and our other equity interests.

Our Certificate of Incorporation also specifies that the Court of Chancery of the State of Delaware shall be the exclusive forum for substantially all disputes between us and our stockholders. Because the applicability of the exclusive forum provision is limited to the extent permitted by applicable law, we do not intend for the exclusive forum provision to apply to suits brought to enforce any duty or liability created by the Exchange Act or any other claim for which the federal courts have exclusive jurisdiction, and acknowledge that federal courts have concurrent jurisdiction over all suits brought to enforce any duty or liability created by the Securities Act of 1933 (Securities Act). We note that there is uncertainty as to whether a court would enforce the provision as it applies to the Securities Act and that investors cannot waive compliance with the federal securities laws and the rules and regulations thereunder. This provision may have the effect of discouraging lawsuits against our directors and officers.

Certain provisions of AAG's Certificate of Incorporation and Bylaws make it difficult for stockholders to change the composition of our Board of Directors and may discourage takeover attempts that some of our stockholders might consider beneficial.

Certain provisions of our Certificate of Incorporation and Bylaws, as currently in effect, may have the effect of delaying or preventing changes in control if our Board of Directors determines that such changes in control are not in our best interest and the best interest of our stockholders. These provisions include, among other things, the following:

- advance notice procedures for stockholder proposals to be considered at stockholders' meetings;
- the ability of our Board of Directors to fill vacancies on the board;
- a prohibition against stockholders taking action by written consent;
- stockholders are restricted from calling a special meeting unless they hold at least 20% of our outstanding shares and follow the procedures provided for in the amended Bylaws;
- a requirement that holders of at least 80% of the voting power of the shares entitled to vote in the election of directors approve any amendment of our Bylaws submitted to stockholders for approval; and
- super-majority voting requirements to modify or amend specified provisions of our Certificate of Incorporation.

These provisions are not intended to prevent a takeover, but are intended to protect and maximize the value of the interests of our stockholders. While these provisions have the effect of encouraging persons seeking to acquire control of our company to negotiate with our Board of Directors, they could enable our Board of Directors to prevent a transaction that some, or a majority, of our stockholders might believe to be in their best interest and, in that case, may prevent or discourage attempts to remove and replace incumbent directors. In addition, we are subject to the provisions of Section 203 of the Delaware General Corporation Law, which prohibits business combinations with interested stockholders. Interested stockholders do not include stockholders whose acquisition of our securities is approved by the Board of Directors prior to the investment under Section 203.

The issuance or sale of shares of our common stock, rights to acquire shares of our common stock, or warrants issued to the U.S. Department of Treasury under the CARES Act, the PSP Extension Law, the ARP, PSP1, PSP2 and PSP3, could depress the trading price of our common stock and the Convertible Notes.

We may conduct future offerings of material amounts of our common stock, preferred stock or other securities that are convertible into or exercisable for our common stock to finance our operations, to fund acquisitions, or for any other purposes at any time and from time to time (including as compensation to the U.S. Government for the proceeds received pursuant to the payroll support program established under the Coronavirus Aid, Relief, and Economic Security Act (CARES Act) (PSP1), the payroll support program established under the Subtitle A of Title IV of Division N of the Consolidated Appropriations Act, 2021 (PSP Extension Law) (PSP2) and the payroll support program established under the American Rescue Plan Act of 2021 (ARP) (PSP3)). If these additional shares or securities are issued or sold, or if it is perceived that they will be sold, into the public market or otherwise, the trading price of our common stock and the 6.50% convertible senior notes due 2025 (the Convertible Notes) could decline substantially. If we issue additional shares of our common stock or rights to acquire shares of our common stock, if any of our existing stockholders sells a substantial amount of our common stock, or if the market perceives that such issuances or sales may occur, then the trading price of our common stock and the Convertible Notes could decline substantially.

ITEM 1B. UNRESOLVED STAFF COMMENTS

We had no unresolved SEC staff comments that were issued 180 days or more preceding December 31, 2023.

ITEM 1C. CYBERSECURITY

Cybersecurity Risk Management and Strategy

The safety and security of our customers and team members is our top priority. This includes working to put in place appropriate administrative, physical and technical cybersecurity safeguards to help protect our assets that keep our operation running and securely store the information in our care. We have developed and implemented a cybersecurity risk management program intended to protect the confidentiality, integrity, and availability of our systems and information.

We have created, and assess our program against, an integrated cybersecurity framework using various National Institute of Standards and Technology (NIST) security standards, guidelines and best practices. This does not imply that we meet any particular technical standards, specifications, or requirements, only that we use various NIST security standards, guidelines and best practices to identify, assess, and manage cybersecurity risks relevant to our business.

Our cybersecurity risk management program is overseen by our Executive Cybersecurity Risk Group (ECRG) which is comprised of our Chief Digital and Information Officer (CDIO), Chief Financial Officer and Chief Legal Officer. The ECRG, working with our Chief Information Security Officer (CISO), assists the Board of Directors and our senior leadership team in fulfilling their responsibilities for cybersecurity governance, approval and oversight through the periodic reporting and review of security strategy and risk management practices. Our cybersecurity risk management program is integrated into our overall risk management processes and shares common reporting channels and governance processes that apply across the enterprise to other legal, compliance, strategic, operational, and financial risk governance programs.

Our cybersecurity risk management program includes:

- risk assessments designed to help identify material cybersecurity risks to our critical systems, information, and our broader enterprise IT environment;
- a cybersecurity team principally responsible for managing our (1) cybersecurity risk assessment processes, (2) security controls, (3) vulnerability management program and (4) detection and response to cybersecurity incidents;

- the use of external service providers, where appropriate, to assess, test or otherwise assist with aspects of our security controls;
- policies, procedures and standards that are utilized to outline expectations, guidelines and best practices for managing cybersecurity risks;
- cybersecurity awareness training for our employees, incident response personnel and senior management;
- a cybersecurity incident response plan that includes procedures for responding to cybersecurity incidents; and
- a third-party risk management process for critical IT service providers, suppliers, and vendors.

We are constantly assessing our environment for cybersecurity threats, and we face risks from cybersecurity threats that, if realized, are reasonably likely to materially affect us, including our operations, business strategy, results of operations or financial condition. At the time of this filing, we have not identified risks from known cybersecurity threats, including as a result of any prior cybersecurity incidents, that have materially affected us, including our operations, business strategy, results of operations or financial condition. See Part I, Item 1A. Risk Factors – "*Evolving cybersecurity and data privacy requirements (in particular, compliance with applicable federal, state and foreign laws relating to handling of personal information about individuals) could increase our costs, and any significant cybersecurity or data privacy incident could disrupt our operations, harm our reputation, expose us to legal risks and otherwise materially adversely affect our business, results of operations and financial condition.*"

Cybersecurity Governance

Our Board of Directors consider cybersecurity risk as part of its risk oversight function and has delegated to the Audit Committee (Committee) oversight of cybersecurity and other information technology risks. The Committee oversees management's implementation of our cybersecurity risk management program.

The Committee receives quarterly reports from management on our cybersecurity risks. In addition, management updates the Committee, as necessary, regarding any material cybersecurity incidents, as well as certain incidents with lesser impact potential.

The Committee reports to the full Board of Directors regarding its activities, including those related to cybersecurity. The full Board of Directors also receives periodic briefings from management on our cyber risk management program. Board of Directors members receive presentations on cybersecurity topics from a combination of our CDIO, CISO, Deputy General Counsel, internal security staff, external counsel or external experts, as part of the Board of Director's continuing education on topics that impact public companies.

Our management team, including our CDIO, CISO, Vice President and Deputy General Counsel – Chief Privacy and Data Protection Officer, Vice President of Infrastructure and Operations and additional members of the ECRG are responsible for assessing and managing our material risks from cybersecurity threats. The team has primary responsibility for our overall cybersecurity risk management program and supervises both our internal cybersecurity personnel and our retained external cybersecurity consultants. Collectively, our management team has extensive information technology experience, as well as cybersecurity incident response, compliance, oversight, and program management experience. Additionally, certain leaders and personnel within the cybersecurity organization hold industry certifications, such as Certified Information Systems Security Professional or Certified Information Security Manager.

Our management team supervises efforts to prevent, detect, mitigate, and remediate cybersecurity risks and incidents through various means, which may include briefings from internal security personnel; threat intelligence and other various sources including external consultants engaged by us.

ITEM 2. PROPERTIES

Flight Equipment

As of December 31, 2023, American operated a mainline fleet of 965 aircraft. During 2023, American accepted delivery of 31 mainline aircraft including 17 Boeing 737-8 MAX, 10 Airbus A321neo and four Boeing 787-8 aircraft and returned nine mainline aircraft to service from temporary storage. We are supported by our wholly-owned and third-party regional carriers that fly under capacity purchase agreements operating as American Eagle. As of December 31, 2023, American Eagle operated 556 regional aircraft. During 2023, we increased our regional fleet by a net of 20 aircraft, including the addition of 83 regional aircraft, the return of 55 regional aircraft to third-party regional carriers and temporarily parking eight regional aircraft.

Mainline

As of December 31, 2023, American's mainline fleet consisted of the following aircraft:

	Average Seating Capacity	Average Age (Years)	Owned	Leased	Total
Airbus A319	128	19.7	21	112	133
Airbus A320	150	22.7	10	38	48
Airbus A321	184	11.4	164	54	218
Airbus A321neo	195	2.9	43	35	78
Boeing 737-800	172	14.1	132	171	303
Boeing 737-8 MAX	172	3.2	26	33	59
Boeing 777-200ER	273	23.0	44	3	47
Boeing 777-300ER	304	9.8	18	2	20
Boeing 787-8	234	5.1	20	17	37
Boeing 787-9	285	6.2	17	5	22
Total		12.9	495	470	965

Regional

As of December 31, 2023, the fleet of our wholly-owned and third-party regional carriers operating as American Eagle consisted of the following aircraft:

	Average Seating Capacity	Owned	Leased	Owned or Leased by Third Party Regional Carrier	Total	Operating Regional Carrier	Number of Aircraft Operated
Bombardier CRJ 200	50	—	—	40	40	Air Wisconsin	40
Bombardier CRJ 700 [1]	65	50	—	90	140	SkyWest	90
						PSA	50
						Total	140
Bombardier CRJ 900 [1]	76	74	—	—	74	PSA	74
Embraer 170 [1]	65	6	23	5	34	Envoy	29
						Republic	5
						Total	34
Embraer 175	76	108	—	102	210	Envoy	108
						Republic	82
						SkyWest	20
						Total	210
Embraer 145 [1]	50	58	—	—	58	Piedmont	58
Total		296	23	237	556		556

(1) Excluded from the total operating aircraft count above are 77 regional aircraft that are being held in temporary storage as follows: 57 owned Embraer 145, seven owned and four leased Bombardier CRJ 700, six owned Bombardier CRJ 900 and three leased Embraer 170.

See Note 11 to AAG's Consolidated Financial Statements in Part II, Item 8A and Note 10 to American's Consolidated Financial Statements in Part II, Item 8B for additional information on our capacity purchase agreements with third-party regional carriers.

Aircraft and Engine Purchase Commitments

As of December 31, 2023, we had definitive purchase agreements for the acquisition of the following new aircraft (1):

	2024	2025	2026	2027	2028	2029 and Thereafter	Total
Airbus							
A320neo Family	3	21	35	5	—	—	64
Boeing							
737 MAX Family	20	33	21	—	—	—	74
787 Family	6	5	4	5	5	5	30
Embraer							
175	12	—	—	—	—	—	12
Total	41	59	60	10	5	5	180

(1) Delivery schedule represents our best estimate as of the date of this report as described in footnote (e) to the " *Contractual Obligations*" table in Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations. Actual delivery dates are subject to change, which could be material, based on various potential factors including production delays by the manufacturer and regulatory concerns.

As of December 31, 2023, we had committed to purchase two used Airbus A321neo aircraft which were delivered in January 2024. We had also committed to purchasing six used Embraer 175 aircraft, which are currently flown under a capacity purchase agreement with a third-party regional carrier and which are already included in our aircraft count. We also have agreements for 44 spare engines to be delivered in 2024 and beyond.

We have financing commitments in place for all aircraft scheduled to be delivered in 2024, except for three Airbus A320neo Family aircraft and two Embraer 175 aircraft. Our ability to draw on the financing commitments we have in place is subject to (1) the satisfaction of various terms and conditions including, in some cases, on our acquisition of the aircraft by a certain date and (2) the performance by the relevant financing counterparty of its obligations thereunder. See Part I, Item 1A. Risk Factors – *"We will need to obtain sufficient financing or other capital to operate successfully"* for additional discussion.

See Note 11 to AAG's Consolidated Financial Statements in Part II, Item 8A and Note 10 to American's Consolidated Financial Statements in Part II, Item 8B for additional information on aircraft and engine acquisition commitments.

Ground Properties

At each airport where we conduct flight operations, we have agreements, generally with a governmental unit or authority, for the use of passenger, operations and baggage handling space as well as runways and taxiways. These agreements, particularly in the U.S., often contain provisions for periodic adjustments to rates and charges applicable under such agreements. These rates and charges also vary with our level of operations and the operations of the airport. Additionally, at our hub locations and in certain other cities we serve, we lease administrative offices, catering, cargo, training, maintenance and other facilities.

We lease or have built on leased property our headquarters and training facilities in Fort Worth, Texas, our principal overhaul and maintenance base in Tulsa, Oklahoma, our regional reservation offices, and administrative offices throughout the U.S. and abroad.

ITEM 3. LEGAL PROCEEDINGS

See Note 11 to AAG's Consolidated Financial Statements in Part II, Item 8A and Note 10 to American's Consolidated Financial Statements in Part II, Item 8B for information on legal proceedings.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR AMERICAN AIRLINES GROUP'S COMMON STOCK, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Stock Exchange Listing

Our common stock is listed on The Nasdaq Global Select Market under the trading symbol "AAL." There is no trading market for the common stock of American, which is a wholly-owned subsidiary of AAG.

As of February 16, 2024, there were approximately 54,000 holders of record of our common stock. However, because many of the shares of our common stock are held by brokers and other institutions on behalf of stockholders, we believe there are substantially more beneficial holders of our common stock than record holders.

Information on securities authorized for issuance under our equity compensation plans will be set forth in our Proxy Statement for the 2024 Annual Meeting of Stockholders of American Airlines Group Inc. (the Proxy Statement) under the caption "Equity Compensation Plan Information" and is incorporated by reference into this Annual Report on Form 10-K.

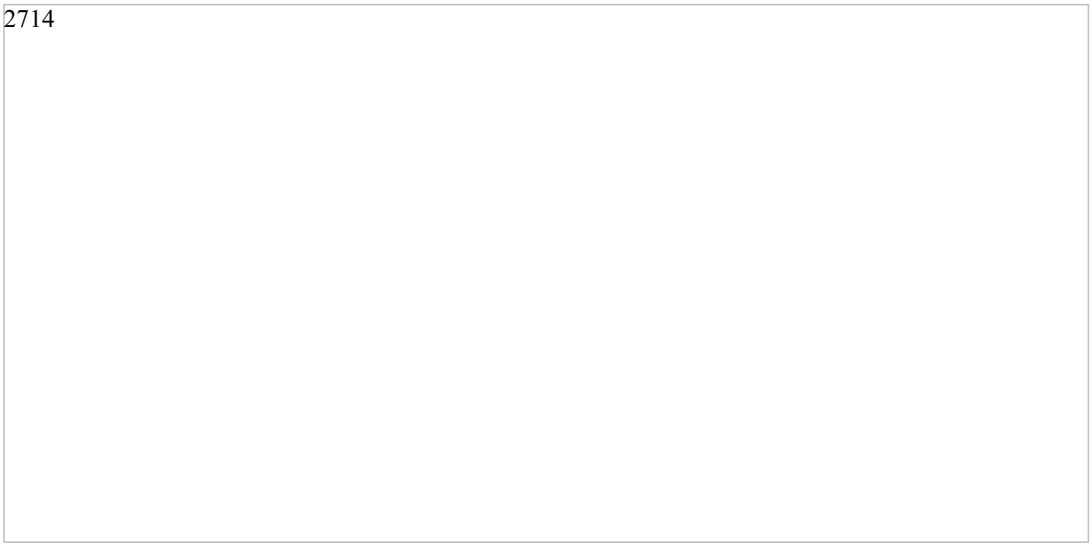
Dividends on Common Stock

There were no cash dividend payments during the years ended December 31, 2023 and 2022. In connection with our receipt of financial assistance under PSP1, PSP2 and PSP3, we agreed not to pay dividends on AAG common stock through September 30, 2022, when this restriction expired. If we determine to make any dividends in the future, such dividends that may be declared and paid from time to time will be subject to market and economic conditions, applicable legal requirements and other relevant factors. We are not obligated to continue a dividend for any fixed period, and the payment of dividends may be suspended or discontinued again at any time at our discretion and without prior notice.

Stock Performance Graph

The following stock performance graph and related information shall not be deemed "soliciting material" or "filed" with the SEC, nor shall such information be incorporated by reference into any future filings under the Securities Act of 1933 or the Exchange Act, each as amended, except to the extent that we specifically incorporate it by reference into such filing.

The following stock performance graph compares the cumulative total stockholder returns during the period from December 31, 2018 to December 31, 2023 of our common stock to the New York Stock Exchange (NYSE) ARCA Airline Index and the Standard and Poor's Financial Services, LLC (S&P) 500 Stock Index. The comparison assumes $100 was invested on December 31, 2018 in our common stock and in each of the foregoing indices and assumes that all dividends were reinvested. The stock performance shown on the following graph represents historical stock performance and is not necessarily indicative of future stock price performance.



	12/31/2018	12/31/2019	12/31/2020	12/31/2021	12/31/2022	12/31/2023
American Airlines Group Inc. (AAL)	$ 100	$ 91	$ 50	$ 57	$ 40	$ 44
NYSE ARCA Airline Index (XAL)	100	121	92	90	58	75
S&P 500 Index (GSPC)	100	129	150	190	153	190

Purchases of Equity Securities by the Issuer and Affiliated Purchasers

The remaining authority under our most recent $2.0 billion share repurchase program expired in December 2020, and in connection with our receipt of financial assistance under PSP1, PSP2 and PSP3, we agreed not to repurchase shares of AAG common stock through September 30, 2022, when this restriction expired. No repurchases of AAG common stock were made in 2023 or 2022 following the lapse of these restrictions. As of December 31, 2023, the Board of Directors of AAG had not authorized another share repurchase program. Any future determination to enter into a share repurchase program will be at the discretion of the Board of Directors, subject to applicable legal limitations, and will depend upon our results of operations, financial condition, contractual restrictions and other factors deemed relevant by the Board of Directors.

See Part I, Item 1A. Risk Factors – "*If we decide to make repurchases of or pay dividends on our common stock, we cannot guarantee that we will continue to do so or that such a capital deployment program will enhance long-term stockholder value.*"

Ownership Restrictions

AAG's Certificate of Incorporation and Bylaws provide that, consistent with the requirements of Subtitle VII of Title 49 of the United States Code, as amended (the Aviation Act), any persons or entities who are not a "citizen of the United States" (as defined under the Aviation Act and administrative interpretations issued by the DOT, its predecessors and successors, from time to time), including any agent, trustee or representative of such persons or entities (a non-citizen), shall not, in the aggregate, own (beneficially or of record) and/or control more than (a) 24.9% of the aggregate votes of all of our outstanding equity securities or (b) 49.0% of our outstanding equity securities. Our Certificate of Incorporation and Bylaws further specify that it is the duty of each stockholder who is a non-citizen to register his, her or its equity securities on our foreign stock record and provide for remedies applicable to stockholders that exceed the voting and ownership caps described above.

In addition, to reduce the risk of a potential adverse effect on our ability to use our NOL carryforwards and certain other tax attributes for federal income tax purposes, and in connection with the expiration in December 2021 of certain transfer restrictions applicable to substantial shareholders contained in our Certificate of Incorporation, the Board of Directors of AAG adopted the Tax Benefit Preservation Plan. The Tax Benefit Preservation Plan was subsequently ratified by our stockholders at the 2022 Annual Meeting of Stockholders of AAG. The Tax Benefit Preservation Plan is designed to reduce the likelihood that we experience an "ownership change" for purposes of Section 382 by deterring certain acquisitions of AAG common stock. There is no assurance, however, that the deterrent mechanism will be effective, and such acquisitions may still occur. In addition, the Tax Benefit Preservation Plan may adversely affect the marketability of AAG common stock by discouraging existing or potential investors from acquiring AAG common stock or additional shares of AAG common stock, because any non-exempt third party that acquires 4.9% or more of the then-outstanding shares of AAG common stock would suffer substantial dilution of its ownership interest in AAG.

See Part I, Item 1A. Risk Factors – *"AAG's Certificate of Incorporation, Bylaws and Tax Benefit Preservation Plan include provisions that limit voting and acquisition and disposition of our equity interests and specify an exclusive forum for certain stockholder disputes"* and "*Our ability to utilize our NOLs and other carryforwards may be limited."* Also see AAG's Certification of Incorporation and Bylaws, which are filed as Exhibits 3.1, 3.2, 3.3 and 3.4 hereto, for the full text of the foregoing restrictions and AAG's Description of the Registrants' Securities Registered Pursuant to Section 12 of the Securities Exchange Act of 1934, which is filed as Exhibit 4.1 hereto, for a more detailed description.

ITEM 6. SELECTED CONSOLIDATED FINANCIAL DATA

Selected Consolidated Financial Data of AAG

The selected consolidated financial data presented below under the captions “Consolidated Statements of Operations data” and “Consolidated Balance Sheet data” for the years ended and as of December 31, 2023, 2022 and 2021, are derived from AAG’s audited consolidated financial statements.

	Year Ended December 31,		
	2023	2022	2021
	(In millions, except share and per share amounts)		
Consolidated Statements of Operations data:			
Total operating revenues	$ 52,788	$ 48,971	$ 29,882
Total operating expenses	49,754	47,364	30,941
Operating income (loss)	3,034	1,607	(1,059)
Net income (loss)	822	127	(1,993)
Earnings (loss) per common share:			
Basic	$ 1.26	$ 0.20	$ (3.09)
Diluted	1.21	0.19	(3.09)
Shares used for computation (in thousands):			
Basic	653,612	650,345	644,015
Diluted	719,669	655,122	644,015
Consolidated Balance Sheet data (at end of period):			
Total assets	$ 63,058	$ 64,716	$ 66,467
Debt and finance leases	32,902	35,663	38,060
Pension and postretirement obligations [1]	3,171	2,926	5,150
Operating lease liabilities	7,761	8,024	8,117
Stockholders’ deficit	(5,202)	(5,799)	(7,340)

[1] Substantially all defined benefit pension plans were frozen effective November 1, 2012. See Note 9 to AAG's Consolidated Financial Statements in Part II, Item 8A for further information on pension and postretirement benefits.

Reconciliation of GAAP to Non-GAAP Financial Measures

We sometimes use financial measures that are derived from the consolidated financial statements but that are not presented in accordance with accounting principles generally accepted in the U.S. (GAAP) to understand and evaluate our current operating performance and to allow for period-to-period comparisons. We believe these non-GAAP financial measures may also provide useful information to investors and others. These non-GAAP measures may not be comparable to similarly titled non-GAAP measures of other companies, and should be considered in addition to, and not as a substitute for or superior to, any measure of performance, cash flow or liquidity prepared in accordance with GAAP. We are providing a reconciliation of reported non-GAAP financial measures to their comparable financial measures on a GAAP basis.

The following table presents the components of our total net special items and the reconciliation of pre-tax income and net income (GAAP measures) to pre-tax income excluding net special items and net income excluding net special items (non-GAAP measures). Management uses these non-GAAP financial measures to evaluate our current operating performance and to allow for period-to-period comparisons. As net special items may vary from period-to-period in nature and amount, the adjustment to exclude net special items allows management an additional tool to understand our core operating performance.

	Year Ended December 31,	
	2023	2022
	(In millions)	
Components of Total Special Items, Net: [1]		
Labor contract expenses [2]	$ 989	$ —
Severance expenses [3]	23	—
Fleet impairment [4]	—	149
Litigation reserve adjustments	—	37
Other operating special items, net	(41)	7
Mainline operating special items, net	971	193
Regional operating special items, net	8	5
Operating special items, net	979	198
Debt refinancing and extinguishment [5]	280	3
Mark-to-market adjustments on equity investments, net [6]	82	71
Nonoperating special items, net	362	74
Pre-tax special items, net	1,341	272
Income tax special items, net	—	(9)
Total special items, net	$ 1,341	$ 263
Reconciliation of Pre-Tax Income Excluding Net Special Items:		
Pre-tax income – GAAP	$ 1,121	$ 186
Adjusted for: Pre-tax special items, net	1,341	272
Pre-tax income excluding net special items	$ 2,462	$ 458
Reconciliation of Net Income Excluding Net Special Items:		
Net income – GAAP	$ 822	$ 127
Adjusted for: Total special items, net	1,341	263
Adjusted for: Net tax effect of net special items	(304)	(62)
Net income excluding net special items	$ 1,859	$ 328

(1) See Note 2 to AAG's Consolidated Financial Statements in Part II, Item 8A for further information on net special items.

(2) Labor contract expenses relate to one-time charges resulting from the ratification of a new collective bargaining agreement with our mainline pilots, including a one-time payment of $754 million as well as adjustments to other benefit-related items of $235 million.

(3) Severance expenses included costs associated with headcount reductions in certain corporate functions.

(4) Fleet impairment included a non-cash impairment charge to write down the carrying value of our retired Airbus A330 fleet to the estimated fair value due to the market conditions for certain used aircraft. We retired our Airbus A330 fleet in 2020 as a result of the decline in demand for air travel due to the COVID-19 pandemic.

(5) Debt refinancing and extinguishment costs in 2023 primarily included cash charges for premiums paid in connection with the early repayment of debt. See Note 4 to AAG's Consolidated Financial Statements in Part II, Item 8A for further information.

(6) Mark-to-market adjustments on equity investments, net included net unrealized gains and losses associated with certain equity investments. See Note 8 to AAG's Consolidated Financial Statements in Part II, Item 8A for further information related to our equity investments.

Additionally, the table below presents the reconciliation of total operating costs (GAAP measure) to total operating costs excluding net special items and fuel (non-GAAP measure) and total operating cost per available seat mile (CASM) to CASM excluding net special items and fuel. Management uses total operating costs excluding net special items and fuel and CASM excluding net special items and fuel to evaluate our current operating performance and for period-to-period comparisons. The price of fuel, over which we have no control, impacts the comparability of period-to-period financial performance. The adjustment to exclude net special items and fuel allows management an additional tool to understand and analyze our non-fuel costs and core operating performance. Amounts may not recalculate due to rounding.

	Year Ended December 31,	
	2023	2022
Reconciliation of CASM Excluding Net Special Items and Fuel:		
(In millions)		
Total operating expenses – GAAP	$ 49,754	$ 47,364
Operating net special items (1):		
Mainline operating special items, net	(971)	(193)
Regional operating special items, net	(8)	(5)
Aircraft fuel and related taxes	(12,257)	(13,791)
Total operating expenses, excluding net special items and fuel	$ 36,518	$ 33,375
(In millions)		
Total Available Seat Miles (ASM)	277,723	260,226
(In cents)		
CASM	17.92	18.20
Operating net special items per ASM (1):		
Mainline operating special items, net	(0.35)	(0.07)
Regional operating special items, net	—	—
Aircraft fuel and related taxes per ASM	(4.41)	(5.30)
CASM, excluding net special items and fuel	13.15	12.83

(1) See Note 2 to AAG's Consolidated Financial Statements in Part II, Item 8A for further information on net special items.

Selected Consolidated Financial Data of American

The selected consolidated financial data presented below under the captions “Consolidated Statements of Operations data” and “Consolidated Balance Sheet data” for the years ended and as of December 31, 2023, 2022 and 2021, are derived from American’s audited consolidated financial statements.

	Year Ended December 31,		
	2023	2022	2021
	(In millions)		
Consolidated Statements of Operations data:			
Total operating revenues	$ 52,784	$ 48,965	$ 29,880
Total operating expenses	49,715	47,312	30,841
Operating income (loss)	3,069	1,653	(961)
Net income (loss)	1,188	338	(1,777)
Consolidated Balance Sheet data (at end of period):			
Total assets	$ 69,074	$ 70,324	$ 71,145
Debt and finance leases	27,675	30,422	32,094
Pension and postretirement obligations [1]	3,148	2,900	5,117
Operating lease liabilities	7,708	7,961	8,074
Stockholder’s equity	6,577	5,593	3,826

[1] Substantially all defined benefit pension plans were frozen effective November 1, 2012. See Note 8 to American's Consolidated Financial Statements in Part II, Item 8B for further information on pension and postretirement benefits.

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

2023 Financial Overview

The selected financial data presented below is derived from AAG's audited consolidated financial statements included in Part II, Item 8A of this report and should be read in conjunction with those financial statements and the related notes thereto.

	Year Ended December 31,		Increase (Decrease)	Percent Increase (Decrease)
	2023	2022		
	(In millions, except percentage changes)			
Passenger revenue	$ 48,512	$ 44,568	$ 3,944	8.8
Cargo revenue	812	1,233	(421)	(34.1)
Other operating revenue	3,464	3,170	294	9.3
Total operating revenues	52,788	48,971	3,817	7.8
Aircraft fuel and related taxes	12,257	13,791	(1,534)	(11.1)
Salaries, wages and benefits	14,580	12,972	1,608	12.4
Total operating expenses	49,754	47,364	2,390	5.0
Operating income	3,034	1,607	1,427	88.8
Pre-tax income	1,121	186	935	nm [2]
Income tax provision	299	59	240	nm
Net income	822	127	695	nm
Pre-tax income – GAAP	$ 1,121	$ 186	$ 935	nm
Adjusted for: pre-tax net special items [1]	1,341	272	1,069	nm
Pre-tax income excluding net special items	$ 2,462	$ 458	$ 2,004	nm

[1] See Part II, Item 6. Selected Consolidated Financial Data – *"Reconciliation of GAAP to Non-GAAP Financial Measures"* and Note 2 to AAG's Consolidated Financial Statements in Part II, Item 8A for details on the components of pre-tax net special items.

[2] Not meaningful or greater than 100% change.

Pre-Tax Income and Net Income

Pre-tax income and net income were $1.1 billion and $822 million, respectively, in 2023. This compares to 2022 pre-tax income and net income of $186 million and $127 million, respectively.

The year-over-year improvement in our pre-tax income on a GAAP basis was driven primarily by higher passenger revenue and lower fuel costs. Continued strength in demand for air travel and a 7.6% increase in revenue passenger miles (RPMs) as compared to 2022 contributed to record passenger revenue in 2023. Aircraft fuel costs were lower primarily due to a 16.3% decrease in the average price per gallon of aircraft fuel as compared to 2022. The increase to pre-tax income was partially offset by increases in certain operating expenses including salaries, wages and benefits costs, primarily as a result of the ratification of a new collective bargaining agreement with our mainline pilots, as described below. The 2023 period also included $1.3 billion of pre-tax net special items, principally related to one-time charges resulting from the new collective bargaining agreement.

Excluding the effects of pre-tax net special items, pre-tax income was $2.5 billion and $458 million in 2023 and 2022, respectively. The year-over-year improvement in our pre-tax income excluding pre-tax net special items was primarily due to record passenger revenue in 2023 and lower fuel costs, offset in part by increases in certain operating expenses including salaries, wages and benefits, as described above.

In May 2023, American and the Allied Pilots Association, the union representing our mainline pilots, reached an agreement in principle on a new collective bargaining agreement, which was ratified in August 2023. This four-year agreement provides wage rate increases, including an initial wage rate increase of 21% effective as of January 1, 2023,

quality-of-life benefits and other benefit-related items. The additional compensation for the 2023 period prior to contract ratification as a result of the higher wage rates was recorded within salaries, wages and benefits in the consolidated statements of operations in the second and third quarters of 2023. The agreement also included a provision for a one-time payment upon ratification. In 2023, one-time charges resulting from the ratification of this new agreement were recorded as mainline operating special items, net in the consolidated statement of operations, including the one-time payment of $754 million as well as adjustments to other benefit-related items of $235 million. The one-time payment and the additional compensation were principally paid in 2023, with remaining payments expected to be paid in the first quarter of 2024.

Revenue

In 2023, we reported total operating revenues of $52.8 billion, an increase of $3.8 billion, or 7.8%, as compared to 2022. Passenger revenue was $48.5 billion, an increase of $3.9 billion, or 8.8%, as compared to 2022. The increase in passenger revenue in 2023 was primarily due to a 7.6% increase in RPMs, driven by the continued strength in demand for air travel, resulting in an 83.5% load factor.

Cargo revenue decreased $421 million, or 34.1%, in 2023 as compared to 2022, primarily due to a 29.4% decrease in cargo yield and a 6.7% decrease in cargo ton miles driven by lower demand and increased air freight capacity globally.

Other operating revenue increased $294 million, or 9.3%, in 2023 as compared to 2022, driven primarily by higher revenue associated with our loyalty program. During 2023 and 2022, cash payments from co-branded credit card and other partners were $5.2 billion and $4.5 billion, respectively.

Our total revenue per available seat mile (TRASM) was 19.01 cents in 2023, a 1.0% increase as compared to 18.82 cents in 2022.

Fuel

In 2023, aircraft fuel expense totaled $12.3 billion, a decrease of $1.5 billion, or 11.1%, as compared to 2022. This decrease was primarily driven by a 16.3% decrease in the average price per gallon of aircraft fuel including related taxes to $2.96 in 2023 from $3.54 in 2022, offset in part by a 6.1% increase in gallons of fuel consumed due to increased capacity.

As of December 31, 2023, we did not have any fuel hedging contracts outstanding to hedge our fuel consumption. Our current policy is not to enter into transactions to hedge our fuel consumption, although we review this policy from time to time based on market conditions and other factors. As such, and assuming we do not enter into any future transactions to hedge our fuel consumption, we will continue to be fully exposed to fluctuations in fuel prices.

Other Costs

We remain committed to actively managing our cost structure, which we believe is necessary in an industry in which economic prospects are heavily dependent upon two variables we cannot control: general economic conditions and the price of fuel.

Our 2023 CASM was 17.92 cents, a decrease of 1.6%, from 18.20 cents in 2022. This decrease in CASM was primarily driven by lower aircraft fuel costs in 2023, as described above, offset by higher salaries, wages and benefits costs associated with the ratification of a new collective bargaining agreement with our mainline pilots, as described above.

Our 2023 CASM excluding net special items and fuel was 13.15 cents, an increase of 2.5%, from 12.83 cents in 2022, which was primarily driven by higher salaries, wages and benefits costs associated with the ratification of a new collective bargaining agreement with our mainline pilots, as described above.

For a reconciliation of total operating CASM to total operating CASM excluding net special items and fuel, see Part II, Item 6. Selected Consolidated Financial Data – *"Reconciliation of GAAP to Non-GAAP Financial Measures."*

Liquidity

As of December 31, 2023, we had $10.4 billion in total available liquidity, consisting of $7.6 billion in unrestricted cash and short-term investments and $2.9 billion in total undrawn capacity under revolving credit and other short-term facilities.

During 2023, we completed the following financing transactions (see Note 4 to AAG's Consolidated Financial Statements in Part II, Item 8A for further information):

- refinanced approximately $1.8 billion in aggregate principal amount of term loans outstanding under the 2013 Term Loan Facility (the 2013 Term Loan Facility Refinancing) through the combination of (i) the issuance of $750 million in aggregate principal amount of 7.25% senior secured notes due 2028 (the 7.25% Senior Secured Notes) and (ii) the entry into the Seventh Amendment to the 2013 Credit Agreement, pursuant to which the maturity of $1.0 billion in term loans under the 2013 Term Loan Facility was extended to February 2028 from June 2025;
- extended the maturity of certain commitments under the 2013 Revolving Facility, 2014 Revolving Facility and April 2016 Revolving Facility to October 2026;
- issued $1.0 billion aggregate principal amount of 8.50% senior secured notes due 2029 (the 8.50% Senior Secured Notes) in a private offering and entered into the 2023 Credit Agreement that provides for a term loan facility (the 2023 Term Loan Facility) in an aggregate principal amount of $1.1 billion. The net proceeds from the offering of the 8.50% Senior Secured Notes, together with net proceeds from borrowings under the 2023 Term Loan Facility and cash on hand were used to redeem all of American's outstanding 11.75% senior secured notes due 2025 (the 11.75% Senior Secured Notes);
- issued $1.1 billion of equipment loans and other notes payable in connection with the financing of certain aircraft; and
- repurchased $552 million of secured and unsecured notes in the open market.

A significant portion of our debt financing agreements contain covenants requiring us to maintain an aggregate of at least $2.0 billion of unrestricted cash and cash equivalents and amounts available to be drawn under revolving credit facilities and/or contain covenants requiring us to meet certain loan to value, collateral coverage and/or peak debt service coverage ratios.

See Note 4 to AAG's Consolidated Financial Statements in Part II, Item 8A for additional information on our debt obligations.

AAG's Results of Operations

For a comparison of the 2022 to 2021 reporting periods, see Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations – "*AAG's Results of Operations*" of our 2022 Form 10-K.

Operating Statistics

The table below sets forth selected operating data for the years ended December 31, 2023 and 2022.

	Year Ended December 31,		Increase (Decrease)
	2023	2022	
Revenue passenger miles (millions) [(a)]	231,926	215,624	7.6%
Available seat miles (millions) [(b)]	277,723	260,226	6.7%
Passenger load factor (percent) [(c)]	83.5	82.9	0.6pts
Yield (cents) [(d)]	20.92	20.67	1.2%
Passenger revenue per available seat mile (cents) [(e)]	17.47	17.13	2.0%
Total revenue per available seat mile (cents) [(f)]	19.01	18.82	1.0%
Fuel consumption (gallons in millions)	4,140	3,901	6.1%
Average aircraft fuel price including related taxes (dollars per gallon)	2.96	3.54	(16.3)%
Total operating cost per available seat mile (cents) [(g)]	17.92	18.20	(1.6)%
Aircraft at end of period [(h)]	1,521	1,461	4.1%
Full-time equivalent employees at end of period	132,100	129,700	1.9%

(a) Revenue passenger mile (RPM) – A basic measure of sales volume. One RPM represents one passenger flown one mile.

(b) Available seat mile (ASM) – A basic measure of production. One ASM represents one seat flown one mile.

(c) Passenger load factor – The percentage of available seats that are filled with revenue passengers.

(d) Yield – A measure of airline revenue derived by dividing passenger revenue by RPMs.

(e) Passenger revenue per available seat mile (PRASM) – Passenger revenue divided by ASMs.

(f) Total revenue per available seat mile (TRASM) – Total revenues divided by ASMs.

(g) Total operating cost per available seat mile (CASM) – Total operating expenses divided by ASMs.

(h) Includes aircraft owned and leased by American as well as aircraft operated by third-party regional carriers under capacity purchase agreements. Excluded from the aircraft count above are 77 regional aircraft in temporary storage as of December 31, 2023 as follows: 57 Embraer 145, 11 Bombardier CRJ 700, six Bombardier CRJ 900 and three Embraer 170.

Operating Revenues

	Year Ended December 31,		Increase (Decrease)	Percent Increase (Decrease)
	2023	2022		
	(In millions, except percentage changes)			
Passenger	$ 48,512	$ 44,568	$ 3,944	8.8
Cargo	812	1,233	(421)	(34.1)
Other	3,464	3,170	294	9.3
Total operating revenues	$ 52,788	$ 48,971	$ 3,817	7.8

This table presents our passenger revenue and the year-over-year change in certain operating statistics:

	Year Ended December 31, 2023	Increase vs. Year Ended December 31, 2022					
		Passenger Revenue	RPMs	ASMs	Load Factor	Passenger Yield	PRASM
	(In millions)						
Passenger revenue	$ 48,512	8.8%	7.6%	6.7%	0.6pts	1.2%	2.0%

Passenger revenue increased $3.9 billion, or 8.8%, in 2023 from 2022 primarily due to continued strength in demand for air travel, resulting in a 7.6% increase in RPMs and an 83.5% load factor in 2023.

Cargo revenue decreased $421 million, or 34.1%, in 2023 from 2022 primarily due to a 29.4% decrease in cargo yield and a 6.7% decrease in cargo ton miles driven by lower demand and increased air freight capacity globally.

Other operating revenue increased $294 million, or 9.3%, in 2023 from 2022 driven primarily by higher revenue associated with our loyalty program. During 2023 and 2022, cash payments from co-branded credit card and other partners were $5.2 billion and $4.5 billion, respectively.

Total operating revenues in 2023 increased $3.8 billion, or 7.8%, from 2022 driven primarily by the increase in passenger revenue as described above. Our TRASM was 19.01 cents in 2023, a 1.0% increase as compared to 18.82 cents in 2022.

Operating Expenses

	Year Ended December 31,		Increase (Decrease)	Percent Increase (Decrease)
	2023	2022		
	(In millions, except percentage changes)			
Aircraft fuel and related taxes	$ 12,257	$ 13,791	$ (1,534)	(11.1)
Salaries, wages and benefits	14,580	12,972	1,608	12.4
Regional expenses	4,643	4,385	258	5.9
Maintenance, materials and repairs	3,265	2,684	581	21.6
Other rent and landing fees	2,928	2,730	198	7.3
Aircraft rent	1,369	1,395	(26)	(1.9)
Selling expenses	1,799	1,815	(16)	(0.9)
Depreciation and amortization	1,936	1,977	(41)	(2.1)
Mainline operating special items, net	971	193	778	nm
Other	6,006	5,422	584	10.8
Total operating expenses	$ 49,754	$ 47,364	$ 2,390	5.0

Additional detail regarding changes in our operating expenses is as follows:

Aircraft fuel and related taxes decreased $1.5 billion, or 11.1%, in 2023 from 2022 primarily due to a 16.3% decrease in the average price per gallon of aircraft fuel including related taxes to $2.96 in 2023 from $3.54 in 2022, offset in part by a 6.1% increase in gallons of fuel consumed due to increased capacity.

Salaries, wages and benefits increased $1.6 billion, or 12.4%, in 2023 from 2022 primarily driven by higher wage rates associated with the ratification of a new collective bargaining agreement with our mainline pilots.

Regional expenses increased $258 million, or 5.9%, in 2023 from 2022 primarily due to increased costs at our wholly-owned regional carriers, including pay rate increases and higher costs for maintenance, materials and repairs driven by an increase in the volume of engine overhauls.

Maintenance, materials and repairs increased $581 million, or 21.6%, in 2023 from 2022 primarily due to increased costs for engine overhauls and airframe heavy checks driven by higher volume, flight hours and cost of materials.

Other rent and landing fees increased $198 million, or 7.3%, in 2023 from 2022 primarily due to rate increases at certain airports, incremental engine leases and higher landing fees.

Selling expenses remained flat in 2023 from 2022 primarily due to a decrease in commissions expense, offset primarily by higher credit card fees driven by the overall increase in passenger revenues.

Other operating expenses increased $584 million, or 10.8%, in 2023 from 2022 primarily driven by the increase in flight operations, including increased costs for onboard food and catering, crew travel, ground handling and airport lounge operations, as well as certain general and administrative expenses.

Operating Special Items, Net

	Year Ended December 31,	
	2023	2022
	(In millions)	
Labor contract expenses [1]	$ 989	$ —
Severance expenses [2]	23	—
Fleet impairment [3]	—	149
Litigation reserve adjustments	—	37
Other operating special items, net	(41)	7
Mainline operating special items, net	971	193
Regional operating special items, net	8	5
Operating special items, net	$ 979	$ 198

[1] Labor contract expenses relate to one-time charges resulting from the ratification of a new collective bargaining agreement with our mainline pilots, including a one-time payment of $754 million as well as adjustments to other benefit-related items of $235 million.

[2] Severance expenses included costs associated with headcount reductions in certain corporate functions.

[3] Fleet impairment included a non-cash impairment charge to write down the carrying value of our retired Airbus A330 fleet to the estimated fair value due to the market conditions for certain used aircraft. We retired our Airbus A330 fleet in 2020 as a result of the decline in demand for air travel due to the COVID-19 pandemic.

Nonoperating Results

	Year Ended December 31,		Increase (Decrease)	Percent Increase (Decrease)
	2023	2022		
	(In millions, except percentage changes)			
Interest income	$ 591	$ 216	$ 375	nm
Interest expense, net	(2,145)	(1,962)	(183)	9.3
Other income (expense), net	(359)	325	(684)	nm
Total nonoperating expense, net	$ (1,913)	$ (1,421)	$ (492)	34.6

Interest income increased $375 million in 2023 compared to 2022 primarily as a result of higher returns on our short-term investments. Interest expense, net increased $183 million, or 9.3%, in 2023 compared to 2022 primarily due to higher interest rates on our variable-rate debt instruments, offset in part by debt repayments.

In 2023, other nonoperating expense, net primarily included $362 million of net special charges principally associated with debt refinancings and extinguishments and mark-to-market net unrealized losses associated with certain equity investments, offset in part by $32 million of non-service related pension and other postretirement benefit plan income.

In 2022, other nonoperating income, net primarily included $424 million of non-service related pension and other postretirement benefit plan income, offset in part by $74 million of net special charges principally for mark-to-market net unrealized losses associated with certain equity investments.

The decrease in non-service related pension and other postretirement benefit plan income in 2023 as compared to 2022 is principally due to an increase in interest cost for the pension and other postretirement benefit obligations driven by higher discount rates and a decrease in expected return on pension plan assets from a reduction in plan assets.

Income Taxes

In 2023, we recorded an income tax provision of $299 million with an effective rate of approximately 27%, which was substantially non-cash. Substantially all of our income before income taxes is attributable to the United States. At December 31, 2023, we had approximately $13.7 billion of gross federal NOLs and $4.7 billion of other carryforwards available to reduce future federal taxable income, of which $3.4 billion will expire beginning in 2029 if unused and $15.0 billion can be carried forward indefinitely. We also had approximately $5.5 billion of NOL carryforwards to reduce future state taxable income at December 31, 2023, which will expire in taxable years 2023 through 2043 if unused.

In 2022, we recorded an income tax provision of $59 million at an effective rate of approximately 32%, which was substantially non-cash.

See Note 6 to AAG's Consolidated Financial Statements in Part II, Item 8A for additional information on income taxes.

American's Results of Operations

For a comparison of the 2022 to 2021 reporting periods, see Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations – "*American's Results of Operations"* of American's 2022 Form 10-K.

Operating Revenues

	Year Ended December 31,			
	2023	2022	Increase (Decrease)	Percent Increase (Decrease)
	(In millions, except percentage changes)			
Passenger	$ 48,512	$ 44,568	$ 3,944	8.8
Cargo	812	1,233	(421)	(34.1)
Other	3,460	3,164	296	9.3
Total operating revenues	$ 52,784	$ 48,965	$ 3,819	7.8

Passenger revenue increased $3.9 billion, or 8.8%, in 2023 from 2022 primarily due to continued strength in demand for air travel, resulting in an increase in RPMs and an increase in load factor in 2023.

Cargo revenue decreased $421 million, or 34.1%, in 2023 from 2022 primarily due to decreases in cargo yield and cargo ton miles driven by lower demand and increased air freight capacity globally.

Other operating revenue increased $296 million, or 9.3%, in 2023 from 2022 driven primarily by higher revenue associated with American's loyalty program. During 2023 and 2022, cash payments from co-branded credit card and other partners were $5.2 billion and $4.5 billion, respectively.

Total operating revenues in 2023 increased $3.8 billion, or 7.8%, from 2022 driven primarily by the increase in passenger revenue as described above.

Operating Expenses

	Year Ended December 31,		Increase (Decrease)	Percent Increase (Decrease)
	2023	2022		
	(In millions, except percentage changes)			
Aircraft fuel and related taxes	$ 12,257	$ 13,791	$ (1,534)	(11.1)
Salaries, wages and benefits	14,572	12,965	1,607	12.4
Regional expenses	4,619	4,345	274	6.3
Maintenance, materials and repairs	3,265	2,684	581	21.6
Other rent and landing fees	2,928	2,730	198	7.3
Aircraft rent	1,369	1,395	(26)	(1.9)
Selling expenses	1,799	1,815	(16)	(0.9)
Depreciation and amortization	1,927	1,969	(42)	(2.2)
Mainline operating special items, net	971	193	778	nm
Other	6,008	5,425	583	10.8
Total operating expenses	$ 49,715	$ 47,312	$ 2,403	5.1

Additional detail regarding changes in American's operating expenses is as follows:

Aircraft fuel and related taxes decreased $1.5 billion, or 11.1%, in 2023 from 2022 primarily due to a 16.3% decrease in the average price per gallon of aircraft fuel including related taxes to $2.96 in 2023 from $3.54 in 2022, offset in part by a 6.1% increase in gallons of fuel consumed due to increased capacity.

Salaries, wages and benefits increased $1.6 billion, or 12.4%, in 2023 from 2022 primarily driven by higher wage rates associated with the ratification of a new collective bargaining agreement with American's mainline pilots.

Regional expenses increased $274 million, or 6.3%, in 2023 from 2022 primarily due to contractual rate increases with American's regional carriers.

Maintenance, materials and repairs increased $581 million, or 21.6%, in 2023 from 2022 primarily due to increased costs for engine overhauls and airframe heavy checks driven by higher volume, flight hours and cost of materials.

Other rent and landing fees increased $198 million, or 7.3%, in 2023 from 2022 primarily due to rate increases at certain airports, incremental engine leases and higher landing fees.

Selling expenses remained flat in 2023 from 2022 primarily due to a decrease in commissions expense, offset primarily by higher credit card fees driven by the overall increase in passenger revenues.

Other operating expenses increased $583 million, or 10.8%, in 2023 from 2022 primarily driven by the increase in flight operations, including increased costs for onboard food and catering, crew travel, ground handling and airport lounge operations, as well as certain general and administrative expenses.

Operating Special Items, Net

	Year Ended December 31,	
	2023	2022
	(In millions)	
Labor contract expenses [1]	$ 989	$ —
Severance expenses [2]	23	—
Fleet impairment [3]	—	149
Litigation reserve adjustments	—	37
Other operating special items, net	(41)	7
Mainline operating special items, net	$ 971	$ 193

(1) Labor contract expenses relate to one-time charges resulting from the ratification of a new collective bargaining agreement with American's mainline pilots, including a one-time payment of $754 million as well as adjustments to other benefit-related items of $235 million.

(2) Severance expenses included costs associated with headcount reductions in certain corporate functions.

(3) Fleet impairment included a non-cash impairment charge to write down the carrying value of American's retired Airbus A330 fleet to the estimated fair value due to the market conditions for certain used aircraft. American retired its Airbus A330 fleet in 2020 as a result of the decline in demand for air travel due to the COVID-19 pandemic.

Nonoperating Results

	Year Ended December 31,		Increase (Decrease)	Percent Increase (Decrease)
	2023	2022		
	(In millions, except percentage changes)			
Interest income	$ 1,078	$ 349	$ 729	nm
Interest expense, net	(2,206)	(1,872)	(334)	17.8
Other income (expense), net	(359)	324	(683)	nm
Total nonoperating expense, net	$ (1,487)	$ (1,199)	$ (288)	24.0

Interest income increased $729 million in 2023 compared to 2022 primarily as a result of higher returns on American's short-term investments and related party receivables from AAG. Interest expense, net increased $334 million, or 17.8%, in 2023 compared to 2022 primarily due to higher interest rates on American's variable-rate debt instruments and related party payables from AAG's wholly-owned subsidiaries, offset in part by debt repayments.

In 2023, other nonoperating expense, net primarily included $362 million of net special charges principally associated with debt refinancings and extinguishments and mark-to-market net unrealized losses associated with certain equity investments, offset in part by $33 million of non-service related pension and other postretirement benefit plan income.

In 2022, other nonoperating income, net primarily included $423 million of non-service related pension and other postretirement benefit plan income, offset in part by $72 million of net special charges principally for mark-to-market net unrealized losses associated with certain equity investments.

The decrease in non-service related pension and other postretirement benefit plan income in 2023 as compared to 2022 is principally due to an increase in interest cost for the pension and other postretirement benefit obligations driven by higher discount rates and a decrease in expected return on pension plan assets from a reduction in plan assets.

Income Taxes

American is a member of AAG's consolidated federal and certain state income tax returns.

In 2023, American recorded an income tax provision of $394 million with an effective rate of approximately 25%, which was substantially non-cash. Substantially all of American's income before income taxes is attributable to the United States. At December 31, 2023, American had approximately $13.7 billion of gross federal NOLs and $3.6 billion of other carryforwards available to reduce future federal taxable income, of which $3.8 billion will expire beginning in 2033 if unused and $13.5 billion can be carried forward indefinitely. American also had approximately $5.3 billion of NOL carryforwards to reduce future state taxable income at December 31, 2023, which will expire in taxable years 2023 through 2043 if unused.

In 2022, American recorded an income tax provision of $116 million at an effective rate of approximately 26%, which was substantially non-cash.

See Note 5 to American's Consolidated Financial Statements in Part II, Item 8B for additional information on income taxes.

Liquidity and Capital Resources

Liquidity

At December 31, 2023, AAG had $10.4 billion in total available liquidity and $910 million in restricted cash and short-term investments. Additional detail regarding our available liquidity is provided in the table below (in millions):

	AAG		American	
	December 31,		December 31,	
	2023	2022	2023	2022
Cash	$ 578	$ 440	$ 567	$ 429
Short-term investments	7,000	8,525	6,998	8,523
Undrawn facilities	2,862	3,033	2,862	3,033
Total available liquidity	$ 10,440	$ 11,998	$ 10,427	$ 11,985

In the ordinary course of our business, we or our affiliates may, at any time and from time to time, seek to prepay, retire or repurchase our outstanding debt through cash purchases and/or exchanges for equity or debt, in open-market purchases, privately negotiated transactions or otherwise. Such repurchases, prepayments, retirements or exchanges, if any, will be conducted on such terms and at such prices as we may determine, and will depend on prevailing market conditions, our liquidity requirements, legal and contractual restrictions and other factors. The amounts involved may be material. For further information regarding our debt repurchases for the year ended December 31, 2023, see Note 4 to AAG's Consolidated Financial Statements in Part II, Item 8A and Note 3 to American's Consolidated Financial Statements in Part II, Item 8B.

Certain Covenants

Our debt agreements contain customary terms and conditions as well as various affirmative, negative and financial covenants that, among other things, may restrict the ability of us and our subsidiaries to incur additional indebtedness, pay dividends or repurchase stock. Our debt agreements also contain customary change of control provisions, which may require us to repay or redeem such indebtedness upon certain events constituting a change of control under the relevant agreement, in certain cases at a premium. Additionally, certain of our debt financing agreements (including our secured notes, term loans, revolving credit facilities and spare engine enhanced equipment trust certificates (EETCs)) contain loan to value (LTV), collateral coverage or peak debt service coverage ratio covenants and certain agreements require us to appraise the related collateral annually or semiannually. Pursuant to such agreements, if the applicable LTV, collateral coverage or peak debt service coverage ratio exceeds or falls below a specified threshold, as the case may be, we will be required, as applicable, to pledge additional qualifying collateral (which in some cases may include cash or investment securities), withhold additional cash in certain accounts, or pay down such financing, in whole or in part, or the interest rate for the relevant financing will be increased. As of the most recent applicable measurement dates, we were in compliance with each of the foregoing LTV, collateral coverage and peak debt service coverage tests. Additionally, a significant portion of our debt financing agreements contain covenants requiring us to maintain an aggregate of at least $2.0 billion of unrestricted cash and cash equivalents and amounts available to be drawn under revolving credit facilities, and our AAdvantage Financing contains a peak debt service coverage ratio, pursuant to which failure to comply with a certain threshold may result in early repayment, in whole or in part, of the AAdvantage Financing. For further information regarding our debt covenants, see Note 4 to AAG's Consolidated Financial Statements in Part II, Item 8A and Note 3 to American's Consolidated Financial Statements in Part II, Item 8B.

Sources and Uses of Cash

For a comparison of the 2022 and 2021 reporting periods, see Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations – "*Sources and Uses of Cash*" of our 2022 Form 10-K.

AAG

Operating Activities

Our net cash provided by operating activities was $3.8 billion and $2.2 billion in 2023 and 2022, respectively, a $1.6 billion year-over-year increase due to higher profitability and cash provided by working capital management.

Investing Activities

Our net cash used in investing activities was $502 million in 2023 as compared to net cash provided by investing activities of $636 million in 2022.

Our principal investing activities in 2023 included $2.6 billion of capital expenditures, which primarily related to the purchase of 17 Boeing 737-8 MAX aircraft, ten Airbus A321neo aircraft, seven Embraer 175 aircraft, seven Bombardier CRJ 900 aircraft and 28 spare engines. These cash outflows were offset in part by $1.5 billion in net sales of short-term investments and $230 million of proceeds from the sale of property and equipment and sale-leaseback transactions.

Our principal investing activities in 2022 included $3.7 billion in net sales of short-term investments. These cash inflows were offset in part by $2.5 billion of capital expenditures, which primarily related to the purchase of 24 Airbus A321neo aircraft and 12 spare engines, and $321 million of equity investments, principally related to GOL.

Financing Activities

Our net cash used in financing activities was $3.2 billion and $2.6 billion in 2023 and 2022, respectively.

Our principal financing activities in 2023 included $2.9 billion in net repayments of debt and finance lease obligations primarily due to scheduled debt repayments. In February 2023, we refinanced approximately $1.8 billion in aggregate principal amount of term loans outstanding under the 2013 Term Loan Facility through the combination of (i) issuing $750 million in aggregate principal amount of the 7.25% Senior Secured Notes and (ii) extending the maturity of $1.0 billion in term loans under the 2013 Term Loan Facility. In December 2023, we issued $1.0 billion aggregate principal amount of the 8.50% Senior Secured Notes and entered into the 2023 Term Loan Facility in an aggregate principal amount of $1.1 billion. The net proceeds of the 8.50% Senior Secured Notes, together with net proceeds from borrowings under the 2023 Term Loan Facility and cash on hand, were used to redeem all of the outstanding 11.75% Senior Secured Notes. In addition, we borrowed $1.1 billion in connection with the financing of certain aircraft and repurchased $552 million of secured and unsecured notes in the open market.

Our principal financing activities in 2022 included $3.8 billion in repayments of debt and finance lease obligations, consisting of $2.2 billion of scheduled debt repayments including the repayment of $401 million in connection with the maturity of our 5.000% unsecured notes, the $1.2 billion prepayment of the December 2016 Term Loan Facility and the repurchase of $349 million of unsecured notes in the open market. These cash outflows were offset in part by $1.1 billion of long-term debt proceeds, consisting of $866 million from the issuance of equipment notes related to EETCs and $205 million in connection with the financing of certain aircraft.

American

Operating Activities

American's net cash provided by operating activities was $3.7 billion and $1.3 billion in 2023 and 2022, respectively, a $2.4 billion year-over-year increase due to higher profitability and cash provided by working capital management.

Investing Activities

American's net cash used in investing activities was $449 million in 2023 as compared to net cash provided by investing activities of $693 million in 2022.

American's principal investing activities in 2023 included $2.5 billion of capital expenditures, which primarily related to the purchase of 17 Boeing 737-8 MAX aircraft, ten Airbus A321neo aircraft, seven Embraer 175 aircraft, seven Bombardier CRJ 900 aircraft and 28 spare engines. These cash outflows were offset in part by $1.5 billion in net sales of short-term investments and $230 million of proceeds from the sale of property and equipment and sale-leaseback transactions.

American's principal investing activities in 2022 included $3.7 billion in net sales of short-term investments. These cash inflows were offset in part by $2.5 billion of capital expenditures, which primarily related to the purchase of 24 Airbus A321neo aircraft and 12 spare engines, and $321 million of equity investments, principally related to GOL.

Financing Activities

American's net cash used in financing activities was $3.2 billion and $1.8 billion in 2023 and 2022, respectively.

American's principal financing activities in 2023 included $2.9 billion in net repayments of debt and finance lease obligations primarily due to scheduled debt repayments. In February 2023, American refinanced approximately $1.8 billion in aggregate principal amount of term loans outstanding under the 2013 Term Loan Facility through the combination of (i) issuing $750 million in aggregate principal amount of the 7.25% Senior Secured Notes and (ii) extending the maturity of $1.0 billion in term loans under the 2013 Term Loan Facility. In December 2023, American issued $1.0 billion aggregate principal amount of the 8.50% Senior Secured Notes and entered into the 2023 Term Loan Facility in an aggregate principal amount of $1.1 billion. The net proceeds of the 8.50% Senior Secured Notes, together with net proceeds from borrowings under the 2023 Term Loan Facility and cash on hand, were used to redeem all of the outstanding 11.75% Senior Secured Notes. In addition, American borrowed $1.1 billion in connection with the financing of certain aircraft and repurchased $539 million of secured notes in the open market.

American's principal financing activities in 2022 included $3.0 billion in repayments of debt and finance lease obligations, consisting of $1.8 billion of scheduled debt repayments and the $1.2 billion prepayment of the December 2016 Term Loan Facility. These cash outflows were offset in part by $1.1 billion of long-term debt proceeds, consisting of $866 million from the issuance of equipment notes related to EETCs and $205 million in connection with the financing of certain aircraft.

Commitments

For further information regarding our commitments, see the Notes to AAG's Consolidated Financial Statements in Part II, Item 8A and the Notes to American's Consolidated Financial Statements in Part II, Item 8B at the referenced footnotes below.

	AAG	American
Debt	Note 4	Note 3
Leases	Note 5	Note 4
Employee Benefit Plans	Note 9	Note 8
Commitments, Contingencies and Guarantees	Note 11	Note 10

Off-Balance Sheet Arrangements

An off-balance sheet arrangement is any transaction, agreement or other contractual arrangement involving an unconsolidated entity under which a company has (1) made guarantees, (2) a retained or a contingent interest in transferred assets, (3) an obligation under derivative instruments classified as equity or (4) any obligation arising out of a material variable interest in an unconsolidated entity that provides financing, liquidity, market risk or credit risk support to us, or that engages in leasing, hedging or research and development arrangements with us.

We have no off-balance sheet arrangements of the types described in the first three categories above that we believe may have a material current or future effect on financial condition, liquidity or results of operations.

Pass-Through Trusts

American currently has 308 owned aircraft and 60 owned spare aircraft engines, which in each case were financed with EETCs issued by pass-through trusts. These trusts are off-balance sheet entities, the primary purpose of which is to finance the acquisition of flight equipment or to permit issuance of debt backed by existing flight equipment. In the case of aircraft EETCs, rather than finance each aircraft separately when such aircraft is purchased, delivered or refinanced, these trusts allow American to raise the financing for a number of aircraft at one time and, if applicable, place such funds in escrow pending a future purchase, delivery or refinancing of the relevant aircraft. Similarly, in the case of the spare engine EETCs, the trusts allow American to use its existing pool of spare engines to raise financing under a single facility. The trusts have also been structured to provide for certain credit enhancements, such as liquidity facilities to cover certain interest payments, that reduce the risks to the purchasers of the trust certificates and, as a result, reduce the cost of aircraft financing to American.

Each trust covers a set number of aircraft or spare engines scheduled to be delivered, financed or refinanced upon the issuance of the EETC or within a specific period of time thereafter. At the time of each covered aircraft or spare engine financing, the relevant trust used the proceeds from the issuance of the EETC (which may have been available at the time of issuance thereof or held in escrow until financing of the applicable aircraft following its delivery) to purchase equipment notes relating to the financed aircraft or engines. The equipment notes are issued, at American's election, in connection with a mortgage financing of the aircraft or spare engines. The equipment notes are secured by a security interest in the aircraft or engines, as applicable. The pass-through trust certificates are not direct obligations of, nor are they guaranteed by, AAG or American. However, the equipment notes issued to the trusts are direct obligations of American and, in certain instances, have been guaranteed by AAG. As of December 31, 2023, $7.7 billion associated with these mortgage financings is reflected as debt in the accompanying consolidated balance sheet.

Letters of Credit and Other

We provide financial assurance, such as letters of credit and surety bonds, primarily to support projected workers' compensation obligations and airport commitments. As of December 31, 2023, we had $318 million of letters of credit and surety bonds securing various obligations, of which $94 million is collateralized with our restricted cash. The letters of credit and surety bonds that are subject to expiration will expire on various dates through 2028.

Contractual Obligations

The following table provides details of our estimated material cash requirements from contractual obligations as of December 31, 2023 (in millions). The table does not include commitments that are contingent on events or other factors that are uncertain or unknown at this time and is subject to other conventions as set forth in the applicable accompanying footnotes.

	Payments Due by Period						
	2024	2025	2026	2027	2028	2029 and Thereafter	Total
American [a]							
Long-term debt:							
Principal amount [b], [d] (See Note 3)	$ 3,502	$ 3,702	$ 4,582	$ 4,618	$ 5,060	$ 6,062	$ 27,526
Interest obligations [c], [d]	1,679	1,341	1,037	737	475	492	5,761
Finance lease obligations (See Note 4)	156	140	114	71	30	89	600
Aircraft and engine purchase commitments [e] (See Note 10(a))	2,410	3,725	3,580	1,118	829	645	12,307
Operating lease commitments (See Note 4)	1,821	1,565	1,347	1,182	1,052	4,051	11,018
Regional capacity purchase agreements [f] (See Note 10(b))	2,038	1,992	1,702	1,473	693	1,332	9,230
Minimum pension obligations [g] (See Note 8)	280	251	244	165	140	65	1,145
Retiree medical and other postretirement benefits (See Note 8)	125	131	137	137	135	606	1,271
Other purchase obligations [h] (See Note 10(a))	4,673	2,044	1,396	150	124	843	9,230
Total American Contractual Obligations	$ 16,684	$ 14,891	$ 14,139	$ 9,651	$ 8,538	$ 14,185	$ 78,088
AAG Parent and Other AAG Subsidiaries [a]							
Long-term debt:							
Principal amount [b] (See Note 4)	$ —	$ 1,487	$ —	$ —	$ —	$ 3,746	$ 5,233
Interest obligations [c]	121	145	156	190	192	399	1,203
Finance lease obligations (See Note 5)	10	—	—	—	—	—	10
Operating lease commitments (See Note 5)	20	12	9	5	3	26	75
Minimum pension obligations [g] (See Note 9)	4	2	2	1	1	3	13
Total AAG Contractual Obligations	$ 16,839	$ 16,537	$ 14,306	$ 9,847	$ 8,734	$ 18,359	$ 84,622

[a] For additional information, see the Notes to AAG's and American's Consolidated Financial Statements in Part II, Items 8A and 8B, respectively, referenced in the table above.

[b] Amounts represent contractual amounts due. Excludes $349 million and $14 million of unamortized debt discount, premium and issuance costs as of December 31, 2023 for American and AAG Parent, respectively.

[c] For variable-rate debt, future interest obligations are estimated using the current forward rates at December 31, 2023.

[d] Includes $7.7 billion of future principal payments and $1.0 billion of future interest payments as of December 31, 2023, related to EETCs associated with mortgage financings of certain aircraft and spare engines.

[e] See Part I, Item 2. Properties – *"Aircraft and Engine Purchase Commitments"* for additional information about the firm commitment aircraft delivery schedule, in particular the footnote to the table thereunder as to potential changes to such delivery schedule. Due to uncertainty surrounding the timing of delivery of certain aircraft, the amounts in the table represent our most current estimate based on contractual delivery schedules adjusted for updates and revisions to such schedules communicated to management by the applicable equipment manufacturer. However, the actual delivery schedule may differ, potentially materially, based on various potential factors including production delays by the manufacturer and regulatory concerns. Additionally, the amounts in the table exclude five Boeing 787 Family

aircraft scheduled to be delivered in 2024, for which we have obtained committed lease financing. This financing is reflected in the operating lease commitments line above.

(f) Represents minimum payments under capacity purchase agreements with third-party regional carriers. These commitments are estimates of costs based on assumed minimum levels of flying under the capacity purchase agreements and American's actual payments could differ materially. Rental payments under operating leases for certain aircraft flown under these capacity purchase agreements are reflected in the operating lease commitments line above.

(g) Represents minimum pension contributions based on actuarially determined estimates as of December 31, 2023 and is based on estimated payments through 2033. In January 2024, we made $280 million of required pension contributions.

(h) Includes purchase commitments for aircraft fuel, flight equipment maintenance and information technology support and excludes obligations under certain fuel offtake agreements or other agreements for which the timing of the related expenditure is uncertain, or which are subject to material contingencies, such as the construction of a production facility.

Capital Raising Activity and Other Possible Actions

In light of our significant financial commitments related to, among other things, the servicing and amortization of existing debt and equipment leasing arrangements and new flight equipment, we and our subsidiaries will regularly consider, and enter into negotiations related to, capital raising and liability management activity, which may include the entry into leasing transactions and future issuances of, and transactions designed to manage the timing and amount of, secured or unsecured debt obligations or additional equity or equity-linked securities in public or private offerings or otherwise. The cash available from operations (if any) and these sources, however, may not be sufficient to cover our cash obligations because economic factors may reduce the amount of cash generated by operations or increase costs. For instance, an economic downturn or general global instability caused by military actions, terrorism, disease outbreaks (such as the COVID-19 pandemic), natural disasters or other causes could reduce the demand for air travel, which would reduce the amount of cash generated by operations. See Part I, Item 1A. Risk Factors – *"Downturns in economic conditions could adversely affect our business"* for additional discussion. An increase in costs, either due to an increase in borrowing costs caused by a reduction in credit ratings or a general increase in interest rates, or due to an increase in the cost of fuel, maintenance, aircraft, aircraft engines or parts, could decrease the amount of cash available to cover cash contractual obligations. Moreover, certain of our financing arrangements contain significant minimum cash balance or similar liquidity requirements. As a result, we cannot use all of our available cash to fund operations, capital expenditures and cash obligations without violating these requirements. See Note 4 to AAG's Consolidated Financial Statements in Part II, Item 8A and Note 3 to American's Consolidated Financial Statements in Part II, Item 8B for information regarding our financing arrangements.

In the past, we have from time to time refinanced, redeemed or repurchased our debt and taken other steps to reduce or otherwise manage the aggregate amount and cost of our debt, lease and other obligations or otherwise improve our balance sheet. Going forward, depending on market conditions, our cash position and other considerations, we may continue to take such actions, and the amounts involved may be material.

OTHER INFORMATION

Basis of Presentation

See Note 1 to each of AAG's and American's Consolidated Financial Statements in Part II, Items 8A and 8B, respectively, for information regarding the basis of presentation.

Critical Accounting Policies and Estimates

The preparation of financial statements in accordance with GAAP requires management to make certain estimates and assumptions that affect the reported amounts of assets and liabilities, revenues and expenses, and the disclosure of contingent assets and liabilities at the date of the financial statements. We believe our estimates and assumptions are reasonable; however, actual results could differ from those estimates. Critical accounting policies are defined as those that are reflective of significant judgments and uncertainties and could potentially result in materially different results under different assumptions and conditions. We have identified the following critical accounting policies that impact the preparation of our consolidated financial statements. See the "*Basis of Presentation and Summary of Significant Accounting Policies* " included in Note 1 to each of AAG's and American's Consolidated Financial Statements in Part II, Items 8A and 8B, respectively, for additional discussion of the application of these estimates and other accounting policies.

Passenger Revenue

We recognize all revenues generated from transportation on American and our regional flights operated under the brand name American Eagle, including associated baggage fees and other inflight services, as passenger revenue when transportation is provided. Ticket and other related sales for transportation that has not yet been provided are initially deferred and recorded as air traffic liability on our consolidated balance sheets. The air traffic liability principally represents tickets sold for future travel on American and partner airlines.

The contract duration of passenger tickets is generally one year. The majority of tickets sold are nonrefundable. A small percentage of tickets, some of which are partially used tickets, expire unused. The estimate for tickets expected to expire unused is generally based on an analysis of our historical data and other current applicable factors such as policy changes. We have consistently applied this accounting method to estimate and recognize revenue from unused tickets at the date of travel. This estimate is periodically evaluated based on subsequent activity to validate its accuracy. Any adjustments resulting from periodic evaluations of the estimated air traffic liability are included in passenger revenue during the period in which the evaluations are completed.

Loyalty Revenue

We currently operate the loyalty program, AAdvantage. This program awards mileage credits to passengers who fly on American, any **one**world airline or other partner airlines, or by using the services of other program participants, such as our co-branded credit cards, and certain hotels and car rental companies. Mileage credits can be redeemed for travel on American and other participating partner airlines, as well as non-air travel awards such as hotels and rental cars. For mileage credits earned by AAdvantage program members, we apply the deferred revenue method.

Mileage credits earned through travel

For mileage credits earned through travel, we apply a relative selling price approach whereby the total amount collected from each passenger ticket sale is allocated between the air transportation and the mileage credits earned. The portion of each passenger ticket sale attributable to mileage credits earned is initially deferred and then recognized in passenger revenue when mileage credits are redeemed and transportation is provided. The estimated selling price of mileage credits is determined using an equivalent ticket value approach, which uses historical data, including award redemption patterns by geographic region and class of service, as well as similar cash fares as those used to settle award redemptions. The estimated selling price of mileage credits is adjusted for an estimate of mileage credits that will not be redeemed using a statistical model based on historical redemption patterns to develop an estimate of the likelihood of future redemption. For the year ended December 31, 2023, a hypothetical 10% increase in the estimated selling price of mileage credits would have decreased revenues by approximately $128 million as a result of additional amounts deferred from passenger ticket sales to be recognized in future periods.

Mileage credits sold to co-branded credit cards and other partners

We sell mileage credits to participating airline partners and non-airline business partners, including our co-branded credit card partners, under contracts with remaining terms generally from one to six years as of December 31, 2023. Consideration received from the sale of mileage credits is variable and payment terms typically are within 30 days subsequent to the month of mileage sale. Sales of mileage credits to non-airline business partners are comprised of two components, transportation and marketing. We allocate the consideration received from these sales of mileage credits based on the relative selling price of each product or service delivered.

Our most significant mileage credit partner agreements are our co-branded credit card agreements with Citi and Barclaycard US. We identified two revenue elements in these co-branded credit card agreements: the transportation component and the marketing component.

The transportation component represents the estimated selling price of future travel awards and is determined using the same equivalent ticket value approach described above. The portion of each mileage credit sold attributable to transportation is initially deferred and then recognized in passenger revenue when mileage credits are redeemed and transportation is provided.

The marketing component includes the use of intellectual property, including the American brand and access to loyalty program member lists, which is the predominant element in these agreements, as well as advertising and other travel-related benefits. We recognize the marketing component in other revenue in the period of the mileage credit sale following the sales-based royalty method.

For the portion of our outstanding mileage credits that we estimate will not be redeemed, we recognize the associated value proportionally as the remaining mileage credits are redeemed. Our estimates use a statistical model based on historical redemption patterns to develop an estimate of the likelihood of future redemption. For the year ended December 31, 2023, a hypothetical 10% increase in our estimate of mileage credits not expected to be redeemed would have increased revenues by approximately $127 million.

Pensions and Retiree Medical and Other Postretirement Benefits

We recognize the funded status (i.e., the difference between the fair value of plan assets and the projected benefit obligations) of our pension and retiree medical and other postretirement benefits plans on the consolidated balance sheets with a corresponding adjustment to accumulated other comprehensive income (loss).

Our pension and retiree medical and other postretirement benefits costs and liabilities are calculated using various actuarial assumptions and methodologies. We use certain assumptions including, but not limited to, the selection of the: (1) discount rate and (2) expected return on plan assets (as discussed below). These assumptions as of December 31 were:

	2023	2022
Pension weighted average discount rate [1]	5.2 %	5.6 %
Retiree medical and other postretirement benefits weighted average discount rate [1]	5.3 %	5.7 %
Expected rate of return on plan assets [2]	8.0 %	8.0 %

[1] When establishing the discount rate to measure our obligations, we match high quality corporate bonds available in the marketplace whose cash flows approximate our projected benefit disbursements. Lowering the discount rate by 50 basis points as of December 31, 2023 would increase our pension and retiree medical and other postretirement benefits obligations by approximately $740 million and $40 million, respectively, and decrease estimated 2024 pension and retiree medical and other postretirement benefits expense by approximately $5 million and $1 million, respectively.

[2] The expected rate of return on plan assets is based upon an evaluation of our historical trends and experience, taking into account current and expected market conditions and our target asset allocation of 30% U.S. fixed income securities, 24% U.S. stocks, 24% private investments, 16% developed international stocks and 6% emerging market stocks. The expected rate of return on plan assets component of our net periodic benefit cost is calculated based on the fair value of plan assets and our target asset allocation. Lowering the expected long-term rate of return on plan assets by 50 basis points as of December 31, 2023 would increase estimated 2024 pension expense and retiree medical and other postretirement benefits expense by approximately $60 million and $1 million, respectively.

Annually, we review and revise certain economic and demographic assumptions including the pension and retiree medical and other postretirement benefits discount rates and health care costs. The net effect of changing these assumptions for the pension plans resulted in an increase of $546 million in the projected benefit obligation at December 31, 2023. The net effect of changing these assumptions for retiree medical and other postretirement benefits plans resulted in an increase of $89 million in the accumulated postretirement benefit obligation at December 31, 2023.

See Note 9 to AAG's Consolidated Financial Statements in Part II, Item 8A and Note 8 to American's Consolidated Financial Statements in Part II, Item 8B for additional information regarding our employee benefit plans.

Income Taxes

Our ability to use our NOLs and other carryforwards depends on the amount of taxable income generated in future periods. We provide a valuation allowance for our deferred tax assets when it is more likely than not that some portion, or all of our deferred tax assets, will not be realized. We consider all available positive and negative evidence and make certain assumptions in evaluating the realizability of our deferred tax assets. Many factors are considered that impact our assessment of future profitability, including conditions which are beyond our control, such as the health of the economy, the availability and price volatility of aircraft fuel and travel demand. We have determined that positive factors outweigh negative factors in the determination of the realizability of our deferred tax assets. There can be no assurance that an additional valuation allowance on our net deferred tax assets will not be required. Such valuation allowance could be material.

Recent Accounting Pronouncements

Accounting Standards Update (ASU) 2023-07: Segment Reporting (Topic 280) Improvements to Reportable Segment Disclosures

This standard improves reportable segment disclosure requirements, primarily through enhanced disclosures about significant segment expenses. The amendments in this update are effective for fiscal years beginning after December 15, 2023 and interim periods within fiscal years beginning after December 15, 2024, and early adoption is permitted. We are currently evaluating how the adoption of this standard will impact our reportable segment disclosures.

ASU 2023-09: Income Taxes (Topic 740) Improvements to Income Tax Disclosures

This standard enhances transparency of income tax information through improvements to income tax disclosures primarily related to the rate reconciliation and income taxes paid information, as well as improvements to the effectiveness and comparability of other income tax disclosures. The amendments in this update are effective for annual periods beginning after December 15, 2024, and early adoption is permitted. We are currently evaluating how the adoption of this standard will impact our income tax disclosures.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

The risk inherent in our market risk sensitive instruments and positions is the potential loss arising from adverse changes in the price of aircraft fuel, foreign currency exchange rates and interest rates as discussed below. The sensitivity analyses presented do not consider the effects that such adverse changes may have on overall economic activity, nor do they consider additional actions we may take to mitigate our exposure to such changes. Therefore, actual results may differ.

Aircraft Fuel

Our operating results are materially impacted by changes in the availability, price volatility and cost of aircraft fuel, which represents one of the largest single cost items in our business. Because of the amount of fuel needed to operate our business, even a relatively small increase or decrease in the price of aircraft fuel can have a material effect on our operating results and liquidity. Market prices for aircraft fuel have fluctuated substantially over the past several years and prices continue to be highly volatile, with market spot prices ranging from a low of approximately $1.32 per gallon to a high of approximately $4.40 per gallon during the period from January 1, 2021 to December 31, 2023.

As of December 31, 2023, we did not have any fuel hedging contracts outstanding to hedge our fuel consumption. Our current policy is not to enter into transactions to hedge our fuel consumption, although we review this policy from time to time based on market conditions and other factors. As such, and assuming we do not enter into any future transactions to hedge our fuel consumption, we will continue to be fully exposed to fluctuations in fuel prices. Based on our 2024 forecasted fuel consumption, we estimate that a one cent per gallon increase in the price of aircraft fuel would increase our 2024 annual fuel expense by approximately $45 million.

Foreign Currency

We are exposed to the effect of foreign exchange rate fluctuations on the U.S. dollar value of foreign currency-denominated transactions. Our largest exposure comes from the Euro, Canadian dollar, British pound sterling and various Latin American currencies (primarily the Brazilian real). We do not currently have a foreign currency hedge program. We estimate a uniform 10% strengthening in the value of the U.S. dollar from 2023 levels relative to each of the currencies in which we have foreign currency exposure would have resulted in a decrease in pre-tax income of approximately $155 million for the year ended December 31, 2023.

Generally, fluctuations in foreign currencies, including devaluations, cannot be predicted by us and can significantly affect the value of our assets located outside the United States. These conditions, as well as any further delays, devaluations or imposition of more stringent repatriation restrictions, may materially adversely affect our business, results of operations and financial condition. See Part I, Item 1A. Risk Factors – *"We operate a global business with international operations that are subject to economic and political instability and have been, and in the future may continue to be, adversely affected by numerous events, circumstances or government actions beyond our control"* for additional discussion of this and other currency risks.

Interest

Our earnings and cash flow are affected by changes in interest rates due to the impact those changes have on our interest expense from variable-rate debt instruments and our interest income from short-term, interest-bearing investments.

Our largest exposure with respect to variable-rate debt comes from changes in the relevant benchmark rate underlying such debt financings, principally SOFR. We had variable-rate debt instruments representing 30% of our total long-term debt at December 31, 2023. We currently do not have an interest rate hedge program to hedge our exposure to floating interest rates on our variable-rate debt obligations. If annual interest rates increase 100 basis points, based on our December 31, 2023 variable-rate debt and short-term investments balances, annual interest expense on variable-rate debt would increase by approximately $100 million and annual interest income on short-term investments would increase by approximately $80 million. Additionally, the fair value of fixed-rate debt would have decreased by approximately $700 million for AAG and $460 million for American.

In connection with the phase-out of LIBOR as a reference rate in June 2023, the U.S. Federal Reserve, in conjunction with the Alternative Reference Rates Committee, has chosen SOFR, and specifically Term SOFR, as the recommended risk-free reference rate for the U.S. (calculated based on repurchase agreements backed by treasury securities). Prior to the discontinuation of LIBOR, we amended substantially all of our LIBOR-based financing arrangements to transition them to successor rates, primarily Term SOFR. We cannot predict the extent to which Term SOFR will gain widespread acceptance as a replacement for LIBOR, the consequences of the replacement of LIBOR on financial markets generally or on our business, financial condition or results of operations specifically, and our transition to successor rates could cause the amount of interest payable on our long-term debt to be different or higher than expected.

ITEM 8A. CONSOLIDATED FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA OF AMERICAN AIRLINES GROUP INC.

Report of Independent Registered Public Accounting Firm

To the Stockholders and Board of Directors
American Airlines Group Inc.:

Opinion on the Consolidated Financial Statements

We have audited the accompanying consolidated balance sheets of American Airlines Group Inc. and subsidiaries (the Company) as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive income (loss), cash flows, and stockholders' equity (deficit) for each of the years in the three-year period ended December 31, 2023, and the related notes (collectively, the consolidated financial statements). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the years in the three-year period ended December 31, 2023, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission, and our report dated February 21, 2024 expressed an unqualified opinion on the effectiveness of the Company's internal control over financial reporting.

Basis for Opinion

These consolidated financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on these consolidated financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the consolidated financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of a critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Sufficiency of audit evidence over the realizability of tax net operating loss and other carryforwards

As discussed in Notes 1(j) and 6 to the consolidated financial statements, the Company had $4.2 billion of tax net operating loss and other carryforwards, which are recorded as deferred tax assets at December 31, 2023. Deferred tax assets are recognized related to tax net operating loss and other carryforwards that will reduce future taxable income. The Company provides a valuation allowance for deferred tax assets when it is more likely than not that some portion, or all of the deferred tax assets, will not be realized. In evaluating the need for a valuation allowance, management considers all available positive and negative evidence.

We identified the evaluation of the sufficiency of audit evidence over the realizability of the federal tax net operating loss and other carryforwards as a critical audit matter. Evaluating the sufficiency of audit evidence required subjective auditor judgment in order to assess the extent of procedures performed in assessing the realizability of the federal tax net operating loss and other carryforwards.

The following are the primary procedures we performed to address this critical audit matter. We evaluated the design and tested the operating effectiveness of certain internal controls related to the Company's deferred tax asset valuation allowance process, including controls related to the realizability of the federal tax net operating loss and other carryforwards. We evaluated positive and negative evidence used in assessing whether the federal tax net operating loss and other carryforwards were more likely than not to be realized in the future. We evaluated the reasonableness of management's projections of future profitability considering historical profitability of the Company, and consistency with industry data. We involved tax professionals with specialized skills and knowledge, who assisted in evaluating the application of tax law. We assessed the sufficiency of audit evidence obtained over the realizability of the federal tax net operating loss and other carryforwards by evaluating the cumulative results of the audit procedures.

/s/ KPMG LLP

We have served as the Company's auditor since 2014.

Dallas, Texas
February 21, 2024

AMERICAN AIRLINES GROUP INC.
CONSOLIDATED STATEMENTS OF OPERATIONS
(In millions, except share and per share amounts)

	Year Ended December 31,		
	2023	2022	2021
Operating revenues:			
Passenger	$ 48,512	$ 44,568	$ 26,063
Cargo	812	1,233	1,314
Other	3,464	3,170	2,505
Total operating revenues	52,788	48,971	29,882
Operating expenses:			
Aircraft fuel and related taxes	12,257	13,791	6,792
Salaries, wages and benefits	14,580	12,972	11,817
Regional expenses	4,643	4,385	3,204
Maintenance, materials and repairs	3,265	2,684	1,979
Other rent and landing fees	2,928	2,730	2,619
Aircraft rent	1,369	1,395	1,425
Selling expenses	1,799	1,815	1,098
Depreciation and amortization	1,936	1,977	2,019
Special items, net	971	193	(4,006)
Other	6,006	5,422	3,994
Total operating expenses	49,754	47,364	30,941
Operating income (loss)	3,034	1,607	(1,059)
Nonoperating income (expense):			
Interest income	591	216	18
Interest expense, net	(2,145)	(1,962)	(1,800)
Other income (expense), net	(359)	325	293
Total nonoperating expense, net	(1,913)	(1,421)	(1,489)
Income (loss) before income taxes	1,121	186	(2,548)
Income tax provision (benefit)	299	59	(555)
Net income (loss)	$ 822	$ 127	$ (1,993)
Earnings (loss) per common share:			
Basic	$ 1.26	$ 0.20	$ (3.09)
Diluted	$ 1.21	$ 0.19	$ (3.09)
Weighted average shares outstanding (in thousands):			
Basic	653,612	650,345	644,015
Diluted	719,669	655,122	644,015

See accompanying notes to consolidated financial statements.

AMERICAN AIRLINES GROUP INC.
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME (LOSS)
(In millions)

	Year Ended December 31,		
	2023	2022	2021
Net income (loss)	$ 822	$ 127	$ (1,993)
Other comprehensive income (loss), net of tax:			
Pension, retiree medical and other postretirement benefits	(312)	1,360	1,161
Investments	3	(3)	—
Total other comprehensive income (loss), net of tax	(309)	1,357	1,161
Total comprehensive income (loss)	$ 513	$ 1,484	$ (832)

See accompanying notes to consolidated financial statements.

AMERICAN AIRLINES GROUP INC.
CONSOLIDATED BALANCE SHEETS
(In millions, except share and par value amounts)

	December 31,	
	2023	2022
ASSETS		
Current assets		
Cash	$ 578	$ 440
Short-term investments	7,000	8,525
Restricted cash and short-term investments	910	995
Accounts receivable, net	2,026	2,138
Aircraft fuel, spare parts and supplies, net	2,400	2,279
Prepaid expenses and other	658	892
Total current assets	13,572	15,269
Operating property and equipment		
Flight equipment	41,794	39,703
Ground property and equipment	10,307	9,913
Equipment purchase deposits	760	613
Total property and equipment, at cost	52,861	50,229
Less accumulated depreciation and amortization	(22,097)	(20,029)
Total property and equipment, net	30,764	30,200
Operating lease right-of-use assets	7,939	8,094
Other assets		
Goodwill	4,091	4,091
Intangibles, net of accumulated amortization of $834 and $827, respectively	2,051	2,059
Deferred tax asset	2,888	3,099
Other assets	1,753	1,904
Total other assets	10,783	11,153
Total assets	$ 63,058	$ 64,716
LIABILITIES AND STOCKHOLDERS' EQUITY (DEFICIT)		
Current liabilities		
Current maturities of long-term debt and finance leases	$ 3,632	$ 3,274
Accounts payable	2,353	2,149
Accrued salaries and wages	2,377	1,713
Air traffic liability	6,200	6,745
Loyalty program liability	3,453	3,169
Operating lease liabilities	1,309	1,465
Other accrued liabilities	2,738	2,981
Total current liabilities	22,062	21,496
Noncurrent liabilities		
Long-term debt and finance leases, net of current maturities	29,270	32,389
Pension and postretirement benefits	3,044	2,837
Loyalty program liability	5,874	5,976
Operating lease liabilities	6,452	6,559
Other liabilities	1,558	1,258
Total noncurrent liabilities	46,198	49,019
Commitments and contingencies (Note 11)		
Stockholders' equity (deficit)		
Common stock, $0.01 par value; 1,750,000,000 shares authorized, 654,273,192 shares issued and outstanding at December 31, 2023; 650,642,461 shares issued and outstanding at December 31, 2022	7	6
Additional paid-in capital	7,374	7,291
Accumulated other comprehensive loss	(4,894)	(4,585)
Retained deficit	(7,689)	(8,511)
Total stockholders' deficit	(5,202)	(5,799)
Total liabilities and stockholders' equity (deficit)	$ 63,058	$ 64,716

See accompanying notes to consolidated financial statements.

AMERICAN AIRLINES GROUP INC.
CONSOLIDATED STATEMENTS OF CASH FLOWS
(In millions)

	Year Ended December 31,		
	2023	2022	2021
Cash flows from operating activities:			
Net income (loss)	$ 822	$ 127	$ (1,993)
Adjustments to reconcile net income (loss) to net cash provided by operating activities:			
Depreciation and amortization	2,254	2,298	2,335
Debt extinguishment costs	267	3	31
Special items, net non-cash	41	226	52
Pension and postretirement	(13)	(405)	(321)
Deferred income tax provision (benefit)	299	65	(555)
Share-based compensation	102	78	98
Other, net	(205)	(37)	16
Changes in operating assets and liabilities:			
Decrease (increase) in accounts receivable	95	(637)	(304)
Increase in other assets	(11)	(775)	(402)
Increase in accounts payable and accrued liabilities	873	585	461
Increase (decrease) in air traffic liability	(545)	658	1,454
Increase (decrease) in loyalty program liability	182	10	(60)
Contributions to pension plans	(73)	(5)	(247)
Increase (decrease) in other liabilities	(285)	(18)	139
Net cash provided by operating activities	3,803	2,173	704
Cash flows from investing activities:			
Capital expenditures, net of aircraft purchase deposit returns	(2,596)	(2,546)	(208)
Proceeds from sale of property and equipment and sale-leaseback transactions	230	147	374
Sales of short-term investments	8,861	14,972	13,923
Purchases of short-term investments	(7,323)	(11,257)	(19,454)
Decrease (increase) in restricted short-term investments	51	1	(401)
Purchase of equity investments	—	(321)	(28)
Other investing activities	275	(360)	(189)
Net cash provided by (used in) investing activities	(502)	636	(5,983)
Cash flows from financing activities:			
Payments on long-term debt and finance leases	(7,718)	(3,752)	(7,343)
Proceeds from issuance of long-term debt	4,822	1,069	12,190
Proceeds from issuance of equity	—	—	460
Other financing activities	(310)	52	(19)
Net cash provided by (used in) financing activities	(3,206)	(2,631)	5,288
Net increase in cash and restricted cash	95	178	9
Cash and restricted cash at beginning of year	586	408	399
Cash and restricted cash at end of year[(a)]	$ 681	$ 586	$ 408

[(a)] The following table provides a reconciliation of cash and restricted cash to amounts reported within the consolidated balance sheets:

Cash	$ 578	$ 440	$ 273
Restricted cash included in restricted cash and short-term investments	103	146	135
Total cash and restricted cash	$ 681	$ 586	$ 408

See accompanying notes to consolidated financial statements.

AMERICAN AIRLINES GROUP INC.
CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY (DEFICIT)
(In millions, except share amounts)

	Common Stock	Additional Paid-in Capital	Accumulated Other Comprehensive Loss	Retained Earnings (Deficit)	Total
Balance at December 31, 2020	$ 6	$ 6,894	$ (7,103)	$ (6,664)	$ (6,867)
Net loss	—	—	—	(1,993)	(1,993)
Other comprehensive income, net	—	—	1,161	—	1,161
Issuance of 24,150,764 shares of AAG common stock pursuant to an at-the-market offering, net of offering costs	—	460	—	—	460
Impact of adoption of Accounting Standards Update (ASU) 2020-06 related to convertible instruments	—	(320)	—	19	(301)
Issuance of PSP2 and PSP3 Warrants (see Note 1(b))	—	121	—	—	121
Issuance of 2,357,187 shares of AAG common stock pursuant to employee stock plans net of shares withheld for cash taxes	—	(18)	—	—	(18)
Settlement of single-dip unsecured claims held in Disputed Claims Reserve (DCR) and retirement of 259,878 shares of AAG common stock	—	(1)	—	—	(1)
Share-based compensation expense	—	98	—	—	98
Balance at December 31, 2021	6	7,234	(5,942)	(8,638)	(7,340)
Net income	—	—	—	127	127
Other comprehensive income, net	—	—	1,357	—	1,357
Issuance of 2,914,866 shares of AAG common stock pursuant to employee stock plans net of shares withheld for cash taxes	—	(21)	—	—	(21)
Share-based compensation expense	—	78	—	—	78
Balance at December 31, 2022	6	7,291	(4,585)	(8,511)	(5,799)
Net income	—	—	—	822	822
Other comprehensive loss, net	—	—	(309)	—	(309)
Issuance of 3,630,731 shares of AAG common stock pursuant to employee stock plans net of shares withheld for cash taxes	1	(23)	—	—	(22)
Share-based compensation expense	—	102	—	—	102
Settlement of single-dip unsecured claims held in DCR	—	4	—	—	4
Balance at December 31, 2023	$ 7	$ 7,374	$ (4,894)	$ (7,689)	$ (5,202)

See accompanying notes to consolidated financial statements.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS OF AMERICAN AIRLINES GROUP INC.

1. Basis of Presentation and Summary of Significant Accounting Policies

(a) Basis of Presentation

American Airlines Group Inc. (we, us, our and similar terms, or AAG), a Delaware corporation, is a holding company whose primary business activity is the operation of a major network air carrier, providing scheduled air transportation for passengers and cargo through its mainline operating subsidiary, American Airlines, Inc. (American) and its wholly-owned regional airline subsidiaries, Envoy Aviation Group Inc., PSA Airlines, Inc. (PSA) and Piedmont Airlines, Inc. (Piedmont), that operate under the brand American Eagle. All significant intercompany transactions have been eliminated.

The preparation of financial statements in accordance with accounting principles generally accepted in the United States (GAAP) requires management to make certain estimates and assumptions that affect the reported amounts of assets and liabilities, revenues and expenses, and the disclosure of contingent assets and liabilities at the date of the financial statements. Actual results could differ from those estimates. The most significant areas of judgment relate to passenger revenue recognition, the loyalty program, deferred tax assets, as well as pension and retiree medical and other postretirement benefits.

(b) Government Assistance

Payroll Support Programs

During 2020 and 2021, American, Envoy Air Inc. (Envoy), Piedmont and PSA (together with American, Envoy and Piedmont, the Subsidiaries) entered into payroll support program agreements (PSP Agreements) with the U.S. Department of Treasury (Treasury) pursuant to the payroll support program established under the Coronavirus Aid, Relief, and Economic Security Act (CARES Act) (PSP1), the payroll support program established under the Subtitle A of Title IV of Division N of the Consolidated Appropriations Act, 2021 (PSP Extension Law) (PSP2) and the payroll support program established under the American Rescue Plan Act of 2021 (ARP) (PSP3). The aggregate amount of financial assistance received was approximately $12.8 billion, and as partial compensation to the U.S. Government for the provision of financial assistance provided under each of these programs, AAG issued promissory notes and warrants to Treasury.

The table below provides a summary of the financial assistance received and the promissory notes and the warrants issued under each program (in millions, except exercise price amounts):

Program	Closing Date	PSP Financial Assistance	Promissory Notes [1]	PSP Warrants	Total	Warrants Issued (Shares) [2]	Exercise Price of Warrants
PSP1	April 20, 2020	$ 4,138	$ 1,757	$ 63	$ 5,958	14.0	$ 12.51
PSP2	January 15, 2021	2,427	1,030	76	3,533	6.6	15.66
PSP3	April 23, 2021	2,290	959	46	3,295	4.4	21.75
Total		$ 8,855	$ 3,746	$ 185	$ 12,786	25.0	

[1] See Note 4 for further information on the promissory notes issued.

[2] The payroll support program warrants (PSP Warrants) are subject to certain anti-dilution provisions, do not have any voting rights and are freely transferable, with registration rights. Each warrant expires on the fifth anniversary of the date of issuance, with expiration dates ranging from April 2025 to June 2026, and will be exercisable either through net share settlement or cash, at our option. The warrants were issued solely as compensation to the U.S. Government related to entry into the PSP Agreements. No separate proceeds (apart from the financial assistance described below) were received upon issuance of the warrants or will be received upon exercise thereof.

In connection with the PSP Agreements entered into with Treasury, we were required to comply with the relevant provisions of the CARES Act, the PSP Extension Law, and the ARP, which included the requirement that funds provided pursuant to these programs be used exclusively for the continuation of payment of eligible employee wages, salaries and benefits, the prohibition against involuntary furloughs and reductions in employee pay rates and benefits, the requirement that certain levels of commercial air service be maintained, provisions that prohibited the repurchase of AAG common stock and the payment of common stock dividends as well as provisions that restrict the payment of certain executive compensation. As of December 31, 2023, all of these provisions have expired.

For accounting purposes, the $12.8 billion of aggregate financial assistance received pursuant to the PSP Agreements was allocated to the promissory notes, warrants and other financial assistance (PSP Financial Assistance). The aggregate principal amount of the promissory notes was recorded as unsecured long-term debt and the total fair value of the warrants, estimated using a Black-Scholes option pricing model, was recorded in stockholders' deficit in the consolidated balance sheets. The remaining amounts were recognized in 2020 and 2021 as a credit to special items, net in the consolidated statements of operations over the period which the continuation of payment of eligible employee wages, salaries and benefits was required.

Treasury Loan Agreement

On September 25, 2020 (the Treasury Loan Closing Date), AAG and American entered into a Loan and Guarantee Agreement (the Treasury Loan Agreement) with Treasury, which provided for a secured term loan facility (the Treasury Term Loan Facility) that permitted American to borrow up to $5.5 billion. Subsequently, on October 21, 2020, AAG and American entered into an amendment to the Treasury Loan Agreement which increased the borrowing amount up to $7.5 billion. In connection with entry into the Treasury Loan Agreement, on the Treasury Loan Closing Date, AAG also entered into a warrant agreement (the Treasury Loan Warrant Agreement) with Treasury.

In September 2020, American borrowed $ 550 million under the Treasury Term Loan Facility and on March 24, 2021, used a portion of the proceeds from the AAdvantage Financing to prepay in full the $550 million of outstanding loans under the Treasury Term Loan Facility and terminated the Treasury Loan Agreement. Pursuant to the Treasury Loan Agreement, AAG issued to Treasury warrants (Treasury Loan Warrants) to purchase up to an aggregate of approximately 4.4 million shares of AAG common stock (the Treasury Loan Warrant Shares), which expire in September 2025. The exercise price of the Treasury Loan Warrant Shares is $ 12.51 per share, subject to certain anti-dilution provisions provided for in the Treasury Loan Warrant Agreement. For accounting purposes, the fair value for the Treasury Loan Warrant Shares, estimated using a Black-Scholes option pricing model, was recorded in stockholders' deficit with an offsetting debt discount to the Treasury Term Loan Facility in the consolidated balance sheet. The provisions of the Treasury Loan Warrants are substantially similar to the PSP Warrants.

(c) Recent Accounting Pronouncements

ASU 2023-07: Segment Reporting (Topic 280) Improvements to Reportable Segment Disclosures

This standard improves reportable segment disclosure requirements, primarily through enhanced disclosures about significant segment expenses. The amendments in this update are effective for fiscal years beginning after December 15, 2023 and interim periods within fiscal years beginning after December 15, 2024, and early adoption is permitted. We are currently evaluating how the adoption of this standard will impact our reportable segment disclosures.

ASU 2023-09: Income Taxes (Topic 740) Improvements to Income Tax Disclosures

This standard enhances transparency of income tax information through improvements to income tax disclosures primarily related to the rate reconciliation and income taxes paid information, as well as improvements to the effectiveness and comparability of other income tax disclosures. The amendments in this update are effective for annual periods beginning after December 15, 2024, and early adoption is permitted. We are currently evaluating how the adoption of this standard will impact our income tax disclosures.

(d) Investments

Short-term investments primarily include debt securities and are classified as available-for-sale and stated at fair value. Realized gains and losses are recorded as interest income in nonoperating expense, net on our consolidated statements of operations. Unrealized gains and losses are recorded as a component of accumulated other comprehensive loss on our consolidated balance sheets. For investments in an unrealized loss position, we determine whether a credit loss exists by considering information about the collectability of the instrument, current market conditions and reasonable and supportable forecasts of economic conditions. There have been no credit losses.

Equity investments are accounted for under the equity method if we are able to exercise significant influence over an investee. Equity investments for which we do not have significant influence are recorded at fair value or at cost, if fair value is not readily determinable, with adjustments for observable changes in price or impairments (referred to as the measurement alternative). Our share of equity method investees' financial results and changes in fair value are recorded in nonoperating other income (expense), net on the consolidated statements of operations. See Note 8 for additional information related to our equity investments.

(e) Restricted Cash and Short-term Investments

We have restricted cash and short-term investments related primarily to collateral held to support workers' compensation obligations and collateral associated with the AAdvantage Financing.

(f) Accounts Receivable, Net

Accounts receivable primarily consist of amounts due from credit card processing companies for tickets sold to individual passengers, amounts due from airline and non-airline business partners, including our co-branded credit card partners and cargo customers. Receivables from ticket sales are short-term, mostly settled within seven days after sale. Receivables from our business partners are typically settled within 30 days. All accounts receivable are reported net of an allowance for credit losses, which was not material as of December 31, 2023 and 2022. We consider past and future financial and qualitative factors, including aging, payment history and other credit monitoring indicators, when establishing the allowance for credit losses.

(g) Aircraft Fuel, Spare Parts and Supplies, Net

Aircraft fuel is recorded on a first-in, first-out basis. Spare parts and supplies are recorded at average costs less an allowance for obsolescence, which is recognized over the weighted average remaining useful life of the related fleet. We also provide an allowance for spare parts and supplies identified as excess or obsolete to reduce the carrying cost to the lower of cost or net realizable value. Aircraft fuel, spare parts and supplies are expensed when used.

(h) Operating Property and Equipment

Operating property and equipment is recorded at cost and depreciated or amortized to residual values over the asset's estimated useful life or the lease term, whichever is less, using the straight-line method. Residual values for aircraft, engines and related rotable parts are generally 5% to 10% of original cost. Costs of major improvements that enhance the usefulness of the asset are capitalized and depreciated or amortized over the estimated useful life of the asset or the lease term, whichever is less. The estimated useful lives for the principal property and equipment classifications are as follows:

Principal Property and Equipment Classification	Estimated Useful Life
Aircraft, engines and related rotable parts	20 – 30 years
Buildings and improvements	5 – 30 years
Furniture, fixtures and other equipment	3 – 15 years
Capitalized software	5 – 10 years

Total mainline and regional depreciation and amortization expense was $ 2.3 billion for each of the years ended December 31, 2023, 2022 and 2021.

We assess impairment of operating property and equipment when events and circumstances indicate that the assets may be impaired. An impairment of an asset or group of assets exists only when the sum of the estimated undiscounted cash flows expected to be generated directly by the assets are less than the carrying value of the assets. We group assets principally by fleet-type when estimating future cash flows, which is generally the lowest level for which identifiable cash flows exist. Estimates of future cash flows are based on historical results adjusted to reflect management's best estimate of future market and operating conditions, including our current fleet plan. If such assets are impaired, the impairment charge recognized is the amount by which the carrying value of the assets exceed their fair value. Fair value reflects management's best estimate including inputs from published pricing guides and bids from third parties as well as contracted sales agreements when applicable.

(i) Leases

We determine if an arrangement is a lease at inception. Operating leases are included in operating lease right-of-use (ROU) assets, current operating lease liabilities and noncurrent operating lease liabilities on our consolidated balance sheets. Finance leases are included in property and equipment, current maturities of long-term debt and finance leases and long-term debt and finance leases, net of current maturities, on our consolidated balance sheets.

ROU assets represent our right to use an underlying asset for the lease term and lease liabilities represent our obligation to make lease payments arising from the lease. ROU assets and liabilities are recognized at the lease commencement date based on the estimated present value of lease payments over the lease term.

We use our estimated incremental borrowing rate, which is derived from information available at the lease commencement date , in determining the present value of lease payments. We give consideration to our recent debt issuances as well as publicly available data for instruments with similar characteristics when calculating our incremental borrowing rates.

Our lease term includes options to extend the lease when it is reasonably certain that we will exercise that option. Leases with a term of 12 months or less are not recorded on our consolidated balance sheets.

Under certain of our capacity purchase agreements with third-party regional carriers, we do not own the underlying aircraft. However, since we control the marketing, scheduling, ticketing, pricing and seat inventories of these aircraft and therefore control the asset, the aircraft is deemed to be leased for accounting purposes. For these capacity purchase agreements, we account for the lease and non-lease components separately. The lease component consists of the aircraft and the non-lease components consist of services, such as the crew and maintenance. Where applicable, we allocate the consideration in the capacity purchase agreements to the lease and non-lease components using their estimated relative standalone prices. See Note 11(b) for additional information on our capacity purchase agreements.

For real estate, we account for the lease and non-lease components as a single lease component.

(j) Income Taxes

Income taxes are accounted for under the asset and liability method. Deferred tax assets and liabilities are recognized for the future tax consequences attributable to differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases and operating loss and tax credit carryforwards. Deferred tax assets and liabilities are recorded net as noncurrent deferred income taxes.

We provide a valuation allowance for our deferred tax assets when it is more likely than not that some portion, or all of our deferred tax assets, will not be realized. The ultimate realization of deferred tax assets is dependent upon the generation of future taxable income. We consider all available positive and negative evidence and make certain assumptions in evaluating the realizability of our deferred tax assets. Many factors are considered that impact our assessment of future profitability, including conditions which are beyond our control, such as the health of the economy, the availability and price volatility of aircraft fuel and travel demand. We have determined that positive factors outweigh negative factors in the determination of the realizability of our deferred tax assets.

(k) Goodwill

Goodwill represents the purchase price in excess of the fair value of the net assets acquired and liabilities assumed in connection with the 2013 merger with US Airways Group, Inc. (US Airways Group). We have one reporting unit. We assess goodwill for impairment annually or more frequently if events or circumstances indicate that the fair value of goodwill may be lower than the carrying value. Our annual assessment date is October 1.

Goodwill is assessed for impairment by initially performing a qualitative assessment. If we determine that it is more likely than not that our goodwill may be impaired, we use a quantitative approach to assess the asset's fair value and the amount of the impairment, if any. Based upon our annual assessment, there was no goodwill impairment in 2023. The carrying value of our goodwill on our consolidated balance sheets was $ 4.1 billion as of December 31, 2023 and 2022.

(l) Other Intangibles, Net

Intangible assets consist primarily of certain domestic airport slots and gate leasehold rights, customer relationships, marketing agreements, commercial agreements, international slots and route authorities and tradenames.

Definite-Lived Intangible Assets

Definite-lived intangible assets are originally recorded at their acquired fair values, subsequently amortized over their respective estimated useful lives and are assessed for impairment whenever events and circumstances indicate that the assets may be impaired.

The following table provides information relating to our amortizable intangible assets as of December 31, 2023 and 2022 (in millions):

	December 31,	
	2023	2022
Domestic airport slots	$ 365	$ 365
Customer relationships	300	300
Marketing agreements	105	105
Tradenames	35	35
Airport gate leasehold rights	137	137
Accumulated amortization	(834)	(827)
Total	$ 108	$ 115

Certain domestic airport slots and airport gate leasehold rights are amortized on a straight-line basis over 25 years. Certain marketing agreements were identified as intangible assets subject to amortization and are amortized on a straight-line basis over approximately 30 years. Customer relationships and tradenames are fully amortized.

We recorded amortization expense related to these intangible assets of $ 7 million for the year ended December 31, 2023 and $ 41 million for each of the years ended December 31, 2022 and 2021. We expect to record annual amortization expense for these intangible assets as follows (in millions):

2024	$ 7
2025	7
2026	6
2027	6
2028	6
2029 and thereafter	76
Total	$ 108

Indefinite-Lived Intangible Assets

Indefinite-lived intangible assets include certain domestic airport slots, international slots and route authorities and our commercial agreement with GOL Linhas Aéreas Inteligentes S.A. (GOL). We assess indefinite-lived intangible assets for impairment annually or more frequently if events or circumstances indicate that the fair values of indefinite-lived intangible assets may be lower than their carrying values. Our annual assessment date is October 1.

Indefinite-lived intangible assets are assessed for impairment by initially performing a qualitative assessment. If we determine that it is more likely than not that our indefinite-lived intangible assets may be impaired, we use a quantitative approach to assess the asset's fair value and the amount of the impairment, if any. Based upon our annual assessment, there were no indefinite-lived intangible asset impairments in 2023. We had $ 1.9 billion of indefinite-lived intangible assets on our consolidated balance sheets as of December 31, 2023 and 2022.

(m) Revenue Recognition

Revenue

The following are the significant categories comprising our operating revenues (in millions):

	Year Ended December 31,		
	2023	2022	2021
Passenger revenue:			
Passenger travel	$ 44,914	$ 41,425	$ 23,896
Loyalty revenue - travel [1]	3,598	3,143	2,167
Total passenger revenue	48,512	44,568	26,063
Cargo	812	1,233	1,314
Other:			
Loyalty revenue - marketing services	2,929	2,657	2,166
Other revenue	535	513	339
Total other revenue	3,464	3,170	2,505
Total operating revenues	$ 52,788	$ 48,971	$ 29,882

[1] Loyalty revenue included in passenger revenue is principally comprised of mileage credit redemptions, which were earned from travel or co-branded credit card and other partners. See "*Loyalty Revenue*" below for further discussion on these mileage credits.

The following is our total passenger revenue by geographic region (in millions):

	Year Ended December 31,		
	2023	2022	2021
Domestic	$ 34,592	$ 32,911	$ 21,453
Latin America	6,719	6,150	3,506
Atlantic	6,205	5,070	965
Pacific	996	437	139
Total passenger revenue	$ 48,512	$ 44,568	$ 26,063

We attribute passenger revenue by geographic region based upon the origin and destination of each flight segment.

Passenger Revenue

We recognize all revenues generated from transportation on American and our regional flights operated under the brand name American Eagle, including associated baggage fees and other inflight services, as passenger revenue when transportation is provided. Ticket and other related sales for transportation that has not yet been provided are initially deferred and recorded as air traffic liability on our consolidated balance sheets. The air traffic liability principally represents tickets sold for future travel on American and partner airlines.

The majority of tickets sold are nonrefundable. A small percentage of tickets, some of which are partially used tickets, expire unused. The estimate for tickets expected to expire unused is generally based on an analysis of our historical data and other current applicable factors such as policy changes. We have consistently applied this accounting method to estimate and recognize revenue from unused tickets at the date of travel. This estimate is periodically evaluated based on subsequent activity to validate its accuracy. Any adjustments resulting from periodic evaluations of the estimated air traffic liability are included in passenger revenue during the period in which the evaluations are completed.

Various taxes and fees assessed on the sale of tickets to end customers are collected by us as an agent and remitted to taxing authorities. These taxes and fees have been presented on a net basis in the accompanying consolidated statements of operations and recorded as a liability until remitted to the appropriate taxing authority.

Loyalty Revenue

We currently operate the loyalty program, AAdvantage. This program awards mileage credits to passengers who fly on American, any **one**world airline or other partner airlines, or by using the services of other program participants, such as our co-branded credit cards, and certain hotels and car rental companies. Mileage credits can be redeemed for travel on American and other participating partner airlines, as well as non-air travel awards such as hotels and rental cars. For mileage credits earned by AAdvantage program members, we apply the deferred revenue method.

Mileage credits earned through travel

For mileage credits earned through travel, we apply a relative selling price approach whereby the total amount collected from each passenger ticket sale is allocated between the air transportation and the mileage credits earned. The portion of each passenger ticket sale attributable to mileage credits earned is initially deferred and then recognized in passenger revenue when mileage credits are redeemed and transportation is provided. The estimated selling price of mileage credits is determined using an equivalent ticket value approach, which uses historical data, including award redemption patterns by geographic region and class of service, as well as similar cash fares as those used to settle award redemptions. The estimated selling price of mileage credits is adjusted for an estimate of mileage credits that will not be redeemed using a statistical model based on historical redemption patterns to develop an estimate of the likelihood of future redemption.

Mileage credits sold to co-branded credit cards and other partners

We sell mileage credits to participating airline partners and non-airline business partners, including our co-branded credit card partners, under contracts with remaining terms generally from one to six years as of December 31, 2023. Consideration received from the sale of mileage credits is variable and payment terms typically are within 30 days subsequent to the month of mileage sale. Sales of mileage credits to non-airline business partners are comprised of two components, transportation and marketing. We allocate the consideration received from these sales of mileage credits based on the relative selling price of each product or service delivered.

Our most significant mileage credit partner agreements are our co-branded credit card agreements with Citi and Barclaycard US. We identified two revenue elements in these co-branded credit card agreements: the transportation component and the marketing component.

The transportation component represents the estimated selling price of future travel awards and is determined using the same equivalent ticket value approach described above. The portion of each mileage credit sold attributable to transportation is initially deferred and then recognized in passenger revenue when mileage credits are redeemed and transportation is provided.

The marketing component includes the use of intellectual property, including the American brand and access to loyalty program member lists, which is the predominant element in these agreements, as well as advertising and other travel-related benefits. We recognize the marketing component in other revenue in the period of the mileage credit sale following the sales-based royalty method.

For the portion of our outstanding mileage credits that we estimate will not be redeemed, we recognize the associated value proportionally as the remaining mileage credits are redeemed. Our estimates use a statistical model based on historical redemption patterns to develop an estimate of the likelihood of future redemption.

Cargo Revenue

Cargo revenue is recognized when we provide the transportation.

Other Revenue

Other revenue includes revenue associated with our loyalty program, which is comprised principally of the marketing component of mileage credit sales to co-branded credit card and other partners and other marketing related payments. The accounting and recognition for the loyalty program marketing services are discussed above in "*Loyalty Revenue*." The remaining amounts included within other revenue relate to airport clubs, other commission revenue, advertising and vacation-related services.

Contract Balances

Our significant contract liabilities are comprised of (1) outstanding loyalty program mileage credits that may be redeemed for future travel and non-air travel awards, reported as loyalty program liability on our consolidated balance sheets and (2) ticket sales for transportation that has not yet been provided, reported as air traffic liability on our consolidated balance sheets.

	December 31,	
	2023	2022
	(In millions)	
Loyalty program liability	$ 9,327	$ 9,145
Air traffic liability	6,200	6,745
Total	$ 15,527	$ 15,890

The balance of the loyalty program liability fluctuates based on seasonal patterns, which impact the volume of mileage credits issued through travel or sold to co-branded credit card and other partners (deferral of revenue) and mileage credits redeemed (recognition of revenue). Changes in loyalty program liability are as follows (in millions):

Balance at December 31, 2022	$ 9,145
Deferral of revenue	3,810
Recognition of revenue [1]	(3,628)
Balance at December 31, 2023 [2]	$ 9,327

[1] Principally relates to revenue recognized from the redemption of mileage credits for both air and non-air travel awards. Mileage credits are combined in one homogenous pool and are not separately identifiable. As such, the revenue is comprised of mileage credits that were part of the loyalty program deferred revenue balance at the beginning of the period, as well as mileage credits that were issued during the period.

[2] Mileage credits can be redeemed at any time and generally do not expire as long as that AAdvantage member has any type of qualifying activity at least every 24 months or if the AAdvantage member is the primary holder of a co-branded credit card. As of December 31, 2023, our current loyalty program liability was $3.5 billion and represents our current estimate of revenue expected to be recognized in the next 12 months based on historical trends, with the balance reflected in long-term loyalty program liability expected to be recognized as revenue in periods thereafter.

The air traffic liability principally represents tickets sold for future travel on American and partner airlines. The balance in our air traffic liability also fluctuates with seasonal travel patterns. The contract duration of passenger tickets is generally one year. Accordingly, any revenue associated with tickets sold for future travel will be recognized within 12 months. For 2023, $5.3 billion of revenue was recognized in passenger revenue that was included in our air traffic liability at December 31, 2022.

(n) Maintenance, Materials and Repairs

Maintenance and repair costs for owned and leased flight equipment are charged to operating expense as incurred, except costs incurred for maintenance and repair under certain power-by-the-hour maintenance agreements, which are charged to operating expense based on contractual terms when an obligation exists.

(o) Selling Expenses

Selling expenses include credit card fees, commissions, third party distribution channel fees and advertising. Selling expenses associated with passenger revenue are expensed when the transportation or service is provided. Advertising costs are expensed as incurred. Advertising expense was $114 million for the year ended December 31, 2023 and $ 105 million for each of the years ended December 31, 2022 and 2021.

(p) Share-based Compensation

We account for our share-based compensation expense based on the fair value of the stock award at the time of grant, which is recognized ratably over the vesting period of the stock award. Certain awards have performance conditions that must be achieved prior to vesting and are expensed based on the expected achievement at each reporting period. The majority of our stock awards are time vested restricted stock units, and the fair value of such awards is based on the market price of the underlying shares of AAG common stock on the date of grant. See Note 14 for further discussion of share-based compensation.

(q) Foreign Currency Gains and Losses

Foreign currency gains and losses are recorded as part of other income (expense), net within total nonoperating expense, net on our consolidated statements of operations. For the years ended December 31, 2023, 2022 and 2021, respectively, foreign currency losses were $30 million, $38 million and $4 million.

(r) Other Operating Expenses

Other operating expenses includes costs associated with onboard food and catering, crew travel, ground and cargo handling, passenger accommodation, international navigation fees, aircraft cleaning, airport lounge operations and certain general and administrative expenses.

(s) Regional Expenses

Our regional carriers provide scheduled air transportation under the brand name “American Eagle.” The American Eagle carriers include our wholly-owned regional carriers as well as third-party regional carriers. Our regional carrier arrangements are in the form of capacity purchase agreements with our third-party regional partners and similar arrangements with our wholly-owned regional affiliates. Expenses associated with American Eagle operations are classified as regional expenses on the consolidated statements of operations.

Regional expenses for the years ended December 31, 2023 , 2022 and 2021 include $ 318 million, $321 million and $ 316 million of depreciation and amortization, respectively, and $7 million, $5 million and $ 6 million of aircraft rent, respectively.

In 2023, 2022 and 2021, we recognized $ 636 million, $592 million and $495 million, respectively, of expense under our capacity purchase agreement with Republic Airways Inc. (Republic). We hold a 25% equity interest in Republic Airways Holdings Inc. (Republic Holdings), the parent company of Republic.

2. Special Items, Net

Special items, net on our consolidated statements of operations consisted of the following (in millions):

	Year Ended December 31,		
	2023	2022	2021
Labor contract expenses [1]	$ 989	$ —	$ —
Severance expenses [2]	23	—	168
Fleet impairment [3]	—	149	—
Litigation reserve adjustments	—	37	(19)
PSP Financial Assistance [4]	—	—	(4,162)
Other operating special items, net	(41)	7	7
Mainline operating special items, net	971	193	(4,006)
PSP Financial Assistance [4]	—	—	(539)
Regional pilot retention program [5]	—	—	61
Fleet impairment [3]	—	—	27
Severance expenses [2]	—	—	2
Other operating special items, net	8	5	—
Regional operating special items, net	8	5	(449)
Operating special items, net	979	198	(4,455)
Debt refinancing, extinguishment and other, net [6]	280	3	29
Mark-to-market adjustments on equity and other investments, net [7]	82	71	31
Nonoperating special items, net	362	74	60
Income tax special items, net	—	(9)	—

[1] Labor contract expenses relate to one-time charges resulting from the ratification of a new collective bargaining agreement with our mainline pilots, including a one-time payment of $754 million as well as adjustments to other benefit-related items of $ 235 million.

[2] Severance expenses for 2023 included costs associated with headcount reductions in certain corporate functions.

Severance expenses for 2021 included salary and medical costs primarily associated with certain team members who opted into voluntary early retirement programs offered as a result of reductions to our operation due to the COVID-19 pandemic.

[3] Fleet impairment for 2022 included a non-cash impairment charge to write down the carrying value of our retired Airbus A330 fleet to the estimated fair value due to the market conditions for certain used aircraft. We retired our Airbus A330 fleet in 2020 as a result of the decline in demand for air travel due to the COVID-19 pandemic.

Fleet impairment for 2021 included a non-cash impairment charge to write down regional aircraft resulting from the retirement of the remaining Embraer 140 fleet earlier than planned.

[4] The PSP Financial Assistance represents recognition of a portion of the financial assistance received from Treasury pursuant to the payroll support programs established by the U.S. Government. See Note 1(b) for further information.

[5] Our regional pilot retention program provides for, among other things, a cash retention bonus paid in the fourth quarter of 2021 to eligible captains at our wholly-owned regional carriers included on the pilot seniority list as of September 1, 2021.

[6] Debt refinancing and extinguishment costs in 2023 primarily included cash charges for premiums paid in connection with the early repayment of debt. See Note 4 for further information.

(7) Mark-to-market adjustments on equity and other investments, net principally included net unrealized gains and losses associated with certain equity investments and certain other investments. See Note 8 for further information related to our equity investments.

3. Earnings (Loss) Per Common Share

The following table provides the computation of basic and diluted earnings (loss) per common share (EPS) (in millions, except share and per share amounts):

	Year Ended December 31,		
	2023	2022	2021
Basic EPS:			
Net income (loss)	$ 822	$ 127	$ (1,993)
Weighted average common shares outstanding (in thousands)	653,612	650,345	644,015
Basic EPS	$ 1.26	$ 0.20	$ (3.09)
Diluted EPS:			
Net income (loss)	$ 822	$ 127	$ (1,993)
Interest expense on 6.50% convertible senior notes	46	—	—
Net income (loss) for purposes of computing diluted EPS	$ 868	$ 127	$ (1,993)
Share computation for diluted EPS (in thousands):			
Basic weighted average common shares outstanding	653,612	650,345	644,015
Dilutive effect of restricted stock unit awards	1,830	1,579	—
Dilutive effect of certain PSP Warrants and Treasury Loan Warrants	2,499	3,198	—
Assumed conversion of 6.50% convertible senior notes	61,728	—	—
Diluted weighted average common shares outstanding	719,669	655,122	644,015
Diluted EPS	$ 1.21	$ 0.19	$ (3.09)

The following were excluded from the calculation of diluted EPS because inclusion of such shares would be antidilutive (in thousands):

	Year Ended December 31,		
	2023	2022	2021
Restricted stock unit awards	4,371	3,987	3,420
6.50% convertible senior notes	—	61,728	61,728

In addition, certain shares underlying our PSP Warrants and Treasury Loan Warrants for the years ended December 31, 2023, 2022 and 2021, were excluded from the calculation of diluted EPS because inclusion of such shares would be antidilutive.

4. Debt

Long-term debt included on our consolidated balance sheets consisted of (in millions):

	December 31, 2023	December 31, 2022
Secured		
2013 Term Loan Facility, variable interest rate of 8.60%, installments through February 2028 [a]	$ 990	$ 1,752
2014 Term Loan Facility, variable interest rate of 7.32%, installments through January 2027 [a]	1,183	1,196
2023 Term Loan Facility, variable interest rate of 8.87%, installments beginning in December 2024 through June 2029 [a]	1,100	—
11.75% senior secured notes, interest only payments until due in July 2025 [b]	—	2,500
10.75% senior secured IP notes, interest only payments until due in February 2026 [b]	1,000	1,000
10.75% senior secured LGA/DCA notes, interest only payments until due in February 2026 [b]	200	200
7.25% senior secured notes, interest only payments until due in February 2028 [b]	750	—
8.50% senior secured notes, interest only payments until due in May 2029 [b]	1,000	—
5.50% senior secured notes, installments through April 2026 [c]	2,917	3,500
5.75% senior secured notes, installments beginning in July 2026 until due in April 2029 [c]	3,000	3,000
AAdvantage Term Loan Facility, variable interest rate of 10.43%, installments through April 2028 [c]	3,150	3,500
Enhanced equipment trust certificates (EETCs), fixed interest rates ranging from 2.88% to 5.90%, averaging 3.60%, maturing from 2024 to 2034	7,657	9,175
Equipment loans and other notes payable, fixed and variable interest rates ranging from 2.55% to 8.90%, averaging 6.98%, maturing from 2024 to 2035 [d]	3,612	3,170
Special facility revenue bonds, fixed interest rates ranging from 2.25% to 5.38%, maturing from 2026 to 2036	967	1,050
	27,526	30,043
Unsecured		
PSP1 Promissory Note, interest only payments until due in April 2030 [e]	1,757	1,757
PSP2 Promissory Note, interest only payments until due in January 2031 [e]	1,030	1,030
PSP3 Promissory Note, interest only payments until due in April 2031 [e]	959	959
6.50% convertible senior notes, interest only payments until due in July 2025 [f]	1,000	1,000
3.75% senior notes, interest only payments until due in March 2025 [g]	487	500
	5,233	5,246
Total long-term debt	32,759	35,289
Less: Total unamortized debt discount, premium and issuance costs	363	386
Less: Current maturities	3,501	3,059
Long-term debt, net of current maturities	$ 28,895	$ 31,844

As of December 31, 2023, the maximum availability under our revolving credit and other facilities is as follows (in millions):

2013 Revolving Facility	$ 736
2014 Revolving Facility	1,631
April 2016 Revolving Facility	446
Other short-term facility	49
Total	$ 2,862

As of December 31, 2023, American had $ 49 million of available borrowing base under a cargo receivables facility that is set to expire in December 2024. As a result of the below amendments to the 2013, 2014 and April 2016 Revolving Facilities, the aggregate commitments under these facilities will be $2.8 billion through October 11, 2024, and thereafter through October 13, 2026, such aggregate commitments will decrease to $ 2.2 billion.

Secured financings, including revolving credit and other facilities, are collateralized by assets, consisting primarily of aircraft, engines, simulators, aircraft spare parts, airport gate leasehold rights, route authorities, airport slots, certain receivables, certain intellectual property and certain loyalty program assets.

At December 31, 2023, the maturities of long-term debt are as follows (in millions):

2024	$	3,501
2025		5,189
2026		4,582
2027		4,618
2028		5,060
2029 and thereafter		9,809
Total	$	32,759

(a) 2013 and 2014 Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility

2013 Credit Facilities

The Amended and Restated Credit and Guaranty Agreement dated as of May 21, 2015, as amended (the 2013 Credit Agreement), includes a revolving credit facility (the 2013 Revolving Facility) and term loan (the 2013 Term Loan Facility), collectively referred to as the 2013 Credit Facilities. In February 2023, American and AAG refinanced approximately $1.8 billion in aggregate principal amount of term loans outstanding under the 2013 Term Loan Facility (the 2013 Term Loan Facility Refinancing) through the combination of (i) the issuance of $ 750 million in aggregate principal amount of 7.25% senior secured notes due 2028 and (ii) the entry into the Seventh Amendment to the 2013 Credit Agreement, pursuant to which the maturity of $1.0 billion in term loans under the 2013 Term Loan Facility was extended to February 2028 from June 2025. The Seventh Amendment also amended certain other terms of the 2013 Credit Agreement, including the interest rate and amortization schedule for the 2013 Term Loan Facility, the requirements for delivery of appraisals and certain covenants relating to dispositions of collateral. Additionally, the Seventh Amendment transitioned the benchmark interest rate from the London Interbank Offered Rate (LIBOR) to the Secured Overnight Financing Rate (SOFR). As a result, the 2013 Term Loan Facility bears interest at a base rate (subject to a floor of 1.00%) plus an applicable margin of 1.75% or, at American's option, the SOFR rate for a tenor of one, three or six months, depending on the interest period selected by American, plus the SOFR adjustment applicable to such interest period (with such SOFR rate plus SOFR adjustment being subject to a floor of 0.00%) and an applicable margin of 2.75%. As of December 31, 2023, the margin elected was 2.75%.

In March 2023, American and AAG entered into the Eighth Amendment to the 2013 Credit Agreement, pursuant to which American extended the maturity of certain commitments under the 2013 Revolving Facility. The Eighth Amendment also amended certain other terms of the 2013 Credit Agreement, including certain covenants and transitioned the benchmark interest rate from LIBOR to SOFR. The 2013 Revolving Facility bears interest at a base rate (subject to a floor of 1.00%) plus an applicable margin of 2.25%, 2.50% or 2.75%, depending on AAG's public corporate rating, or, at American's option, the SOFR rate for a tenor of one, three or six months, depending on the interest period selected by American, plus the SOFR adjustment applicable to such interest period (with such SOFR rate plus SOFR adjustment being subject to a floor of 0.00%) plus an applicable margin of 3.25%, 3.50% or 3.75%, depending on AAG's public corporate rating. Additionally, as a result of the Eighth Amendment, through October 11, 2024, the aggregate commitments under the 2013 Revolving Facility will be $736 million, and thereafter through October 13, 2026, such aggregate commitments will decrease to $563 million. As of December 31, 2023, there were no borrowings or letters of credit outstanding under the 2013 Revolving Facility.

2014 Credit Facilities

The Amended and Restated Credit and Guaranty Agreement, dated as of April 20, 2015, as amended (the 2014 Credit Agreement), includes a revolving credit facility (the 2014 Revolving Facility) and term loan (the 2014 Term Loan Facility), collectively referred to as the 2014 Credit Facilities. In March 2023, American and AAG entered into the Ninth Amendment to the 2014 Credit Agreement, pursuant to which American extended the maturity of certain commitments under the 2014 Revolving Facility. The Ninth Amendment also amended certain other terms of the 2014 Credit Agreement including the requirements for delivery of appraisals and certain other covenants and transitioned the benchmark interest rate for the 2014 Revolving Facility and the 2014 Term Loan Facility from LIBOR to SOFR. The 2014 Revolving Facility bears interest at the same base rate and applicable margin as the 2013 Revolving Facility, as noted above in "*2013 Credit Facilities.*" The 2014 Term Loan Facility bears interest at a base rate (subject to a floor of 1.00%) plus an applicable margin of 0.75% or, at American's option, the SOFR rate for a tenor of one, three or six months, depending on the interest period selected by American, plus the SOFR adjustment applicable to such interest period (with such SOFR rate plus SOFR adjustment being subject to a floor of 0.00%) plus an applicable margin of 1.75%. As of December 31, 2023, the margin elected was 1.75%. Additionally, as a result of the Ninth Amendment, through October 11, 2024, the aggregate commitments under the 2014 Revolving Facility will be $1.6 billion, and thereafter through October 13, 2026, such aggregate commitments will decrease to $1.2 billion. As of December 31, 2023, there were no borrowings or letters of credit outstanding under the 2014 Revolving Facility.

April 2016 Revolving Facility

In March 2023, American and AAG entered into the Sixth Amendment to the Credit and Guaranty Agreement, dated as of April 29, 2016 (the April 2016 Credit Agreement), which includes a revolving credit facility (the April 2016 Revolving Facility). Pursuant to the Sixth Amendment, American extended the maturity of certain commitments under the April 2016 Revolving Facility. The Sixth Amendment also amended certain other terms under the April 2016 Credit Agreement including the requirements for delivery of appraisals and certain other covenants and transitioned the benchmark interest rate for the April 2016 Revolving Facility from LIBOR to SOFR. The April 2016 Revolving Facility bears interest at the same base rate and applicable margin as the 2013 Revolving Facility, as noted above in "*2013 Credit Facilities.*" Additionally, as a result of the Sixth Amendment, through October 11, 2024, the aggregate commitments under the April 2016 Revolving Facility will be $446 million, and thereafter through October 13, 2026, such aggregate commitments will decrease to $342 million. As of December 31, 2023, there were no borrowings outstanding under the April 2016 Revolving Facility.

2023 Term Loan Facility

In December 2023, American and AAG entered into a credit and guaranty agreement (the 2023 Credit Agreement) that provided for a term loan facility (the 2023 Term Loan Facility) in an aggregate principal amount of $1.1 billion, maturing in June 2029. Loans made under the 2023 Term Loan Facility bear interest at a base rate (subject to a floor of 1.00%) plus an applicable margin of 2.50% or, at American's option, the SOFR rate for a tenor of one, three or six months (or if agreed by the relevant lenders, any other tenor), depending on the interest period selected by American (subject to a floor of 0.00%), plus an applicable margin of 3.50%. As of December 31, 2023, the margin elected was 3.50%. The net proceeds from the 2023 Term Loan Facility, together with the net proceeds from the private offering of the 8.50% Senior Secured Notes (as defined below) and cash on hand, were used to redeem all of the outstanding 11.75% Senior Secured Notes in December 2023.

Other Terms of the 2013 and 2014 Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility

The term loans under the 2013 Credit Facilities and 2014 Credit Facilities (collectively referred to as the Credit Facilities) and the 2023 Term Loan Facility are repayable in annual installments, in an amount equal to 1.00% of the aggregate principal amount issued, with any unpaid balance due on the respective maturity dates. Voluntary prepayments may be made by American at any time.

The 2013 Revolving Facility, 2014 Revolving Facility and April 2016 Revolving Facility provide that American may from time to time borrow, repay and reborrow loans thereunder. The 2013 Revolving Facility and 2014 Revolving Facility have the ability to issue letters of credit thereunder in an aggregate amount outstanding at any time up to $150 million and $300 million, respectively. The 2013 Revolving Facility, 2014 Revolving Facility and April 2016 Revolving Facility are each subject to an undrawn annual fee of 0.750%.

Subject to certain limitations and exceptions, the Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility are secured by collateral, including certain spare parts, slots, route authorities, simulators and leasehold rights. American has the ability to make future modifications to the collateral pledged, subject to certain restrictions. American's obligations under the Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility are guaranteed by AAG, and such guarantee is AAG's senior unsecured obligations (all of the collateral is owned by American, and AAG has not granted a security interest in any assets to secure any of the foregoing obligations). The Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility contain events of default customary for similar financings, including cross default and cross-acceleration to other material indebtedness.

(b) Senior Secured Notes

11.75% Senior Secured Notes

In June 2020, American issued $ 2.5 billion aggregate principal amount of 11.75% senior secured notes due 2025 (the 11.75% Senior Secured Notes) at a price equal to 99% of their aggregate principal amount. In December 2023, American redeemed all of its outstanding 11.75% Senior Secured Notes using net proceeds from the offering of the 8.50% Senior Secured Notes (as defined below), together with net proceeds from borrowings under the 2023 Term Loan Facility and cash on hand. In connection with the early redemption of the 11.75% Senior Secured Notes, in the fourth quarter of 2023, American recorded a $186 million cash special charge for the make-whole premium paid and a $ 19 million non-cash special charge to write off unamortized debt issuance costs and debt discount.

10.75% Senior Secured Notes

On September 25, 2020 (the 10.75% Senior Secured Notes Closing Date), American issued $ 1.0 billion in initial principal amount of senior secured IP notes (the IP Notes) and $200 million in initial principal amount of senior secured LGA/DCA notes (the LGA/DCA Notes and together with the IP Notes, the 10.75% Senior Secured Notes). The obligations of American under the 10.75% Senior Secured Notes are fully and unconditionally guaranteed (the 10.75% Senior Secured Notes Guarantees) on a senior unsecured basis by AAG. The 10.75% Senior Secured Notes bear interest at a rate of 10.75% per annum in cash. Interest on the 10.75% Senior Secured Notes is payable semiannually in arrears on September 1 and March 1 of each year, which began on March 1, 2021. The 10.75% Senior Secured Notes will mature on February 15, 2026.

The IP Notes are secured by a first lien security interest on certain intellectual property of American, including the "American Airlines" trademark and the "aa.com" domain name in the United States and certain foreign jurisdictions (the IP Collateral), and a second lien on certain slots related to American's operations at New York LaGuardia and Ronald Reagan Washington National airports and certain other assets (the LGA/DCA Collateral and together with the IP Collateral, the 10.75% Senior Secured Notes Collateral). LGA/DCA Notes are secured by a first lien security interest in the LGA/DCA Collateral.

On or prior to the fourth anniversary of the 10.75% Senior Secured Notes Closing Date, American may redeem all or any part of the 10.75% Senior Secured Notes, at its option, at a redemption price equal to 100% of the principal amount of the 10.75% Senior Secured Notes redeemed plus a "make-whole" premium, together with accrued and unpaid interest thereon, if any. After the fourth anniversary of the 10.75% Senior Secured Notes Closing Date and on or prior to the fifth anniversary of the 10.75% Senior Secured Notes Closing Date, American may redeem all or any part of the 10.75% Senior Secured Notes, at its option, at a redemption price equal to 105.375% of the principal amount of the 10.75% Senior Secured Notes redeemed, together with accrued and unpaid interest thereon, if any. After the fifth anniversary of the 10.75% Senior Secured Notes Closing Date, American may redeem all or any part of the 10.75% Senior Secured Notes, at its option, at par, together with accrued and unpaid interest thereon, if any.

7.25% Senior Secured Notes

On February 15, 2023, as part of the 2013 Term Loan Facility Refinancing, American issued $ 750 million aggregate principal amount of 7.25% senior secured notes due 2028 (the 7.25% Senior Secured Notes) in a private offering. The 7.25% Senior Secured Notes were issued at par and bear interest at a rate of 7.25% per annum (subject to increase if the collateral coverage ratio described below is not met). Interest on the 7.25% Senior Secured Notes is payable semiannually in arrears on February 15 and August 15 of each year, which began on August 15, 2023. The 7.25% Senior Secured Notes will mature on February 15, 2028. The obligations of American under the 7.25% Senior Secured Notes are fully and unconditionally guaranteed on a senior unsecured basis by AAG. American used the proceeds from the offering of the 7.25% Senior Secured Notes, together with cash on hand, to repay a portion of the term loans then outstanding under the 2013 Term Loan Facility and to pay related fees and expenses.

The 7.25% Senior Secured Notes were issued pursuant to an indenture, dated as of February 15, 2023 (the 7.25% Senior Secured Notes Indenture), by and among American, AAG and Wilmington Trust, National Association, as trustee and collateral agent. The 7.25% Senior Secured Notes are American's senior secured obligations and are secured on a first lien basis by security interests in certain assets, rights and properties that American uses to provide non-stop scheduled air carrier services between (a) certain airports in the United States and (b) airports in certain countries in South America and New Zealand (collectively, the 7.25% Senior Secured Notes Collateral). The 7.25% Senior Secured Notes Collateral also secures, on a first lien, pari passu basis with the 7.25% Senior Secured Notes, the 2013 Credit Facilities under the 2013 Credit Agreement.

American may redeem the 7.25% Senior Secured Notes, in whole at any time or in part from time to time prior to February 15, 2025, at a redemption price equal to 100% of the principal amount of the 7.25% Senior Secured Notes to be redeemed, plus a "make-whole" premium, plus any accrued and unpaid interest thereon to but excluding the date of redemption. At any time on or after February 15, 2025, American may redeem all or any of the 7.25% Senior Secured Notes in whole at any time, or in part from time to time, at the redemption prices described in the 7.25% Senior Secured Notes Indenture, plus any accrued and unpaid interest thereon to but excluding the date of redemption. In addition, at any time prior to February 15, 2025, American may redeem up to 40% of the original aggregate principal amount of the 7.25% Senior Secured Notes (calculated after giving effect to any issuance of additional notes) with the net cash proceeds of certain equity offerings, at a redemption price equal to 107.250% of the aggregate principal amount of the 7.25% Senior Secured Notes to be redeemed, plus any accrued and unpaid interest thereon to but excluding the date of redemption.

Twice per year, American is required to deliver an appraisal of the 7.25% Senior Secured Notes Collateral and an officer's certificate demonstrating the calculation of a collateral coverage ratio in relation to the 7.25% Senior Secured Notes Collateral (the 7.25% Senior Secured Notes Collateral Coverage Ratio) as of the date of delivery of the appraisal for the applicable period. If the 7.25% Senior Secured Notes Collateral Coverage Ratio is less than 1.6 to 1.0 as of the date of delivery of the appraisal for the applicable period, then, subject to a cure period in which additional collateral can be provided or debt repaid such that American meets the required 7.25% Senior Secured Notes Collateral Coverage Ratio, American will be required to pay special interest in an additional amount equal to 2.0% per annum of the principal amount of the 7.25% Senior Secured Notes until the 7.25% Senior Secured Notes Collateral Coverage Ratio is established to be at least 1.6 to 1.0.

8.50% Senior Secured Notes

On December 4, 2023, American issued $ 1.0 billion aggregate principal amount of 8.50% senior secured notes due 2029 (the 8.50% Senior Secured Notes) in a private offering. The 8.50% Senior Secured Notes were issued at par and bear interest at a rate of 8.50% per annum (subject to increase if the collateral coverage ratio described below is not met). Interest on the 8.50% Senior Secured Notes is payable semiannually in arrears on May 15 and November 15 of each year, beginning on May 15, 2024. The 8.50% Senior Secured Notes will mature on May 15, 2029. The obligations of American under the 8.50% Senior Secured Notes are fully and unconditionally guaranteed on a senior unsecured basis by AAG. The net proceeds from the 8.50% Senior Secured Notes, together with borrowings under the 2023 Term Loan Facility and cash on hand, were used to redeem all of the outstanding 11.75% Senior Secured Notes in December 2023.

The 8.50% Senior Secured Notes were issued pursuant to an indenture, dated as of December 4, 2023 (the 8.50% Senior Secured Notes Indenture), by and among American, AAG and Wilmington Trust, National Association, as trustee and collateral agent. The 8.50% Senior Secured Notes are American's senior secured obligations and are secured on a first lien basis by security interests in certain assets, rights and properties that American uses to provide non-stop scheduled air carrier services between (a) certain airports in the United States and (b) certain airports in Australia, Canada, the Caribbean, Central America, China, Hong Kong, Japan, Mexico, South Korea and Switzerland (collectively, the 8.50% Senior Secured Notes Collateral). The 8.50% Senior Secured Notes Collateral also secures, on a first lien, pari passu basis with the 8.50% Senior Secured Notes, the 2023 Term Loan Facility.

American may redeem the 8.50% Senior Secured Notes, in whole at any time or in part from time to time prior to November 15, 2025, at a redemption price equal to 100% of the principal amount of the 8.50% Senior Secured Notes to be redeemed, plus a "make-whole" premium, plus any accrued and unpaid interest thereon to but excluding the date of redemption. At any time on or after November 15, 2025, American may redeem all or any of the 8.50% Senior Secured Notes in whole at any time, or in part from time to time, at the redemption prices described in the 8.50% Senior Secured Notes Indenture, plus any accrued and unpaid interest thereon to but excluding the date of redemption. In addition, at any time prior to November 15, 2025, American may redeem up to 40% of the original aggregate principal amount of the 8.50% Senior Secured Notes (calculated after giving effect to any issuance of additional notes) with the net cash proceeds of certain equity offerings, at a redemption price equal to 108.50% of the aggregate principal amount of the 8.50% Senior

Secured Notes to be redeemed, plus any accrued and unpaid interest thereon to but excluding the date of redemption. In addition, during each twelve-month period beginning on December 4, 2023 and ending on or prior to November 15, 2025, American may redeem up to 10% of the original aggregate principal amount of the 8.50% Senior Secured Notes at a redemption price of 103% of the principal amount thereof, plus any accrued and unpaid interest thereon to, but excluding, the applicable date of redemption.

Twice per year, American is required to deliver an appraisal of the 8.50% Senior Secured Notes Collateral and an officer's certificate demonstrating the calculation of a collateral coverage ratio in relation to the 8.50% Senior Secured Notes Collateral (the 8.50% Senior Secured Notes Collateral Coverage Ratio) as of the date of delivery of the appraisal for the applicable period. If the 8.50% Senior Secured Notes Collateral Coverage Ratio is less than 1.6 to 1.0 as of the date of delivery of the appraisal for the applicable period, then, subject to a cure period in which additional collateral can be provided or debt repaid such that American meets the required 8.50% Senior Secured Notes Collateral Coverage Ratio, American will be required to pay special interest in an additional amount equal to 2.0% per annum of the principal amount of the 8.50% Senior Secured Notes until the 8.50% Senior Secured Notes Collateral Coverage Ratio is established to be at least 1.6 to 1.0.

(c) AAdvantage Financing

On March 24, 2021 (the AAdvantage Financing Closing Date), American and AAdvantage Loyalty IP Ltd., a Cayman Islands exempted company incorporated with limited liability and an indirect wholly-owned subsidiary of American (Loyalty Issuer and, together with American, the AAdvantage Issuers), completed the offering of $3.5 billion aggregate principal amount of 5.50% Senior Secured Notes due 2026 (the 2026 Notes) and $ 3.0 billion aggregate principal amount of 5.75% Senior Secured Notes due 2029 (the 2029 Notes, and together with the 2026 Notes, the AAdvantage Notes). The AAdvantage Notes are fully and unconditionally guaranteed by the SPV Guarantors and AAG.

Concurrent with the issuance of the AAdvantage Notes, the AAdvantage Issuers, as co-borrowers, entered into a term loan credit and guaranty agreement, dated March 24, 2021, providing for a $3.5 billion term loan facility (the AAdvantage Term Loan Facility and collectively with the AAdvantage Notes, the AAdvantage Financing) and pursuant to which the full $3.5 billion of term loans (the AAdvantage Loans) were drawn on the AAdvantage Financing Closing Date. The AAdvantage Loans are fully and unconditionally guaranteed (together with the AAdvantage Note Guarantees, the AAdvantage Guarantees) by the SPV Guarantors and AAG.

Subject to certain permitted liens and other exceptions, the AAdvantage Notes, AAdvantage Loans and AAdvantage Guarantees provided by the SPV Guarantors are secured by a first-priority security interest in, and pledge of, various agreements with respect to the AAdvantage program (the AAdvantage Agreements) (including all payments thereunder) and certain intellectual property licenses, certain deposit accounts that will receive cash under the AAdvantage Agreements, certain reserve accounts, the equity of each of Loyalty Issuer and the SPV Guarantors and substantially all other assets of Loyalty Issuer and the SPV Guarantors including American's rights to certain data and other intellectual property used in the AAdvantage program (subject to certain exceptions) (collectively, the AAdvantage Collateral).

Payment Terms of the AAdvantage Notes and AAdvantage Loans under the AAdvantage Term Loan Facility

Interest on the AAdvantage Notes is payable in cash, quarterly in arrears on the 20th day of each January, April, July and October (each, an AAdvantage Payment Date), which began on July 20, 2021. The 2026 Notes will mature on April 20, 2026, and the 2029 Notes will mature on April 20, 2029. The outstanding principal on the 2026 Notes will be repaid in quarterly installments of $292 million on each AAdvantage Payment Date, which began in July 2023. The outstanding principal on the 2029 Notes will be repaid in quarterly installments of $250 million on each AAdvantage Payment Date, beginning on July 20, 2026.

The AAdvantage Issuers may redeem the AAdvantage Notes, at their option, in whole at any time or in part from time to time, at a redemption price equal to 100% of the principal amount of the AAdvantage Notes redeemed plus a "make-whole" premium, together with accrued and unpaid interest to the date of redemption.

The scheduled maturity date of the AAdvantage Loans under the AAdvantage Term Loan Facility is April 20, 2028. The outstanding principal on the AAdvantage Loans will be repaid in quarterly installments of $175 million, on each AAdvantage Payment Date, which began in July 2023. These amortization payments (as well as those for the AAdvantage Notes) will be subject to the occurrence of certain early amortization events, including the failure to satisfy a minimum debt service coverage ratio at specified determination dates.

Prepayment of some or all of the AAdvantage Loans outstanding under the AAdvantage Term Loan Facility is permitted, although payment of an applicable premium is required as specified in the AAdvantage Term Loan Facility.

The AAdvantage Indenture and the AAdvantage Term Loan Facility contain mandatory prepayment provisions triggered upon (i) the issuance or incurrence by Loyalty Issuer or the SPV Guarantors of certain indebtedness or (ii) the receipt by American or its subsidiaries of net proceeds from pre-paid frequent flyer (i.e., AAdvantage) mile sales exceeding $505 million. Each of these prepayments would also require payment of an applicable premium. Certain other events, including the occurrence of a change of control with respect to AAG and certain AAdvantage Collateral sales exceeding a specified threshold, will also trigger mandatory repurchase or mandatory prepayment provisions under the AAdvantage Indenture and the AAdvantage Term Loan Facility, respectively.

In June 2023, American and AAdvantage Loyalty IP Ltd. entered into the First Amendment to the AAdvantage Term Loan Facility pursuant to which the benchmark interest rate transitioned from LIBOR to SOFR, effective July 1, 2023. As a result, the AAdvantage Term Loan Facility bears interest at a base rate (subject to a floor of 0.00%) plus an applicable margin of 3.75% or, at American's option, the SOFR rate for a tenor of three months, plus a 0.26161% credit spread adjustment (with such SOFR rate plus SOFR adjustment being subject to a floor of 0.75%) and an applicable margin of 4.75%. As of December 31, 2023, the margin elected was 4.75%. Other than the foregoing, the terms of the AAdvantage Term Loan Facility remain substantially unchanged.

(d) Equipment Loans and Other Notes Payable Issued in 2023

In 2023, American entered into agreements under which it borrowed $ 1.1 billion in connection with the financing of certain aircraft. Debt incurred under these agreements matures in 2032 through 2035 and bears interest at fixed and variable rates (comprised of SOFR plus an applicable margin) averaging 7.15% as of December 31, 2023.

(e) PSP Promissory Notes

As partial compensation to the U.S. Government for the provision of financial assistance under the PSP Agreements, AAG issued promissory notes to Treasury (PSP1 Promissory Note, PSP2 Promissory Note and PSP3 Promissory Note, collectively the PSP Promissory Notes), in the aggregate principal sum of $3.7 billion which provides for the guarantee of our obligations under the PSP Promissory Notes by the Subsidiaries.

The PSP Promissory Notes bear interest on the outstanding principal amount at a rate equal to 1.00% per annum until the fifth anniversary of the applicable PSP closing date and 2.00% plus an interest rate based on SOFR per annum or other benchmark replacement rate consistent with customary market conventions (but not to be less than 0.00%) thereafter until the tenth anniversary of the applicable PSP closing date, and interest accrued thereon will be payable in arrears on the last business day of March and September of each year. The aggregate principal amount outstanding under the PSP Promissory Notes, together with all accrued and unpaid interest thereon and all other amounts payable under the PSP Promissory Notes, will be due and payable on the maturity date.

The PSP Promissory Notes are our senior unsecured obligation and each guarantee of the PSP Promissory Notes is the senior unsecured obligation of each of the Subsidiaries, respectively.

We may, at any time and from time to time, voluntarily prepay amounts outstanding under the PSP Promissory Notes, in whole or in part, without penalty or premium. Within 30 days of the occurrence of certain change of control triggering events, we are required to prepay the aggregate outstanding principal amount of the PSP Promissory Notes at such time, together with any accrued interest or other amounts owing under the PSP Promissory Notes at such time.

(f) 6.50% Convertible Senior Notes

In June 2020, AAG completed the public offering of $ 1.0 billion aggregate principal amount of AAG's 6.50% convertible senior notes due 2025 (the Convertible Notes). The Convertible Notes are fully and unconditionally guaranteed by American on a senior unsecured basis (the Convertible Notes Guarantee). The net proceeds from the Convertible Notes were approximately $970 million, after deducting the underwriters' discounts and commissions and our estimated offering expenses.

The Convertible Notes bear interest at a rate of 6.50% per annum. Interest on the Convertible Notes is payable semiannually in arrears on January 1 and July 1 of each year, which began on January 1, 2021. The Convertible Notes will mature on July 1, 2025, unless earlier converted, redeemed or repurchased by us.

Upon conversion, AAG will pay or deliver, as the case may be, cash, shares of AAG common stock or a combination of cash and shares of AAG common stock, at AAG's election. The initial conversion rate is 61.7284 shares of AAG common stock per $1,000 principal amount of Convertible Notes (equivalent to an initial conversion price of approximately $16.20 per share of AAG common stock). The conversion rate is subject to adjustment in some events as described in the Convertible Notes Indenture.

Holders may convert their Convertible Notes at their option only in the following circumstances: (1) during any calendar quarter (and only during such calendar quarter) commencing after the calendar quarter ending on September 30, 2020, if the last reported sale price per share of AAG common stock exceeds 130% of the conversion price for each of at least 20 trading days (whether or not consecutive) during the 30 consecutive trading days ending on, and including, the last trading day of the immediately preceding calendar quarter; (2) during the five consecutive business days immediately after any 10 consecutive trading day period (such 10 consecutive trading day period, the measurement period) in which the trading price per $1,000 principal amount of Convertible Notes for each trading day of the measurement period was less than 98% of the product of the last reported sale price per share of AAG common stock on such trading day and the conversion rate on such trading day; (3) upon the occurrence of certain corporate events or distributions on AAG common stock; (4) if AAG calls such Convertible Notes for redemption; and (5) at any time from, and including, April 1, 2025 until the close of business on the scheduled trading day immediately before the maturity date of the Convertible Notes.

In addition, following certain corporate events that occur prior to the maturity date or upon AAG's issuance of a notice of redemption, AAG will increase the conversion rate for a holder who elects to convert its Convertible Notes in connection with such corporate event or during the related redemption period in certain circumstances by a specified number of shares of AAG common stock as described in the Convertible Notes Indenture.

AAG did not have the right to redeem the Convertible Notes prior to July 5, 2023. On or after July 5, 2023 and on or before the 20th scheduled trading day immediately before the maturity date, AAG may redeem the Convertible Notes, in whole or in part, if the last reported sale price of AAG common stock has been at least 130% of the conversion price then in effect on (1) each of at least 20 trading days (whether or not consecutive) during the 30 consecutive trading days ending on, and including, the trading day immediately before the date AAG sends the related redemption notice; and (2) the trading day immediately before the date AAG sends such notice. In the case of any optional redemption, AAG will redeem the Convertible Notes at a redemption price equal to 100% of the principal amount of such Convertible Notes to be redeemed, plus accrued and unpaid interest to, but excluding, the redemption date.

The following table provides information relating to the Convertible Notes as of December 31, 2023 and 2022 (in millions):

	December 31,	
	2023	2022
Principal amount	$ 1,000	$ 1,000
Unamortized debt discount	(10)	(16)
Net carrying amount	$ 990	$ 984

The effective interest rate for the Convertible Notes was 7% for each of the years ended December 31, 2023, 2022, and 2021. Interest recognized for the Convertible Notes is as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Contractual coupon interest	$ 65	$ 65	$ 65
Non-cash amortization of debt discount	6	6	5
Total interest expense	$ 71	$ 71	$ 70

At December 31, 2023, the if-converted value of the Convertible Notes did not exceed the principal amount. The last reported sale price per share of our common stock (as defined in the Convertible Notes Indenture) did not exceed 130% of the conversion price of the Convertible Notes for at least 20 of the 30 consecutive trading days ending on December 31, 2023. Accordingly, pursuant to the terms of the Convertible Notes Indenture, the holders of the Convertible Notes cannot convert at their option at any time during the quarter ending March 31, 2024. Each $1,000 principal amount of Convertible Notes is convertible at a rate of 61.7284 shares of our common stock, subject to adjustment as provided in the Convertible Notes Indenture. We may settle conversions by paying or delivering, as applicable, cash, shares of our common stock or a combination of cash and shares of our common stock, at our election.

(g) Unsecured Senior Notes

3.75% Senior Notes

In February 2020, AAG issued $500 million aggregate principal amount of 3.75% senior notes due 2025 (the 3.75% Senior Notes). The 3.75% Senior Notes bear interest at a rate of 3.75% per annum, payable semiannually in arrears in March and September of each year, which began in September 2020. The 3.75% Senior Notes mature in March 2025.

The 3.75% Senior Notes are senior unsecured obligations of AAG. These Senior Notes are fully and unconditionally guaranteed by American. The indentures for these Senior Notes contain covenants and events of default generally customary for similar financings.

Other Financing Activities

During the year ended December 31, 2023, we repurchased $552 million of secured and unsecured notes in the open market. In connection with the repurchase of these secured and unsecured notes in the open market, American recorded $57 million of cash special charges for premiums paid and $6 million of non-cash special charges to write off unamortized debt issuance costs and debt discounts.

Guarantees

As of December 31, 2023, AAG had issued guarantees covering approximately $ 17.5 billion of American's secured debt (and interest thereon), including the Credit Facilities, 2023 Term Loan Facility, the AAdvantage Financing, certain EETC financings and special facility revenue bonds.

Certain Covenants

Our debt agreements contain customary terms and conditions as well as various affirmative, negative and financial covenants that, among other things, may restrict the ability of us and our subsidiaries to incur additional indebtedness, pay dividends or repurchase stock. Our debt agreements also contain customary change of control provisions, which may require us to repay or redeem such indebtedness upon certain events constituting a change of control under the relevant agreement, in certain cases at a premium. Certain of our debt financing agreements (including our secured notes, term loans, revolving credit facilities and spare engine EETCs) contain loan to value (LTV), collateral coverage or peak debt service coverage ratio covenants and certain agreements require us to appraise the related collateral annually or semiannually. Pursuant to such agreements, if the applicable LTV, collateral coverage or peak debt service coverage ratio exceeds or falls below a specified threshold, as the case may be, we will be required, as applicable, to pledge additional qualifying collateral (which in some cases may include cash or investment securities), withhold additional cash in certain accounts, or pay down such financing, in whole or in part, or the interest rate for the relevant financing will be increased. Additionally, a significant portion of our debt financing agreements contain covenants requiring us to maintain an aggregate of at least $2.0 billion of unrestricted cash and cash equivalents and amounts available to be drawn under revolving credit facilities, and our AAdvantage Financing contains a peak debt service coverage ratio, pursuant to which failure to comply with a certain threshold may result in early repayment, in whole or in part, of the AAdvantage Financing.

Specifically, we are required to meet certain collateral coverage tests for our Credit Facilities, April 2016 Revolving Facility, 2023 Term Loan Facility, 7.25% Senior Secured Notes, 8.50% Senior Secured Notes and 10.75% Senior Secured Notes, as described below:

	2013 Credit Facilities	7.25% Senior Secured Notes	2014 Credit Facilities	April 2016 Revolving Facility	2023 Term Loan Facility	8.50% Senior Secured Notes	10.75% Senior Secured Notes
LTV Requirement	1.6x Collateral valuation to amount of debt outstanding (62.5% LTV)						
LTV as of Last Measurement Date	34.2%		16.4%	Not Applicable	25.9%		6.9%
Frequency of Appraisals of Appraised Collateral	Semi-Annual						Annual
Collateral Description	Generally, certain slots, route authorities and airport gate leasehold rights used by American to operate certain services between the U.S. and South America and New Zealand		Generally, certain slots, route authorities and airport gate leasehold rights used by American to operate certain services between the U.S. and European Union (including London Heathrow)	Generally, certain spare parts	Generally, certain slots, route authorities and airport gate leasehold rights used by American to operate certain services between the U.S. and Australia, Canada, the Caribbean, Central America, China, Hong Kong, Japan, Mexico, South Korea and Switzerland		Generally, certain DCA slots, certain LGA slots, certain simulators and certain leasehold rights and, in the case of the IP Notes, certain intellectual property of American

At December 31, 2023, we were in compliance with the applicable collateral coverage tests as of the most recent measurement dates.

5. Leases

We lease certain aircraft and engines, including aircraft under capacity purchase agreements. As of December 31, 2023, we operated 737 leased aircraft, including seven aircraft in temporary storage and 237 aircraft leased under capacity purchase agreements, with remaining terms ranging from less than one year to 10 years.

At each airport where we conduct flight operations, we have agreements, generally with a governmental unit or authority, for the use of passenger, operations and baggage handling space as well as runways and taxiways. These agreements, particularly in the U.S., often contain provisions for periodic adjustments to rates and charges applicable under such agreements. These rates and charges also vary with our level of operations and the operations of the airport. Because of the variable nature of these rates, these leases are not recorded on our consolidated balance sheets as a ROU asset or a lease liability. Additionally, at our hub locations and in certain other cities we serve, we lease administrative offices, catering, cargo, training, maintenance and other facilities.

The components of lease expense were as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Operating lease cost	$ 2,016	$ 2,007	$ 2,012
Finance lease cost:			
Amortization of assets	128	143	107
Interest on lease liabilities	45	47	44
Variable lease cost	2,720	2,580	2,471
Total net lease cost	$ 4,909	$ 4,777	$ 4,634

Included in the table above is $274 million, $242 million and $190 million of operating lease cost under our capacity purchase agreement with Republic for the years ended December 31, 2023, 2022 and 2021, respectively. We hold a 25% equity interest in Republic Holdings, the parent company of Republic.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS OF AMERICAN AIRLINES GROUP INC .

Supplemental balance sheet information related to leases was as follows (in millions, except lease term and discount rate):

	December 31,	
	2023	2022
Operating leases:		
Operating lease ROU assets	$ 7,939	$ 8,094
Current operating lease liabilities	$ 1,309	$ 1,465
Noncurrent operating lease liabilities	6,452	6,559
Total operating lease liabilities	$ 7,761	$ 8,024
Finance leases:		
Property and equipment, at cost	$ 1,380	$ 1,364
Accumulated amortization	(891)	(779)
Property and equipment, net	$ 489	$ 585
Current finance lease liabilities	$ 131	$ 216
Noncurrent finance lease liabilities	375	545
Total finance lease liabilities	$ 506	$ 761
Weighted average remaining lease term (in years):		
Operating leases	8.4	8.3
Finance leases	5.8	5.1
Weighted average discount rate:		
Operating leases	7.6 %	7.4 %
Finance leases	7.2 %	7.2 %

Supplemental cash flow and other information related to leases was as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Cash paid for amounts included in the measurement of lease liabilities:			
Operating cash flows from operating leases	$ 2,033	$ 1,990	$ 2,053
Operating cash flows from finance leases	48	47	37
Financing cash flows from finance leases	265	190	126
Gain on sale leaseback transactions, net	12	2	25

Maturities of lease liabilities were as follows (in millions):

	December 31, 2023	
	Operating Leases	Finance Leases
2024	$ 1,809	$ 166
2025	1,510	140
2026	1,289	114
2027	1,120	71
2028	988	30
2029 and thereafter	3,708	89
Total lease payments	10,424	610
Less: Imputed interest	(2,663)	(104)
Total lease obligations	7,761	506
Less: Current obligations	(1,309)	(131)
Long-term lease obligations	$ 6,452	$ 375

As of December 31, 2023, we had additional operating lease commitments that have not yet commenced of approximately $ 669 million for five Boeing 787 Family aircraft scheduled to be delivered in 2024 with lease terms of 10 years.

6. Income Taxes

The significant components of the income tax provision (benefit) were (in millions):

	Year Ended December 31,		
	2023	2022	2021
Current income tax benefit:			
State, local and foreign	$ —	$ (6)	$ —
Deferred income tax provision (benefit):			
Federal	268	59	(508)
State and local	31	6	(47)
Deferred income tax provision (benefit)	299	65	(555)
Total income tax provision (benefit)	$ 299	$ 59	$ (555)

The income tax provision (benefit) differed from amounts computed at the statutory federal income tax rate as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Statutory income tax provision (benefit)	$ 235	$ 39	$ (535)
State, local and foreign income tax provision (benefit), net of federal tax effect	22	—	(37)
Book expenses not deductible for tax purposes	38	22	23
Change in valuation allowance	3	—	—
Other, net	1	(2)	(6)
Income tax provision (benefit)	$ 299	$ 59	$ (555)

The components of our deferred tax assets and liabilities were (in millions):

	December 31,	
	2023	2022
Deferred tax assets:		
Net operating loss and other carryforwards	$ 4,238	$ 4,679
Loyalty program liability	1,774	1,809
Leases	1,758	1,819
Pension benefits	434	474
Postretirement benefits other than pension benefits	274	179
Rent expense	84	130
Other	902	742
Total deferred tax assets	9,464	9,832
Valuation allowance	(22)	(19)
Net deferred tax assets	9,442	9,813
Deferred tax liabilities:		
Accelerated depreciation and amortization	(4,503)	(4,630)
Leases	(1,798)	(1,832)
Other	(262)	(262)
Total deferred tax liabilities	(6,563)	(6,724)
Net deferred tax asset	$ 2,879	$ 3,089

At December 31, 2023, we had approximately $ 13.7 billion of gross federal net operating losses (NOLs) and $ 4.7 billion of other carryforwards available to reduce future federal taxable income, of which $3.4 billion will expire beginning in 2029 if unused and $ 15.0 billion can be carried forward indefinitely. We also had approximately $5.5 billion of NOL carryforwards to reduce future state taxable income at December 31, 2023, which will expire in taxable years 2023 through 2043 if unused.

Our ability to use our NOLs and other carryforwards depends on the amount of taxable income generated in future periods. We provide a valuation allowance for our deferred tax assets, which include our NOLs, when it is more likely than not that some portion, or all of our deferred tax assets, will not be realized. We consider all available positive and negative evidence and make certain assumptions in evaluating the realizability of our deferred tax assets. Many factors are considered that impact our assessment of future profitability, including conditions which are beyond our control, such as the health of the economy, the availability and price volatility of aircraft fuel and travel demand. We have determined that positive factors outweigh negative factors in the determination of the realizability of our deferred tax assets. There can be no assurance that an additional valuation allowance on our net deferred tax assets will not be required. Such valuation allowance could be material.

Our ability to deduct our NOL carryforwards and to utilize certain other available tax attributes can be substantially constrained under the general annual limitation rules of Section 382 where an “ownership change” has occurred. Substantially all of our remaining federal NOL carryforwards attributable to US Airways Group are subject to limitation under Section 382; however, our ability to utilize such NOL carryforwards is not anticipated to be effectively constrained as a result of such limitation. Similar limitations may apply for state income tax purposes. Our ability to utilize any new NOL carryforwards arising after the ownership changes is not affected by the annual limitation rules imposed by Section 382 unless another ownership change occurs. Under the Section 382 limitation, cumulative stock ownership changes among material stockholders exceeding 50% during a rolling three-year period can potentially limit our future use of NOLs and tax credits.

In 2023, we recorded an income tax provision of $ 299 million, with an effective rate of approximately 27%, which was substantially non-cash. Substantially all of our income before income taxes is attributable to the United States.

We file our tax returns as prescribed by the tax laws of the jurisdictions in which we operate. Our 2020 through 2022 tax years are still subject to examination by the Internal Revenue Service. Various state and foreign jurisdiction tax years remain open to examination, and we are under examination, in administrative appeals or engaged in tax litigation in certain jurisdictions. We believe that the effect of any assessments will not be material to our consolidated financial statements.

The amount of, and changes to, our uncertain tax positions were not material in any of the years presented. We accrue interest and penalties related to unrecognized tax benefits in interest expense and operating expense, respectively.

7. Fair Value Measurements

Assets Measured at Fair Value on a Recurring Basis

Fair value is defined as the price that would be received from the sale of an asset or paid to transfer a liability (i.e., an exit price) on the measurement date in an orderly transaction between market participants in the principal or most advantageous market for the asset or liability. Accounting standards include disclosure requirements around fair values used for certain financial instruments and establish a fair value hierarchy. The hierarchy prioritizes valuation inputs into three levels based on the extent to which inputs used in measuring fair value are observable in the market. Each fair value measurement is reported in one of three levels:

- Level 1 – Observable inputs such as quoted prices in active markets;
- Level 2 – Inputs, other than quoted prices in active markets, that are observable either directly or indirectly; and
- Level 3 – Unobservable inputs in which there is little or no market data, which require the reporting entity to develop its own assumptions.

When available, we use quoted market prices to determine the fair value of our financial assets. If quoted market prices are not available, we measure fair value using valuation techniques that use, when possible, current market-based or independently-sourced market parameters, such as interest rates and currency rates.

We utilize the market approach to measure the fair value of our financial assets. The market approach uses prices and other relevant information generated by market transactions involving identical or comparable assets. Our short-term investments, restricted cash and restricted short-term investments classified as Level 2 utilize significant observable inputs, other than quoted prices in active markets, for valuation of these securities. No changes in valuation techniques or inputs occurred during the year ended December 31, 2023.

Assets measured at fair value on a recurring basis are summarized below (in millions):

	Fair Value Measurements as of December 31, 2023			
	Total	Level 1	Level 2	Level 3
Short-term investments [1], [2]:				
Money market funds	$ 818	$ 818	$ —	$ —
Corporate obligations	4,046	—	4,046	—
Bank notes/certificates of deposit/time deposits	1,586	—	1,586	—
Repurchase agreements	450	—	450	—
U.S. government and agency obligations	100	—	100	—
	7,000	818	6,182	—
Restricted cash and short-term investments [1], [3]	910	459	451	—
Long-term investments [4]	163	163	—	—
Total	$ 8,073	$ 1,440	$ 6,633	$ —

	Fair Value Measurements as of December 31, 2022			
	Total	Level 1	Level 2	Level 3
Short-term investments [1]:				
Money market funds	$ 732	$ 732	$ —	$ —
Corporate obligations	3,688	—	3,688	—
Bank notes/certificates of deposit/time deposits	3,655	—	3,655	—
Repurchase agreements	450	—	450	—
	8,525	732	7,793	—
Restricted cash and short-term investments [1], [3]	995	535	460	—
Long-term investments [4]	245	245	—	—
Total	$ 9,765	$ 1,512	$ 8,253	$ —

[1] All short-term investments are classified as available-for-sale and stated at fair value. Unrealized gains and losses are recorded in accumulated other comprehensive loss at each reporting period. There were no credit losses.

[2] Our short-term investments as of December 31, 2023 mature in one year or less.

[3] Restricted cash and short-term investments primarily include collateral held to support workers' compensation obligations and collateral associated with the payment of interest for the AAdvantage Financing. Restricted short-term investments mature in one year or less except for $218 million as of December 31, 2023.

[4] Long-term investments include our equity investments in China Southern Airlines Company Limited (China Southern Airlines), GOL and Vertical Aerospace Ltd. (Vertical). See Note 8 for further information on our equity investments.

Fair Value of Debt

The fair value of our long-term debt was estimated using quoted market prices or discounted cash flow analyses based on our current estimated incremental borrowing rates for similar types of borrowing arrangements. If our long-term debt was measured at fair value, it would have been classified as Level 2 except for $3.7 billion as of December 31, 2023 and December 31, 2022, which would have been classified as Level 3 in the fair value hierarchy. The fair value of the Convertible Notes, which would have been classified as Level 2, was $1.1 billion as of December 31, 2023 and December 31, 2022.

The carrying value and estimated fair value of our long-term debt, including current maturities, were as follows (in millions):

	December 31, 2023		December 31, 2022	
	Carrying Value	Fair Value	Carrying Value	Fair Value
Long-term debt, including current maturities	$ 32,396	$ 32,310	$ 34,903	$ 32,569

8. Investments

To help expand our network and as part of our ongoing commitment to sustainability, we enter into various commercial relationships or other strategic partnerships, including equity investments, with other airlines and companies. Our equity investments are reflected in other assets on our consolidated balance sheets. Our share of equity method investees' financial results and changes in fair value are recorded in nonoperating other income (expense), net on the consolidated statements of operations.

Our equity investments ownership interest and carrying value were as follows:

		Ownership Interest		Carrying Value (in millions)	
		December 31,		December 31,	
	Accounting Treatment	2023	2022	2023	2022
Republic Holdings	Equity Method	25.0 %	25.0 %	$ 240	$ 222
China Southern Airlines	Fair Value	1.5 %	1.5 %	115	176
Other investments [1]	Various			186	212
Total				$ 541	$ 610

[1] Primarily includes our investment in JetSMART Holdings Limited, which is accounted for under the equity method, and our investments in GOL and Vertical, which are each accounted for at fair value.

9. Employee Benefit Plans

We sponsor defined benefit and defined contribution pension plans for eligible employees. The defined benefit pension plans provide benefits for participating employees based on years of service and average compensation for a specified period of time before retirement. Effective November 1, 2012, substantially all of our defined benefit pension plans were frozen and we began providing enhanced benefits under our defined contribution pension plans for certain employee groups. We use a December 31 measurement date for all of our defined benefit pension plans. We also provide certain retiree medical and other postretirement benefits, including health care and life insurance benefits, to retired employees.

Benefit Obligations, Fair Value of Plan Assets and Funded Status

The following tables provide a reconciliation of the changes in the pension and retiree medical and other postretirement benefits obligations, fair value of plan assets and funded status as of December 31, 2023 and 2022:

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
	(In millions)			
Benefit obligation at beginning of period	$ 14,037	$ 18,910	$ 906	$ 1,098
Service cost	2	3	17	16
Interest cost	758	556	55	30
Actuarial loss (gain) [1], [2]	507	(4,563)	92	(167)
Plan amendments [3]	—	—	339	—
Other	—	—	—	3
Benefit payments	(894)	(869)	(84)	(74)
Benefit obligation at end of period	$ 14,410	$ 14,037	$ 1,325	$ 906
Fair value of plan assets at beginning of period	$ 11,884	$ 14,691	$ 133	$ 167
Actual return (loss) on plan assets	1,368	(1,943)	14	(18)
Employer contributions [4]	73	5	70	58
Benefit payments	(894)	(869)	(84)	(74)
Fair value of plan assets at end of period	$ 12,431	$ 11,884	$ 133	$ 133
Funded status at end of period	$ (1,979)	$ (2,153)	$ (1,192)	$ (773)

[1] The 2023 and 2022 pension actuarial loss (gain) primarily relates to the change in our weighted average discount rate assumption.

[2] The 2023 and 2022 retiree medical and other postretirement benefits actuarial loss (gain) primarily relates to the change in our weighted average discount rate assumption and, in 2023, the change in health care cost assumptions.

(3) As of September 30, 2023, we remeasured our retiree medical and other postretirement benefits to account for enhanced retirement benefits provided to our mainline pilots pursuant to the new collective bargaining agreement ratified in August 2023. As a result, we increased our postretirement benefits obligation by $339 million, which was included as a component of prior service cost in accumulated other comprehensive loss.

(4) In 2023, we made required contributions of $ 69 million to our defined benefit pension plans.

Balance Sheet Position

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
	(In millions)			
As of December 31,				
Current liability	$ 5	$ 4	$ 122	$ 85
Noncurrent liability	1,974	2,149	1,070	688
Total liabilities	$ 1,979	$ 2,153	$ 1,192	$ 773

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
	(In millions)			
Net actuarial loss (gain)	$ 3,566	$ 3,613	$ (383)	$ (505)
Prior service cost (benefit)	—	18	197	(148)
Total accumulated other comprehensive loss (income), pre-tax	$ 3,566	$ 3,631	$ (186)	$ (653)

Plans with Projected Benefit Obligations Exceeding Fair Value of Plan Assets

	Pension Benefits	
	2023	2022
	(In millions)	
Projected benefit obligation	$ 14,410	$ 14,037
Fair value of plan assets	12,431	11,884

Plans with Accumulated Benefit Obligations Exceeding Fair Value of Plan Assets

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
	(In millions)			
Accumulated benefit obligation	$ 14,403	$ 14,030	$ —	$ —
Accumulated postretirement benefit obligation	—	—	1,325	906
Fair value of plan assets	12,431	11,884	133	133

Net Periodic Benefit Cost (Income)

	Pension Benefits			Retiree Medical and Other Postretirement Benefits		
	2023	2022	2021	2023	2022	2021
	(In millions)					
Defined benefit plans:						
Service cost	$ 2	$ 3	$ 4	$ 17	$ 16	$ 12
Interest cost	758	556	526	55	30	30
Expected return on assets	(918)	(1,138)	(1,084)	(11)	(12)	(12)
Special termination benefits	—	—	—	—	—	139
Amortization of:						
Prior service cost (benefit)	18	28	28	(6)	(14)	(13)
Unrecognized net loss (gain)	106	156	212	(34)	(30)	(24)
Net periodic benefit cost (income)	$ (34)	$ (395)	$ (314)	$ 21	$ (10)	$ 132

The service cost component of net periodic benefit cost (income) is included in operating expenses, the cost for the special termination benefits is included in special items, net and the other components of net periodic benefit cost (income) are included in nonoperating other income (expense), net on our consolidated statements of operations.

Assumptions

The following actuarial assumptions were used to determine our benefit obligations and net periodic benefit cost (income) for the periods presented:

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
Benefit obligations:				
Weighted average discount rate	5.2%	5.6%	5.3%	5.7%

	Pension Benefits			Retiree Medical and Other Postretirement Benefits		
	2023	2022	2021	2023	2022	2021
Net periodic benefit cost (income):						
Weighted average discount rate	5.6%	3.0%	2.7%	5.7%	2.8%	2.4%
Weighted average expected rate of return on plan assets	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%
Weighted average health care cost trend rate assumed for next year [1]	N/A	N/A	N/A	6.5%	5.8%	4.8%

[1] The weighted average health care cost trend rate at December 31, 2023 is assumed to decline gradually to 4.5% by 2033 and remain level thereafter.

As of December 31, 2023, our estimate of the long-term rate of return on plan assets was 8.0% based on the target asset allocation. Expected returns on long duration bonds are based on yields to maturity of the bonds held at year-end. Expected returns on other assets are based on a combination of long-term historical returns, actual returns on plan assets achieved over the last 10 years, current and expected market conditions, and expected value to be generated through active management and securities lending programs.

Minimum Contributions

We are required to make minimum contributions to our defined benefit pension plans under the minimum funding requirements of the Employee Retirement Income Security Act of 1974 (ERISA) and various other laws for U.S. based plans as well as underfunding rules specific to countries where we maintain defined benefit pension plans. Based on current funding assumptions, we have minimum required contributions of $284 million for 2024 including contributions to defined benefit pension plans for our wholly-owned subsidiaries. Our future funding obligations will depend on the performance of our investments held in a trust by the pension plans, interest rates for determining funding targets, the amount of and timing of any supplemental contributions and our actuarial experience.

Benefit Payments

The following benefit payments, which reflect expected future service as appropriate, are expected to be paid (approximately, in millions):

	2024	2025	2026	2027	2028	2029-2033
Pension benefits	$ 952	$ 977	$ 1,002	$ 1,023	$ 1,036	$ 5,265
Retiree medical and other postretirement benefits	138	144	150	150	148	674

Plan Assets

The objectives of our investment policies are to: maintain sufficient income and liquidity to pay retirement benefits; produce a long-term rate of return that meets or exceeds the assumed rate of return for plan assets; limit the volatility of asset performance and funded status; and diversify assets among asset classes and investment managers.

Based on these investment objectives, a long-term strategic asset allocation has been established. This strategic allocation seeks to balance the potential benefit of improving the funded position with the potential risk that the funded position would decline. The current strategic target asset allocation with the corresponding allowed range is as follows:

Asset Class/Sub-Class	Target Allocation	Allowed Range
Equity	61 %	30% - 85%
Public:		
U.S. Large	19 %	10% - 40%
U.S. Small/Mid	5 %	0% - 10%
International Large	13 %	5% - 25%
International Small/Mid	3 %	0% - 10%
Emerging Markets	6 %	0% - 15%
Private Equity	15 %	5% - 30%
Fixed Income	39 %	15% - 70%
Public U.S. Fixed Income	30 %	15% - 60%
Private Income	9 %	0% - 20%
Other	— %	0% - 5%
Cash Equivalents	— %	0% - 20%

Public equity investments are intended to provide a real return over a full market cycle and, therefore, to contribute to the pension plan's long-term objective. Public fixed income investments are intended to provide income to the plan and offer the potential for long term capital appreciation. Private investments, such as private equity and private income, are used to provide higher expected returns than public markets over the long-term by assuming reduced levels of liquidity and higher levels of risk. The pension plan's master trust participates in securities lending programs to generate additional income by loaning plan assets to borrowers on a fully collateralized basis. The pension plan's master trust will also engage in derivative instruments to equitize residual levels of cash as well as hedge the pension plan's exposure to interest rates. Such programs are subject to market risk and counterparty risk.

Investments in securities traded on recognized securities exchanges are valued at the last reported sales price on the last business day of the year. Securities traded in the over-the-counter market are valued at the last bid price. Investments in limited partnerships are carried at estimated net asset value (NAV) as determined by and reported by the general partners of the partnerships and represent the proportionate share of the estimated fair value of the underlying assets of the limited partnerships. Mutual funds are valued once daily through a NAV calculation provided at the end of each trade day. Common/collective trusts are valued at NAV based on the fair values of the underlying investments of the trusts as determined by the sponsor of the trusts. No changes in valuation techniques or inputs occurred during the year.

Benefit Plan Assets Measured at Fair Value on a Recurring Basis

The fair value of our pension plan assets at December 31, 2023 and 2022, by asset category, were as follows (in millions) [1]:

	December 31, 2023				December 31, 2022			
	Level 1	Level 2	Level 3	Total	Level 1	Level 2	Level 3	Total
Equity [2]	$ 3,182	$ —	$ —	$ 3,182	$ 3,097	$ —	$ —	$ 3,097
Fixed income [3]	260	3,238	—	3,498	227	2,917	—	3,144
Other, net [4]	(6)	348	84	426	74	278	75	427
Measured at NAV [5]:								
Common collective trusts [6]	—	—	—	1,244	—	—	—	1,694
Private investments [7]	—	—	—	4,081	—	—	—	3,522
Total plan assets	$ 3,436	$ 3,586	$ 84	$ 12,431	$ 3,398	$ 3,195	$ 75	$ 11,884

[1] See Note 7 for a description of the levels within the fair value hierarchy.

[2] Equity investments include domestic and international common stock, preferred stock and mutual funds invested in equity securities.

[3] Fixed income investments include corporate, government and U.S. municipal bonds, as well as mutual funds invested in fixed income securities.

[4] Other primarily includes a short-term investment fund, net receivables and payables of the master trust for dividends, interest and amounts due to or from the sale and purchase of securities and cash and cash equivalents.

[5] Includes investments that were measured at NAV per share (or its equivalent) as a practical expedient that have not been classified in the fair value hierarchy.

[6] Common collective trusts include commingled funds primarily invested in equity securities. For some trusts, requests for withdrawals must meet specific requirements with advance notice of redemption preferred.

[7] Private investments include limited partnerships that invest primarily in domestic private equity and private income opportunities. The pension plan's master trust does not have the right to redeem its limited partnership investment at its NAV, but rather receives distributions as the underlying assets are liquidated. It is estimated that the underlying assets of these funds will be gradually liquidated over the next 10 years. As of December 31, 2023, the pension plan's master trust has future funding commitments to these limited partnerships of approximately $1.3 billion, most of which are expected to be called over the next five years.

Changes in fair value measurements of Level 3 investments during the years ended December 31, 2023 and 2022, were as follows (in millions):

	2023	2022
Balance at beginning of year	$ 75	$ 58
Actual gain (loss) on plan assets:		
Relating to assets still held at the reporting date	(9)	1
Purchases	20	29
Sales	(2)	(9)
Transfers out	—	(4)
Balance at end of year	$ 84	$ 75

Plan assets in the retiree medical and other postretirement benefits plans are primarily Level 2 mutual funds valued by quoted prices on the active market, which is fair value, and represents the NAV of the shares of such funds as of the close of business at the end of the period. NAV is based on the fair market value of the funds' underlying assets and liabilities at the date of determination.

Defined Contribution and Multiemployer Plans

The costs associated with our defined contribution plans were $ 1.1 billion, $949 million and $920 million for the years ended December 31, 2023, 2022 and 2021, respectively.

We participate in the International Association of Machinists & Aerospace Workers (IAM) National Pension Fund, Employer Identification No. 51-6031295 and Plan No. 002 (the IAM Pension Fund). Our contributions to the IAM Pension Fund were $52 million, $46 million and $43 million for the years ended December 31, 2023, 2022 and 2021, respectively. The IAM Pension Fund reported $533 million in employers' contributions for the year ended December 31, 2022, which is the most recent year for which such information is available. For 2022, our contributions represented more than 5% of total contributions to the IAM Pension Fund.

On March 29, 2019, the actuary for the IAM Pension Fund certified that the fund was in "endangered" status despite reporting a funded status of over 80%. Additionally, the IAM Pension Fund's Board voluntarily elected to enter into "critical" status on April 17, 2019. Upon entry into critical status, the IAM Pension Fund was required by law to adopt a rehabilitation plan aimed at restoring the financial health of the pension plan and did so on April 17, 2019 (the Rehabilitation Plan). Under the Rehabilitation Plan, we were subject to an immaterial contribution surcharge, which ceased to apply June 14, 2019 upon our mandatory adoption of a contribution schedule under the Rehabilitation Plan. The contribution schedule requires 2.5% annual increases to our contribution rate. This contribution schedule will remain in effect through the earlier of December 31, 2031 or the date the IAM Pension Fund emerges from critical status.

Profit Sharing Program

We accrue a percentage of our pre-tax income excluding net special items for our profit sharing program. For the year ended December 31, 2023, we accrued $261 million for this program, which will be distributed to employees in the first quarter of 2024.

10. Accumulated Other Comprehensive Loss

The components of AOCI are as follows (in millions):

	Pension, Retiree Medical and Other Postretirement Benefits	Unrealized Gain (Loss) on Investments	Income Tax Benefit (Provision) [1]	Total
Balance at December 31, 2021	$ (4,736)	$ (2)	$ (1,204)	$ (5,942)
Other comprehensive income (loss) before reclassifications	1,618	(4)	(365)	1,249
Amounts reclassified from AOCI	140	—	(32) [2]	108
Net current-period other comprehensive income (loss)	1,758	(4)	(397)	1,357
Balance at December 31, 2022	(2,978)	(6)	(1,601)	(4,585)
Other comprehensive income (loss) before reclassifications	(486)	4	108	(374)
Amounts reclassified from AOCI	84	—	(19) [2]	65
Net current-period other comprehensive income (loss)	(402)	4	89	(309)
Balance at December 31, 2023	$ (3,380)	$ (2)	$ (1,512)	$ (4,894)

(1) Relates principally to pension, retiree medical and other postretirement benefits obligations that will not be recognized in net income (loss) until the obligations are fully extinguished.

(2) Relates to pension, retiree medical and other postretirement benefits obligations and is recognized within the income tax provision (benefit) on our consolidated statements of operations.

Reclassifications out of AOCI for the years ended December 31, 2023 and 2022 are as follows (in millions):

	Amounts reclassified from AOCI Year Ended December 31,		
AOCI Components	2023	2022	Affected line items on the consolidated statements of operations
Amortization of pension, retiree medical and other postretirement benefits:			
Prior service cost	$ 9	$ 11	Nonoperating other income (expense), net
Actuarial loss	56	97	Nonoperating other income (expense), net
Total reclassifications for the period, net of tax	$ 65	$ 108	

11. Commitments, Contingencies and Guarantees

(a) Aircraft, Engine and Other Purchase Commitments

Under all of our aircraft and engine purchase agreements, our total future commitments as of December 31, 2023 are expected to be as follows (approximately, in millions):

	2024	2025	2026	2027	2028	2029 and Thereafter	Total
Payments for aircraft and engine commitments [1]	$ 2,410	$ 3,725	$ 3,580	$ 1,118	$ 829	$ 645	$ 12,307

(1) These amounts are net of purchase deposits currently held by the manufacturers. Our purchase deposits held by all manufacturers totaled $ 760 million and $613 million as of December 31, 2023 and 2022, respectively.

Due to uncertainty surrounding the timing of delivery of certain aircraft, the amounts in the table represent our most current estimate based on contractual delivery schedules adjusted for updates and revisions to such schedules communicated to management by the applicable equipment manufacturer. However, the actual delivery schedule may differ, potentially materially, based on various potential factors including production delays by the manufacturer and regulatory concerns.

Additionally, the amounts in the table exclude five Boeing 787 Family aircraft scheduled to be delivered in 2024, for which we have obtained committed lease financing. See Note 5 for information regarding this operating lease commitment.

Additionally, we have other purchase commitments primarily related to aircraft fuel, flight equipment maintenance and information technology support as follows (approximately): $4.7 billion in 2024, $2.0 billion in 2025, $1.4 billion in 2026, $150 million in 2027, $124 million in 2028 and $843 million in 2029 and thereafter. These amounts exclude obligations under certain fuel offtake agreements or other agreements for which the timing of the related expenditure is uncertain, or which are subject to material contingencies, such as the construction of a production facility.

(b) Capacity Purchase Agreements with Third-Party Regional Carriers

American has capacity purchase agreements with third-party regional carriers. The capacity purchase agreements provide that all revenues, including passenger, in-flight, ancillary, mail and freight revenues, go to American. American controls marketing, scheduling, ticketing, pricing and seat inventories. In return, American agrees to pay predetermined fees to these airlines for operating an agreed-upon number of aircraft, without regard to the number of passengers on board. In addition, these agreements provide that American either reimburses or pays 100% of certain variable costs, such as airport landing fees, fuel and passenger liability insurance.

As of December 31, 2023, American's capacity purchase agreements with third-party regional carriers had expiration dates ranging from 2024 to 2032, with rights of American to extend the respective terms of certain agreements.

As of December 31, 2023, American's minimum obligations under its capacity purchase agreements with third-party regional carriers are expected to be as follows (approximately, in millions):

	2024	2025	2026	2027	2028	2029 and Thereafter	Total
Minimum obligations under capacity purchase agreements with third-party regional carriers [1]	$ 2,038	$ 1,992	$ 1,702	$ 1,473	$ 693	$ 1,332	$ 9,230

[1] Represents minimum payments under capacity purchase agreements with third-party regional carriers. These commitments are estimates of costs based on assumed minimum levels of flying under the capacity purchase agreements and American's actual payments could differ materially. Rental payments under operating leases for certain aircraft flown under these capacity purchase agreements are reflected in the operating lease commitments in Note 5.

(c) Airport Redevelopment

Los Angeles International Airport (LAX)

From time to time, airports engage in construction projects, often substantial, that result in new or improved facilities that are ultimately funded through increases in the rent and other occupancy costs payable by airlines operating at the airport. Unlike this construction and funding model, we are managing a project at LAX where we have legal title to the assets during construction. In 2018, we executed a lease agreement with Los Angeles World Airports (LAWA), which owns and operates LAX, in connection with a $1.6 billion modernization project related to LAX Terminals 4 and 5. Construction, which started in October 2018 and is expected to be completed in 2028, will occur in a phased approach. The modernization project will include a unified departure hall to the entranceway of Terminals 4 and 5, reconfigured ticket counter and check-in areas with seamless access to security screening areas, 10 new security screening lanes with automated technology in addition to the existing Terminal 5 lanes, and a new Terminal 4 South concourse with more open and upgraded amenities at gate areas. The project will also include renovated break rooms, multi-use meeting rooms and team gathering spaces throughout the terminals to support our team members at LAX.

As each phase is completed and ready for use, the assets will be sold and transferred to LAWA, including the site improvements and other non-proprietary improvements. As we control the assets during construction, they are recognized on our consolidated balance sheets within operating property and equipment until the assets are sold and transferred to LAWA. As of December 31, 2023, we have incurred $862 million in costs relating to the LAX modernization project, of which $283 million were incurred in 2023. Cash paid for non-proprietary improvements are included within other investing activities on our consolidated statements of cash flows. In addition, as of December 31, 2023, we have sold and transferred $346 million of non-proprietary improvements to LAWA, of which $170 million occurred during 2023. For non-proprietary improvements which are not yet ready for use, any cash payments received from LAWA will be reflected as a financial liability included within noncurrent other liabilities on our consolidated balance sheets and reflected as other

financing activities on our consolidated statements of cash flows. As of December 31, 2023, $ 53 million of cash proceeds received for non-proprietary improvements were not yet ready for use, and therefore have not been sold and transferred back to LAWA.

(d) Off-Balance Sheet Arrangements

Pass-Through Trusts

American currently has 308 owned aircraft and 60 owned spare aircraft engines, which in each case were financed with EETCs issued by pass-through trusts. These trusts are off-balance sheet entities, the primary purpose of which is to finance the acquisition of flight equipment or to permit issuance of debt backed by existing flight equipment. In the case of aircraft EETCs, rather than finance each aircraft separately when such aircraft is purchased, delivered or refinanced, these trusts allow American to raise the financing for a number of aircraft at one time and, if applicable, place such funds in escrow pending a future purchase, delivery or refinancing of the relevant aircraft. Similarly, in the case of the spare engine EETCs, the trusts allow American to use its existing pool of spare engines to raise financing under a single facility. The trusts have also been structured to provide for certain credit enhancements, such as liquidity facilities to cover certain interest payments, that reduce the risks to the purchasers of the trust certificates and, as a result, reduce the cost of aircraft financing to American.

Each trust covers a set number of aircraft or spare engines scheduled to be delivered, financed or refinanced upon the issuance of the EETC or within a specific period of time thereafter. At the time of each covered aircraft or spare engine financing, the relevant trust used the proceeds from the issuance of the EETC (which may have been available at the time of issuance thereof or held in escrow until financing of the applicable aircraft following its delivery) to purchase equipment notes relating to the financed aircraft or engines. The equipment notes are issued, at American's election, in connection with a mortgage financing of the aircraft or spare engines. The equipment notes are secured by a security interest in the aircraft or engines, as applicable. The pass-through trust certificates are not direct obligations of, nor are they guaranteed by, AAG or American. However, the equipment notes issued to the trusts are direct obligations of American and, in certain instances, have been guaranteed by AAG. As of December 31, 2023, $7.7 billion associated with these mortgage financings is reflected as debt in the accompanying consolidated balance sheet.

Letters of Credit and Other

We provide financial assurance, such as letters of credit and surety bonds, primarily to support projected workers' compensation obligations and airport commitments. As of December 31, 2023, we had $318 million of letters of credit and surety bonds securing various obligations, of which $ 94 million is collateralized with our restricted cash. The letters of credit and surety bonds that are subject to expiration will expire on various dates through 2028.

(e) Legal Proceedings

Government Antitrust Action Related to the Northeast Alliance. On September 21, 2021, the United States Department of Justice, joined by Attorneys General from six states and the District of Columbia, filed an antitrust complaint against American and JetBlue Airways Corporation (JetBlue) in the U.S. District Court for the District of Massachusetts alleging that American and JetBlue violated U.S. antitrust law in connection with the previously disclosed Northeast Alliance arrangement (NEA).

On May 19, 2023, the U.S. District Court for the District of Massachusetts issued an order permanently enjoining American and JetBlue from continuing and further implementing the NEA. In June 2023, JetBlue delivered a notice of termination of the NEA, effective July 29, 2023, and the carriers have commenced wind-down activities to accommodate mutual customers. Following written submissions by the parties and a hearing on July 26, 2023, the U.S. District Court for the District of Massachusetts entered a Final Judgment and Order Entering Permanent Injunction on July 28, 2023. The parties are complying with the terms of the Final Judgment and Order Entering Permanent Injunction, including winding down activities related to the NEA. American filed a notice of appeal to the U.S. Court of Appeals for the First Circuit on September 25, 2023, and American's opening brief was filed on December 6, 2023.

Private Party Antitrust Actions Related to the Northeast Alliance. On December 5, 2022 and December 7, 2022, two private party plaintiffs filed putative class action antitrust complaints against American and JetBlue in the U.S. District Court for the Eastern District of New York alleging that American and JetBlue violated U.S. antitrust law in connection with the previously disclosed NEA. These actions were consolidated on January 10, 2023. The private party plaintiffs filed an amended consolidated complaint on February 3, 2023. On February 2, 2023 and February 15, 2023, private party plaintiffs filed two additional putative class action antitrust complaints against American and JetBlue in the U.S. District

Court for the District of Massachusetts and the U.S. District Court for the Eastern District of New York, respectively. In March 2023, American filed a motion in the U.S. District Court for the District of Massachusetts case asking to transfer the case to the U.S. District Court for the Eastern District of New York and consolidate it with the cases pending in that venue. The U.S. District Court for the District of Massachusetts granted that motion. The remaining cases were consolidated with the other actions in the Eastern District of New York. In June 2023, the private party plaintiffs filed a second amended consolidated complaint, followed by a third amended complaint filed in August 2023. In September 2023, American, together with JetBlue, filed a motion to dismiss the third amended complaint, and that motion remains pending. We believe these lawsuits are without merit and are defending against them vigorously.

General. In addition to the specifically identified legal proceedings, we and our subsidiaries are also engaged in other legal proceedings from time to time. Legal proceedings can be complex and take many months, or even years, to reach resolution, with the final outcome depending on a number of variables, some of which are not within our control. Therefore, although we will vigorously defend ourselves in each of the actions described above and such other legal proceedings, their ultimate resolution and potential financial and other impacts on us are uncertain but could be material.

(f) Guarantees and Indemnifications

We are party to many routine contracts in which we provide general indemnities in the normal course of business to third parties for various risks. We are not able to estimate the potential amount of any liability resulting from the indemnities. These indemnities are discussed in the following paragraphs.

In our aircraft financing agreements, we generally indemnify the financing parties, trustees acting on their behalf and other relevant parties against liabilities (including certain taxes) resulting from the financing, manufacture, design, ownership, operation and maintenance of the aircraft regardless of whether these liabilities (including certain taxes) relate to the negligence of the indemnified parties.

Our loan agreements and certain other financing transactions may obligate us to reimburse the applicable lender for incremental costs due to a change in law that imposes (i) any reserve or special deposit requirement against assets of, deposits with or credit extended by such lender related to the loan, (ii) any tax, duty or other charge with respect to the loan (except standard income tax) or (iii) capital adequacy requirements. In addition, our loan agreements and other financing arrangements typically contain a withholding tax provision that requires us to pay additional amounts to the applicable lender or other financing party, generally if withholding taxes are imposed on such lender or other financing party as a result of a change in the applicable tax law.

In certain transactions, including certain aircraft financing leases and loans, the lessors, lenders and/or other parties have rights to terminate the transaction based on changes in foreign tax law, illegality or certain other events or circumstances. In such a case, we may be required to make a lump sum payment to terminate the relevant transaction.

We have general indemnity clauses in many of our airport and other real estate leases where we as lessee indemnify the lessor (and related parties) against liabilities related to our use of the leased property. Generally, these indemnifications cover liabilities resulting from the negligence of the indemnified parties, but not liabilities resulting from the gross negligence or willful misconduct of the indemnified parties. In addition, we provide environmental indemnities in many of these leases for contamination related to our use of the leased property.

Under certain contracts with third parties, we indemnify the third-party against legal liability arising out of an action by the third-party, or certain other parties. The terms of these contracts vary and the potential exposure under these indemnities cannot be determined. We have liability insurance protecting us from some of the obligations we have undertaken under these indemnities.

American is required to make principal and interest payments for certain special facility revenue bonds issued by municipalities primarily to build or improve airport facilities and purchase equipment, which are leased to American. The payment of principal and interest of certain special facility revenue bonds is guaranteed by AAG. As of December 31, 2023, the remaining lease payments through 2035 guaranteeing the principal and interest on these bonds are $520 million and the current carrying amount of the associated operating lease liability in the accompanying consolidated balance sheet is $ 321 million.

As of December 31, 2023, AAG had issued guarantees covering approximately $ 17.5 billion of American’s secured debt (and interest thereon), including the Credit Facilities, 2023 Term Loan Facility, the AAdvantage Financing, certain EETC financings and special facility revenue bonds.

(g) Credit Card Processing Agreements

We have agreements with companies that process customer credit card transactions for the sale of air travel and other services. Our agreements allow these credit card processing companies, under certain conditions, to hold an amount of our cash (referred to as a holdback) equal to all or a portion of advance ticket sales that have been processed by that company, but for which we have not yet provided the air transportation. These holdback requirements can be implemented at the discretion of the credit card processing companies upon the occurrence of specific events, including material adverse changes in our financial condition or the triggering of a liquidity covenant. These credit card processing companies are not currently entitled to maintain any holdbacks. The imposition of holdback requirements would reduce our liquidity.

(h) Labor Contracts

In May 2023, American and the Allied Pilots Association, the union representing our mainline pilots, reached an agreement in principle on a new collective bargaining agreement (CBA), which was ratified in August 2023. This four-year agreement provides wage rate increases, including an initial wage rate increase of 21% effective as of January 1, 2023, quality-of-life benefits and other benefit-related items. The additional compensation for the 2023 period prior to contract ratification as a result of the higher wage rates was recorded within salaries, wages and benefits in the consolidated statements of operations in the second and third quarters of 2023. The agreement also included a provision for a one-time payment upon ratification. In 2023, one-time charges resulting from the ratification of this new agreement were recorded as mainline operating special items, net in the consolidated statement of operations, including the one-time payment of $754 million as well as adjustments to other benefit-related items of $ 235 million. The one-time payment and the additional compensation were principally paid in 2023, with remaining payments expected to be paid in the first quarter of 2024.

As of December 31, 2023, we employed approximately 132,100 active full-time equivalent (FTE) employees, of which 28,900 were employed by our wholly-owned regional subsidiaries. Of the total active FTE employees, 87% are covered by CBAs with various labor unions and 34% are covered by CBAs that are currently amendable or that will become amendable within one year. In January 2024, mainline passenger service employees represented by the CWA-IBT ratified a new five-year agreement. The CBA covering our flight attendants is now amendable. The CBAs covering certain employee groups at our wholly-owned regional subsidiaries are also amendable.

12. Supplemental Cash Flow Information

Supplemental disclosure of cash flow information and non-cash investing and financing activities are as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Non-cash investing and financing activities:			
ROU assets acquired through operating leases	$ 1,180	$ 1,483	$ 1,386
Property and equipment acquired through debt, finance leases and other	317	46	180
Finance leases converted to operating leases	42	3	—
Operating leases converted to finance leases	5	107	102
Settlement of bankruptcy obligations	4	—	(1)
Equity investments	—	12	88
Supplemental information:			
Interest paid, net	2,180	1,852	1,632
Income taxes paid	6	2	3

13. Operating Segments and Related Disclosures

We are managed as a single business unit that provides air transportation for passengers and cargo. This allows us to benefit from an integrated revenue pricing and route network that includes American and our wholly-owned and third-party regional carriers that fly under capacity purchase agreements operating as American Eagle. The flight equipment of all these carriers is combined to form one fleet that is deployed through a single route scheduling system. Financial information and annual operational plans and forecasts are prepared and reviewed by the chief operating decision maker at the consolidated level. When making operational decisions, the chief operating decision maker evaluates flight profitability data, which considers aircraft type and route economics, but is indifferent to the results of the individual regional carriers. The objective in making operational decisions is to maximize consolidated financial results, not the individual results of American or American Eagle.

See Note 1(m) for our passenger revenue by geographic region. Our tangible assets consist primarily of flight equipment, which are mobile across geographic markets and, therefore, have not been allocated.

14. Share-based Compensation

In May 2023, the stockholders of AAG approved the 2023 Incentive Award Plan (the 2023 Plan). The 2023 Plan replaces and supersedes AAG's 2013 Incentive Award Plan (the 2013 Plan). No further awards will be granted under the 2013 Plan; however, the terms and conditions of the 2013 Plan will continue to govern any outstanding awards granted thereunder. The 2023 Plan provides that an award may be in the form of a stock option, including an incentive stock option and nonqualified stock option, stock appreciation right, restricted stock, restricted stock unit, performance bonus award, performance stock unit, other stock or cash-based award and dividend equivalent to eligible individuals.

The 2023 Plan authorizes the grant of awards for the issuance of 17.2 million shares less any shares granted under the 2013 Plan after March 22, 2023, the date the Board of Directors of AAG approved the 2023 Plan. Any shares underlying awards granted under the 2023 Plan or 2013 Plan that are forfeited, terminate or are settled in cash (in whole or in part) without the delivery of shares will again be available for grant under the 2023 Plan.

For the years ended December 31, 2023, 2022 and 2021, we recorded $ 102 million, $78 million and $98 million, respectively, of share-based compensation costs principally in salaries, wages and benefits expense on our consolidated statements of operations.

During 2023, 2022 and 2021, we withheld approximately 1.5 million, 1.2 million and 1.0 million shares of AAG common stock, respectively, and paid approximately $23 million, $21 million and $ 18 million, respectively, in satisfaction of certain tax withholding obligations associated with employee equity awards.

Restricted Stock Unit Awards (RSUs)

We have granted RSUs with service conditions (time vested primarily over three years) and performance conditions. The grant-date fair value of these RSUs is equal to the market price of the underlying shares of AAG common stock on the date of grant. For time vested awards, the expense is recognized on a straight-line basis over the vesting period for the entire award. For awards with performance conditions, the expense is recognized based on the expected achievement at each reporting period. RSUs are classified as equity awards as the vesting results in the issuance of shares of AAG common stock.

RSU award activity for all plans for the years ended December 31, 2023, 2022 and 2021 is as follows:

	Number of Shares	Weighted Average Grant Date Fair Value
	(In thousands)	
Outstanding at December 31, 2020	7,882	$ 23.66
Granted	5,525	18.34
Vested and released	(3,314)	25.58
Forfeited	(692)	18.78
Outstanding at December 31, 2021	9,401	$ 20.17
Granted	5,882	15.93
Vested and released	(4,131)	21.04
Forfeited	(889)	18.04
Outstanding at December 31, 2022	10,263	$ 17.51
Granted	9,834	14.54
Vested and released	(5,161)	17.81
Forfeited	(701)	20.49
Outstanding at December 31, 2023	14,235	$ 15.18

As of December 31, 2023, there was $ 127 million of unrecognized compensation cost related to RSUs. These costs are expected to be recognized over a weighted average period of one year. The total fair value of RSUs vested during the years ended December 31, 2023, 2022 and 2021 was $ 78 million, $70 million and $ 62 million, respectively.

15. Valuation and Qualifying Accounts (in millions)

	Balance at Beginning of Year	Additions Charged to Statement of Operations Accounts	Deductions and Other	Balance at End of Year
Allowance for obsolescence of spare parts				
Year ended December 31, 2023	$ 616	$ 98	$ 14	$ 728
Year ended December 31, 2022	634	96	(114)	616
Year ended December 31, 2021	490	177	(33)	634

ITEM 8B. CONSOLIDATED FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA OF AMERICAN AIRLINES, INC.

Report of Independent Registered Public Accounting Firm

To the Stockholder and Board of Directors
American Airlines, Inc.:

Opinion on the Consolidated Financial Statements

We have audited the accompanying consolidated balance sheets of American Airlines, Inc. and subsidiaries (American) as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive income (loss), cash flows, and stockholder's equity for each of the years in the three-year period ended December 31, 2023, and the related notes (collectively, the consolidated financial statements). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of American as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the years in the three-year period ended December 31, 2023, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), American's internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission, and our report dated February 21, 2024 expressed an unqualified opinion on the effectiveness of American's internal control over financial reporting.

Basis for Opinion

These consolidated financial statements are the responsibility of American's management. Our responsibility is to express an opinion on these consolidated financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to American in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the consolidated financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of a critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Sufficiency of audit evidence over the realizability of tax net operating loss and other carryforwards

As discussed in Notes 1(j) and 5 to the consolidated financial statements, American had $4.0 billion of tax net operating loss and other carryforwards, which are recorded as deferred tax assets at December 31, 2023. Deferred tax assets are recognized related to tax net operating loss and other carryforwards that will reduce future taxable income. American provides a valuation allowance for deferred tax assets when it is more likely than not that some portion, or all of the deferred tax assets, will not be realized. In evaluating the need for a valuation allowance, management considers all available positive and negative evidence.

We identified the evaluation of the sufficiency of audit evidence over the realizability of the federal tax net operating loss and other carryforwards as a critical audit matter. Evaluating the sufficiency of audit evidence required subjective auditor judgment in order to assess the extent of procedures performed in assessing the realizability of the federal tax net operating loss and other carryforwards.

The following are the primary procedures we performed to address this critical audit matter. We evaluated the design and tested the operating effectiveness of certain internal controls related to American's deferred tax asset valuation allowance process, including controls related to the realizability of the federal tax net operating loss and other carryforwards. We evaluated positive and negative evidence used in assessing whether the federal tax net operating loss and other carryforwards were more likely than not to be realized in the future. We evaluated the reasonableness of management's projections of future profitability considering historical profitability of American, and consistency with industry data. We involved tax professionals with specialized skills and knowledge, who assisted in evaluating the application of tax law. We assessed the sufficiency of audit evidence obtained over the realizability of the federal tax net operating loss and other carryforwards by evaluating the cumulative results of the audit procedures.

/s/ KPMG LLP

We have served as American's auditor since 2014.

Dallas, Texas
February 21, 2024

AMERICAN AIRLINES, INC.
CONSOLIDATED STATEMENTS OF OPERATIONS
(In millions)

	Year Ended December 31,		
	2023	2022	2021
Operating revenues:			
Passenger	$ 48,512	$ 44,568	$ 26,063
Cargo	812	1,233	1,314
Other	3,460	3,164	2,503
Total operating revenues	52,784	48,965	29,880
Operating expenses:			
Aircraft fuel and related taxes	12,257	13,791	6,792
Salaries, wages and benefits	14,572	12,965	11,811
Regional expenses	4,619	4,345	3,111
Maintenance, materials and repairs	3,265	2,684	1,979
Other rent and landing fees	2,928	2,730	2,619
Aircraft rent	1,369	1,395	1,425
Selling expenses	1,799	1,815	1,098
Depreciation and amortization	1,927	1,969	2,019
Special items, net	971	193	(4,006)
Other	6,008	5,425	3,993
Total operating expenses	49,715	47,312	30,841
Operating income (loss)	3,069	1,653	(961)
Nonoperating income (expense):			
Interest income	1,078	349	34
Interest expense, net	(2,206)	(1,872)	(1,642)
Other income (expense), net	(359)	324	292
Total nonoperating expense, net	(1,487)	(1,199)	(1,316)
Income (loss) before income taxes	1,582	454	(2,277)
Income tax provision (benefit)	394	116	(500)
Net income (loss)	$ 1,188	$ 338	$ (1,777)

See accompanying notes to consolidated financial statements.

AMERICAN AIRLINES, INC.
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME (LOSS)
(In millions)

	Year Ended December 31,		
	2023	2022	2021
Net income (loss)	$ 1,188	$ 338	$ (1,777)
Other comprehensive income (loss), net of tax:			
Pension, retiree medical and other postretirement benefits	(312)	1,354	1,153
Investments	3	(3)	—
Total other comprehensive income (loss), net of tax	(309)	1,351	1,153
Total comprehensive income (loss)	$ 879	$ 1,689	$ (624)

See accompanying notes to consolidated financial statements.

AMERICAN AIRLINES, INC.
CONSOLIDATED BALANCE SHEETS
(In millions, except share and par value amounts)

	December 31,	
	2023	2022
ASSETS		
Current assets		
Cash	$ 567	$ 429
Short-term investments	6,998	8,523
Restricted cash and short-term investments	910	995
Accounts receivable, net	1,995	2,117
Receivables from related parties, net	7,070	6,588
Aircraft fuel, spare parts and supplies, net	2,266	2,157
Prepaid expenses and other	561	798
Total current assets	20,367	21,607
Operating property and equipment		
Flight equipment	41,440	39,359
Ground property and equipment	9,848	9,479
Equipment purchase deposits	760	613
Total property and equipment, at cost	52,048	49,451
Less accumulated depreciation and amortization	(21,588)	(19,569)
Total property and equipment, net	30,460	29,882
Operating lease right-of-use assets	7,886	8,033
Other assets		
Goodwill	4,091	4,091
Intangibles, net of accumulated amortization of $834 and $827, respectively	2,051	2,059
Deferred tax asset	2,589	2,893
Other assets	1,630	1,759
Total other assets	10,361	10,802
Total assets	$ 69,074	$ 70,324
LIABILITIES AND STOCKHOLDER'S EQUITY		
Current liabilities		
Current maturities of long-term debt and finance leases	$ 3,625	$ 3,267
Accounts payable	2,232	2,071
Accrued salaries and wages	2,210	1,529
Air traffic liability	6,200	6,745
Loyalty program liability	3,453	3,169
Operating lease liabilities	1,292	1,449
Other accrued liabilities	2,605	2,852
Total current liabilities	21,617	21,082
Noncurrent liabilities		
Long-term debt and finance leases, net of current maturities	24,050	27,155
Pension and postretirement benefits	3,020	2,811
Loyalty program liability	5,874	5,976
Operating lease liabilities	6,416	6,512
Other liabilities	1,520	1,195
Total noncurrent liabilities	40,880	43,649
Commitments and contingencies (Note 10)		
Stockholder's equity		
Common stock, $1.00 par value; 1,000 shares authorized, issued and outstanding	—	—
Additional paid-in capital	17,335	17,230
Accumulated other comprehensive loss	(4,999)	(4,690)
Retained deficit	(5,759)	(6,947)
Total stockholder's equity	6,577	5,593
Total liabilities and stockholder's equity	$ 69,074	$ 70,324

See accompanying notes to consolidated financial statements.

AMERICAN AIRLINES, INC.
CONSOLIDATED STATEMENTS OF CASH FLOWS
(In millions)

	Year Ended December 31,		
	2023	2022	2021
Cash flows from operating activities:			
Net income (loss)	$ 1,188	$ 338	$ (1,777)
Adjustments to reconcile net income (loss) to net cash provided by operating activities:			
Depreciation and amortization	2,198	2,238	2,282
Debt extinguishment costs	267	1	31
Special items, net non-cash	41	226	52
Pension and postretirement	(14)	(404)	(320)
Deferred income tax provision (benefit)	394	122	(500)
Share-based compensation	97	75	95
Other, net	(216)	(48)	(2)
Changes in operating assets and liabilities:			
Decrease (increase) in accounts receivable	104	(636)	(290)
Increase in other assets	(2)	(744)	(370)
Increase in accounts payable and accrued liabilities	828	549	335
Increase (decrease) in air traffic liability	(545)	658	1,454
Decrease (increase) in receivables from related parties, net	(482)	(1,044)	1,857
Increase (decrease) in loyalty program liability	182	10	(60)
Contributions to pension plans	(71)	(4)	(247)
Increase (decrease) in other liabilities	(263)	(8)	650
Net cash provided by operating activities	3,706	1,329	3,190
Cash flows from investing activities:			
Capital expenditures, net of aircraft purchase deposit returns	(2,542)	(2,489)	(169)
Proceeds from sale of property and equipment and sale-leaseback transactions	230	147	373
Sales of short-term investments	8,861	14,972	13,923
Purchases of short-term investments	(7,324)	(11,257)	(19,454)
Decrease (increase) in restricted short-term investments	51	1	(401)
Purchase of equity investments	—	(321)	(28)
Other investing activities	275	(360)	(189)
Net cash provided by (used in) investing activities	(449)	693	(5,945)
Cash flows from financing activities:			
Payments on long-term debt and finance leases	(7,697)	(2,991)	(7,320)
Proceeds from issuance of long-term debt	4,822	1,069	10,209
Other financing activities	(287)	75	(119)
Net cash provided by (used in) financing activities	(3,162)	(1,847)	2,770
Net increase in cash and restricted cash	95	175	15
Cash and restricted cash at beginning of year	575	400	385
Cash and restricted cash at end of year[(a)]	$ 670	$ 575	$ 400

[(a)] The following table provides a reconciliation of cash and restricted cash to amounts reported within the consolidated balance sheets:

Cash	$ 567	$ 429	$ 265
Restricted cash included in restricted cash and short-term investments	103	146	135
Total cash and restricted cash	$ 670	$ 575	$ 400

See accompanying notes to consolidated financial statements.

AMERICAN AIRLINES, INC.
CONSOLIDATED STATEMENTS OF STOCKHOLDER'S EQUITY
(In millions)

	Common Stock	Additional Paid-in Capital	Accumulated Other Comprehensive Loss	Retained Earnings (Deficit)	Total
Balance at December 31, 2020	$ —	$ 17,050	$ (7,194)	$ (5,508)	$ 4,348
Net loss	—	—	—	(1,777)	(1,777)
Other comprehensive income, net	—	—	1,153	—	1,153
Share-based compensation expense	—	95	—	—	95
Intercompany equity transfer	—	7	—	—	7
Balance at December 31, 2021	—	17,152	(6,041)	(7,285)	3,826
Net income	—	—	—	338	338
Other comprehensive income, net	—	—	1,351	—	1,351
Share-based compensation expense	—	75	—	—	75
Intercompany equity transfer	—	3	—	—	3
Balance at December 31, 2022	—	17,230	(4,690)	(6,947)	5,593
Net income	—	—	—	1,188	1,188
Other comprehensive loss, net	—	—	(309)	—	(309)
Share-based compensation expense	—	97	—	—	97
Intercompany equity transfer	—	8	—	—	8
Balance at December 31, 2023	$ —	$ 17,335	$ (4,999)	$ (5,759)	$ 6,577

See accompanying notes to consolidated financial statements.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS OF AMERICAN AIRLINES, INC.

1. Basis of Presentation and Summary of Significant Accounting Policies

(a) Basis of Presentation

American Airlines, Inc. (American) is a Delaware corporation whose primary business activity is the operation of a major network air carrier, providing scheduled air transportation for passengers and cargo. American is the principal wholly-owned subsidiary of American Airlines Group Inc. (AAG), which owns all of American's outstanding common stock, par value $1.00 per share. All significant intercompany transactions have been eliminated.

The preparation of financial statements in accordance with accounting principles generally accepted in the United States (GAAP) requires management to make certain estimates and assumptions that affect the reported amounts of assets and liabilities, revenues and expenses, and the disclosure of contingent assets and liabilities at the date of the financial statements. Actual results could differ from those estimates. The most significant areas of judgment relate to passenger revenue recognition, the loyalty program, deferred tax assets, as well as pension and retiree medical and other postretirement benefits.

(b) Government Assistance

Payroll Support Programs

During 2020 and 2021, American, Envoy Air Inc. (Envoy), Piedmont Airlines, Inc. (Piedmont) and PSA Airlines, Inc. (PSA and together with American, Envoy and Piedmont, the Subsidiaries) entered into payroll support program agreements (PSP Agreements) with the U.S. Department of Treasury (Treasury) pursuant to the payroll support program established under the Coronavirus Aid, Relief, and Economic Security Act (CARES Act) (PSP1), the payroll support program established under the Subtitle A of Title IV of Division N of the Consolidated Appropriations Act, 2021 (PSP Extension Law) (PSP2) and the payroll support program established under the American Rescue Plan Act of 2021 (ARP) (PSP3). The aggregate amount of financial assistance received was approximately $12.8 billion, and as partial compensation to the U.S. Government for the provision of financial assistance provided under each of these programs, AAG issued promissory notes and warrants to Treasury.

The table below provides a summary of the financial assistance received and the promissory notes and the warrants issued under each program (in millions, except exercise price amounts):

Program	Closing Date	PSP Financial Assistance	Promissory Notes [1]	PSP Warrants	Total	Warrants Issued (Shares) [2]	Exercise Price of Warrants
PSP1	April 20, 2020	$ 4,138	$ 1,757	$ 63	$ 5,958	14.0	$ 12.51
PSP2	January 15, 2021	2,427	1,030	76	3,533	6.6	15.66
PSP3	April 23, 2021	2,290	959	46	3,295	4.4	21.75
Total		$ 8,855	$ 3,746	$ 185	$ 12,786	25.0	

[1] As partial compensation to the U.S. Government for the provision of financial assistance under the PSP Agreements, AAG issued promissory notes to Treasury (PSP1 Promissory Note, PSP2 Promissory Note and PSP3 Promissory Note, collectively the PSP Promissory Notes), in the aggregate principal sum of $3.7 billion which provides for the guarantee of AAG's obligations under the PSP Promissory Notes by the Subsidiaries.

[2] The payroll support program warrants (PSP Warrants) are subject to certain anti-dilution provisions, do not have any voting rights and are freely transferable, with registration rights. Each warrant expires on the fifth anniversary of the date of issuance, with expiration dates ranging from April 2025 to June 2026, and will be exercisable either through net share settlement or cash, at AAG's option. The warrants were issued solely as compensation to the U.S. Government related to entry into the PSP Agreements. No separate proceeds (apart from the financial assistance described below) were received upon issuance of the warrants or will be received upon exercise thereof.

In connection with the PSP Agreements entered into with Treasury, AAG and the Subsidiaries were required to comply with the relevant provisions of the CARES Act, the PSP Extension Law, and the ARP, which included the requirement that funds provided pursuant to these programs be used exclusively for the continuation of payment of eligible employee wages, salaries and benefits, the prohibition against involuntary furloughs and reductions in employee pay rates and benefits, the requirement that certain levels of commercial air service be maintained, provisions that prohibited the

repurchase of AAG common stock and the payment of common stock dividends as well as provisions that restrict the payment of certain executive compensation. As of December 31, 2023, all of these provisions have expired.

For accounting purposes, the $12.8 billion of aggregate financial assistance received pursuant to the PSP Agreements was allocated to the promissory notes, warrants and other financial assistance (PSP Financial Assistance). The aggregate principal amount of the promissory notes was recorded as unsecured long-term debt and the total fair value of the warrants, estimated using a Black-Scholes option pricing model, was recorded in stockholders' deficit in AAG's consolidated balance sheets. The remaining amounts were recognized in 2020 and 2021 as a credit to special items, net in the consolidated statements of operations over the period which the continuation of payment of eligible employee wages, salaries and benefits was required.

Treasury Loan Agreement

On September 25, 2020 (the Treasury Loan Closing Date), AAG and American entered into a Loan and Guarantee Agreement (the Treasury Loan Agreement) with Treasury, which provided for a secured term loan facility (the Treasury Term Loan Facility) that permitted American to borrow up to $5.5 billion. Subsequently, on October 21, 2020, AAG and American entered into an amendment to the Treasury Loan Agreement which increased the borrowing amount up to $7.5 billion. In connection with AAG's entry into the Treasury Loan Agreement, on the Treasury Loan Closing Date, AAG also entered into a warrant agreement (the Treasury Loan Warrant Agreement) with Treasury.

In September 2020, American borrowed $ 550 million under the Treasury Term Loan Facility and on March 24, 2021, used a portion of the proceeds from the AAdvantage Financing to prepay in full the $550 million of outstanding loans under the Treasury Term Loan Facility and terminated the Treasury Loan Agreement. Pursuant to the Treasury Loan Agreement, AAG issued to Treasury warrants (Treasury Loan Warrants) to purchase up to an aggregate of approximately 4.4 million shares of AAG common stock (the Treasury Loan Warrant Shares), which expire in September 2025. The exercise price of the Treasury Loan Warrant Shares is $12.51 per share, subject to certain anti-dilution provisions provided for in the Treasury Loan Warrant Agreement. For accounting purposes, the fair value for the Treasury Loan Warrant Shares, estimated using a Black-Scholes option pricing model, was recorded in stockholders' deficit in AAG's consolidated balance sheet with an offsetting debt discount to the Treasury Term Loan Facility in American's consolidated balance sheet. The provisions of the Treasury Loan Warrants are substantially similar to the PSP Warrants.

(c) Recent Accounting Pronouncements

Accounting Standards Update (ASU) 2023-07: Segment Reporting (Topic 280) Improvements to Reportable Segment Disclosures

This standard improves reportable segment disclosure requirements, primarily through enhanced disclosures about significant segment expenses. The amendments in this update are effective for fiscal years beginning after December 15, 2023 and interim periods within fiscal years beginning after December 15, 2024, and early adoption is permitted. American is currently evaluating how the adoption of this standard will impact its reportable segment disclosures.

ASU 2023-09: Income Taxes (Topic 740) Improvements to Income Tax Disclosures

This standard enhances transparency of income tax information through improvements to income tax disclosures primarily related to the rate reconciliation and income taxes paid information, as well as improvements to the effectiveness and comparability of other income tax disclosures. The amendments in this update are effective for annual periods beginning after December 15, 2024, and early adoption is permitted. American is currently evaluating how the adoption of this standard will impact its income tax disclosures.

(d) Investments

Short-term investments primarily include debt securities and are classified as available-for-sale and stated at fair value. Realized gains and losses are recorded as interest income in nonoperating expense, net on American's consolidated statements of operations. Unrealized gains and losses are recorded as a component of accumulated other comprehensive loss on American's consolidated balance sheets. For investments in an unrealized loss position, American determines whether a credit loss exists by considering information about the collectability of the instrument, current market conditions and reasonable and supportable forecasts of economic conditions. There have been no credit losses.

Equity investments are accounted for under the equity method if American is able to exercise significant influence over an investee. Equity investments for which American does not have significant influence are recorded at fair value or at cost, if fair value is not readily determinable, with adjustments for observable changes in price or impairments (referred to as the measurement alternative). American's share of equity method investees' financial results and changes in fair value are recorded in nonoperating other income (expense), net on the consolidated statements of operations. See Note 7 for additional information related to American's equity investments.

(e) Restricted Cash and Short-term Investments

American has restricted cash and short-term investments related primarily to collateral held to support workers' compensation obligations and collateral associated with the AAdvantage Financing.

(f) Accounts Receivable, Net

Accounts receivable primarily consist of amounts due from credit card processing companies for tickets sold to individual passengers, amounts due from airline and non-airline business partners, including American's co-branded credit card partners and cargo customers. Receivables from ticket sales are short-term, mostly settled within seven days after sale. Receivables from American's business partners are typically settled within 30 days. All accounts receivable are reported net of an allowance for credit losses, which was not material as of December 31, 2023 and 2022. American considers past and future financial and qualitative factors, including aging, payment history and other credit monitoring indicators, when establishing the allowance for credit losses.

(g) Aircraft Fuel, Spare Parts and Supplies, Net

Aircraft fuel is recorded on a first-in, first-out basis. Spare parts and supplies are recorded at average costs less an allowance for obsolescence, which is recognized over the weighted average remaining useful life of the related fleet. American also provides an allowance for spare parts and supplies identified as excess or obsolete to reduce the carrying cost to the lower of cost or net realizable value. Aircraft fuel, spare parts and supplies are expensed when used.

(h) Operating Property and Equipment

Operating property and equipment is recorded at cost and depreciated or amortized to residual values over the asset's estimated useful life or the lease term, whichever is less, using the straight-line method. Residual values for aircraft, engines and related rotable parts are generally 5% to 10% of original cost. Costs of major improvements that enhance the usefulness of the asset are capitalized and depreciated or amortized over the estimated useful life of the asset or the lease term, whichever is less. The estimated useful lives for the principal property and equipment classifications are as follows:

Principal Property and Equipment Classification	Estimated Useful Life
Aircraft, engines and related rotable parts	20 – 30 years
Buildings and improvements	5 – 30 years
Furniture, fixtures and other equipment	3 – 15 years
Capitalized software	5 – 10 years

Total mainline and regional depreciation and amortization expense was $ 2.2 billion for each of the years ended December 31, 2023 and 2022 and $ 2.3 billion for the year ended December 31, 2021.

American assesses impairment of operating property and equipment when events and circumstances indicate that the assets may be impaired. An impairment of an asset or group of assets exists only when the sum of the estimated undiscounted cash flows expected to be generated directly by the assets are less than the carrying value of the assets. American groups assets principally by fleet-type when estimating future cash flows, which is generally the lowest level for which identifiable cash flows exist. Estimates of future cash flows are based on historical results adjusted to reflect management's best estimate of future market and operating conditions, including American's current fleet plan. If such assets are impaired, the impairment charge recognized is the amount by which the carrying value of the assets exceed their fair value. Fair value reflects management's best estimate including inputs from published pricing guides and bids from third parties as well as contracted sales agreements when applicable.

(i) Leases

American determines if an arrangement is a lease at inception. Operating leases are included in operating lease right-of-use (ROU) assets, current operating lease liabilities and noncurrent operating lease liabilities on American's consolidated balance sheets. Finance leases are included in property and equipment, current maturities of long-term debt and finance leases and long-term debt and finance leases, net of current maturities, on American's consolidated balance sheets.

ROU assets represent American's right to use an underlying asset for the lease term and lease liabilities represent its obligation to make lease payments arising from the lease. ROU assets and liabilities are recognized at the lease commencement date based on the estimated present value of lease payments over the lease term.

American uses its estimated incremental borrowing rate, which is derived from information available at the lease commencement date , in determining the present value of lease payments. American gives consideration to its recent debt issuances as well as publicly available data for instruments with similar characteristics when calculating its incremental borrowing rates.

American's lease term includes options to extend the lease when it is reasonably certain that it will exercise that option. Leases with a term of 12 months or less are not recorded on its consolidated balance sheets.

Under certain of American's capacity purchase agreements with third-party regional carriers, American does not own the underlying aircraft. However, since American controls the marketing, scheduling, ticketing, pricing and seat inventories of these aircraft and therefore control the asset, the aircraft is deemed to be leased for accounting purposes. For these capacity purchase agreements, American accounts for the lease and non-lease components separately. The lease component consists of the aircraft and the non-lease components consist of services, such as the crew and maintenance. Where applicable, American allocates the consideration in the capacity purchase agreements to the lease and non-lease components using their estimated relative standalone prices. See Note 10(b) for additional information on its capacity purchase agreements.

For real estate, American accounts for the lease and non-lease components as a single lease component.

(j) Income Taxes

Income taxes are accounted for under the asset and liability method. Deferred tax assets and liabilities are recognized for the future tax consequences attributable to differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases and operating loss and tax credit carryforwards. Deferred tax assets and liabilities are recorded net as noncurrent deferred income taxes.

American provides a valuation allowance for its deferred tax assets when it is more likely than not that some portion, or all of its deferred tax assets, will not be realized. The ultimate realization of deferred tax assets is dependent upon the generation of future taxable income. American considers all available positive and negative evidence and makes certain assumptions in evaluating the realizability of its deferred tax assets. Many factors are considered that impact American's assessment of future profitability, including conditions which are beyond its control, such as the health of the economy, the availability and price volatility of aircraft fuel and travel demand. American has determined that positive factors outweigh negative factors in the determination of the realizability of its deferred tax assets.

(k) Goodwill

Goodwill represents the purchase price in excess of the fair value of the net assets acquired and liabilities assumed in connection with the 2013 merger with US Airways Group, Inc. (US Airways Group). American has one reporting unit. American assesses goodwill for impairment annually or more frequently if events or circumstances indicate that the fair value of goodwill may be lower than the carrying value. American's annual assessment date is October 1.

Goodwill is assessed for impairment by initially performing a qualitative assessment. If American determines that it is more likely than not that its goodwill may be impaired, it uses a quantitative approach to assess the asset's fair value and the amount of the impairment, if any. Based upon American's annual assessment, there was no goodwill impairment in 2023. The carrying value of American's goodwill on its consolidated balance sheets was $4.1 billion as of December 31, 2023 and 2022.

(l) Other Intangibles, Net

Intangible assets consist primarily of certain domestic airport slots and gate leasehold rights, customer relationships, marketing agreements, commercial agreements, international slots and route authorities and tradenames.

Definite-Lived Intangible Assets

Definite-lived intangible assets are originally recorded at their acquired fair values, subsequently amortized over their respective estimated useful lives and are assessed for impairment whenever events and circumstances indicate that the assets may be impaired.

The following table provides information relating to American's amortizable intangible assets as of December 31, 2023 and 2022 (in millions):

	December 31,	
	2023	2022
Domestic airport slots	$ 365	$ 365
Customer relationships	300	300
Marketing agreements	105	105
Tradenames	35	35
Airport gate leasehold rights	137	137
Accumulated amortization	(834)	(827)
Total	$ 108	$ 115

Certain domestic airport slots and airport gate leasehold rights are amortized on a straight-line basis over 25 years. Certain marketing agreements were identified as intangible assets subject to amortization and are amortized on a straight-line basis over approximately 30 years. Customer relationships and tradenames are fully amortized.

American recorded amortization expense related to these intangible assets of $ 7 million for the year ended December 31, 2023 and $ 41 million for each of the years ended December 31, 2022 and 2021. American expects to record annual amortization expense for these intangible assets as follows (in millions):

2024	$ 7
2025	7
2026	6
2027	6
2028	6
2029 and thereafter	76
Total	$ 108

Indefinite-Lived Intangible Assets

Indefinite-lived intangible assets include certain domestic airport slots, international slots and route authorities and American's commercial agreement with GOL Linhas Aéreas Inteligentes S.A. (GOL). American assesses indefinite-lived intangible assets for impairment annually or more frequently if events or circumstances indicate that the fair values of indefinite-lived intangible assets may be lower than their carrying values. American's annual assessment date is October 1.

Indefinite-lived intangible assets are assessed for impairment by initially performing a qualitative assessment. If American determines that it is more likely than not that its indefinite-lived intangible assets may be impaired, American uses a quantitative approach to assess the asset's fair value and the amount of the impairment, if any. Based upon American's annual assessment, there were no indefinite-lived intangible asset impairments in 2023. American had $1.9 billion of indefinite-lived intangible assets on its consolidated balance sheets as of December 31, 2023 and 2022.

(m) Revenue Recognition

Revenue

The following are the significant categories comprising American's operating revenues (in millions):

	Year Ended December 31,		
	2023	2022	2021
Passenger revenue:			
Passenger travel	$ 44,914	$ 41,425	$ 23,896
Loyalty revenue - travel [1]	3,598	3,143	2,167
Total passenger revenue	48,512	44,568	26,063
Cargo	812	1,233	1,314
Other:			
Loyalty revenue - marketing services	2,929	2,657	2,166
Other revenue	531	507	337
Total other revenue	3,460	3,164	2,503
Total operating revenues	$ 52,784	$ 48,965	$ 29,880

[1] Loyalty revenue included in passenger revenue is principally comprised of mileage credit redemptions, which were earned from travel or co-branded credit card and other partners. See "*Loyalty Revenue"* below for further discussion on these mileage credits.

The following is American's total passenger revenue by geographic region (in millions):

	Year Ended December 31,		
	2023	2022	2021
Domestic	$ 34,592	$ 32,911	$ 21,453
Latin America	6,719	6,150	3,506
Atlantic	6,205	5,070	965
Pacific	996	437	139
Total passenger revenue	$ 48,512	$ 44,568	$ 26,063

American attributes passenger revenue by geographic region based upon the origin and destination of each flight segment.

Passenger Revenue

American recognizes all revenues generated from transportation on American and its regional flights operated under the brand name American Eagle, including associated baggage fees and other inflight services, as passenger revenue when transportation is provided. Ticket and other related sales for transportation that has not yet been provided are initially deferred and recorded as air traffic liability on American's consolidated balance sheets. The air traffic liability principally represents tickets sold for future travel on American and partner airlines.

The majority of tickets sold are nonrefundable. A small percentage of tickets, some of which are partially used tickets, expire unused. The estimate for tickets expected to expire unused is generally based on an analysis of American's historical data and other current applicable factors such as policy changes. American has consistently applied this accounting method to estimate and recognize revenue from unused tickets at the date of travel. This estimate is periodically evaluated based on subsequent activity to validate its accuracy. Any adjustments resulting from periodic evaluations of the estimated air traffic liability are included in passenger revenue during the period in which the evaluations are completed.

Various taxes and fees assessed on the sale of tickets to end customers are collected by American as an agent and remitted to taxing authorities. These taxes and fees have been presented on a net basis in the accompanying consolidated statements of operations and recorded as a liability until remitted to the appropriate taxing authority.

Loyalty Revenue

American currently operates the loyalty program, AAdvantage. This program awards mileage credits to passengers who fly on American, any **one**world airline or other partner airlines, or by using the services of other program participants, such as American's co-branded credit cards, and certain hotels and car rental companies. Mileage credits can be redeemed for travel on American and other participating partner airlines, as well as non-air travel awards such as hotels and rental cars. For mileage credits earned by AAdvantage program members, American applies the deferred revenue method.

Mileage credits earned through travel

For mileage credits earned through travel, American applies a relative selling price approach whereby the total amount collected from each passenger ticket sale is allocated between the air transportation and the mileage credits earned. The portion of each passenger ticket sale attributable to mileage credits earned is initially deferred and then recognized in passenger revenue when mileage credits are redeemed and transportation is provided. The estimated selling price of mileage credits is determined using an equivalent ticket value approach, which uses historical data, including award redemption patterns by geographic region and class of service, as well as similar cash fares as those used to settle award redemptions. The estimated selling price of mileage credits is adjusted for an estimate of mileage credits that will not be redeemed using a statistical model based on historical redemption patterns to develop an estimate of the likelihood of future redemption.

Mileage credits sold to co-branded credit cards and other partners

American sells mileage credits to participating airline partners and non-airline business partners, including American's co-branded credit card partners, under contracts with remaining terms generally from one to six years as of December 31, 2023. Consideration received from the sale of mileage credits is variable and payment terms typically are within 30 days subsequent to the month of mileage sale. Sales of mileage credits to non-airline business partners are comprised of two components, transportation and marketing. American allocates the consideration received from these sales of mileage credits based on the relative selling price of each product or service delivered.

American's most significant mileage credit partner agreements are its co-branded credit card agreements with Citi and Barclaycard US. American identified two revenue elements in these co-branded credit card agreements: the transportation component and the marketing component.

The transportation component represents the estimated selling price of future travel awards and is determined using the same equivalent ticket value approach described above. The portion of each mileage credit sold attributable to transportation is initially deferred and then recognized in passenger revenue when mileage credits are redeemed and transportation is provided.

The marketing component includes the use of intellectual property, including the American brand and access to loyalty program member lists, which is the predominant element in these agreements, as well as advertising and other travel-related benefits. American recognizes the marketing component in other revenue in the period of the mileage credit sale following the sales-based royalty method.

For the portion of American's outstanding mileage credits that it estimates will not be redeemed, American recognizes the associated value proportionally as the remaining mileage credits are redeemed. American's estimates use a statistical model based on historical redemption patterns to develop an estimate of the likelihood of future redemption.

Cargo Revenue

Cargo revenue is recognized when American provides the transportation.

Other Revenue

Other revenue includes revenue associated with American's loyalty program, which is comprised principally of the marketing component of mileage credit sales to co-branded credit card and other partners and other marketing related payments. The accounting and recognition for the loyalty program marketing services are discussed above in "*Loyalty Revenue*." The remaining amounts included within other revenue relate to airport clubs, other commission revenue, advertising and vacation-related services.

Contract Balances

American's significant contract liabilities are comprised of (1) outstanding loyalty program mileage credits that may be redeemed for future travel and non-air travel awards, reported as loyalty program liability on American's consolidated balance sheets and (2) ticket sales for transportation that has not yet been provided, reported as air traffic liability on American's consolidated balance sheets.

	December 31,	
	2023	2022
	(In millions)	
Loyalty program liability	$ 9,327	$ 9,145
Air traffic liability	6,200	6,745
Total	$ 15,527	$ 15,890

The balance of the loyalty program liability fluctuates based on seasonal patterns, which impact the volume of mileage credits issued through travel or sold to co-branded credit card and other partners (deferral of revenue) and mileage credits redeemed (recognition of revenue). Changes in loyalty program liability are as follows (in millions):

Balance at December 31, 2022	$ 9,145
Deferral of revenue	3,810
Recognition of revenue [1]	(3,628)
Balance at December 31, 2023 [2]	$ 9,327

[1] Principally relates to revenue recognized from the redemption of mileage credits for both air and non-air travel awards. Mileage credits are combined in one homogenous pool and are not separately identifiable. As such, the revenue is comprised of mileage credits that were part of the loyalty program deferred revenue balance at the beginning of the period, as well as mileage credits that were issued during the period.

[2] Mileage credits can be redeemed at any time and generally do not expire as long as that AAdvantage member has any type of qualifying activity at least every 24 months or if the AAdvantage member is the primary holder of a co-branded credit card. As of December 31, 2023, American's current loyalty program liability was $3.5 billion and represents American's current estimate of revenue expected to be recognized in the next 12 months based on historical trends, with the balance reflected in long-term loyalty program liability expected to be recognized as revenue in periods thereafter.

The air traffic liability principally represents tickets sold for future travel on American and partner airlines. The balance in American's air traffic liability also fluctuates with seasonal travel patterns. The contract duration of passenger tickets is generally one year. Accordingly, any revenue associated with tickets sold for future travel will be recognized within 12 months. For 2023, $5.3 billion of revenue was recognized in passenger revenue that was included in American's air traffic liability at December 31, 2022.

(n) Maintenance, Materials and Repairs

Maintenance and repair costs for owned and leased flight equipment are charged to operating expense as incurred, except costs incurred for maintenance and repair under certain power-by-the-hour maintenance agreements, which are charged to operating expense based on contractual terms when an obligation exists.

(o) Selling Expenses

Selling expenses include credit card fees, commissions, third party distribution channel fees and advertising. Selling expenses associated with passenger revenue are expensed when the transportation or service is provided. Advertising costs are expensed as incurred. Advertising expense was $114 million for the year ended December 31, 2023 and $ 105 million for each of the years ended December 31, 2022 and 2021.

(p) Share-based Compensation

American accounts for its share-based compensation expense based on the fair value of the stock award at the time of grant, which is recognized ratably over the vesting period of the stock award. Certain awards have performance conditions that must be achieved prior to vesting and are expensed based on the expected achievement at each reporting period. The majority of American's stock awards are time vested restricted stock units, and the fair value of such awards is based on the market price of the underlying shares of AAG common stock on the date of grant. See Note 13 for further discussion of share-based compensation.

(q) Foreign Currency Gains and Losses

Foreign currency gains and losses are recorded as part of other income (expense), net within total nonoperating expense, net on American's consolidated statements of operations. For the years ended December 31, 2023 , 2022 and 2021, respectively, foreign currency losses were $ 30 million, $38 million and $ 4 million.

(r) Other Operating Expenses

Other operating expenses includes costs associated with onboard food and catering, crew travel, ground and cargo handling, passenger accommodation, international navigation fees, aircraft cleaning, airport lounge operations and certain general and administrative expenses.

(s) Regional Expenses

American's regional carriers provide scheduled air transportation under the brand name "American Eagle." The American Eagle carriers include AAG's wholly-owned regional carriers as well as third-party regional carriers. American's regional carrier arrangements are in the form of capacity purchase agreements with its third-party regional partners and similar arrangements with AAG's wholly-owned regional affiliates. Expenses associated with American Eagle operations are classified as regional expenses on the consolidated statements of operations.

Regional expenses for the years ended December 31, 2023 , 2022 and 2021 include $ 271 million, $269 million and $ 263 million of depreciation and amortization, respectively, and $7 million, $5 million and $ 6 million of aircraft rent, respectively.

In 2023, 2022 and 2021, American recognized $ 636 million, $592 million and $ 495 million, respectively, of expense under its capacity purchase agreement with Republic Airways Inc. (Republic). American holds a 25% equity interest in Republic Airways Holdings Inc. (Republic Holdings), the parent company of Republic.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS OF AMERICAN AIRLINES, INC.

2. Special Items, Net

Special items, net on American's consolidated statements of operations consisted of the following (in millions):

	Year Ended December 31,		
	2023	2022	2021
Labor contract expenses [1]	$ 989	$ —	$ —
Severance expenses [2]	23	—	168
Fleet impairment [3]	—	149	—
Litigation reserve adjustments	—	37	(19)
PSP Financial Assistance [4]	—	—	(4,162)
Other operating special items, net	(41)	7	7
Mainline operating special items, net	971	193	(4,006)
PSP Financial Assistance [4]	—	—	(539)
Fleet impairment [3]	—	—	27
Regional operating special items, net	—	—	(512)
Operating special items, net	971	193	(4,518)
Debt refinancing, extinguishment and other, net [5]	280	1	29
Mark-to-market adjustments on equity and other investments, net [6]	82	71	31
Nonoperating special items, net	362	72	60
Income tax special items, net	—	(9)	—

[1] Labor contract expenses relate to one-time charges resulting from the ratification of a new collective bargaining agreement with American's mainline pilots, including a one-time payment of $754 million as well as adjustments to other benefit-related items of $ 235 million.

[2] Severance expenses for 2023 included costs associated with headcount reductions in certain corporate functions.

Severance expenses for 2021 included salary and medical costs primarily associated with certain team members who opted into voluntary early retirement programs offered as a result of reductions to American's operation due to the COVID-19 pandemic.

[3] Fleet impairment for 2022 included a non-cash impairment charge to write down the carrying value of American's retired Airbus A330 fleet to the estimated fair value due to the market conditions for certain used aircraft. American retired its Airbus A330 fleet in 2020 as a result of the decline in demand for air travel due to the COVID-19 pandemic.

Fleet impairment for 2021 included a non-cash impairment charge to write down regional aircraft resulting from the retirement of the remaining Embraer 140 fleet earlier than planned.

[4] The PSP Financial Assistance represents recognition of a portion of the financial assistance received from Treasury pursuant to the payroll support programs established by the U.S. Government. See Note 1(b) for further information.

[5] Debt refinancing and extinguishment costs in 2023 primarily included cash charges for premiums paid in connection with the early repayment of debt. See Note 3 for further information.

[6] Mark-to-market adjustments on equity and other investments, net principally included net unrealized gains and losses associated with certain equity investments and certain other investments. See Note 7 for further information related to American's equity investments.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS OF AMERICAN AIRLINES, INC.

3. Debt

Long-term debt included on American's consolidated balance sheets consisted of (in millions):

	December 31, 2023	December 31, 2022
Secured		
2013 Term Loan Facility, variable interest rate of 8.60%, installments through February 2028 [a]	$ 990	$ 1,752
2014 Term Loan Facility, variable interest rate of 7.32%, installments through January 2027 [a]	1,183	1,196
2023 Term Loan Facility, variable interest rate of 8.87%, installments beginning in December 2024 through June 2029 [a]	1,100	—
11.75% senior secured notes, interest only payments until due in July 2025 [b]	—	2,500
10.75% senior secured IP notes, interest only payments until due in February 2026 [b]	1,000	1,000
10.75% senior secured LGA/DCA notes, interest only payments until due in February 2026 [b]	200	200
7.25% senior secured notes, interest only payments until due in February 2028 [b]	750	—
8.50% senior secured notes, interest only payments until due in May 2029 [b]	1,000	—
5.50% senior secured notes, installments through April 2026 [c]	2,917	3,500
5.75% senior secured notes, installments beginning in July 2026 until due in April 2029 [c]	3,000	3,000
AAdvantage Term Loan Facility, variable interest rate of 10.43%, installments through April 2028 [c]	3,150	3,500
Enhanced equipment trust certificates (EETCs), fixed interest rates ranging from 2.88% to 5.90%, averaging 3.60%, maturing from 2024 to 2034	7,657	9,175
Equipment loans and other notes payable, fixed and variable interest rates ranging from 2.55% to 8.90%, averaging 6.98%, maturing from 2024 to 2035 [d]	3,612	3,170
Special facility revenue bonds, fixed interest rates ranging from 2.25% to 5.38%, maturing from 2026 to 2036	967	1,050
Total long-term debt	27,526	30,043
Less: Total unamortized debt discount, premium and issuance costs	349	364
Less: Current maturities	3,501	3,059
Long-term debt, net of current maturities	$ 23,676	$ 26,620

As of December 31, 2023, the maximum availability under American's revolving credit and other facilities is as follows (in millions):

2013 Revolving Facility	$ 736
2014 Revolving Facility	1,631
April 2016 Revolving Facility	446
Other short-term facility	49
Total	$ 2,862

As of December 31, 2023, American had $ 49 million of available borrowing base under a cargo receivables facility that is set to expire in December 2024. As a result of the below amendments to the 2013, 2014 and April 2016 Revolving Facilities, the aggregate commitments under these facilities will be $2.8 billion through October 11, 2024, and thereafter through October 13, 2026, such aggregate commitments will decrease to $ 2.2 billion.

Secured financings, including revolving credit and other facilities, are collateralized by assets, consisting primarily of aircraft, engines, simulators, aircraft spare parts, airport gate leasehold rights, route authorities, airport slots, certain receivables, certain intellectual property and certain loyalty program assets.

At December 31, 2023, the maturities of long-term debt are as follows (in millions):

2024	$	3,501
2025		3,702
2026		4,582
2027		4,618
2028		5,060
2029 and thereafter		6,063
Total	$	27,526

(a) 2013 and 2014 Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility

2013 Credit Facilities

The Amended and Restated Credit and Guaranty Agreement dated as of May 21, 2015, as amended (the 2013 Credit Agreement), includes a revolving credit facility (the 2013 Revolving Facility) and term loan (the 2013 Term Loan Facility), collectively referred to as the 2013 Credit Facilities. In February 2023, American and AAG refinanced approximately $1.8 billion in aggregate principal amount of term loans outstanding under the 2013 Term Loan Facility (the 2013 Term Loan Facility Refinancing) through the combination of (i) the issuance of $750 million in aggregate principal amount of 7.25% senior secured notes due 2028 and (ii) the entry into the Seventh Amendment to the 2013 Credit Agreement, pursuant to which the maturity of $1.0 billion in term loans under the 2013 Term Loan Facility was extended to February 2028 from June 2025. The Seventh Amendment also amended certain other terms of the 2013 Credit Agreement, including the interest rate and amortization schedule for the 2013 Term Loan Facility, the requirements for delivery of appraisals and certain covenants relating to dispositions of collateral. Additionally, the Seventh Amendment transitioned the benchmark interest rate from the London Interbank Offered Rate (LIBOR) to the Secured Overnight Financing Rate (SOFR). As a result, the 2013 Term Loan Facility bears interest at a base rate (subject to a floor of 1.00%) plus an applicable margin of 1.75% or, at American's option, the SOFR rate for a tenor of one, three or six months, depending on the interest period selected by American, plus the SOFR adjustment applicable to such interest period (with such SOFR rate plus SOFR adjustment being subject to a floor of 0.00%) and an applicable margin of 2.75%. As of December 31, 2023, the margin elected was 2.75%.

In March 2023, American and AAG entered into the Eighth Amendment to the 2013 Credit Agreement, pursuant to which American extended the maturity of certain commitments under the 2013 Revolving Facility. The Eighth Amendment also amended certain other terms of the 2013 Credit Agreement, including certain covenants and transitioned the benchmark interest rate from LIBOR to SOFR. The 2013 Revolving Facility bears interest at a base rate (subject to a floor of 1.00%) plus an applicable margin of 2.25%, 2.50% or 2.75%, depending on AAG's public corporate rating, or, at American's option, the SOFR rate for a tenor of one, three or six months, depending on the interest period selected by American, plus the SOFR adjustment applicable to such interest period (with such SOFR rate plus SOFR adjustment being subject to a floor of 0.00%) plus an applicable margin of 3.25%, 3.50% or 3.75%, depending on AAG's public corporate rating. Additionally, as a result of the Eighth Amendment, through October 11, 2024, the aggregate commitments under the 2013 Revolving Facility will be $736 million, and thereafter through October 13, 2026, such aggregate commitments will decrease to $563 million. As of December 31, 2023, there were no borrowings or letters of credit outstanding under the 2013 Revolving Facility.

2014 Credit Facilities

The Amended and Restated Credit and Guaranty Agreement, dated as of April 20, 2015, as amended (the 2014 Credit Agreement), includes a revolving credit facility (the 2014 Revolving Facility) and term loan (the 2014 Term Loan Facility), collectively referred to as the 2014 Credit Facilities. In March 2023, American and AAG entered into the Ninth Amendment to the 2014 Credit Agreement, pursuant to which American extended the maturity of certain commitments under the 2014 Revolving Facility. The Ninth Amendment also amended certain other terms of the 2014 Credit Agreement including the requirements for delivery of appraisals and certain other covenants and transitioned the benchmark interest rate for the 2014 Revolving Facility and the 2014 Term Loan Facility from LIBOR to SOFR. The 2014 Revolving Facility bears interest at the same base rate and applicable margin as the 2013 Revolving Facility, as noted above in "2013 Credit Facilities." The 2014 Term Loan Facility bears interest at a base rate (subject to a floor of 1.00%) plus an applicable margin of 0.75% or, at American's option, the SOFR rate for a tenor of one, three or six months, depending on the interest period selected by American, plus the SOFR adjustment applicable to such interest period (with such SOFR rate plus SOFR adjustment

being subject to a floor of 0.00%) plus an applicable margin of 1.75%. As of December 31, 2023, the margin elected was 1.75%. Additionally, as a result of the Ninth Amendment, through October 11, 2024, the aggregate commitments under the 2014 Revolving Facility will be $1.6 billion, and thereafter through October 13, 2026, such aggregate commitments will decrease to $1.2 billion. As of December 31, 2023, there were no borrowings or letters of credit outstanding under the 2014 Revolving Facility.

April 2016 Revolving Facility

In March 2023, American and AAG entered into the Sixth Amendment to the Credit and Guaranty Agreement, dated as of April 29, 2016 (the April 2016 Credit Agreement), which includes a revolving credit facility (the April 2016 Revolving Facility). Pursuant to the Sixth Amendment, American extended the maturity of certain commitments under the April 2016 Revolving Facility. The Sixth Amendment also amended certain other terms under the April 2016 Credit Agreement including the requirements for delivery of appraisals and certain other covenants and transitioned the benchmark interest rate for the April 2016 Revolving Facility from LIBOR to SOFR. The April 2016 Revolving Facility bears interest at the same base rate and applicable margin as the 2013 Revolving Facility, as noted above in "2013 Credit Facilities." Additionally, as a result of the Sixth Amendment, through October 11, 2024, the aggregate commitments under the April 2016 Revolving Facility will be $446 million, and thereafter through October 13, 2026, such aggregate commitments will decrease to $342 million. As of December 31, 2023, there were no borrowings outstanding under the April 2016 Revolving Facility.

2023 Term Loan Facility

In December 2023, American and AAG entered into a credit and guaranty agreement (the 2023 Credit Agreement) that provided for a term loan facility (the 2023 Term Loan Facility) in an aggregate principal amount of $1.1 billion, maturing in June 2029. Loans made under the 2023 Term Loan Facility bear interest at a base rate (subject to a floor of 1.00%) plus an applicable margin of 2.50% or, at American's option, the SOFR rate for a tenor of one, three or six months (or if agreed by the relevant lenders, any other tenor), depending on the interest period selected by American (subject to a floor of 0.00%), plus an applicable margin of 3.50%. As of December 31, 2023, the margin elected was 3.50%. The net proceeds from the 2023 Term Loan Facility, together with the net proceeds from the private offering of the 8.50% Senior Secured Notes (as defined below) and cash on hand, were used to redeem all of the outstanding 11.75% Senior Secured Notes in December 2023.

Other Terms of the 2013 and 2014 Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility

The term loans under the 2013 Credit Facilities and 2014 Credit Facilities (collectively referred to as the Credit Facilities) and the 2023 Term Loan Facility are repayable in annual installments, in an amount equal to 1.00% of the aggregate principal amount issued, with any unpaid balance due on the respective maturity dates. Voluntary prepayments may be made by American at any time.

The 2013 Revolving Facility, 2014 Revolving Facility and April 2016 Revolving Facility provide that American may from time to time borrow, repay and reborrow loans thereunder. The 2013 Revolving Facility and 2014 Revolving Facility have the ability to issue letters of credit thereunder in an aggregate amount outstanding at any time up to $150 million and $300 million, respectively. The 2013 Revolving Facility, 2014 Revolving Facility and April 2016 Revolving Facility are each subject to an undrawn annual fee of 0.750%.

Subject to certain limitations and exceptions, the Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility are secured by collateral, including certain spare parts, slots, route authorities, simulators and leasehold rights. American has the ability to make future modifications to the collateral pledged, subject to certain restrictions. American's obligations under the Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility are guaranteed by AAG, and such guarantee is AAG's senior unsecured obligations (all of the collateral is owned by American, and AAG has not granted a security interest in any assets to secure any of the foregoing obligations). The Credit Facilities, April 2016 Revolving Facility and 2023 Term Loan Facility contain events of default customary for similar financings, including cross default and cross-acceleration to other material indebtedness.

(b) Senior Secured Notes

11.75% Senior Secured Notes

In June 2020, American issued $2.5 billion aggregate principal amount of 11.75% senior secured notes due 2025 (the 11.75% Senior Secured Notes) at a price equal to 99% of their aggregate principal amount. In December 2023, American redeemed all of its outstanding 11.75% Senior Secured Notes using net proceeds from the offering of the 8.50% Senior Secured Notes (as defined below), together with net proceeds from borrowings under the 2023 Term Loan Facility and cash on hand. In connection with the early redemption of the 11.75% Senior Secured Notes, in the fourth quarter of 2023, American recorded a $186 million cash special charge for the make-whole premium paid and a $19 million non-cash special charge to write off unamortized debt issuance costs and debt discount.

10.75% Senior Secured Notes

On September 25, 2020 (the 10.75% Senior Secured Notes Closing Date), American issued $1.0 billion in initial principal amount of senior secured IP notes (the IP Notes) and $200 million in initial principal amount of senior secured LGA/DCA notes (the LGA/DCA Notes and together with the IP Notes, the 10.75% Senior Secured Notes). The obligations of American under the 10.75% Senior Secured Notes are fully and unconditionally guaranteed (the 10.75% Senior Secured Notes Guarantees) on a senior unsecured basis by AAG. The 10.75% Senior Secured Notes bear interest at a rate of 10.75% per annum in cash. Interest on the 10.75% Senior Secured Notes is payable semiannually in arrears on September 1 and March 1 of each year, which began on March 1, 2021. The 10.75% Senior Secured Notes will mature on February 15, 2026.

The IP Notes are secured by a first lien security interest on certain intellectual property of American, including the “American Airlines” trademark and the “aa.com” domain name in the United States and certain foreign jurisdictions (the IP Collateral), and a second lien on certain slots related to American’s operations at New York LaGuardia and Ronald Reagan Washington National airports and certain other assets (the LGA/DCA Collateral and together with the IP Collateral, the 10.75% Senior Secured Notes Collateral). LGA/DCA Notes are secured by a first lien security interest in the LGA/DCA Collateral.

On or prior to the fourth anniversary of the 10.75% Senior Secured Notes Closing Date, American may redeem all or any part of the 10.75% Senior Secured Notes, at its option, at a redemption price equal to 100% of the principal amount of the 10.75% Senior Secured Notes redeemed plus a “make-whole” premium, together with accrued and unpaid interest thereon, if any. After the fourth anniversary of the 10.75% Senior Secured Notes Closing Date and on or prior to the fifth anniversary of the 10.75% Senior Secured Notes Closing Date, American may redeem all or any part of the 10.75% Senior Secured Notes, at its option, at a redemption price equal to 105.375% of the principal amount of the 10.75% Senior Secured Notes redeemed, together with accrued and unpaid interest thereon, if any. After the fifth anniversary of the 10.75% Senior Secured Notes Closing Date, American may redeem all or any part of the 10.75% Senior Secured Notes, at its option, at par, together with accrued and unpaid interest thereon, if any.

7.25% Senior Secured Notes

On February 15, 2023, as part of the 2013 Term Loan Facility Refinancing, American issued $750 million aggregate principal amount of 7.25% senior secured notes due 2028 (the 7.25% Senior Secured Notes) in a private offering. The 7.25% Senior Secured Notes were issued at par and bear interest at a rate of 7.25% per annum (subject to increase if the collateral coverage ratio described below is not met). Interest on the 7.25% Senior Secured Notes is payable semiannually in arrears on February 15 and August 15 of each year, which began on August 15, 2023. The 7.25% Senior Secured Notes will mature on February 15, 2028. The obligations of American under the 7.25% Senior Secured Notes are fully and unconditionally guaranteed on a senior unsecured basis by AAG. American used the proceeds from the offering of the 7.25% Senior Secured Notes, together with cash on hand, to repay a portion of the term loans then outstanding under the 2013 Term Loan Facility and to pay related fees and expenses.

The 7.25% Senior Secured Notes were issued pursuant to an indenture, dated as of February 15, 2023 (the 7.25% Senior Secured Notes Indenture), by and among American, AAG and Wilmington Trust, National Association, as trustee and collateral agent. The 7.25% Senior Secured Notes are American’s senior secured obligations and are secured on a first lien basis by security interests in certain assets, rights and properties that American uses to provide non-stop scheduled air carrier services between (a) certain airports in the United States and (b) airports in certain countries in South America and New Zealand (collectively, the 7.25% Senior Secured Notes Collateral). The 7.25% Senior Secured Notes Collateral also secures, on a first lien, pari passu basis with the 7.25% Senior Secured Notes, the 2013 Credit Facilities under the 2013 Credit Agreement.

American may redeem the 7.25% Senior Secured Notes, in whole at any time or in part from time to time prior to February 15, 2025, at a redemption price equal to 100% of the principal amount of the 7.25% Senior Secured Notes to be redeemed, plus a "make-whole" premium, plus any accrued and unpaid interest thereon to but excluding the date of redemption. At any time on or after February 15, 2025, American may redeem all or any of the 7.25% Senior Secured Notes in whole at any time, or in part from time to time, at the redemption prices described in the 7.25% Senior Secured Notes Indenture, plus any accrued and unpaid interest thereon to but excluding the date of redemption. In addition, at any time prior to February 15, 2025, American may redeem up to 40% of the original aggregate principal amount of the 7.25% Senior Secured Notes (calculated after giving effect to any issuance of additional notes) with the net cash proceeds of certain equity offerings, at a redemption price equal to 107.250% of the aggregate principal amount of the 7.25% Senior Secured Notes to be redeemed, plus any accrued and unpaid interest thereon to but excluding the date of redemption.

Twice per year, American is required to deliver an appraisal of the 7.25% Senior Secured Notes Collateral and an officer's certificate demonstrating the calculation of a collateral coverage ratio in relation to the 7.25% Senior Secured Notes Collateral (the 7.25% Senior Secured Notes Collateral Coverage Ratio) as of the date of delivery of the appraisal for the applicable period. If the 7.25% Senior Secured Notes Collateral Coverage Ratio is less than 1.6 to 1.0 as of the date of delivery of the appraisal for the applicable period, then, subject to a cure period in which additional collateral can be provided or debt repaid such that American meets the required 7.25% Senior Secured Notes Collateral Coverage Ratio, American will be required to pay special interest in an additional amount equal to 2.0% per annum of the principal amount of the 7.25% Senior Secured Notes until the 7.25% Senior Secured Notes Collateral Coverage Ratio is established to be at least 1.6 to 1.0.

8.50% Senior Secured Notes

On December 4, 2023, American issued $ 1.0 billion aggregate principal amount of 8.50% senior secured notes due 2029 (the 8.50% Senior Secured Notes) in a private offering. The 8.50% Senior Secured Notes were issued at par and bear interest at a rate of 8.50% per annum (subject to increase if the collateral coverage ratio described below is not met). Interest on the 8.50% Senior Secured Notes is payable semiannually in arrears on May 15 and November 15 of each year, beginning on May 15, 2024. The 8.50% Senior Secured Notes will mature on May 15, 2029. The obligations of American under the 8.50% Senior Secured Notes are fully and unconditionally guaranteed on a senior unsecured basis by AAG. The net proceeds from the 8.50% Senior Secured Notes, together with borrowings under the 2023 Term Loan Facility and cash on hand, were used to redeem all of the outstanding 11.75% Senior Secured Notes in December 2023.

The 8.50% Senior Secured Notes were issued pursuant to an indenture, dated as of December 4, 2023 (the 8.50% Senior Secured Notes Indenture), by and among American, AAG and Wilmington Trust, National Association, as trustee and collateral agent. The 8.50% Senior Secured Notes are American's senior secured obligations and are secured on a first lien basis by security interests in certain assets, rights and properties that American uses to provide non-stop scheduled air carrier services between (a) certain airports in the United States and (b) certain airports in Australia, Canada, the Caribbean, Central America, China, Hong Kong, Japan, Mexico, South Korea and Switzerland (collectively, the 8.50% Senior Secured Notes Collateral). The 8.50% Senior Secured Notes Collateral also secures, on a first lien, pari passu basis with the 8.50% Senior Secured Notes, the 2023 Term Loan Facility.

American may redeem the 8.50% Senior Secured Notes, in whole at any time or in part from time to time prior to November 15, 2025, at a redemption price equal to 100% of the principal amount of the 8.50% Senior Secured Notes to be redeemed, plus a "make-whole" premium, plus any accrued and unpaid interest thereon to but excluding the date of redemption. At any time on or after November 15, 2025, American may redeem all or any of the 8.50% Senior Secured Notes in whole at any time, or in part from time to time, at the redemption prices described in the 8.50% Senior Secured Notes Indenture, plus any accrued and unpaid interest thereon to but excluding the date of redemption. In addition, at any time prior to November 15, 2025, American may redeem up to 40% of the original aggregate principal amount of the 8.50% Senior Secured Notes (calculated after giving effect to any issuance of additional notes) with the net cash proceeds of certain equity offerings, at a redemption price equal to 108.50% of the aggregate principal amount of the 8.50% Senior Secured Notes to be redeemed, plus any accrued and unpaid interest thereon to but excluding the date of redemption. In addition, during each twelve-month period beginning on December 4, 2023 and ending on or prior to November 15, 2025, American may redeem up to 10% of the original aggregate principal amount of the 8.50% Senior Secured Notes at a redemption price of 103% of the principal amount thereof, plus any accrued and unpaid interest thereon to, but excluding, the applicable date of redemption.

Twice per year, American is required to deliver an appraisal of the 8.50% Senior Secured Notes Collateral and an officer's certificate demonstrating the calculation of a collateral coverage ratio in relation to the 8.50% Senior Secured Notes Collateral (the 8.50% Senior Secured Notes Collateral Coverage Ratio) as of the date of delivery of the appraisal

for the applicable period. If the 8.50% Senior Secured Notes Collateral Coverage Ratio is less than 1.6 to 1.0 as of the date of delivery of the appraisal for the applicable period, then, subject to a cure period in which additional collateral can be provided or debt repaid such that American meets the required 8.50% Senior Secured Notes Collateral Coverage Ratio, American will be required to pay special interest in an additional amount equal to 2.0% per annum of the principal amount of the 8.50% Senior Secured Notes until the 8.50% Senior Secured Notes Collateral Coverage Ratio is established to be at least 1.6 to 1.0.

(c) AAdvantage Financing

On March 24, 2021 (the AAdvantage Financing Closing Date), American and AAdvantage Loyalty IP Ltd., a Cayman Islands exempted company incorporated with limited liability and an indirect wholly-owned subsidiary of American (Loyalty Issuer and, together with American, the AAdvantage Issuers), completed the offering of $3.5 billion aggregate principal amount of 5.50% Senior Secured Notes due 2026 (the 2026 Notes) and $ 3.0 billion aggregate principal amount of 5.75% Senior Secured Notes due 2029 (the 2029 Notes, and together with the 2026 Notes, the AAdvantage Notes). The AAdvantage Notes are fully and unconditionally guaranteed by the SPV Guarantors and AAG.

Concurrent with the issuance of the AAdvantage Notes, the AAdvantage Issuers, as co-borrowers, entered into a term loan credit and guaranty agreement, dated March 24, 2021, providing for a $3.5 billion term loan facility (the AAdvantage Term Loan Facility and collectively with the AAdvantage Notes, the AAdvantage Financing) and pursuant to which the full $3.5 billion of term loans (the AAdvantage Loans) were drawn on the AAdvantage Financing Closing Date. The AAdvantage Loans are fully and unconditionally guaranteed (together with the AAdvantage Note Guarantees, the AAdvantage Guarantees) by the SPV Guarantors and AAG.

Subject to certain permitted liens and other exceptions, the AAdvantage Notes, AAdvantage Loans and AAdvantage Guarantees provided by the SPV Guarantors are secured by a first-priority security interest in, and pledge of, various agreements with respect to the AAdvantage program (the AAdvantage Agreements) (including all payments thereunder) and certain intellectual property licenses, certain deposit accounts that will receive cash under the AAdvantage Agreements, certain reserve accounts, the equity of each of Loyalty Issuer and the SPV Guarantors and substantially all other assets of Loyalty Issuer and the SPV Guarantors including American's rights to certain data and other intellectual property used in the AAdvantage program (subject to certain exceptions) (collectively, the AAdvantage Collateral).

Payment Terms of the AAdvantage Notes and AAdvantage Loans under the AAdvantage Term Loan Facility

Interest on the AAdvantage Notes is payable in cash, quarterly in arrears on the 20th day of each January, April, July and October (each, an AAdvantage Payment Date), which began on July 20, 2021. The 2026 Notes will mature on April 20, 2026, and the 2029 Notes will mature on April 20, 2029. The outstanding principal on the 2026 Notes will be repaid in quarterly installments of $292 million on each AAdvantage Payment Date, which began in July 2023. The outstanding principal on the 2029 Notes will be repaid in quarterly installments of $250 million on each AAdvantage Payment Date, beginning on July 20, 2026.

The AAdvantage Issuers may redeem the AAdvantage Notes, at their option, in whole at any time or in part from time to time, at a redemption price equal to 100% of the principal amount of the AAdvantage Notes redeemed plus a "make-whole" premium, together with accrued and unpaid interest to the date of redemption.

The scheduled maturity date of the AAdvantage Loans under the AAdvantage Term Loan Facility is April 20, 2028. The outstanding principal on the AAdvantage Loans will be repaid in quarterly installments of $175 million, on each AAdvantage Payment Date, which began in July 2023. These amortization payments (as well as those for the AAdvantage Notes) will be subject to the occurrence of certain early amortization events, including the failure to satisfy a minimum debt service coverage ratio at specified determination dates.

Prepayment of some or all of the AAdvantage Loans outstanding under the AAdvantage Term Loan Facility is permitted, although payment of an applicable premium is required as specified in the AAdvantage Term Loan Facility.

The AAdvantage Indenture and the AAdvantage Term Loan Facility contain mandatory prepayment provisions triggered upon (i) the issuance or incurrence by Loyalty Issuer or the SPV Guarantors of certain indebtedness or (ii) the receipt by American or its subsidiaries of net proceeds from pre-paid frequent flyer (i.e., AAdvantage) mile sales exceeding $505 million. Each of these prepayments would also require payment of an applicable premium. Certain other events, including the occurrence of a change of control with respect to AAG and certain AAdvantage Collateral sales exceeding a specified threshold, will also trigger mandatory repurchase or mandatory prepayment provisions under the AAdvantage Indenture and the AAdvantage Term Loan Facility, respectively.

In June 2023, American and AAdvantage Loyalty IP Ltd. entered into the First Amendment to the AAdvantage Term Loan Facility pursuant to which the benchmark interest rate transitioned from LIBOR to SOFR, effective July 1, 2023. As a result, the AAdvantage Term Loan Facility bears interest at a base rate (subject to a floor of 0.00%) plus an applicable margin of 3.75% or, at American's option, the SOFR rate for a tenor of three months, plus a 0.26161% credit spread adjustment (with such SOFR rate plus SOFR adjustment being subject to a floor of 0.75%) and an applicable margin of 4.75%. As of December 31, 2023, the margin elected was 4.75%. Other than the foregoing, the terms of the AAdvantage Term Loan Facility remain substantially unchanged.

(d) Equipment Loans and Other Notes Payable Issued in 2023

In 2023, American entered into agreements under which it borrowed $ 1.1 billion in connection with the financing of certain aircraft. Debt incurred under these agreements matures in 2032 through 2035 and bears interest at fixed and variable rates (comprised of SOFR plus an applicable margin) averaging 7.15% as of December 31, 2023.

Other Financing Activities

During the year ended December 31, 2023, American repurchased $ 539 million of secured notes in the open market. In connection with the repurchase of these secured notes in the open market, American recorded $57 million of cash special charges for premiums paid and $ 6 million of non-cash special charges to write off unamortized debt issuance costs and debt discounts.

Guarantees

As of December 31, 2023, American had issued guarantees covering AAG's $ 1.8 billion aggregate principal amount of the PSP1 Promissory Note due April 2030, $1.0 billion aggregate principal amount of the PSP2 Promissory Note due January 2031, $ 959 million aggregate principal amount of the PSP3 Promissory Note due April 2031, $1.0 billion aggregate principal amount of 6.50% convertible senior notes due July 2025 and $ 487 million of 3.75% senior notes due March 2025.

Certain Covenants

American's debt agreements contain customary terms and conditions as well as various affirmative, negative and financial covenants that, among other things, may restrict the ability of American to incur additional indebtedness. American's debt agreements also contain customary change of control provisions, which may require it to repay or redeem such indebtedness upon certain events constituting a change of control under the relevant agreement, in certain cases at a premium. Certain of American's debt financing agreements (including its secured notes, term loans, revolving credit facilities and spare engine EETCs) contain loan to value (LTV), collateral coverage or peak debt service coverage ratio covenants and certain agreements require American to appraise the related collateral annually or semiannually. Pursuant to such agreements, if the applicable LTV, collateral coverage or peak debt service coverage ratio exceeds or falls below a specified threshold, as the case may be, American will be required, as applicable, to pledge additional qualifying collateral (which in some cases may include cash or investment securities), withhold additional cash in certain accounts, or pay down such financing, in whole or in part, or the interest rate for the relevant financing will be increased. Additionally, a significant portion of American's debt financing agreements contain covenants requiring it to maintain an aggregate of at least $2.0 billion of unrestricted cash and cash equivalents and amounts available to be drawn under revolving credit facilities, and its AAdvantage Financing contains a peak debt service coverage ratio, pursuant to which failure to comply with a certain threshold may result in early repayment, in whole or in part, of the AAdvantage Financing.

Specifically, American is required to meet certain collateral coverage tests for its Credit Facilities, April 2016 Revolving Facility, 2023 Term Loan Facility, 7.25% Senior Secured Notes, 8.50% Senior Secured Notes and 10.75% Senior Secured Notes, as described below:

	2013 Credit Facilities	7.25% Senior Secured Notes	2014 Credit Facilities	April 2016 Revolving Facility	2023 Term Loan Facility	8.50% Senior Secured Notes	10.75% Senior Secured Notes
LTV Requirement	1.6x Collateral valuation to amount of debt outstanding (62.5% LTV)						
LTV as of Last Measurement Date	34.2%		16.4%	Not Applicable	25.9%		6.9%
Frequency of Appraisals of Appraised Collateral	Semi-Annual						Annual
Collateral Description	Generally, certain slots, route authorities and airport gate leasehold rights used by American to operate certain services between the U.S. and South America and New Zealand		Generally, certain slots, route authorities and airport gate leasehold rights used by American to operate certain services between the U.S. and European Union (including London Heathrow)	Generally, certain spare parts	Generally, certain slots, route authorities and airport gate leasehold rights used by American to operate certain services between the U.S. and Australia, Canada, the Caribbean, Central America, China, Hong Kong, Japan, Mexico, South Korea and Switzerland		Generally, certain DCA slots, certain LGA slots, certain simulators and certain leasehold rights and, in the case of the IP Notes, certain intellectual property of American

At December 31, 2023, American was in compliance with the applicable collateral coverage tests as of the most recent measurement dates.

4. Leases

American leases certain aircraft and engines, including aircraft under capacity purchase agreements. As of December 31, 2023, American operated 737 leased aircraft, including seven aircraft in temporary storage and 237 aircraft leased under capacity purchase agreements, with remaining terms ranging from less than one year to 10 years.

At each airport where American conducts flight operations, American has agreements, generally with a governmental unit or authority, for the use of passenger, operations and baggage handling space as well as runways and taxiways. These agreements, particularly in the U.S., often contain provisions for periodic adjustments to rates and charges applicable under such agreements. These rates and charges also vary with American's level of operations and the operations of the airport. Because of the variable nature of these rates, these leases are not recorded on American's consolidated balance sheets as a ROU asset or a lease liability. Additionally, at American's hub locations and in certain other cities it serves, American leases administrative offices, catering, cargo, training, maintenance and other facilities.

The components of lease expense were as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Operating lease cost	$ 1,992	$ 1,987	$ 1,998
Finance lease cost:			
Amortization of assets	119	135	107
Interest on lease liabilities	44	46	44
Variable lease cost	2,703	2,572	2,461
Total net lease cost	$ 4,858	$ 4,740	$ 4,610

Included in the table above is $274 million, $242 million and $190 million of operating lease cost under American's capacity purchase agreement with Republic for the years ended December 31, 2023, 2022 and 2021, respectively. American holds a 25% equity interest in Republic Holdings, the parent company of Republic.

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Supplemental balance sheet information related to leases was as follows (in millions, except lease term and discount rate):

	December 31,	
	2023	2022
Operating leases:		
Operating lease ROU assets	$ 7,886	$ 8,033
Current operating lease liabilities	$ 1,292	$ 1,449
Noncurrent operating lease liabilities	6,416	6,512
Total operating lease liabilities	$ 7,708	$ 7,961
Finance leases:		
Property and equipment, at cost	$ 1,352	$ 1,336
Accumulated amortization	(870)	(767)
Property and equipment, net	$ 482	$ 569
Current finance lease liabilities	$ 124	$ 209
Noncurrent finance lease liabilities	374	535
Total finance lease liabilities	$ 498	$ 744
Weighted average remaining lease term (in years):		
Operating leases	8.4	8.3
Finance leases	5.8	5.0
Weighted average discount rate:		
Operating leases	7.6 %	7.4 %
Finance leases	7.1 %	7.1 %

Supplemental cash flow and other information related to leases was as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Cash paid for amounts included in the measurement of lease liabilities:			
Operating cash flows from operating leases	$ 2,011	$ 1,968	$ 2,040
Operating cash flows from finance leases	47	46	37
Financing cash flows from finance leases	255	179	126
Gain on sale leaseback transactions, net	12	2	25

Maturities of lease liabilities were as follows (in millions):

	December 31, 2023	
	Operating Leases	Finance Leases
2024	$ 1,789	$ 156
2025	1,498	140
2026	1,280	114
2027	1,115	71
2028	985	30
2029 and thereafter	3,682	89
Total lease payments	10,349	600
Less: Imputed interest	(2,641)	(102)
Total lease obligations	7,708	498
Less: Current obligations	(1,292)	(124)
Long-term lease obligations	$ 6,416	$ 374

As of December 31, 2023, American had additional operating lease commitments that have not yet commenced of approximately $ 669 million for five Boeing 787 Family aircraft scheduled to be delivered in 2024 with lease terms of 10 years.

5. Income Taxes

The significant components of the income tax provision (benefit) were (in millions):

	Year Ended December 31,		
	2023	2022	2021
Current income tax benefit:			
State, local and foreign	$ —	$ (6)	$ —
Deferred income tax provision (benefit):			
Federal	361	112	(453)
State and local	33	10	(47)
Deferred income tax provision (benefit)	394	122	(500)
Total income tax provision (benefit)	$ 394	$ 116	$ (500)

The income tax provision (benefit) differed from amounts computed at the statutory federal income tax rate as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Statutory income tax provision (benefit)	$ 332	$ 95	$ (478)
State, local and foreign income tax provision (benefit), net of federal tax effect	25	3	(37)
Book expenses not deductible for tax purposes	35	20	21
Change in valuation allowance	3	—	—
Other, net	(1)	(2)	(6)
Income tax provision (benefit)	$ 394	$ 116	$ (500)

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS OF AMERICAN AIRLINES, INC.

The components of American's deferred tax assets and liabilities were (in millions):

	December 31,	
	2023	2022
Deferred tax assets:		
Net operating loss and other carryforwards	$ 3,960	$ 4,492
Loyalty program liability	1,774	1,809
Leases	1,746	1,804
Pension benefits	428	467
Postretirement benefits other than pension benefits	273	179
Rent expense	84	130
Other	846	689
Total deferred tax assets	9,111	9,570
Valuation allowance	(12)	(9)
Net deferred tax assets	9,099	9,561
Deferred tax liabilities:		
Accelerated depreciation and amortization	(4,479)	(4,603)
Leases	(1,786)	(1,817)
Other	(254)	(256)
Total deferred tax liabilities	(6,519)	(6,676)
Net deferred tax asset	$ 2,580	$ 2,885

At December 31, 2023, American had approximately $ 13.7 billion of gross federal net operating losses (NOLs) and $ 3.6 billion of other carryforwards available to reduce future federal taxable income, of which $3.8 billion will expire beginning in 2033 if unused and $ 13.5 billion can be carried forward indefinitely. American is a member of AAG's consolidated federal and certain state income tax returns. American also had approximately $ 5.3 billion of NOL carryforwards to reduce future state taxable income at December 31, 2023, which will expire in taxable years 2023 through 2043 if unused.

American's ability to use its NOLs and other carryforwards depends on the amount of taxable income generated in future periods. American provides a valuation allowance for its deferred tax assets, which include the NOLs, when it is more likely than not that some portion, or all of its deferred tax assets, will not be realized. American considers all available positive and negative evidence and makes certain assumptions in evaluating the realizability of its deferred tax assets. Many factors are considered that impact American's assessment of future profitability, including conditions which are beyond its control, such as the health of the economy, the availability and price volatility of aircraft fuel and travel demand. American has determined that positive factors outweigh negative factors in the determination of the realizability of its deferred tax assets. There can be no assurance that an additional valuation allowance on American's net deferred tax assets will not be required. Such valuation allowance could be material.

American's ability to deduct its NOL carryforwards and to utilize certain other available tax attributes can be substantially constrained under the general annual limitation rules of Section 382 where an "ownership change" has occurred. Substantially all of American's remaining federal NOL carryforwards attributable to US Airways Group are subject to limitation under Section 382; however, American's ability to utilize such NOL carryforwards is not anticipated to be effectively constrained as a result of such limitation. Similar limitations may apply for state income tax purposes. American's ability to utilize any new NOL carryforwards arising after the ownership changes is not affected by the annual limitation rules imposed by Section 382 unless another ownership change occurs. Under the Section 382 limitation, cumulative stock ownership changes among material stockholders exceeding 50% during a rolling three-year period can potentially limit American's future use of NOLs and tax credits.

In 2023, American recorded an income tax provision of $ 394 million, with an effective rate of approximately 25%, which was substantially non-cash. Substantially all of American's income before income taxes is attributable to the United States.

American files its tax returns as prescribed by the tax laws of the jurisdictions in which it operates. American's 2020 through 2022 tax years are still subject to examination by the Internal Revenue Service. Various state and foreign jurisdiction tax years remain open to examination, and American is under examination, in administrative appeals or engaged in tax litigation in certain jurisdictions. American believes that the effect of any assessments will not be material to its consolidated financial statements.

The amount of, and changes to, American's uncertain tax positions were not material in any of the years presented. American accrues interest and penalties related to unrecognized tax benefits in interest expense and operating expense, respectively.

6. Fair Value Measurements

Assets Measured at Fair Value on a Recurring Basis

Fair value is defined as the price that would be received from the sale of an asset or paid to transfer a liability (i.e., an exit price) on the measurement date in an orderly transaction between market participants in the principal or most advantageous market for the asset or liability. Accounting standards include disclosure requirements around fair values used for certain financial instruments and establish a fair value hierarchy. The hierarchy prioritizes valuation inputs into three levels based on the extent to which inputs used in measuring fair value are observable in the market. Each fair value measurement is reported in one of three levels:

- Level 1 – Observable inputs such as quoted prices in active markets;
- Level 2 – Inputs, other than quoted prices in active markets, that are observable either directly or indirectly; and
- Level 3 – Unobservable inputs in which there is little or no market data, which require the reporting entity to develop its own assumptions.

When available, American uses quoted market prices to determine the fair value of its financial assets. If quoted market prices are not available, American measures fair value using valuation techniques that use, when possible, current market-based or independently-sourced market parameters, such as interest rates and currency rates.

American utilizes the market approach to measure the fair value of its financial assets. The market approach uses prices and other relevant information generated by market transactions involving identical or comparable assets. American's short-term investments, restricted cash and restricted short-term investments classified as Level 2 utilize significant observable inputs, other than quoted prices in active markets, for valuation of these securities. No changes in valuation techniques or inputs occurred during the year ended December 31, 2023.

Assets measured at fair value on a recurring basis are summarized below (in millions):

	Fair Value Measurements as of December 31, 2023			
	Total	Level 1	Level 2	Level 3
Short-term investments [1], [2]:				
Money market funds	$ 817	$ 817	$ —	$ —
Corporate obligations	4,046	—	4,046	—
Bank notes/certificates of deposit/time deposits	1,585	—	1,585	—
Repurchase agreements	450	—	450	—
U.S. government and agency obligations	100	—	100	—
	6,998	817	6,181	—
Restricted cash and short-term investments [1], [3]	910	459	451	—
Long-term investments [4]	163	163	—	—
Total	$ 8,071	$ 1,439	$ 6,632	$ —

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS OF AMERICAN AIRLINES, INC.

	Fair Value Measurements as of December 31, 2022			
	Total	Level 1	Level 2	Level 3
Short-term investments [1]:				
Money market funds	$ 731	$ 731	$ —	$ —
Corporate obligations	3,688	—	3,688	—
Bank notes/certificates of deposit/time deposits	3,654	—	3,654	—
Repurchase agreements	450	—	450	—
	8,523	731	7,792	—
Restricted cash and short-term investments [1], [3]	995	535	460	—
Long-term investments [4]	245	245	—	—
Total	$ 9,763	$ 1,511	$ 8,252	$ —

(1) All short-term investments are classified as available-for-sale and stated at fair value. Unrealized gains and losses are recorded in accumulated other comprehensive loss at each reporting period. There were no credit losses.

(2) American's short-term investments as of December 31, 2023 mature in one year or less.

(3) Restricted cash and short-term investments primarily include collateral held to support workers' compensation obligations and collateral associated with the payment of interest for the AAdvantage Financing. Restricted short-term investments mature in one year or less except for $218 million as of December 31, 2023.

(4) Long-term investments include American's equity investments in China Southern Airlines Company Limited (China Southern Airlines), GOL and Vertical Aerospace Ltd. (Vertical). See Note 7 for further information on American's equity investments.

Fair Value of Debt

The fair value of American's long-term debt was estimated using quoted market prices or discounted cash flow analyses based on American's current estimated incremental borrowing rates for similar types of borrowing arrangements. If American's long-term debt was measured at fair value, it would have been classified as Level 2 in the fair value hierarchy.

The carrying value and estimated fair value of American's long-term debt, including current maturities, were as follows (in millions):

	December 31, 2023		December 31, 2022	
	Carrying Value	Fair Value	Carrying Value	Fair Value
Long-term debt, including current maturities	$ 27,177	$ 27,008	$ 29,679	$ 28,453

7. Investments

To help expand American's network and as part of its ongoing commitment to sustainability, American enters into various commercial relationships or other strategic partnerships, including equity investments, with other airlines and companies. American's equity investments are reflected in other assets on its consolidated balance sheets. American's share of equity method investees' financial results and changes in fair value are recorded in nonoperating other income (expense), net on the consolidated statements of operations.

American's equity investments ownership interest and carrying value were as follows:

		Ownership Interest		Carrying Value (in millions)	
		December 31,		December 31,	
	Accounting Treatment	2023	2022	2023	2022
Republic Holdings	Equity Method	25.0 %	25.0 %	$ 240	$ 222
China Southern Airlines	Fair Value	1.5 %	1.5 %	115	176
Other investments [1]	Various			186	212
Total				$ 541	$ 610

[1] Primarily includes American's investment in JetSMART Holdings Limited, which is accounted for under the equity method, and American's investments in GOL and Vertical, which are each accounted for at fair value.

8. Employee Benefit Plans

American sponsors defined benefit and defined contribution pension plans for eligible employees. The defined benefit pension plans provide benefits for participating employees based on years of service and average compensation for a specified period of time before retirement. Effective November 1, 2012, substantially all of American's defined benefit pension plans were frozen and American began providing enhanced benefits under its defined contribution pension plans for certain employee groups. American uses a December 31 measurement date for all of its defined benefit pension plans. American also provides certain retiree medical and other postretirement benefits, including health care and life insurance benefits, to retired employees.

Benefit Obligations, Fair Value of Plan Assets and Funded Status

The following tables provide a reconciliation of the changes in the pension and retiree medical and other postretirement benefits obligations, fair value of plan assets and funded status as of December 31, 2023 and 2022:

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
	(In millions)			
Benefit obligation at beginning of period	$ 13,948	$ 18,791	$ 906	$ 1,098
Service cost	2	3	17	16
Interest cost	753	552	55	30
Actuarial loss (gain) [1], [2]	501	(4,534)	92	(167)
Plan amendments [3]	—	—	339	—
Other	—	—	—	3
Benefit payments	(890)	(864)	(84)	(74)
Benefit obligation at end of period	$ 14,314	$ 13,948	$ 1,325	$ 906
Fair value of plan assets at beginning of period	$ 11,821	$ 14,605	$ 133	$ 167
Actual return (loss) on plan assets	1,356	(1,924)	14	(18)
Employer contributions [4]	71	4	70	58
Benefit payments	(890)	(864)	(84)	(74)
Fair value of plan assets at end of period	$ 12,358	$ 11,821	$ 133	$ 133
Funded status at end of period	$ (1,956)	$ (2,127)	$ (1,192)	$ (773)

[1] The 2023 and 2022 pension actuarial loss (gain) primarily relates to the change in American's weighted average discount rate assumption.

[2] The 2023 and 2022 retiree medical and other postretirement benefits actuarial loss (gain) primarily relates to the change in American's weighted average discount rate assumption and, in 2023, the change in health care cost assumptions.

(3) As of September 30, 2023, American remeasured its retiree medical and other postretirement benefits to account for enhanced retirement benefits provided to its mainline pilots pursuant to the new collective bargaining agreement ratified in August 2023. As a result, American increased its postretirement benefits obligation by $339 million, which was included as a component of prior service cost in accumulated other comprehensive loss.

(4) In 2023, American made required contributions of $ 67 million to its defined benefit pension plans.

Balance Sheet Position

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
	(In millions)			
As of December 31,				
Current liability	$ 6	$ 4	$ 122	$ 85
Noncurrent liability	1,950	2,123	1,070	688
Total liabilities	$ 1,956	$ 2,127	$ 1,192	$ 773

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
	(In millions)			
Net actuarial loss (gain)	$ 3,561	$ 3,609	$ (382)	$ (505)
Prior service cost (benefit)	—	18	197	(148)
Total accumulated other comprehensive loss (income), pre-tax	$ 3,561	$ 3,627	$ (185)	$ (653)

Plans with Projected Benefit Obligations Exceeding Fair Value of Plan Assets

	Pension Benefits	
	2023	2022
	(In millions)	
Projected benefit obligation	$ 14,314	$ 13,948
Fair value of plan assets	12,358	11,821

Plans with Accumulated Benefit Obligations Exceeding Fair Value of Plan Assets

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
	(In millions)			
Accumulated benefit obligation	$ 14,307	$ 13,941	$ —	$ —
Accumulated postretirement benefit obligation	—	—	1,325	906
Fair value of plan assets	12,358	11,821	133	133

Net Periodic Benefit Cost (Income)

	Pension Benefits			Retiree Medical and Other Postretirement Benefits		
	2023	2022	2021	2023	2022	2021
	(In millions)					
Defined benefit plans:						
Service cost	$ 2	$ 3	$ 3	$ 17	$ 16	$ 12
Interest cost	753	552	523	55	30	30
Expected return on assets	(914)	(1,133)	(1,078)	(11)	(12)	(12)
Special termination benefits	—	—	—	—	—	139
Amortization of:						
Prior service cost (benefit)	18	28	28	(6)	(14)	(13)
Unrecognized net loss (gain)	106	156	211	(34)	(30)	(24)
Net periodic benefit cost (income)	$ (35)	$ (394)	$ (313)	$ 21	$ (10)	$ 132

The service cost component of net periodic benefit cost (income) is included in operating expenses, the cost for the special termination benefits is included in special items, net and the other components of net periodic benefit cost (income) are included in nonoperating other income (expense), net on American's consolidated statements of operations.

Assumptions

The following actuarial assumptions were used to determine American's benefit obligations and net periodic benefit cost (income) for the periods presented:

	Pension Benefits		Retiree Medical and Other Postretirement Benefits	
	2023	2022	2023	2022
Benefit obligations:				
Weighted average discount rate	5.2%	5.6%	5.3%	5.7%

	Pension Benefits			Retiree Medical and Other Postretirement Benefits		
	2023	2022	2021	2023	2022	2021
Net periodic benefit cost (income):						
Weighted average discount rate	5.6%	3.0%	2.7%	5.7%	2.8%	2.4%
Weighted average expected rate of return on plan assets	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%
Weighted average health care cost trend rate assumed for next year [1]	N/A	N/A	N/A	6.5%	5.8%	4.8%

[1] The weighted average health care cost trend rate at December 31, 2023 is assumed to decline gradually to 4.5% by 2033 and remain level thereafter.

As of December 31, 2023, American's estimate of the long-term rate of return on plan assets was 8.0% based on the target asset allocation. Expected returns on long duration bonds are based on yields to maturity of the bonds held at year-end. Expected returns on other assets are based on a combination of long-term historical returns, actual returns on plan assets achieved over the last 10 years, current and expected market conditions, and expected value to be generated through active management and securities lending programs.

Minimum Contributions

American is required to make minimum contributions to its defined benefit pension plans under the minimum funding requirements of the Employee Retirement Income Security Act of 1974 (ERISA) and various other laws for U.S. based plans as well as underfunding rules specific to countries where American maintains defined benefit pension plans. Based on current funding assumptions, American has minimum required contributions of $280 million for 2024. American's future funding obligations will depend on the performance of American's investments held in a trust by the pension plans, interest rates for determining funding targets, the amount of and timing of any supplemental contributions and American's actuarial experience.

Benefit Payments

The following benefit payments, which reflect expected future service as appropriate, are expected to be paid (approximately, in millions):

	2024	2025	2026	2027	2028	2029-2033
Pension benefits	$ 947	$ 972	$ 996	$ 1,017	$ 1,030	$ 5,232
Retiree medical and other postretirement benefits	138	144	150	150	148	674

Plan Assets

The objectives of American's investment policies are to: maintain sufficient income and liquidity to pay retirement benefits; produce a long-term rate of return that meets or exceeds the assumed rate of return for plan assets; limit the volatility of asset performance and funded status; and diversify assets among asset classes and investment managers.

Based on these investment objectives, a long-term strategic asset allocation has been established. This strategic allocation seeks to balance the potential benefit of improving the funded position with the potential risk that the funded position would decline. The current strategic target asset allocation with the corresponding allowed range is as follows:

Asset Class/Sub-Class	Target Allocation	Allowed Range
Equity	61 %	30% - 85%
Public:		
U.S. Large	19 %	10% - 40%
U.S. Small/Mid	5 %	0% - 10%
International Large	13 %	5% - 25%
International Small/Mid	3 %	0% - 10%
Emerging Markets	6 %	0% - 15%
Private Equity	15 %	5% - 30%
Fixed Income	39 %	15% - 70%
Public U.S. Fixed Income	30 %	15% - 60%
Private Income	9 %	0% - 20%
Other	— %	0% - 5%
Cash Equivalents	— %	0% - 20%

Public equity investments are intended to provide a real return over a full market cycle and, therefore, to contribute to the pension plan's long-term objective. Public fixed income investments are intended to provide income to the plan and offer the potential for long term capital appreciation. Private investments, such as private equity and private income, are used to provide higher expected returns than public markets over the long-term by assuming reduced levels of liquidity and higher levels of risk. The pension plan's master trust participates in securities lending programs to generate additional income by loaning plan assets to borrowers on a fully collateralized basis. The pension plan's master trust will also engage in derivative instruments to equitize residual levels of cash as well as hedge the pension plan's exposure to interest rates. Such programs are subject to market risk and counterparty risk.

Investments in securities traded on recognized securities exchanges are valued at the last reported sales price on the last business day of the year. Securities traded in the over-the-counter market are valued at the last bid price. Investments in limited partnerships are carried at estimated net asset value (NAV) as determined by and reported by the general partners of the partnerships and represent the proportionate share of the estimated fair value of the underlying assets of the limited partnerships. Mutual funds are valued once daily through a NAV calculation provided at the end of each trade day. Common/collective trusts are valued at NAV based on the fair values of the underlying investments of the trusts as determined by the sponsor of the trusts. No changes in valuation techniques or inputs occurred during the year.

Benefit Plan Assets Measured at Fair Value on a Recurring Basis

The fair value of American's pension plan assets at December 31, 2023 and 2022, by asset category, were as follows (in millions) [1]:

	December 31, 2023				December 31, 2022			
	Level 1	Level 2	Level 3	Total	Level 1	Level 2	Level 3	Total
Equity [2]	$ 3,134	$ —	$ —	$ 3,134	$ 3,055	$ —	$ —	$ 3,055
Fixed income [3]	235	3,238	—	3,473	206	2,917	—	3,123
Other, net [4]	(6)	348	84	426	74	278	75	427
Measured at NAV [5]:								
Common collective trusts [6]	—	—	—	1,244	—	—	—	1,694
Private investments [7]	—	—	—	4,081	—	—	—	3,522
Total plan assets	$ 3,363	$ 3,586	$ 84	$ 12,358	$ 3,335	$ 3,195	$ 75	$ 11,821

[1] See Note 6 for a description of the levels within the fair value hierarchy.

[2] Equity investments include domestic and international common stock and preferred stock.

[3] Fixed income investments include corporate, government and U.S. municipal bonds, as well as mutual funds invested in fixed income securities.

[4] Other primarily includes a short-term investment fund, net receivables and payables of the master trust for dividends, interest and amounts due to or from the sale and purchase of securities and cash and cash equivalents.

[5] Includes investments that were measured at NAV per share (or its equivalent) as a practical expedient that have not been classified in the fair value hierarchy.

[6] Common collective trusts include commingled funds primarily invested in equity securities. For some trusts, requests for withdrawals must meet specific requirements with advance notice of redemption preferred.

[7] Private investments include limited partnerships that invest primarily in domestic private equity and private income opportunities. The pension plan's master trust does not have the right to redeem its limited partnership investment at its NAV, but rather receives distributions as the underlying assets are liquidated. It is estimated that the underlying assets of these funds will be gradually liquidated over the next 10 years. As of December 31, 2023, the pension plan's master trust has future funding commitments to these limited partnerships of approximately $1.3 billion, most of which are expected to be called over the next five years.

Changes in fair value measurements of Level 3 investments during the years ended December 31, 2023 and 2022, were as follows (in millions):

	2023	2022
Balance at beginning of year	$ 75	$ 58
Actual gain (loss) on plan assets:		
Relating to assets still held at the reporting date	(9)	1
Purchases	20	29
Sales	(2)	(9)
Transfers out	—	(4)
Balance at end of year	$ 84	$ 75

Plan assets in the retiree medical and other postretirement benefits plans are primarily Level 2 mutual funds valued by quoted prices on the active market, which is fair value, and represents the NAV of the shares of such funds as of the close of business at the end of the period. NAV is based on the fair market value of the funds' underlying assets and liabilities at the date of determination.

Defined Contribution and Multiemployer Plans

The costs associated with American's defined contribution plans were $ 1.1 billion, $916 million and $893 million for the years ended December 31, 2023, 2022 and 2021, respectively.

American participates in the International Association of Machinists & Aerospace Workers (IAM) National Pension Fund, Employer Identification No. 51-6031295 and Plan No. 002 (the IAM Pension Fund). American's contributions to the IAM Pension Fund were $52 million, $46 million and $43 million for the years ended December 31, 2023, 2022 and 2021, respectively. The IAM Pension Fund reported $533 million in employers' contributions for the year ended December 31, 2022, which is the most recent year for which such information is available. For 2022, American's contributions represented more than 5% of total contributions to the IAM Pension Fund.

On March 29, 2019, the actuary for the IAM Pension Fund certified that the fund was in "endangered" status despite reporting a funded status of over 80%. Additionally, the IAM Pension Fund's Board voluntarily elected to enter into "critical" status on April 17, 2019. Upon entry into critical status, the IAM Pension Fund was required by law to adopt a rehabilitation plan aimed at restoring the financial health of the pension plan and did so on April 17, 2019 (the Rehabilitation Plan). Under the Rehabilitation Plan, American was subject to an immaterial contribution surcharge, which ceased to apply June 14, 2019 upon American's mandatory adoption of a contribution schedule under the Rehabilitation Plan. The contribution schedule requires 2.5% annual increases to its contribution rate. This contribution schedule will remain in effect through the earlier of December 31, 2031 or the date the IAM Pension Fund emerges from critical status.

Profit Sharing Program

American accrues a percentage of its pre-tax income excluding net special items for its profit sharing program. For the year ended December 31, 2023, American accrued $261 million for this program, which will be distributed to employees in the first quarter of 2024.

9. Accumulated Other Comprehensive Loss

The components of AOCI are as follows (in millions):

	Pension, Retiree Medical and Other Postretirement Benefits	Unrealized Gain (Loss) on Investments	Income Tax Benefit (Provision) [1]	Total
Balance at December 31, 2021	$ (4,724)	$ (2)	$ (1,315)	$ (6,041)
Other comprehensive income (loss) before reclassifications	1,610	(4)	(363)	1,243
Amounts reclassified from AOCI	140	—	(32) [2]	108
Net current-period other comprehensive income (loss)	1,750	(4)	(395)	1,351
Balance at December 31, 2022	(2,974)	(6)	(1,710)	(4,690)
Other comprehensive income (loss) before reclassifications	(486)	4	108	(374)
Amounts reclassified from AOCI	84	—	(19) [2]	65
Net current-period other comprehensive income (loss)	(402)	4	89	(309)
Balance at December 31, 2023	$ (3,376)	$ (2)	$ (1,621)	$ (4,999)

[1] Relates principally to pension, retiree medical and other postretirement benefits obligations that will not be recognized in net income (loss) until the obligations are fully extinguished.

[2] Relates to pension, retiree medical and other postretirement benefits obligations and is recognized within the income tax provision (benefit) on American's consolidated statements of operations.

Reclassifications out of AOCI for the years ended December 31, 2023 and 2022 are as follows (in millions):

AOCI Components	Amounts reclassified from AOCI Year Ended December 31, 2023	2022	Affected line items on the consolidated statements of operations
Amortization of pension, retiree medical and other postretirement benefits:			
Prior service cost	$ 9	$ 11	Nonoperating other income (expense), net
Actuarial loss	56	97	Nonoperating other income (expense), net
Total reclassifications for the period, net of tax	$ 65	$ 108	

10. Commitments, Contingencies and Guarantees

(a) Aircraft, Engine and Other Purchase Commitments

Under all of American's aircraft and engine purchase agreements, its total future commitments as of December 31, 2023 are expected to be as follows (approximately, in millions):

	2024	2025	2026	2027	2028	2029 and Thereafter	Total
Payments for aircraft and engine commitments [1]	$ 2,410	$ 3,725	$ 3,580	$ 1,118	$ 829	$ 645	$ 12,307

[1] These amounts are net of purchase deposits currently held by the manufacturers. American's purchase deposits held by all manufacturers totaled $760 million and $613 million as of December 31, 2023 and 2022, respectively.

Due to uncertainty surrounding the timing of delivery of certain aircraft, the amounts in the table represent American's most current estimate based on contractual delivery schedules adjusted for updates and revisions to such schedules communicated to management by the applicable equipment manufacturer. However, the actual delivery schedule may differ, potentially materially, based on various potential factors including production delays by the manufacturer and regulatory concerns.

Additionally, the amounts in the table exclude five Boeing 787 Family aircraft scheduled to be delivered in 2024, for which American has obtained committed lease financing. See Note 4 for information regarding this operating lease commitment.

Additionally, American has other purchase commitments primarily related to aircraft fuel, flight equipment maintenance and information technology support as follows (approximately): $4.7 billion in 2024, $2.0 billion in 2025, $1.4 billion in 2026, $150 million in 2027, $124 million in 2028 and $843 million in 2029 and thereafter. These amounts exclude obligations under certain fuel offtake agreements or other agreements for which the timing of the related expenditure is uncertain, or which are subject to material contingencies, such as the construction of a production facility.

(b) Capacity Purchase Agreements with Third-Party Regional Carriers

American has capacity purchase agreements with third-party regional carriers. The capacity purchase agreements provide that all revenues, including passenger, in-flight, ancillary, mail and freight revenues, go to American. American controls marketing, scheduling, ticketing, pricing and seat inventories. In return, American agrees to pay predetermined fees to these airlines for operating an agreed-upon number of aircraft, without regard to the number of passengers on board. In addition, these agreements provide that American either reimburses or pays 100% of certain variable costs, such as airport landing fees, fuel and passenger liability insurance.

As of December 31, 2023, American's capacity purchase agreements with third-party regional carriers had expiration dates ranging from 2024 to 2032, with rights of American to extend the respective terms of certain agreements.

As of December 31, 2023, American's minimum obligations under its capacity purchase agreements with third-party regional carriers are expected to be as follows (approximately, in millions):

	2024	2025	2026	2027	2028	2029 and Thereafter	Total
Minimum obligations under capacity purchase agreements with third-party regional carriers [1]	$ 2,038	$ 1,992	$ 1,702	$ 1,473	$ 693	$ 1,332	$ 9,230

[1] Represents minimum payments under capacity purchase agreements with third-party regional carriers. These commitments are estimates of costs based on assumed minimum levels of flying under the capacity purchase agreements and American's actual payments could differ materially. Rental payments under operating leases for certain aircraft flown under these capacity purchase agreements are reflected in the operating lease commitments in Note 4.

(c) Airport Redevelopment

Los Angeles International Airport (LAX)

From time to time, airports engage in construction projects, often substantial, that result in new or improved facilities that are ultimately funded through increases in the rent and other occupancy costs payable by airlines operating at the airport. Unlike this construction and funding model, American is managing a project at LAX where it has legal title to the assets during construction. In 2018, American executed a lease agreement with Los Angeles World Airports (LAWA), which owns and operates LAX, in connection with a $1.6 billion modernization project related to LAX Terminals 4 and 5. Construction, which started in October 2018 and is expected to be completed in 2028, will occur in a phased approach. The modernization project will include a unified departure hall to the entranceway of Terminals 4 and 5, reconfigured ticket counter and check-in areas with seamless access to security screening areas, 10 new security screening lanes with automated technology in addition to the existing Terminal 5 lanes, and a new Terminal 4 South concourse with more open and upgraded amenities at gate areas. The project will also include renovated break rooms, multi-use meeting rooms and team gathering spaces throughout the terminals to support American's team members at LAX.

As each phase is completed and ready for use, the assets will be sold and transferred to LAWA, including the site improvements and other non-proprietary improvements. As American controls the assets during construction, they are recognized on its consolidated balance sheets within operating property and equipment until the assets are sold and transferred to LAWA. As of December 31, 2023, American has incurred $862 million in costs relating to the LAX modernization project, of which $283 million were incurred in 2023. Cash paid for non-proprietary improvements are included within other investing activities on American's consolidated statements of cash flows. In addition, as of December 31, 2023, American has sold and transferred $346 million of non-proprietary improvements to LAWA, of which $170 million occurred during 2023. For non-proprietary improvements which are not yet ready for use, any cash payments received from LAWA will be reflected as a financial liability included within noncurrent other liabilities on American's consolidated balance sheets and reflected as other financing activities on its consolidated statements of cash flows. As of December 31, 2023, $53 million of cash proceeds received for non-proprietary improvements were not yet ready for use, and therefore have not been sold and transferred back to LAWA.

(d) Off-Balance Sheet Arrangements

Pass-Through Trusts

American currently has 308 owned aircraft and 60 owned spare aircraft engines, which in each case were financed with EETCs issued by pass-through trusts. These trusts are off-balance sheet entities, the primary purpose of which is to finance the acquisition of flight equipment or to permit issuance of debt backed by existing flight equipment. In the case of aircraft EETCs, rather than finance each aircraft separately when such aircraft is purchased, delivered or refinanced, these trusts allow American to raise the financing for a number of aircraft at one time and, if applicable, place such funds in escrow pending a future purchase, delivery or refinancing of the relevant aircraft. Similarly, in the case of the spare engine EETCs, the trusts allow American to use its existing pool of spare engines to raise financing under a single facility. The trusts have also been structured to provide for certain credit enhancements, such as liquidity facilities to cover certain interest payments, that reduce the risks to the purchasers of the trust certificates and, as a result, reduce the cost of aircraft financing to American.

Each trust covers a set number of aircraft or spare engines scheduled to be delivered, financed or refinanced upon the issuance of the EETC or within a specific period of time thereafter. At the time of each covered aircraft or spare engine financing, the relevant trust used the proceeds from the issuance of the EETC (which may have been available at the time

of issuance thereof or held in escrow until financing of the applicable aircraft following its delivery) to purchase equipment notes relating to the financed aircraft or engines. The equipment notes are issued, at American's election, in connection with a mortgage financing of the aircraft or spare engines. The equipment notes are secured by a security interest in the aircraft or engines, as applicable. The pass-through trust certificates are not direct obligations of, nor are they guaranteed by, AAG or American. However, the equipment notes issued to the trusts are direct obligations of American and, in certain instances, have been guaranteed by AAG. As of December 31, 2023, $7.7 billion associated with these mortgage financings is reflected as debt in the accompanying consolidated balance sheet.

Letters of Credit and Other

American provides financial assurance, such as letters of credit and surety bonds, primarily to support projected workers' compensation obligations and airport commitments. As of December 31, 2023, American had $318 million of letters of credit and surety bonds securing various obligations, of which $ 94 million is collateralized with American's restricted cash. The letters of credit and surety bonds that are subject to expiration will expire on various dates through 2028.

(e) Legal Proceedings

Government Antitrust Action Related to the Northeast Alliance. On September 21, 2021, the United States Department of Justice, joined by Attorneys General from six states and the District of Columbia, filed an antitrust complaint against American and JetBlue Airways Corporation (JetBlue) in the U.S. District Court for the District of Massachusetts alleging that American and JetBlue violated U.S. antitrust law in connection with the previously disclosed Northeast Alliance arrangement (NEA).

On May 19, 2023, the U.S. District Court for the District of Massachusetts issued an order permanently enjoining American and JetBlue from continuing and further implementing the NEA. In June 2023, JetBlue delivered a notice of termination of the NEA, effective July 29, 2023, and the carriers have commenced wind-down activities to accommodate mutual customers. Following written submissions by the parties and a hearing on July 26, 2023, the U.S. District Court for the District of Massachusetts entered a Final Judgment and Order Entering Permanent Injunction on July 28, 2023. The parties are complying with the terms of the Final Judgment and Order Entering Permanent Injunction, including winding down activities related to the NEA. American filed a notice of appeal to the U.S. Court of Appeals for the First Circuit on September 25, 2023, and American's opening brief was filed on December 6, 2023.

Private Party Antitrust Actions Related to the Northeast Alliance. On December 5, 2022 and December 7, 2022, two private party plaintiffs filed putative class action antitrust complaints against American and JetBlue in the U.S. District Court for the Eastern District of New York alleging that American and JetBlue violated U.S. antitrust law in connection with the previously disclosed NEA. These actions were consolidated on January 10, 2023. The private party plaintiffs filed an amended consolidated complaint on February 3, 2023. On February 2, 2023 and February 15, 2023, private party plaintiffs filed two additional putative class action antitrust complaints against American and JetBlue in the U.S. District Court for the District of Massachusetts and the U.S. District Court for the Eastern District of New York, respectively. In March 2023, American filed a motion in the U.S. District Court for the District of Massachusetts case asking to transfer the case to the U.S. District Court for the Eastern District of New York and consolidate it with the cases pending in that venue. The U.S. District Court for the District of Massachusetts granted that motion. The remaining cases were consolidated with the other actions in the Eastern District of New York. In June 2023, the private party plaintiffs filed a second amended consolidated complaint, followed by a third amended complaint filed in August 2023. In September 2023, American, together with JetBlue, filed a motion to dismiss the third amended complaint, and that motion remains pending. American believes these lawsuits are without merit and is defending against them vigorously.

General. In addition to the specifically identified legal proceedings, American and its subsidiaries are also engaged in other legal proceedings from time to time. Legal proceedings can be complex and take many months, or even years, to reach resolution, with the final outcome depending on a number of variables, some of which are not within American's control. Therefore, although American will vigorously defend itself in each of the actions described above and such other legal proceedings, their ultimate resolution and potential financial and other impacts on American are uncertain but could be material.

(f) Guarantees and Indemnifications

American is a party to many routine contracts in which it provides general indemnities in the normal course of business to third parties for various risks. American is not able to estimate the potential amount of any liability resulting from the indemnities. These indemnities are discussed in the following paragraphs.

In its aircraft financing agreements, American generally indemnifies the financing parties, trustees acting on their behalf and other relevant parties against liabilities (including certain taxes) resulting from the financing, manufacture, design, ownership, operation and maintenance of the aircraft regardless of whether these liabilities (including certain taxes) relate to the negligence of the indemnified parties.

American's loan agreements and certain other financing transactions may obligate American to reimburse the applicable lender for incremental costs due to a change in law that imposes (i) any reserve or special deposit requirement against assets of, deposits with or credit extended by such lender related to the loan, (ii) any tax, duty or other charge with respect to the loan (except standard income tax) or (iii) capital adequacy requirements. In addition, American's loan agreements and other financing arrangements typically contain a withholding tax provision that requires American to pay additional amounts to the applicable lender or other financing party, generally if withholding taxes are imposed on such lender or other financing party as a result of a change in the applicable tax law.

In certain transactions, including certain aircraft financing leases and loans, the lessors, lenders and/or other parties have rights to terminate the transaction based on changes in foreign tax law, illegality or certain other events or circumstances. In such a case, American may be required to make a lump sum payment to terminate the relevant transaction.

American has general indemnity clauses in many of its airport and other real estate leases where American as lessee indemnifies the lessor (and related parties) against liabilities related to American's use of the leased property. Generally, these indemnifications cover liabilities resulting from the negligence of the indemnified parties, but not liabilities resulting from the gross negligence or willful misconduct of the indemnified parties. In addition, American provides environmental indemnities in many of these leases for contamination related to American's use of the leased property.

Under certain contracts with third parties, American indemnifies the third-party against legal liability arising out of an action by the third-party, or certain other parties. The terms of these contracts vary and the potential exposure under these indemnities cannot be determined. American has liability insurance protecting American from some of the obligations it has undertaken under these indemnities.

American is required to make principal and interest payments for certain special facility revenue bonds issued by municipalities primarily to build or improve airport facilities and purchase equipment, which are leased to American. The payment of principal and interest of certain special facility revenue bonds is guaranteed by American. As of December 31, 2023, the remaining lease payments through 2035 guaranteeing the principal and interest on these bonds are $520 million and the current carrying amount of the associated operating lease liability in the accompanying consolidated balance sheet is $ 321 million.

As of December 31, 2023, American had issued guarantees covering AAG's $ 1.8 billion aggregate principal amount of the PSP1 Promissory Note due April 2030, $1.0 billion aggregate principal amount of the PSP2 Promissory Note due January 2031, $ 959 million aggregate principal amount of the PSP3 Promissory Note due April 2031, $1.0 billion aggregate principal amount of 6.50% convertible senior notes due July 2025 and $ 487 million of 3.75% senior notes due March 2025.

(g) Credit Card Processing Agreements

American has agreements with companies that process customer credit card transactions for the sale of air travel and other services. American's agreements allow these credit card processing companies, under certain conditions, to hold an amount of its cash (referred to as a holdback) equal to all or a portion of advance ticket sales that have been processed by that company, but for which American has not yet provided the air transportation. These holdback requirements can be implemented at the discretion of the credit card processing companies upon the occurrence of specific events, including material adverse changes in American's financial condition or the triggering of a liquidity covenant. These credit card processing companies are not currently entitled to maintain any holdbacks. The imposition of holdback requirements would reduce American's liquidity.

(h) Labor Contracts

In May 2023, American and the Allied Pilots Association, the union representing American's mainline pilots, reached an agreement in principle on a new collective bargaining agreement (CBA), which was ratified in August 2023. This four-year agreement provides wage rate increases, including an initial wage rate increase of 21% effective as of January 1, 2023, quality-of-life benefits and other benefit-related items. The additional compensation for the 2023 period prior to contract ratification as a result of the higher wage rates was recorded within salaries, wages and benefits in the consolidated statements of operations in the second and third quarters of 2023. The agreement also included a provision for a one-time payment upon ratification. In 2023, one-time charges resulting from the ratification of this new agreement were recorded as mainline operating special items, net in the consolidated statement of operations, including the one-time payment of $754 million as well as adjustments to other benefit-related items of $ 235 million. The one-time payment and the additional compensation were principally paid in 2023, with remaining payments expected to be paid in the first quarter of 2024.

As of December 31, 2023, American employed approximately 103,200 active full-time equivalent (FTE) employees. Of the total active FTE employees, 87% are covered by CBAs with various labor unions and 38% are covered by CBAs that are currently amendable or that will become amendable within one year. In January 2024, mainline passenger service employees represented by the CWA-IBT ratified a new five-year agreement. The CBA covering American's flight attendants is now amendable.

11. Supplemental Cash Flow Information

Supplemental disclosure of cash flow information and non-cash investing and financing activities are as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Non-cash investing and financing activities:			
ROU assets acquired through operating leases	$ 1,172	$ 1,448	$ 1,381
Property and equipment acquired through debt, finance leases and other	317	46	180
Finance leases converted to operating leases	42	3	—
Operating leases converted to finance leases	5	107	102
Settlement of bankruptcy obligations	4	—	4
Equity investments	—	12	88
Supplemental information:			
Interest paid, net	2,058	1,716	1,481
Income taxes paid	6	2	2

12. Operating Segments and Related Disclosures

American is managed as a single business unit that provides air transportation for passengers and cargo. This allows it to benefit from an integrated revenue pricing and route network that includes American and AAG's wholly-owned and third-party regional carriers that fly under capacity purchase agreements operating as American Eagle. The flight equipment of all these carriers is combined to form one fleet that is deployed through a single route scheduling system. Financial information and annual operational plans and forecasts are prepared and reviewed by the chief operating decision maker at the consolidated level. When making operational decisions, the chief operating decision maker evaluates flight profitability data, which considers aircraft type and route economics, but is indifferent to the results of the individual regional carriers. The objective in making operational decisions is to maximize consolidated financial results, not the individual results of American or American Eagle.

See Note 1(m) for American's passenger revenue by geographic region. American's tangible assets consist primarily of flight equipment, which are mobile across geographic markets and, therefore, have not been allocated.

13. Share-based Compensation

In May 2023, the stockholders of AAG approved the 2023 Incentive Award Plan (the 2023 Plan). The 2023 Plan replaces and supersedes AAG's 2013 Incentive Award Plan (the 2013 Plan). No further awards will be granted under the 2013 Plan; however, the terms and conditions of the 2013 Plan will continue to govern any outstanding awards granted thereunder. The 2023 Plan provides that an award may be in the form of a stock option, including an incentive stock option and nonqualified stock option, stock appreciation right, restricted stock, restricted stock unit, performance bonus award, performance stock unit, other stock or cash-based award and dividend equivalent to eligible individuals.

The 2023 Plan authorizes the grant of awards for the issuance of 17.2 million shares less any shares granted under the 2013 Plan after March 22, 2023, the date the Board of Directors of AAG approved the 2023 Plan. Any shares underlying awards granted under the 2023 Plan or 2013 Plan that are forfeited, terminate or are settled in cash (in whole or in part) without the delivery of shares will again be available for grant under the 2023 Plan.

For the years ended December 31, 2023, 2022 and 2021, American recorded $ 97 million, $75 million and $95 million, respectively, of share-based compensation costs principally in salaries, wages and benefits expense on its consolidated statements of operations.

During 2023, 2022 and 2021, AAG withheld approximately 1.5 million, 1.2 million and 1.0 million shares of AAG common stock, respectively, and paid approximately $23 million, $21 million and $18 million, respectively, in satisfaction of certain tax withholding obligations associated with employee equity awards.

Restricted Stock Unit Awards (RSUs)

AAG has granted RSUs with service conditions (time vested primarily over three years) and performance conditions. The grant-date fair value of these RSUs is equal to the market price of the underlying shares of AAG common stock on the date of grant. For time vested awards, the expense is recognized on a straight-line basis over the vesting period for the entire award. For awards with performance conditions, the expense is recognized based on the expected achievement at each reporting period. RSUs are classified as equity awards as the vesting results in the issuance of shares of AAG common stock.

RSU award activity for all plans for the years ended December 31, 2023, 2022 and 2021 is as follows:

	Number of Shares	Weighted Average Grant Date Fair Value
	(In thousands)	
Outstanding at December 31, 2020	7,882	$ 23.66
Granted	5,525	18.34
Vested and released	(3,314)	25.58
Forfeited	(692)	18.78
Outstanding at December 31, 2021	9,401	$ 20.17
Granted	5,882	15.93
Vested and released	(4,131)	21.04
Forfeited	(889)	18.04
Outstanding at December 31, 2022	10,263	$ 17.51
Granted	9,834	14.54
Vested and released	(5,161)	17.81
Forfeited	(701)	20.49
Outstanding at December 31, 2023	14,235	$ 15.18

As of December 31, 2023, there was $120 million of unrecognized compensation cost related to RSUs. These costs are expected to be recognized over a weighted average period of one year. The total fair value of RSUs vested during the years ended December 31, 2023, 2022 and 2021 was $ 78 million, $70 million and $62 million, respectively.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS OF AMERICAN AIRLINES, INC.

14. Valuation and Qualifying Accounts (in millions)

	Balance at Beginning of Year	Additions Charged to Statement of Operations Accounts	Deductions and Other	Balance at End of Year
Allowance for obsolescence of spare parts				
Year ended December 31, 2023	$ 566	$ 83	$ 26	$ 675
Year ended December 31, 2022	588	82	(104)	566
Year ended December 31, 2021	442	165	(19)	588

15. Transactions with Related Parties

The following represents the net receivables (payables) from or to related parties (in millions):

	December 31,	
	2023	2022
AAG	$ 9,144	$ 8,692
AAG's wholly-owned subsidiaries [1]	(2,074)	(2,104)
Total	$ 7,070	$ 6,588

[1] The net payable to AAG's wholly-owned subsidiaries consists primarily of amounts due under regional capacity purchase agreements with AAG's wholly-owned regional airlines operating under the brand name of American Eagle.

Pursuant to a capacity purchase agreement between American and AAG's wholly-owned regional airlines operating as American Eagle, American purchases all of the capacity from these carriers and recognizes passenger revenue from flights operated by American Eagle. In 2023, 2022 and 2021, American recognized expense of approximately $2.7 billion, $2.5 billion and $2.1 billion, respectively, related to wholly-owned regional airline capacity purchase agreements.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Management's Evaluation of Disclosure Controls and Procedures

The term "disclosure controls and procedures" is defined in Rules 13a-15(e) and 15d-15(e) of the Securities Exchange Act of 1934, as amended (the Exchange Act). This term refers to the controls and procedures of a company that are designed to ensure that information required to be disclosed by a company in the reports that it files under the Exchange Act is recorded, processed, summarized and reported, within the time periods specified by the SEC's rules and forms, and is accumulated and communicated to the company's management, including the principal executive and principal financial officers, or persons performing similar functions, as appropriate to allow timely decisions regarding required disclosure. An evaluation of the effectiveness of AAG's and American's disclosure controls and procedures as of December 31, 2023 was performed under the supervision and with the participation of AAG's and American's management, including AAG's and American's principal executive officer, the Chief Executive Officer (CEO), and principal financial officer, the Chief Financial Officer (CFO). Based on that evaluation, AAG's and American's management, including AAG's and American's CEO and CFO, concluded that AAG's and American's disclosure controls and procedures were effective as of December 31, 2023 at the reasonable assurance level.

Changes in Internal Control over Financial Reporting

During the quarter ended December 31, 2023, there have been no changes in AAG's or American's internal control over financial reporting that has materially affected, or is reasonably likely to materially affect, AAG's and American's internal control over financial reporting.

Limitation on the Effectiveness of Controls

We believe that a controls system, no matter how well designed and operated, cannot provide absolute assurance that the objectives of the controls system are met, and no evaluation of controls can provide absolute assurance that all control issues and instances of fraud, if any, within a company have been detected. Our disclosure controls and procedures are designed to provide reasonable assurance of achieving their objectives, and, as noted above, the CEO and CFO of AAG and American believe that our disclosure controls and procedures were effective at the reasonable assurance level as of December 31, 2023.

Management's Annual Report on Internal Control over Financial Reporting

Management of AAG and American is responsible for establishing and maintaining adequate internal control over financial reporting as defined in Rules 13a-15(f) and 15d-15(f) under the Exchange Act. AAG's and American's internal control over financial reporting is designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with GAAP. AAG's and American's internal control over financial reporting includes policies and procedures that:

- pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of AAG or American, respectively;
- provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with GAAP, and that receipts and expenditures of AAG or American are being made only in accordance with authorizations of management and directors of AAG or American, respectively; and
- provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of AAG's or American's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Management assessed the effectiveness of AAG's and American's internal control over financial reporting as of December 31, 2023. In making this assessment, management used the criteria set forth by the Committee of Sponsoring Organizations of the Treadway Commission in its Internal Control – Integrated Framework (2013 Framework).

Based on our assessment and those criteria, AAG's and American's management concludes that AAG and American, respectively, maintained effective internal control over financial reporting as of December 31, 2023.

AAG's and American's independent registered public accounting firm has issued an attestation report on the effectiveness of AAG's and American's internal control over financial reporting. That report has been included herein.

Report of Independent Registered Public Accounting Firm

To the Stockholders and Board of Directors
American Airlines Group Inc.:

Opinion on Internal Control Over Financial Reporting

We have audited American Airlines Group Inc. and subsidiaries' (the Company) internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission. In our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive income (loss), cash flows, and stockholders' equity (deficit) for each of the years in the three-year period ended December 31, 2023, and the related notes (collectively, the consolidated financial statements), and our report dated February 21, 2024 expressed an unqualified opinion on those consolidated financial statements.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Annual Report on Internal Control Over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audit also included performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ KPMG LLP

Dallas, Texas
February 21, 2024

Report of Independent Registered Public Accounting Firm

To the Stockholder and Board of Directors
American Airlines, Inc.:

Opinion on Internal Control Over Financial Reporting

We have audited American Airlines, Inc. and subsidiaries' (American) internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission. In our opinion, American maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of American as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive income (loss), cash flows, and stockholder's equity for each of the years in the three-year period ended December 31, 2023, and the related notes (collectively, the consolidated financial statements), and our report dated February 21, 2024 expressed an unqualified opinion on those consolidated financial statements.

Basis for Opinion

American's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Annual Report on Internal Control Over Financial Reporting. Our responsibility is to express an opinion on American's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to American in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audit also included performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ KPMG LLP

Dallas, Texas
February 21, 2024

ITEM 9B. OTHER INFORMATION

Securities Trading Plans of Directors and Executive Officers

During the quarter ended December 31, 2023, none of our directors or executive officers adopted or terminated any contract, instruction or written plan for the purchase or sale of AAG securities that was intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) or any "non-Rule 10b5-1 trading arrangement."

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not Applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

Except as stated below, the information required by this Item will be set forth in the Proxy Statement under the captions "Proposal 1 – Election of Directors," "Executive Officers," "Board Composition" and "Information About the Board of Directors and Corporate Governance" and is incorporated by reference into this Annual Report on Form 10-K.

AAG and American have adopted Standards of Business Conduct (the Ethics Standards) within the meaning of Item 406(b) of Regulation S-K. The Ethics Standards apply to all officers and employees of AAG and its subsidiaries, including American. The Ethics Standards are available on our website at *www.aa.com*. If we make substantive amendments to the Ethics Standards or grant any waiver, including any implicit waiver, to our principal executive officer, principal financial officer, principal accounting officer or controller, or persons performing similar functions, we will disclose the nature of such amendment or waiver on our website or in a Current Report on Form 8-K in accordance with applicable rules and regulations.

ITEM 11. EXECUTIVE COMPENSATION

The information required by this Item will be set forth in the Proxy Statement under the captions "Information About the Board of Directors and Corporate Governance - Risk Assessment with Respect to Compensation Practices," "Director Compensation," "Compensation Discussion and Analysis," "Executive Compensation" and "Compensation Committee Report" and is incorporated by reference into this Annual Report on Form 10-K.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information required by this Item will be set forth in the Proxy Statement under the captions "Security Ownership of Certain Beneficial Owners and Management" and "Equity Compensation Plan Information" and is incorporated by reference into this Annual Report on Form 10-K.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

The information required by this Item will be set forth in the Proxy Statement under the captions "Certain Relationships and Related Party Transactions" and "Information About the Board of Directors and Corporate Governance" and is incorporated by reference into this Annual Report on Form 10-K.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

The information required by this Item will be set forth in the Proxy Statement under the caption "Proposal 2 – Ratification of Appointment of Independent Registered Public Accounting Firm" and is incorporated by reference into this Annual Report on Form 10-K.

PART IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

Consolidated Financial Statements

The following consolidated financial statements of American Airlines Group Inc. and Independent Auditors' Report are filed as part of this report:

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Report of Independent Registered Public Accounting Firm (KPMG LLP, Dallas, TX, Auditor Firm ID: 185)	80
Consolidated Statements of Operations for the Years Ended December 31, 2023, 2022 and 2021	82
Consolidated Statements of Comprehensive Income (Loss) for the Years Ended December 31, 2023, 2022 and 2021	83
Consolidated Balance Sheets at December 31, 2023 and 2022	84
Consolidated Statements of Cash Flows for the Years Ended December 31, 2023, 2022 and 2021	85
Consolidated Statements of Stockholders' Equity (Deficit) for the Years Ended December 31, 2023, 2022 and 2021	86
Notes to Consolidated Financial Statements	87

The following consolidated financial statements of American Airlines, Inc. and Independent Auditors' Report are filed as part of this report:

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Report of Independent Registered Public Accounting Firm (KPMG LLP, Dallas, TX, Auditor Firm ID: 185)	126
Consolidated Statements of Operations for the Years Ended December 31, 2023, 2022 and 2021	128
Consolidated Statements of Comprehensive Income (Loss) for the Years Ended December 31, 2023, 2022 and 2021	129
Consolidated Balance Sheets at December 31, 2023 and 2022	130
Consolidated Statements of Cash Flows for the Years Ended December 31, 2023, 2022 and 2021	131
Consolidated Statements of Stockholder's Equity for the Years Ended December 31, 2023, 2022 and 2021	132
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Schedules not included have been omitted because they are not applicable or because the required information is included in the Consolidated Financial Statements or notes thereto.

Exhibits

Exhibits required to be filed by Item 601 of Regulation S-K: Where the amount of securities authorized to be issued under any of our long-term debt agreements does not exceed 10% of our assets, pursuant to paragraph (b)(4) of Item 601 of Regulation S-K, in lieu of filing such as an exhibit, we hereby agree to furnish to the Commission upon request a copy of any agreement with respect to such long-term debt.

Exhibit Number	Description
2.1	Confirmation Order and Plan (incorporated by reference to Exhibit 2.1 to AMR's Current Report on Form 8-K filed on October 23, 2013 (Commission File No. 1-8400)).
2.2	Agreement and Plan of Merger, dated as of December 28, 2015, between American Airlines, Inc. and US Airways, Inc. (incorporated by reference to Exhibit 2.1 to AAG's Current Report on Form 8-K filed on December 31, 2015 (Commission File No. 1-8400)).
3.1	Restated Certificate of Incorporation of American Airlines Group Inc., including the Certificate of Designations, Powers, Preferences and Rights of the American Airlines Group Inc. Series A Convertible Preferred Stock attached as Annex I thereto (incorporated by reference to Exhibit 3.1 to AAG's Current Report on Form 8-K filed on December 9, 2013 (Commission File No. 1-8400)).
3.2	Certificate of Amendment of Restated Certificate of Incorporation of American Airlines Group Inc. (incorporated by reference to Exhibit 3.1 to AAG's Current Report on Form 8-K filed on June 13, 2018 (Commission File No. 1-8400)).
3.3	Third Amended and Restated Bylaws of American Airlines Group Inc . (incorporated by reference to Exhibit 3.3 to AAG 's Annual Report on Form 10-K for the year ended December 31, 2022 (Commission File No. 1-8400)) .
3.4	Amendment to the Third Amended and Restated Bylaws of American Airlines Group Inc. (incorporated by reference to Exhibit 3.1 to AAG's Current Report on Form 8-K filed on February 3, 2023 (Commission File No. 1-8400)).
3.5	Amended and Restated Certificate of Incorporation of American Airlines, Inc. (incorporated by reference to Exhibit 3.3 to AAG's Annual Report on Form 10-K for the year ended December 31, 2013 (Commission File No. 1-8400)).
3.6	Amended and Restated Bylaws of American Airlines, Inc. (incorporated by reference to Exhibit 3.4 to AAG's Annual Report on Form 10-K for the year ended December 31, 2013 (Commission File No. 1-8400)).
3.7	Certificate of Designations of Series B Junior Participating Preferred Stock of American Airlines Group Inc., filed with the Secretary of State of the State of Delaware on December 21, 2021 (incorporated by reference to Exhibit 3.1 to AAG's Current Report on Form 8-K filed on December 22, 2021 (Commission File No. 1-8400)).
4.1	Description of securities registered under Section 12 of the Exchange Act (incorporated by reference to Exhibit 4.1 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).
4.2	Pass Through Trust Agreement, dated as of September 16, 2014, between American Airlines, Inc. and Wilmington Trust Company, as Trustee (incorporated by reference to Exhibit 4.1 to American's Current Report on Form 8-K filed on September 17, 2014 (Commission File No. 1-2691)).
4.3	Trust Supplement No. 2014-1A, dated as of September 16, 2014, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on September 17, 2014 (Commission File No. 1-2691)).
4.4	Trust Supplement No. 2014-1B, dated as of September 16, 2014, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on September 17, 2014 (Commission File No. 1-2691)).
4.5	Intercreditor Agreement (2014-1), dated as of September 16, 2014, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2014-1A and as Trustee of the American Airlines Pass Through Trust 2014-1B, Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on September 17, 2014 (Commission File No. 1-2691)).
4.6	Amendment No. 1 to Intercreditor Agreement (2014-1), dated as of June 24, 2015, among American Airlines, Inc., Credit Agricole Corporate and Investment Bank, as Class A and Class B liquidity provider and Wilmington Trust Company, as subordination agent and trustee (incorporated by reference to Exhibit 10.6 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2015 (Commission File No. 1-8400)).

Exhibit Number	Description
4.7	Note Purchase Agreement, dated as of September 16, 2014, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.9 to American's Current Report on Form 8-K filed on September 17, 2014 (Commission File No. 1-2691)).
4.8	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (Exhibit B to Note Purchase Agreement) (incorporated by reference to Exhibit 4.10 to American's Current Report on Form 8-K filed on September 17, 2014 (Commission File No. 1-2691)).
4.9	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (Exhibit C to Note Purchase Agreement) (incorporated by reference to Exhibit 4.11 to American's Current Report on Form 8-K filed on September 17, 2014 (Commission File No. 1-2691)).
4.10	Revolving Credit Agreement (2014-1A), dated as of September 16, 2014, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2014-1A, as Borrower, and Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Liquidity Provider (incorporated by reference to Exhibit 4.14 to American's Current Report on Form 8-K filed on September 17, 2014 (Commission File No. 1-2691)).
4.11	Revolving Credit Agreement (2014-1B), dated as of September 16, 2014, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2014-1B, as Borrower, and Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Liquidity Provider (incorporated by reference to Exhibit 4.15 to American's Current Report on Form 8-K filed on September 17, 2014 (Commission File No. 1-2691)).
4.12	First Supplemental Indenture, dated as of December 30, 2015, among American Airlines Group Inc., American Airlines, Inc. and Wilmington Trust, National Association, as trustee, to the Indenture dated as of March 5, 2015 (incorporated by reference to Exhibit 4.3 to AAG's Current Report on Form 8-K filed on December 31, 2015 (Commission File No. 1-8400)).
4.13	Trust Supplement No. 2015-1A, dated as of March 16, 2015, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).
4.14	Trust Supplement No. 2015-1B, dated as of March 16, 2015, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).
4.15	Intercreditor Agreement (2015-1), dated as of March 16, 2015, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2015-1A and as Trustee of the American Airlines Pass Through Trust 2015-1B, Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).
4.16	Note Purchase Agreement, dated as of March 16, 2015, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.9 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).
4.17	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit 4.10 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).
4.18	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit 4.11 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).
4.19	Form of Pass Through Trust Certificate, Series 2015-1A (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).
4.20	Form of Pass Through Trust Certificate, Series 2015-1B (incorporated by reference to Exhibit A to Exhibit 4.3 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).

Exhibit Number	Description
4.21	Revolving Credit Agreement (2015-1A), dated as of March 16, 2015, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2015-1A, as Borrower, and Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Liquidity Provider (incorporated by reference to Exhibit 4.14 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).
4.22	Revolving Credit Agreement (2015-1B), dated as of March 16, 2015, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2015-1B, as Borrower, and Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Liquidity Provider (incorporated by reference to Exhibit 4.15 to American's Current Report on Form 8-K filed on March 16, 2015 (Commission File No. 1-2691)).
4.23	Trust Supplement No. 2015-2AA, dated as of September 24, 2015, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.24	Trust Supplement No. 2015-2A, dated as of September 24, 2015, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.25	Trust Supplement No. 2015-2B, dated as of September 24, 2015, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.26	Intercreditor Agreement (2015-2), dated as of September 24, 2015, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2015-2AA, as Trustee of the American Airlines Pass Through Trust 2015-2A and as Trustee of the American Airlines Pass Through Trust 2015-2B, Commonwealth Bank of Australia, New York Branch, as Class AA Liquidity Provider, Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.5 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.27	Note Purchase Agreement, dated as of September 24, 2015, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.6 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.28	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.6 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.29	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.6 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.30	Form of Pass Through Trust Certificate, Series 2015-2AA (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.31	Form of Pass Through Trust Certificate, Series 2015-2A (incorporated by reference to Exhibit A to Exhibit 4.3 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.32	Form of Pass Through Trust Certificate, Series 2015-2B (incorporated by reference to Exhibit A to Exhibit 4.4 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.33	Revolving Credit Agreement (2015-2AA), dated as of September 24, 2015, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2015-2AA, as Borrower, and Commonwealth Bank of Australia, New York Branch, as Liquidity Provider (incorporated by reference to Exhibit 4.12 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.34	Revolving Credit Agreement (2015-2A), dated as of September 24, 2015, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2015-2A, as Borrower, and Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Liquidity Provider (incorporated by reference to Exhibit 4.13 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).

Exhibit Number	Description
4.35	Revolving Credit Agreement (2015-2B), dated as of September 24, 2015, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2015-2B, as Borrower, and Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Liquidity Provider (incorporated by reference to Exhibit 4.14 to American's Current Report on Form 8-K filed on September 24, 2015 (Commission File No. 1-2691)).
4.36	Note Purchase Agreement, dated as of April 24, 2013, among American Airlines, Inc. (as successor in interest to US Airways, Inc.) Wilmington Trust Company, as Pass Through Trustee, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.12 to US Airways Group's Current Report on Form 8-K filed on April 25, 2013 (Commission File No. 1-8444)).
4.37	Assumption Agreement, dated as of December 30, 2015, by American Airlines, Inc. for the benefit of Wilmington Trust Company, as pass through trustee, subordination agent, and paying agent, and Wilmington Trust, National Association, as escrow agent, in each case, under the Note Purchase Agreement, dated as of April 24, 2013, among American Airlines, Inc. (as successor in interest to US Airways, Inc.), Wilmington Trust Company, Wilmington Trust, National Association and Wilmington Trust Company (incorporated by reference to Exhibit 10.2 to AAG's Current Report on Form 8-K filed on December 31, 2015 (Commission File No. 1-8400)).
4.38	Form of Participation Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee, Subordination Agent and Pass Through Trustee (incorporated by reference to Exhibit 4.13 to US Airways Group's Current Report on Form 8-K filed on April 25, 2013 (Commission File No. 1-8444)).
4.39	Form of Trust Indenture and Security Agreement among American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, Wilmington Trust, National Association, as Securities Intermediary, and Wilmington Trust Company, as Indenture Trustee (incorporated by reference to Exhibit 4.14 to US Airways Group's Current Report on Form 8-K filed on April 25, 2013 (Commission File No. 1-8444)).
4.40	Form of Amendment No. 1 to Participation Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee, Subordination Agent and Pass Through Trustee (Exhibit A to Note Purchase Agreement) (incorporated by reference to Exhibit 4.8 to US Airways Group's Current Report on Form 8-K filed on June 6, 2013 (Commission File No. 1-8444)).
4.41	Form of Amendment No. 1 to Trust Indenture and Security Agreement among American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, Wilmington Trust, National Association, as Securities Intermediary, and Wilmington Trust Company, as Indenture Trustee (Exhibit B to Note Purchase Agreement) (incorporated by reference to Exhibit 4.9 to US Airways Group's Current Report on Form 8-K filed on June 6, 2013 (Commission File No. 1-8444)).
4.42	Amended and Restated Guarantee, dated as of March 31, 2014, from American Airlines Group Inc. (as successor in interest to US Airways Group, Inc.) relating to obligations of US Airways under the equipment notes relating to its Series 2013-1 Pass Through Certificates (incorporated by reference to Exhibit 10.5 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2014 (Commission File No. 1-8400)).
4.43	Form of Participation Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee, Subordination Agent and Pass Through Trustee (Schedule I to Amendment No. 1 to Note Purchase Agreement (2012-2)) (incorporated by reference to Exhibit 4.10 to US Airways Group's Current Report on Form 8-K filed on June 6, 2013 (Commission File No. 1-8444)).
4.44	Form of Trust Indenture and Security Agreement among American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, Wilmington Trust, National Association, as Securities Intermediary, and Wilmington Trust Company, as Indenture Trustee (Exhibit A to Amendment No. 1 to Note Purchase Agreement (2012-2)) (incorporated by reference to Exhibit 4.11 to US Airways Group's Current Report on Form 8-K filed on June 6, 2013 (Commission File No. 1-8444)).
4.45	Form of Participation Agreement (Participation Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee and Subordination Agent) (incorporated by reference to Exhibit 4.14 to US Airways Group's Current Report on Form 8-K filed on December 23, 2010 (Commission File No. 1-8444)).
4.46	Form of Indenture (Trust Indenture and Security Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee) (incorporated by reference to Exhibit 4.15 to US Airways Group's Current Report on Form 8-K filed on December 23, 2010 (Commission File No. 1-8444)).
4.47	Amended and Restated Guarantee, dated as of March 31, 2014, from American Airlines Group Inc. (as successor in interest to US Airways Group, Inc.) relating to obligations of US Airways under the equipment notes relating to its Series 2010-1 Pass Through Certificates (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2014 (Commission File No. 1-8400)).

Exhibit Number	Description
4.48	Form of Participation Agreement (Participation Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee and Subordination Agent) (incorporated by reference to Exhibit 4.18 to US Airways Group's Current Report on Form 8-K filed on July 1, 2011 (Commission File No. 1-08444)).
4.49	Form of Indenture (Trust Indenture and Security Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee) (incorporated by reference to Exhibit 4.19 to US Airways Group's Current Report on Form 8-K filed on July 1, 2011 (Commission File No. 1-08444)).
4.50	Guarantee, dated as of June 28, 2011, from American Airlines Group Inc. (as successor in interest to US Airways Group, Inc.) (incorporated by reference to Exhibit 4.23 to US Airways Group's Current Report on Form 8-K filed on July 1, 2011 (Commission File No. 1-08444)).
4.51	Amended and Restated Guarantee, dated as of March 31, 2014, from American Airlines Group Inc. (as successor in interest to US Airways Group, Inc.) relating to obligations of US Airways under the equipment notes relating to its Series 2011-1 Pass Through Certificates (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2014 (Commission File No. 1-8400)).
4.52	Form of Participation Agreement (Participation Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee and Subordination Agent) (incorporated by reference to Exhibit 4.18 to US Airways Group's Current Report on Form 8-K filed on May 16, 2012 (Commission File No. 1-08444)).
4.53	Form of Indenture (Trust Indenture and Security Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee) (incorporated by reference to Exhibit 4.19 to US Airways Group's Current Report on Form 8-K filed on May 16, 2012 (Commission File No. 1-08444)).
4.54	Amended and Restated Guarantee, dated as of March 31, 2014, from American Airlines Group Inc. (as successor in interest to US Airways Group, Inc.) relating to obligations of US Airways under the equipment notes relating to its Series 2012-1 Pass Through Certificates (incorporated by reference to Exhibit 10.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2014 (Commission File No. 1-8400)).
4.55	Form of Participation Agreement (Participation Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee and Subordination Agent) (incorporated by reference to Exhibit B to Exhibit 4.12 to US Airways Group's Current Report on Form 8-K filed on December 13, 2012 (Commission File No. 1-08444)).
4.56	Form of Indenture (Trust Indenture and Security Agreement between American Airlines, Inc. (as successor in interest to US Airways, Inc.), as Owner, and Wilmington Trust Company, as Indenture Trustee) (incorporated by reference to Exhibit C to Exhibit 4.12 to US Airways Group's Current Report on Form 8-K filed on December 13, 2012 (Commission File No. 1-08444)).
4.57	Amended and Restated Guarantee, dated as of March 31, 2014, from American Airlines Group Inc. (as successor in interest to US Airways Group, Inc.) relating to obligations of US Airways under the equipment notes relating to its Series 2012-2 Pass Through Certificates (incorporated by reference to Exhibit 10.4 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2014 (Commission File No. 1-8400)).
4.58	Form of Assumption Agreement, dated as of December 30, 2015, by American Airlines, Inc. for the benefit of Wilmington Trust Company, as Indenture Trustee, to (i) each Participation Agreement between, among others, American Airlines, Inc. (as successor in interest to US Airways, Inc.) and Wilmington Trust Company, as Indenture Trustee, entered into pursuant to the 2010-1, 2011-1, 2012-1, 2012-2 and 2013-1 EETC note purchase agreements and (ii) each Trust Indenture and Security Agreement, between, among others, American Airlines, Inc. (as successor in interest to US Airways, Inc.), and Wilmington Trust Company, as Indenture Trustee entered into pursuant to the 2010-1, 2011-1, 2012-1, 2012-2 and 2013-1 EETC note purchase agreements (incorporated by reference to Exhibit 10.3 to AAG's Current Report on Form 8-K filed on December 31, 2015 (Commission File No. 1-8400)).
4.59	Trust Supplement No. 2016-1AA, dated as of January 19, 2016, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.60	Trust Supplement No. 2016-1A, dated as of January 19, 2016, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.61	Trust Supplement No. 2016-1B, dated as of January 19, 2016, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).

Exhibit Number	Description
4.62	Intercreditor Agreement (2016-1), dated as of January 19, 2016, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2016-1AA, as Trustee of the American Airlines Pass Through Trust 2016-1A and as Trustee of the American Airlines Pass Through Trust 2016-1B, KfW IPEX-Bank GmbH, as Class AA Liquidity Provider, Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to exhibit 4.5 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.63	Note Purchase Agreement, dated as of January 19, 2016, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.6 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.64	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.6 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.65	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.6 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.66	Form of Pass Through Trust Certificate, Series 2016-1AA (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.67	Form of Pass Through Trust Certificate, Series 2016-1A (incorporated by reference to Exhibit A to Exhibit 4.3 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.68	Form of Pass Through Trust Certificate, Series 2016-1B (incorporated by reference to Exhibit A to Exhibit 4.4 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.69	Revolving Credit Agreement (2016-1AA), dated as of January 19, 2016, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2016-1AA, as Borrower, and KfW IPEX-Bank GmbH, as Liquidity Provider (incorporated by reference to Exhibit 4.12 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.70	Revolving Credit Agreement (2016-1A), dated as of January 19, 2016, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2016-1A, as Borrower, and KfW IPEX-Bank GmbH, as Liquidity Provider (incorporated by reference to Exhibit 4.13 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.71	Revolving Credit Agreement (2016-1B), dated as of January 19, 2016, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2016-1B, as Borrower, and KfW IPEX-Bank GmbH, as Liquidity Provider (incorporated by reference to Exhibit 4.14 to American's Current Report on Form 8-K filed on January 21, 2016 (Commission File No. 1-2691)).
4.72	Trust Supplement No. 2016-2AA, dated as of May 16, 2016, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).
4.73	Trust Supplement No. 2016-2A, dated as of May 16, 2016, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).
4.74	Intercreditor Agreement (2016-2), dated as of May 16, 2016, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2016-2AA and as Trustee of the American Airlines Pass Through Trust 2016-2A, KfW IPEX-Bank GmbH, as Class AA Liquidity Provider and Class A Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).
4.75	Note Purchase Agreement, dated as of May 16, 2016, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.9 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).
4.76	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.9 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).

Exhibit Number	Description
4.77	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.9 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).
4.78	Form of Pass Through Trust Certificate, Series 2016-2AA (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).
4.79	Form of Pass Through Trust Certificate, Series 2016-2A (incorporated by reference to Exhibit A to Exhibit 4.3 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).
4.80	Revolving Credit Agreement (2016-2AA), dated as of May 16, 2016, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2016-2AA, as Borrower, and KfW IPEX-Bank GmbH, as Liquidity Provider (incorporated by reference to Exhibit 4.14 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).
4.81	Revolving Credit Agreement (2016-2A), dated as of May 16, 2016, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2016-2A, as Borrower, and KfW IPEX-Bank GmbH, as Liquidity Provider (incorporated by reference to Exhibit 4.15 to American's Current Report on Form 8-K filed on May 17, 2016 (Commission File No. 1-2691)).
4.82	Trust Supplement No. 2016-2B, dated as of July 8, 2016, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on July 12, 2016 (Commission File No. 1-2691)).
4.83	Amended and Restated Intercreditor Agreement (2016-2), dated as of July 8, 2016, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2016-2AA, as Trustee of the American Airlines Pass Through Trust 2016-2A and as Trustee of the American Airlines Pass Through Trust 2016-2B, KfW IPEX-Bank GmbH, as Class AA Liquidity Provider, Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on July 12, 2016 (Commission File No. 1-2691)).
4.84	Amended and Restated Note Purchase Agreement, dated as of July 8, 2016, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.6 to American's Current Report on Form 8-K filed on July 12, 2016 (Commission File No. 1-2691)).
4.85	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.6 to American's Current Report on Form 8-K filed on July 12, 2016 (Commission File No. 1-2691)).
4.86	Form of First Amendment to Participation Agreement (First Amendment to Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit D to Exhibit 4.6 to American's Current Report on Form 8-K filed on July 12, 2016 (Commission File No. 1-2691)).
4.87	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.6 to American's Current Report on Form 8-K filed on July 12, 2016 (Commission File No. 1-2691)).
4.88	Form of First Amendment to Indenture and Security Agreement (First Amendment to Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit E to Exhibit 4.6 to American's Current Report on Form 8-K filed on July 12, 2016 (Commission File No. 1-2691)).
4.89	Form of Pass Through Trust Certificate, Series 2016-2B (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on July 12, 2016 (Commission File No. 1-2691)).
4.90	Revolving Credit Agreement (2016-2B), dated as of July 8, 2016, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2016-2B, as Borrower, and KfW IPEX Bank GmbH, as Liquidity Provider (incorporated by reference to Exhibit 4.12 to American's Current Report on Form 8-K filed on July 12, 2016 (Commission File No. 1-2691)).
4.91	Trust Supplement No. 2016-3AA, dated as of October 3, 2016, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).

Exhibit Number	Description
4.92	Trust Supplement No. 2016-3A, dated as of October 3, 2016, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).
4.93	Intercreditor Agreement (2016-3), dated as of October 3, 2016, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2016-3AA and as Trustee of the American Airlines Pass Through Trust 2016-3A, KfW IPEX-Bank GmbH, as Class AA Liquidity Provider and Class A Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).
4.94	Amended and Restated Intercreditor Agreement (2016-3), dated as of October 4, 2017, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2016-3AA, as Trustee of the American Airlines Pass Through Trust 2016-3A and as Trustee of the American Airlines Pass Through Trust 2016-3B, KfW IPEX-Bank GmbH, as Class AA Liquidity Provider, Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on October 5, 2017 (Commission File No. 1-2691)).
4.95	Note Purchase Agreement, dated as of October 3, 2016, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.9 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).
4.96	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.9 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).
4.97	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.9 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).
4.98	Form of Pass Through Trust Certificate, Series 2016-3AA (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).
4.99	Form of Pass Through Trust Certificate, Series 2016-3A (incorporated by reference to Exhibit A to Exhibit 4.3 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).
4.100	Revolving Credit Agreement (2016-3AA), dated as of October 3, 2016, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2016-3AA, as Borrower, and KfW IPEX-Bank GmbH, as Liquidity Provider (incorporated by reference to Exhibit 4.14 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).
4.101	Revolving Credit Agreement (2016-3A), dated as of October 3, 2016, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2016-3A, as Borrower, and KfW IPEX-Bank GmbH, as Liquidity Provider (incorporated by reference to Exhibit 4.15 to American's Current Report on Form 8-K filed on October 4, 2016 (Commission File No. 1-2691)).
4.102	Trust Supplement No. 2017-1AA, dated as of January 13, 2017, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.103	Trust Supplement No. 2017-1A, dated as of January 13, 2017, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014, (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.104	Trust Supplement No. 2017-1B, dated as of January 13, 2017, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.105	Intercreditor Agreement (2017-1), dated as of January 13, 2017, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2017-1AA, as Trustee of the American Airlines Pass Through Trust 2017-1A and as Trustee of the American Airlines Pass Through Trust 2017-1B, Citibank N.A., as Class AA Liquidity Provider, Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.5 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).

Exhibit Number	Description
4.106	Note Purchase Agreement, dated as of January 13, 2017, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.12 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.107	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.12 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.108	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.12 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.109	Form of Pass Through Trust Certificate, Series 2017-1AA (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.110	Form of Pass Through Trust Certificate, Series 2017-1A (incorporated by reference to Exhibit A to Exhibit 4.3 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.111	Form of Pass Through Trust Certificate, Series 2017-1B (incorporated by reference to Exhibit A to Exhibit 4.4 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.112	Revolving Credit Agreement (2017-1AA), dated as of January 13, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2017-1AA, as Borrower, and Citibank N.A., as Liquidity Provider (incorporated by reference to Exhibit 4.18 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.113	Revolving Credit Agreement (2017-1A), dated as of January 13, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2017-1A, as Borrower, and Citibank N.A., as Liquidity Provider (incorporated by reference to Exhibit 4.19 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.114	Revolving Credit Agreement (2017-1B), dated as of January 13, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2017-1B, as Borrower, and Citibank N.A., as Liquidity Provider (incorporated by reference to Exhibit 4.20 to American's Current Report on Form 8-K filed on January 17, 2017 (Commission File No. 1-02691)).
4.115	Acknowledgment and Agreement (2017-1), dated as of March 31, 2017, by and among American Airlines Inc., Citibank, N.A., as initial Liquidity Provider, National Australia Bank Limited, as Replacement Liquidity Provider, and Wilmington Trust Company, as Subordination Agent and trustee (incorporated by reference to Exhibit 4.20 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2017 (Commission File No. 1-8400)).
4.116	Revolving Credit Agreement (2017-1AA), dated as of March 31, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2017-1AA, as Borrower, and National Australia Bank Limited, as Liquidity Provider (incorporated by reference to Exhibit 4.21 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2017 (Commission File No. 1-8400)).
4.117	Revolving Credit Agreement (2017-1A), dated as of March 31, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2017-1A, as Borrower, and National Australia Bank Limited, as Liquidity Provider (incorporated by reference to Exhibit 4.22 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2017 (Commission File No. 1-8400)).
4.118	Revolving Credit Agreement (2017-1B), dated as of March 31, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2017-1B, as Borrower, and National Australia Bank Limited, as Liquidity Provider (incorporated by reference to Exhibit 4.23 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2017 (Commission File No. 1-8400)).
4.119	Form of American Airlines Group Inc. Indenture for Debt Securities (incorporated by reference to Exhibit 4.1 to AAG's Registration Statement on Form S-3ASR filed on February 22, 2017 (Commission File No. 333-216167).
4.120	Form of American Airlines, Inc. Indenture for Debt Securities (incorporated by reference to Exhibit 4.2 to AAG's Registration Statement on Form S-3ASR filed on February 22, 2017 (Commission File No. 333-216167).
4.121	Trust Supplement No. 2017-2AA, dated as of August 14, 2017, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).

Exhibit Number	Description
4.122	Trust Supplement No. 2017-2A, dated as of August 14, 2017, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).
4.123	Intercreditor Agreement (2017-2), dated as of August 14, 2017, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2017-2AA and as Trustee of the American Airlines Pass Through Trust 2017-2A, National Australia Bank Limited, as Class AA Liquidity Provider and Class A Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).
4.124	Note Purchase Agreement, dated as of August 14, 2017, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.9 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).
4.125	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.9 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).
4.126	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.9 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).
4.127	Form of Pass Through Trust Certificate, Series 2017-2AA (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).
4.128	Form of Pass Through Trust Certificate, Series 2017-2A (incorporated by reference to Exhibit A to Exhibit 4.3 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).
4.129	Revolving Credit Agreement (2017-2AA), dated as of August 14, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2017-2AA, as Borrower, and National Australia Bank Limited, as Liquidity Provider (incorporated by reference to Exhibit 4.14 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).
4.130	Revolving Credit Agreement (2017-2A), dated as of August 14, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2017-2A, as Borrower, and National Australia Bank Limited, as Liquidity Provider (incorporated by reference to Exhibit 4.15 to American's Current Report on Form 8-K filed on August 14, 2017 (Commission File No. 1-2691)).
4.131	Trust Supplement No. 2016-3B, dated as of October 4, 2017, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on October 5, 2017 (Commission File No. 1-2691)).
4.132	Amended and Restated Note Purchase Agreement, dated as of October 4, 2017, amending the Note Purchase Agreement, dated as of October 3, 2016, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on October 5, 2017 (Commission File No. 1-2691)).
4.133	Form of First Amendment to Participation Agreement (First Amendment to Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit A to Exhibit 4.4 to American's Current Report on Form 8-K filed on October 5, 2017 (Commission File No. 1-2691)).
4.134	Form of First Amendment to Indenture and Security Agreement (First Amendment to Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit E to Exhibit 4.6 to American's Current Report on Form 8-K filed on October 6, 2017 (Commission File No. 1-2691)).
4.135	Form of First Amendment to Indenture and Security Agreement (First Amendment to Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit B to Exhibit 4.4 to American's Current Report on Form 8-K filed on October 5, 2017 (Commission File No. 1-2691)).
4.136	Form of Pass Through Trust Certificate, Series 2016-3B (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on October 5, 2017 (Commission File No. 1-2691)).

Exhibit Number	Description
4.137	Revolving Credit Agreement (2016-3B), dated as of October 4, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2016-3B, as Borrower, and KfW IPEX-Bank GmbH, as Liquidity Provider 3B (incorporated by reference to Exhibit 4.8 to American's Current Report on Form 8-K filed on October 5, 2017 (Commission File No. 1-2691)).
4.138	Trust Supplement No. 2017-2B, dated as of October 5, 2017, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on October 6, 2017 (Commission File No. 1-2691)).
4.139	Amended and Restated Intercreditor Agreement (2017-2), dated as of October 5, 2017, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2017-2AA, as Trustee of the American Airlines Pass Through Trust 2017-2A and as Trustee of the American Airlines Pass Through Trust 2017-2B, National Australia Bank Limited, as Class AA Liquidity Provider, Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on October 6, 2017 (Commission File No. 1-2691)).
4.140	Amended and Restated Note Purchase Agreement, dated as of October 5, 2017, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.6 to American's Current Report on Form 8-K filed on October 6, 2017 (Commission File No. 1-2691)).
4.141	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.6 to American's Current Report on Form 8-K filed on October 6, 2017 (Commission File No. 1-2691)).
4.142	Form of First Amendment to Participation Agreement (First Amendment to Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit D to Exhibit 4.6 to American's Current Report on Form 8-K filed on October 6, 2017 (Commission File No. 1-2691)).
4.143	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.6 to American's Current Report on Form 8-K filed on October 6, 2017 (Commission File No. 1-2691)).
4.144	Form of Pass Through Trust Certificate, Series 2017-2B (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on October 6, 2017 (Commission File No. 1-2691)).
4.145	Revolving Credit Agreement (2017-2B), dated as of October 5, 2017, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2017-2B, as Borrower, and National Australia Bank Limited, as Liquidity Provider (incorporated by reference to Exhibit 4.12 to American's Current Report on Form 8-K filed on October 6, 2017 (Commission File No. 1-2691)).
4.146	Trust Supplement No. 2012-2C(R), dated as of May 15, 2018, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on May 16, 2018 (Commission File No. 1-2691)).
4.147	Form of Amendment No. 2 to Intercreditor Agreement (2012-2C(R)) among Wilmington Trust Company, not in its individual capacity but solely as Trustee of the American Airlines, Inc. Pass Through Trust 2012-2C(R), American Airlines, Inc. and Wilmington Trust Company, not in its individual capacity but solely as Subordination Agent and Trustee (incorporated by reference to Exhibit C to Exhibit 4.6 to American's Current Report on Form 8-K filed on May 16, 2018 (Commission File No. 1-2691)).
4.148	Note Purchase Agreement, dated as of May 15, 2018, among American Airlines, Inc., Wilmington Trust Company, not in its individual capacity, but solely as Pass Through Trustee under the Class C(R) Pass Through Trust Agreement, as Subordination Agent and as Indenture Trustee, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.6 to American's Current Report on Form 8-K filed on May 16, 2018 (Commission File No. 1- 2691)).
4.149	Form of Amendment to Participation Agreement (Amendment to Participation Agreement among American Airlines, Inc., Wilmington Trust Company, not in its individual capacity, but solely as Subordination Agent and as Indenture Trustee, and Wilmington Trust Company, not in its individual capacity, but solely as Pass Through Trustee under each of the Pass Through Trust Agreements) (incorporated by reference to Exhibit A to Exhibit 4.6 to American's Current Report on Form 8-K filed on May 16, 2018 (Commission File No. 1- 2691)).
4.150	Form of Pass Through Trust Certificate, Series 2021-1A (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).

Exhibit Number	Description
4.151	Form of Pass Through Trust Certificate, Series 2021-1B (incorporated by reference to Exhibit A to Exhibit 4.3 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).
4.152	Revolving Credit Agreement (2021-1A), dated as of November 8, 2021, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2021-1A, as Borrower, and Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Liquidity Provider (incorporated by reference to Exhibit 4.14 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).
4.153	Revolving Credit Agreement (2021-1B), dated as of November 8, 2021, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2021-1B, as Borrower, and Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Liquidity Provider (incorporated by reference to Exhibit 4.15 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).
4.154	Tax Benefit Preservation Plan, dated as of December 21, 2021, between American Airlines Group Inc. and American Stock Transfer & Trust Company, LLC, which includes the Form of Certificate of Designations of Series B Junior Participating Preferred Stock as Exhibit A, the Form of Right Certificate as Exhibit B and the Summary of Rights to Purchase Preferred Shares as Exhibit C (incorporated by reference to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on December 22, 2021 (Commission File No. 1-8400)).
4.155	Form of Amendment to Trust Indenture and Security Agreement (Amendment to Trust Indenture and Security Agreement between American Airlines, Inc., Wilmington Trust Company, not in its individual capacity, but solely as Indenture Trustee, and Wilmington Trust, National Association, as Securities Intermediary) (incorporated by reference to Exhibit B to Exhibit 4.6 to American's Current Report on Form 8-K filed on May 16, 2018 (Commission File No. 1-2691)).
4.156	Form of Pass Through Trust Certificate, Series 2012-2C(R) (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on May 16, 2018 (Commission File No. 1-2691)).
4.157	Trust Supplement No. 2019-1AA (Aircraft EETC), dated as of August 15, 2019, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.158	Trust Supplement No. 2019-1A (Aircraft EETC), dated as of August 15, 2019, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.159	Trust Supplement No. 2019-1B (Aircraft EETC), dated as of August 15, 2019, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.160	Intercreditor Agreement (2019-1), dated as of August 15, 2019, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2019-1AA (Aircraft EETC), as Trustee of the American Airlines Pass Through Trust 2019-1A (Aircraft EETC) and as Trustee of the American Airlines Pass Through Trust 2019-1B (Aircraft EETC), National Australia Bank Limited, as Class AA Liquidity Provider, Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.5 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.161	Note Purchase Agreement, dated as of August 15, 2019, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.12 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.162	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.12 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.163	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.12 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.164	Form of Pass Through Trust Certificate, Series 2019-1AA (Aircraft EETC) (incorporated by reference to Exhibit A to Exhibit 4.2 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).

Exhibit Number	Description
4.165	Form of Pass Through Trust Certificate, Series 2019-1A (Aircraft EETC) (incorporated by reference to Exhibit A to Exhibit 4.3 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.166	Form of Pass Through Trust Certificate, Series 2019-1B (Aircraft EETC) (incorporated by reference to Exhibit A to Exhibit 4.4 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.167	Revolving Credit Agreement (2019-1AA), dated as of August 15, 2019, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2019-1AA (Aircraft EETC), as Borrower, and National Australia Bank Limited, as Liquidity Provider (incorporated by reference to Exhibit 4.18 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.168	Revolving Credit Agreement (2019-1A), dated as of August 15, 2019, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2019-1A (Aircraft EETC), as Borrower, and National Australia Bank Limited, as Liquidity Provider (incorporated by reference to Exhibit 4.19 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.169	Trust Supplement No. 2021-1A, dated as of November 8, 2021, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.2 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).
4.170	Trust Supplement No. 2021-1B, dated as of November 8, 2021, between American Airlines, Inc. and Wilmington Trust Company, as Trustee, to the Pass Through Trust Agreement, dated as of September 16, 2014 (incorporated by reference to Exhibit 4.3 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).
4.171	Intercreditor Agreement (2021-1), dated as of November 8, 2021, among Wilmington Trust Company, as Trustee of the American Airlines Pass Through Trust 2021-1A and as Trustee of the American Airlines Pass Through Trust 2021-1B, Crédit Agricole Corporate and Investment Bank, acting through its New York Branch, as Class A Liquidity Provider and Class B Liquidity Provider, and Wilmington Trust Company, as Subordination Agent (incorporated by reference to Exhibit 4.4 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).
4.172	Note Purchase Agreement, dated as of November 8, 2021, among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust, National Association, as Escrow Agent, and Wilmington Trust Company, as Paying Agent (incorporated by reference to Exhibit 4.9 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).
4.173	Form of Participation Agreement (Participation Agreement among American Airlines, Inc., Wilmington Trust Company, as Pass Through Trustee under each of the Pass Through Trust Agreements, Wilmington Trust Company, as Subordination Agent, Wilmington Trust Company, as Loan Trustee, and Wilmington Trust Company, in its individual capacity as set forth therein) (incorporated by reference to Exhibit B to Exhibit 4.9 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).
4.174	Form of Indenture and Security Agreement (Indenture and Security Agreement between American Airlines, Inc., and Wilmington Trust Company, as Loan Trustee) (incorporated by reference to Exhibit C to Exhibit 4.9 to American's Current Report on Form 8-K filed on November 12, 2021 (Commission File No. 1-02691)).
4.175	Revolving Credit Agreement (2019-1B), dated as of August 15, 2019, between Wilmington Trust Company, as Subordination Agent, as agent and trustee for the trustee of the American Airlines Pass Through Trust 2019-1B (Aircraft EETC), as Borrower, and National Australia Bank Limited, as Liquidity Provider (incorporated by reference to Exhibit 4.20 to American's Current Report on Form 8-K filed on August 15, 2019 (Commission File No. 1-02691)).
4.176	Indenture, dated as of May 20, 2019, by and among American Airlines Group Inc., the Guarantor (as defined therein) and Wilmington Trust, National Association, as trustee (incorporated by reference to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on May 21, 2019 (Commission File No. 1-8400)).
4.177	Form of 5.000% Senior Notes due 2022 (incorporated by reference to Exhibit A to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on May 21, 2019 (Commission File No. 1-8400)).
4.178	Indenture, dated as of February 25, 2020, by and among American Airlines Group Inc., the Guarantor (as defined therein) and Wilmington Trust, National Association, as trustee (incorporated by reference to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on February 26, 2020 (Commission File No. 1-8400)).
4.179	Form of 3.75% Senior Notes due 2025 (incorporated by reference to Exhibit A to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on February 26, 2020 (Commission File No. 1-8400)).

Exhibit Number	Description
4.180	Indenture, dated as of June 25, 2020, by and between American Airlines Group Inc. and Wilmington Trust, National Association, as trustee (incorporated by reference to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on June 25, 2020 (Commission File No. 1-8400)).
4.181	First Supplemental Indenture, dated as of June 25, 2020, by and among American Airlines Group Inc., American Airlines, Inc. and Wilmington Trust, National Association, as trustee (incorporated by reference to Exhibit 4.2 to AAG's Current Report on Form 8-K filed on June 25, 2020 (Commission File No. 1-8400)).
4.182	Form of 6.50% Convertible Senior Notes due 2025 (incorporated by reference to Exhibit A to Exhibit 4.2 to AAG's Current Report on Form 8-K filed on June 25, 2020 (Commission File No. 1-8400)).
4.183	Indenture, dated as of June 30, 2020, by and among American Airlines, Inc., American Airlines Group, Inc. and Wilmington Trust, National Association, as trustee (incorporated by reference to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on July 2, 2020 (Commission File No. 1-8400)).
4.184	Form of 11.75% Senior Notes due 2025 (incorporated by reference to Exhibit A to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on July 2, 2020 (Commission File Number 1-8400)).
4.185	Indenture (IP Notes), dated as of September 25, 2020, by and among American Airlines, Inc., American Airlines Group Inc. and Wilmington Trust, National Association, as trustee and as collateral trustee (incorporated by reference to Exhibit 4.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2020 (Commission File No. 1-8400)).#
4.186	Form of 10.75%/12.00% PIK Senior Secured IP Notes due 2026 (incorporated by reference to Exhibit A to Exhibit 4.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2020 (Commission File No. 1-8400)).
4.187	Indenture (LGA/DCA Notes), dated as of September 25, 2020, by and among American Airlines, Inc., American Airlines Group Inc. and Wilmington Trust, National Association, as trustee and as collateral trustee (incorporated by reference to Exhibit 4.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2020 (Commission File No. 1-8400)).#
4.188	Form of 10.75%/12.00% PIK Senior Secured Notes due 2026 (incorporated by reference to Exhibit A to Exhibit 4.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2020 (Commission File No. 1-8400)).
4.189	Warrant Agreement, dated as of April 20, 2020, between American Airlines Group, Inc. and the United States Department of the Treasury (incorporated by reference to Exhibit 4.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2020 (Commission File No. 1-8400)).
4.190	Form of PSP1 Warrant (incorporated by reference to Annex B to Exhibit 4.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2020 (Commission File No. 1-8400)).
4.191	Warrant Agreement, dated as of September 25, 2020, between American Airlines Group, Inc. and the United States Department of the Treasury (incorporated by reference to Exhibit 4.5 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2020 (Commission File No. 1-8400)).
4.192	Form of Treasury Loan Warrant (incorporated by reference to Annex B to Exhibit 4.5 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2020 (Commission File No. 1-8400)).
4.193	Warrant Agreement, dated as of January 15, 2021, between American Airlines Group, Inc. and the United States Department of the Treasury (incorporated by reference to Exhibit 4.182 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).
4.194	Form of PSP2 Warrant (incorporated by reference to Annex B to Exhibit 4.182).
4.195	Warrant Agreement, dated as of April 23, 2021, between American Airlines Group Inc. and the United States Department of the Treasury (incorporated by reference to Exhibit 4.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2021 (Commission File No. 1-8400)).
4.196	Form of PSP3 Warrant (incorporated by reference to Annex B to Exhibit 4.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2021 (Commission File No. 1-8400)).
4.197	Indenture, dated as of March 24, 2021, by and among American Airlines, Inc., AAdvantage Loyalty IP Ltd., American Airlines Group Inc., AAdvantage Holdings 1, Ltd. and AAdvantage Holdings 2, Ltd. and Wilmington Trust, National Association, as trustee (incorporated by reference to Exhibit 4.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).
4.198	Form of 5.50% Senior Secured Notes due 2026 (incorporated by reference as Exhibit A-1 to Exhibit 4.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).
4.199	Form of 5.75% Senior Secured Notes due 2029 (incorporated by reference as Exhibit A-2 to Exhibit 4.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).
4.200	Indenture, dated as of February 15, 2023, by and among American Airlines, Inc., American Airlines Group Inc. and Wilmington Trust, National Association, as trustee and as collateral agent (incorporated by reference to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on February 15, 2023 (Commission File No. 1-8400)).

Exhibit Number	Description
4.201	Form of 7.25% Senior Secured Notes due 2028 (incorporated by reference to Exhibit A to Exhibit 4.1 of AAG's Current Report on Form 8-K filed on February 15, 2023 (Commission File No. 1-8400)).
4.202	Indenture, dated as of December 4, 2023, by and among American Airlines, Inc., American Airlines Group Inc. and Wilmington Trust, National Association, as trustee and as collateral trustee (incorporated by reference to Exhibit 4.1 to AAG's Current Report on Form 8-K filed on December 4, 2023 (Commission File No. 1-8400)).
4.203	Form of 8.50% Senior Secured Notes due 2029 (incorporated by reference to Exhibit A to Exhibit 4.1 of AAG's Current Report on Form 8-K filed on December 4, 2023 (Commission File No. 1-8400)).
10.1	Payroll Support Program Agreement, dated as of April 20, 2020, between American Airlines, Inc. and the United States Department of the Treasury (incorporated by reference to Exhibit 10.5 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2020 (Commission File No. 1-8400)).
10.2	Promissory Note, dated as of April 20, 2020, issued by American Airlines Group Inc. in the name of the United States Department of the Treasury and guaranteed by American Airlines, Inc., Envoy Air Inc., Piedmont Airlines, Inc. and PSA Airlines, Inc. (incorporated by reference to Exhibit 10.6 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2020 (Commission File No. 1-8400)).
10.3	Payroll Support Program Extension Agreement, dated as of January 15, 2021, between American Airlines, Inc. and the United States Department of the Treasury (incorporated by reference to Exhibit 10.3 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).
10.4	Promissory Note, dated as of January 15, 2021, issued by American Airlines Group Inc. in the name of the United States Department of the Treasury and guaranteed by American Airlines, Inc., Envoy Air Inc., Piedmont Airlines, Inc. and PSA Airlines, Inc. (incorporated by reference to Exhibit 10.4 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).
10.5	Amendment, dated as of June 30, 2022, to the Promissory Notes and the Warrants issued by American Airlines Group Inc. to the United States Department of the Treasury (incorporated by reference to Exhibit 10.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2022 (Commission File No. 1-8400)).
10.6	Loan and Guarantee Agreement, dated as of September 25, 2020, among American Airlines, Inc., American Airlines Group Inc., the other guarantors party thereto from time to time, the United States Department of the Treasury and the Bank of New York Mellon, as administrative and collateral agent (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2020 (Commission File No. 1-8400).
10.7	Restatement Agreement, dated as of October 21, 2020, to Loan and Guarantee Agreement, dated as of September 25, 2020, among American Airlines, Inc., American Airlines Group Inc., the other guarantors party thereto from time to time, the United States Department of the Treasury and the Bank of New York Mellon, as administrative and collateral agent (incorporated by reference to Exhibit 10.6 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.8	Letter Agreement, dated as of January 15, 2021, to Loan and Guarantee Agreement, dated as of September 25, 2020, among American Airlines, Inc., American Airlines Group Inc., the other guarantors party thereto from time to time, the United States Department of the Treasury and the Bank of New York Mellon, as administrative and collateral agent (incorporated by reference to Exhibit 10.7 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).
10.9	Amended and Restated Credit and Guaranty Agreement, dated as of December 15, 2016, amending the Loan Agreement, dated as of May 23, 2013, among American Airlines, Inc. (as successor in interest to US Airways, Inc., as borrower), as the borrower, American Airlines Group Inc., as parent and guarantor (as successor in interest to US Airways Group, Inc., as parent and guarantor), the lenders from time to time party thereto, Citibank N.A., as administrative agent and collateral agent (as successor in interest to Citicorp North America Inc., as administrative agent and collateral agent), and certain other parties thereto. (incorporated by reference to Exhibit 10.1 to AAG's Annual Report on Form 10-K for the year ended December 31, 2016 (Commission File No. 1-8400)).
10.10	First Amendment to Amended and Restated Credit and Guaranty Agreement, dated as of November 14, 2017, amending the Amended and Restated Credit and Guaranty Agreement, dated as of December 15, 2016, amending the Loan Agreement, dated as of May 23, 2013, among American Airlines, Inc. (as successor in interest to US Airways, Inc., as borrower), as the borrower, American Airlines Group Inc., as parent and guarantor (as successor in interest to US Airways Group, Inc., as parent and guarantor), the lenders from time to time party thereto, Citibank N.A., as administrative agent and collateral agent (as successor in interest to Citicorp North America Inc., as administrative agent and collateral agent), and certain other parties thereto (incorporated by reference to Exhibit 10.2 to AAG's Annual Report on Form 10-K for the year ended December 31, 2017 (Commission File No. 1-8400)).

Exhibit Number	Description
10.11	First Amendment and Restatement Agreement, dated as of April 20, 2015, in relation to the Credit and Guaranty Agreement, dated as of October 10, 2014 (as amended), among American Airlines Group Inc. (as successor in interest to US Airways Group, Inc.), American Airlines, Inc. (as successor in interest to US Airways, Inc.), the Revolving Lenders (as defined therein) party thereto, the 2015 Term Loan Lenders (as defined therein) party thereto and Citibank N.A., as administrative agent and collateral agent (incorporated by reference to Exhibit 10.4 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2015 (Commission File No. 1-8400)).
10.12	First Amendment to Amended and Restated Credit and Guaranty Agreement, dated as of October 26, 2015, amending the Amended and Restated Credit and Guaranty Agreement, dated as of April 20, 2015, among American Airlines, Inc. (as successor in interest to US Airways, Inc.), American Airlines Group Inc. (as successor in interest to US Airways Group, Inc.), the lenders from time to time party thereto, Citibank N.A., as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.6 to AAG's Annual Report on Form 10-K for the year ended December 31, 2015 (Commission File No. 1-8400)).
10.13	Second Amendment to Amended and Restated Credit and Guaranty Agreement, dated as of September 22, 2016, amending the Amended and Restated Credit and Guaranty Agreement, dated as of April 20, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Citibank N.A., as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2016 (Commission File No. 1-8400)).
10.14	Letter Agreement No. AAL-LA-2100511, dated as of March 9, 2021, to Purchase Agreement No. 3219 by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.8 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).**
10.15	Amendment 1, dated as of March 25, 2021, to the Letter Agreement No. AAL-LA-2100511, dated as of March 9, 2021, to Purchase Agreement No. 3219 by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.9 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).**
10.16	Amendment 2, dated as of June 28, 2021, to the Letter Agreement No. AAL-LA-2100511, dated as of March 9, 2021, to Purchase Agreement No. 3219 by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2021 (Commission File No. 1-8400)).**
10.17	Amendment No. 3, dated as of September 24, 2021, to the Letter Agreement No. AAL-LA-2100511, dated as of March 9, 2021, to Purchase Agreement No. 3219 by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2021 (Commission File No. 1-8400)).**
10.18	Letter Agreement No. AAL LA 2100530, dated as of March 9, 2021, to Purchase Agreement No. 3219 by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.10 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).**
10.19	Amendment No. 4, dated as of December 15, 2021, to the Letter Agreement No. AAL-LA-2100511, dated as of March 9, 2021, to Purchase Agreement No. 3219 by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.19 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.20	Letter Agreement, dated as of January 19, 2023, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2023 (Commission File No. 1-8400)).**
10.21	Third Amendment to the Amended and Restated Credit and Guaranty Agreement, dated as of June 14, 2017, amending the Amended and Restated Credit and Guaranty Agreement, dated as of April 20, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Citibank N.A., as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2017 (Commission File No. 1-8400)).
10.22	Fourth Amendment to the Amended and Restated Credit and Guaranty Agreement, dated as of August 21, 2017, amending the Amended and Restated Credit and Guaranty Agreement, dated as of April 20, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Citibank N.A., as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2018 (Commission File No. 1-8400)).*

Exhibit Number	Description
10.23	Fifth Amendment to the Amended and Restated Credit and Guaranty Agreement, dated as of September 17, 2018, amending the Amended and Restated Credit and Guaranty Agreement, dated as of April 20, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Citibank N.A., as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2018 (Commission File No. 1-8400)).
10.24	Sixth Amendment to the Amended and Restated Credit and Guaranty Agreement, dated as of December 10, 2018, amending the Amended and Restated Credit and Guaranty Agreement, dated as of April 20, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Citibank N.A., as administrative agent, and certain other parties thereto.**
10.25	Seventh Amendment to the Amended and Restated Credit and Guaranty Agreement, dated as of November 8, 2019, amending the Amended and Restated Credit and Guaranty Agreement, dated as of April 20, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Citibank N.A., as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.10 to AAG's Annual Report on Form 10-K for the year ended December 31, 2019 (Commission File No. 1-8400)).**
10.26	First Amendment and Restatement Agreement, dated as of May 21, 2015, in relation to the Credit and Guaranty Agreement, dated as of June 27, 2013 (as amended), among American Airlines Group Inc. (as successor in interest to US Airways Group, Inc.), American Airlines, Inc. (as successor in interest to US Airways, Inc.), the Revolving Lenders (as defined therein) party thereto, the 2015 Term Loan Lenders (as defined therein) party thereto and Deutsche Bank AG New York Branch, as administrative agent and collateral agent (incorporated by reference to Exhibit 10.5 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2015 (Commission File No. 1-8400)).
10.27	First Amendment to Amended and Restated Credit and Guaranty Agreement, dated as of October 26, 2015, amending the Amended and Restated Credit and Guaranty Agreement, dated as of May 21, 2015, among American Airlines, Inc. (as successor in interest to US Airways, Inc.), American Airlines Group Inc., (as successor in interest to US Airways Group, Inc.), the lenders from time to time party thereto, Deutsche Bank AG New York Branch, as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.8 to AAG's Annual Report on Form 10-K for the year ended December 31, 2015 (Commission File No. 1-8400)).
10.28	Second Amendment to Amended and Restated Credit and Guaranty Agreement, dated as of March 14, 2017, amending the Amended and Restated Credit and Guaranty Agreement, dated as of May 21, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Deutsche Bank AG New York Branch, as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2017 (Commission File No. 1-8400)).
10.29	Third Amendment to the Amended and Restated Credit And Guaranty Agreement, dated as of August 21, 2017, amending the Amended and Restated Credit and Guaranty Agreement, dated as of May 21, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Deutsche Bank AG New York Branch, as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.11 to AAG's Annual Report on Form 10-K for the year ended December 31, 2017 (Commission File No. 1-8400)).*
10.30	Fourth Amendment to Amended and Restated Credit and Guaranty Agreement, dated as of May 15, 2018, amending the Amended and Restated Credit and Guaranty Agreement, dated as of May 21, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Deutsche Bank AG New York Branch, as administrative agent, and Barclays Bank PLC, as designated replacement term lender (incorporated by reference to Exhibit 10.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2018 (Commission File No. 1-8400)).
10.31	Fifth Amendment to Amended and Restated Credit and Guaranty Agreement, dated as of December 10, 2018, amending the Amended and Restated Credit and Guaranty Agreement, dated as of May 21, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Deutsche Bank AG New York Branch, as administrative agent, and Barclays Bank PLC, as designated replacement term lender.**
10.32	Sixth Amendment to Amended and Restated Credit and Guaranty Agreement, dated as of November 8, 2019, amending the Amended and Restated Credit and Guaranty Agreement, dated as of May 21, 2015, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Deutsche Bank AG New York Branch, as administrative agent, and Barclays Bank PLC, as designated replacement term lender (incorporated by reference to Exhibit 10.17 to AAG's Annual Report on Form 10-K for the year ended December 31, 2019 (Commission File No. 1-8400)).**

Exhibit Number	Description
10.33	Credit and Guaranty Agreement, dated as of April 29, 2016, among American Airlines, Inc. as borrower, American Airlines Group Inc., as parent and guarantor, certain other subsidiaries of American Airlines Group Inc., as guarantors, the lenders party thereto, Barclays Bank PLC, as administrative agent and collateral agent, and certain other parties thereto (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q filed on July 22, 2016 (Commission File No. 1-8400)).
10.34	First Amendment to Credit and Guaranty Agreement, dated as of October 31, 2016, amending the Credit and Guaranty Agreement, dated as of April 29, 2016, among American Airlines, Inc. as borrower, American Airlines Group Inc., as parent and guarantor, the lenders party thereto, Barclays Bank PLC, as administrative agent (incorporated by reference to Exhibit 10.81 to AAG's Annual Report on Form 10-K for the year ended December 31, 2016 (Commission File No. 1-8400)).
10.35	Second Amendment to the Credit and Guaranty Agreement, dated as of August 21, 2017, amending the Credit and Guaranty Agreement, dated as of April 29, 2016, among American Airlines, Inc., American Airlines Group Inc., the lenders from time to time party thereto, Barclays Bank PLC, as administrative agent, and certain other parties thereto (incorporated by reference to Exhibit 10.15 to AAG's Annual Report on Form 10-K for the year ended December 31, 2017 (Commission File No. 1-8400)).*
10.36	Third Amendment to Credit and Guaranty Agreement, dated as of November 1, 2017, amending the Credit and Guaranty Agreement, dated as of April 29, 2016, among American Airlines, Inc. as borrower, American Airlines Group Inc., as parent and guarantor, the lenders party thereto, Barclays Bank PLC, as administrative agent (incorporated by reference to Exhibit 10.16 to AAG's Annual Report on Form 10-K for the year ended December 31, 2017 (Commission File No. 1-8400)).
10.37	Fourth Amendment to Credit and Guaranty Agreement, dated as of December 10, 2018, amending the Credit and Guaranty Agreement, dated as of April 29, 2016, among American Airlines, Inc. as borrower, American Airlines Group Inc., as parent and guarantor, the lenders party thereto, Barclays Bank PLC, as administrative agent.**
10.38	Fifth Amendment to Credit and Guaranty Agreement, dated as of November 8, 2019, amending the Credit and Guaranty Agreement, dated as of April 29, 2016, among American Airlines, Inc. as borrower, American Airlines Group Inc., as parent and guarantor, the lenders party thereto, Barclays Bank PLC, as administrative agent (incorporated by reference to Exhibit 10.23 to AAG's Annual Report on Form 10-K for the year ended December 31, 2019 (Commission File No. 1-8400)). *
10.39	Purchase Agreement No. 3219, dated as of October 15, 2008, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.38 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.40	Supplemental Agreement No. 2, dated as of July 21, 2010, to Purchase Agreement No. 3219 between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.39 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.41	Supplemental Agreement No. 3, dated as of February 1, 2013, to Purchase Agreement No. 3219 between American Airlines, Inc., and The Boeing Company (incorporated by reference to Exhibit 10.40 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.42	Supplemental Agreement No. 4, dated as of June 9, 2014, to Purchase Agreement No. 3219 between The Boeing Company and American Airlines, Inc. dated as of October 15, 2008, relating to Boeing Model 787 Aircraft, as amended, restated, amended and restated, supplemented or otherwise modified (incorporated by reference to Exhibit 10.41 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.43	Supplemental Agreement No. 5, dated as of January 20, 2015, to Purchase Agreement No. 3219 between The Boeing Company and American Airlines, Inc., dated as of October 15, 2008, relating to Boeing Model 787 Aircraft, as amended, restated, amended and restated, supplemented or otherwise modified (incorporated by reference to Exhibit 10.42 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.44	Supplemental Agreement No. 6, dated as of April 21, 2015, to Purchase Agreement No. 3219 between American Airlines, Inc. and The Boeing Company, dated as of October 15, 2008, as amended, restated, amended and restated, supplemented or otherwise modified (incorporated by reference to Exhibit 10.43 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.45	Supplemental Agreement No. 7, dated as of September 12, 2016, to Purchase Agreement No. 3219 dated as of October 15, 2008, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.44 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**

Exhibit Number	Description
10.46	Supplemental Agreement No. 8, dated as of January 26, 2017, to Purchase Agreement No. 3219 dated as of October 15, 2008, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.45 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.47	Supplemental Agreement No. 9, dated as of April 24, 2017, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.46 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.48	Supplemental Agreement No. 10, dated as of May 11, 2017, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.47 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.49	Supplemental Agreement No. 11, dated as of April 6, 2018, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2018 (Commission File No. 1-8400)).*
10.50	Supplemental Agreement No. 12, dated as of May 29, 2019, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2019 (Commission File No. 1-8400)).**
10.51	Supplemental Agreement No. 13, dated as of August 20, 2019, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2019 (Commission File No. 1-8400)).**
10.52	Supplemental Agreement No. 14, dated as of February 24, 2020, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2020 (Commission File No. 1-8400)).**
10.53	Supplemental Agreement No. 15, dated as of March 16, 2020, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2020 (Commission File No. 1-8400)).**
10.54	Supplemental Agreement No. 16, dated as of May 21, 2021, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.4 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2021 (Commission File No. 1-8400)).**
10.55	Supplemental Agreement No. 17, dated as of January 31, 2022, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2022 (Commission File No. 1-8400)).**
10.56	Supplemental Agreement No. 18, dated as of February 9, 2023, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2023 (Commission File No. 1-8400)).**
10.57	Supplemental Agreement No. 19, dated as of April 23, 2023, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2023 (Commission File No. 1-8400)).**
10.58	Supplemental Agreement No. 20, dated as of August 31, 2023, to Purchase Agreement No. 3219 dated as of October 15, 2008, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2023 (Commission File No. 1-8400)).**
10.59	A320 Family Aircraft Purchase Agreement, dated as of July 20, 2011, between American Airlines, Inc. and Airbus S.A.S (incorporated by reference to Exhibit 10.46 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.60	Amendment No. 1, dated as of January 11, 2013, to A320 Family Aircraft Purchase Agreement between American Airlines, Inc. and Airbus S.A.S., dated as of July 20, 2011 (incorporated by reference to Exhibit 10.47 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**

Exhibit Number	Description
10.61	Amendment No. 2, dated as of May 30, 2013, to A320 Family Aircraft Purchase Agreement between American Airlines, Inc. and Airbus S.A.S, dated as of July 20, 2011 (incorporated by reference to Exhibit 10.48 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.62	Amendment No. 3, dated as of November 20, 2013, to A320 Family Aircraft Purchase Agreement between American Airlines, Inc. and Airbus S.A.S., dated as of July 20, 2011 (incorporated by reference to Exhibit 10.49 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.63	Amendment No. 4, dated as of June 18, 2014, to the A320 Family Aircraft Purchase Agreement between Airbus S.A.S., as seller, and American Airlines, Inc., as buyer, dated as of July 20, 2011, as amended, restated, amended and restated, supplemented or otherwise modified (incorporated by reference to Exhibit 10.50 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.64	Amendment No. 5, dated as of June 24, 2014, to the A320 Family Aircraft Purchase Agreement between Airbus S.A.S., as seller, and American Airlines, Inc., as buyer, dated as of July 20, 2011, as amended, restated, amended and restated, supplemented or otherwise modified (incorporated by reference to Exhibit 10.51 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.65	Amendment No. 6, dated as of July 1, 2014, to the A320 Family Aircraft Purchase Agreement between Airbus S.A.S., as seller, and American Airlines, Inc., as buyer, dated as of July 20, 2011, as amended, restated, amended and restated, supplemented or otherwise modified (incorporated by reference to Exhibit 10.52 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.66	Amendment No. 7, dated as of November 25, 2014, to the A320 Family Aircraft Purchase Agreement between Airbus S.A.S., as seller, and American Airlines, Inc., as buyer, dated as of July 20, 2011, as amended, restated, amended and restated, supplemented or otherwise (incorporated by reference to Exhibit 10.53 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.67	Amendment No. 8, dated as of June 11, 2015, to the A320 Family Aircraft Purchase Agreement between American Airlines, Inc. and Airbus S.A.S., dated as of July 20, 2011, as amended, restated, amended and restated, supplemented or otherwise modified (incorporated by reference to Exhibit 10.54 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.68	Amendment No. 9, dated as of September 23, 2015, to the A320 Family Aircraft Purchase Agreement, dated as of July 20, 2011, between American Airlines, Inc. and Airbus S.A.S. (incorporated by reference to Exhibit 10.55 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.69	Amendment No. 10, dated as of July 16, 2018, to the A320 Family Aircraft Purchase Agreement, dated as of July 20, 2011, between American Airlines, Inc. and Airbus S.A.S. (incorporated by reference to Exhibit 10.56 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.70	Amendment No. 11, dated as of June 19, 2019, to the A320 Family Aircraft Purchase Agreement, dated as of July 20, 2011, between American Airlines, Inc. and Airbus S.A.S. (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2019 (Commission File No. 1-8400)).**
10.71	Amendment No. 12, dated as of June 26, 2020, to the A320 Family Aircraft Purchase Agreement between Airbus S.A.S., as seller, and American Airlines, Inc. as buyer, dated as of July 20, 2011, as amended, restated, amended and restated, supplemented or otherwise (incorporated by reference to Exhibit 10.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2020 (Commission File No. 1-8400)).**
10.72	Amendment No. 13, dated as of July 13, 2020, to the A320 Family Aircraft Purchase Agreement between Airbus S.A.S., as seller, and American Airlines, Inc. as buyer, dated as of July 20, 2011, as amended, restated, amended and restated, supplemented or otherwise (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2020 (Commission File No. 1-8400)).**
10.73	Amendment No. 15, dated as of June 30, 2021, to the A320 Family Aircraft Purchase Agreement between Airbus S.A.S., as seller, and American Airlines, Inc. as buyer, dated as of July 20, 2011, as amended, restated, amended and restated, supplemented or otherwise (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2021 (Commission File No. 1-8400)).**
10.74	Amendment No. 14, dated as of October 8, 2020, to the A320 Family Aircraft Purchase Agreement between Airbus S.A.S., as seller, and American Airlines, Inc. as buyer, dated as of July 20, 2011, as amended, restated, amended and restated, supplemented or otherwise (incorporated by reference to Exhibit 10.60 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.75	Purchase Agreement No. 03735, dated as of February 1, 2013, between American Airlines, Inc., and The Boeing Company (incorporated by reference to Exhibit 10.69 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**

Exhibit Number	Description
10.76	Supplemental Agreement No. 1, dated as of April 15, 2013, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.70 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.77	Supplemental Agreement No. 2, dated as of March 6, 2015, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.71 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.78	Supplemental Agreement No. 3, dated as of May 22, 2015, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.72 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.79	Letter Agreement, dated as of January 14, 2016, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.73 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.80	Supplemental Agreement No. 4, dated as of June 6, 2016, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.74 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.81	Supplemental Agreement No. 5, dated as of August 8, 2016, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company.**
10.82	Supplemental Agreement No. 6, dated as of November 15, 2016, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.76 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.83	Supplemental Agreement No. 7, dated as of March 2, 2017, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.77 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.84	Supplemental Agreement No. 8, dated as of December 7, 2017, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.45 to AAG's Annual Report on Form 10-K for the year ended December 31, 2017 (Commission File No. 1-8400)).*
10.85	Supplemental Agreement No. 9, dated as of April 6, 2018, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2018 (Commission File No. 1-8400)).*
10.86	Supplemental Agreement No. 10, dated as of March 26, 2019, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2019 (Commission File No. 1-8400)).**
10.87	Letter Agreement, dated as of September 4, 2020, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2020 (Commission File No. 1-8400)).**
10.88	Supplemental Agreement No. 11, dated as of October 9, 2020, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.74 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.89	Supplemental Agreement No. 12, dated as of October 22, 2020, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.75 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**

Exhibit Number	Description
10.90	Supplemental Agreement No. 13, dated as of November 17, 2020, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.76 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.91	Supplemental Agreement No. 14, dated as of November 25, 2020, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.77 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.92	Supplemental Agreement No. 15, dated as of December 15, 2020, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.78 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.93	Amendment 1, dated as of December 31, 2020, to the Letter Agreement, dated as of September 4, 2020, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.79 to AAG's Annual Report on Form 10-K for the year ended December 31, 2020 (Commission File No. 1-8400)).**
10.94	Supplemental Agreement No. 16, dated as of January 14, 2021, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.5 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).**
10.95	Supplemental Agreement No. 17, dated as of February 11, 2021, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.6 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).**
10.96	Supplemental Agreement No. 18, dated as of March 12, 2021, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.7 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).**
10.97	Supplemental Agreement No. 19, dated as of April 8, 2021, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.5 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2021 (Commission File No. 1-8400)).**
10.98	Supplemental Agreement No. 20, dated as of November 12, 2021, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.92 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400)).**
10.99	Supplemental Agreement No. 21, dated as of January 14, 2022, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2022 (Commission File No. 1-8400)).**
10.100	Supplemental Agreement No. 22, dated as of January 31, 2022, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2022 (Commission File No. 1-8400)).**
10.101	Supplemental Agreement No. 23, dated as of May 5, 2022, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2022 (Commission File No. 1-8400)).**
10.102	Supplemental Agreement No. 24, dated as of June 6, 2022, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2022 (Commission File No. 1-8400)).**
10.103	Supplemental Agreement No. 25, dated as of July 1, 2022, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2022 (Commission File No. 1-8400)).**

Exhibit Number	Description
10.104	Supplemental Agreement No. 26, dated as of August 1, 2022, to Purchase Agreement No. 03735 dated as of February 1, 2013, by and between American Airlines, Inc. and The Boeing Company (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2022 (Commission File No. 1-8400)).**
10.105	Supplemental Agreement No. 27, dated as of October 3, 2022, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and the Boeing Company (incorporated by reference to Exhibit 10.101 to AAG 's Annual Report on Form 10-K for the year ended December 31, 2022 (Commission File No. 1-8400)).**
10.106	Supplemental Agreement No. 28, dated as of November 4, 2022, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and the Boeing Company (incorporated by reference to Exhibit 10.102 to AAG 's Annual Report on Form 10-K for the year ended December 31, 2022 (Commission File No. 1-8400)).**
10.107	Supplemental Agreement No. 29, dated as of May 23, 2023, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and the Boeing Company (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2023 (Commission File No. 1-8400)).**
10.108	Supplemental Agreement No. 30, dated as of June 2, 2023, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and the Boeing Company (incorporated by reference to Exhibit 10.3 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2023 (Commission File No. 1-8400)).**
10.109	Supplemental Agreement No. 31, dated as of July 5, 2023, to Purchase Agreement No. 03735 dated as of February 1, 2013, between American Airlines, Inc. and the Boeing Company (incorporated by reference to Exhibit 10. 1 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2023 (Commission File No. 1-8400)).**
10.110	Consent Agreement, dated as of October 5, 2015, between American Airlines, Inc. (as successor in interest to US Airways, Inc.), American Airlines, Inc. and Airbus S.A.S. (incorporated by reference to Exhibit 10.95 to AAG's Annual Report on Form 10-K for the year ended December 31, 2021 (Commission File No. 1-8400))**
10.111	Supplemental Executive Retirement Program for Officers of American Airlines, Inc., as amended and restated as of January 1, 2005 (incorporated by reference to Exhibit 10.127 to AMR's Annual Report on Form 10-K for the year ended December 31, 2008 (Commission File No. 1-8400)).†
10.112	Trust Agreement Under Supplemental Retirement Program for Officers of American Airlines, Inc., as amended and restated as of June 1, 2007 (incorporated by reference to Exhibit 10.128 to AMR's Annual Report on Form 10-K for the year ended December 31, 2008 (Commission File No. 1-8400)).†
10.113	Trust Agreement Under Supplemental Executive Retirement Program for Officers of American Airlines, Inc. Participating in the Super Saver Plus Plan, as amended and restated as of June 1, 2007 (incorporated by reference to Exhibit 10.129 to AMR's Annual Report on Form 10-K for the year ended December 31, 2008 (Commission File No. 1-8400)).†
10.114	American Airlines Group Inc. 2013 Incentive Award Plan (incorporated by reference to Exhibit 4.1 of AAG's Form S-8 Registration Statement, filed on December 4, 2013 (Registration No. 333-192660)).†
10.115	First Amendment to the American Airlines Group Inc. 2013 Incentive Award Plan (incorporated by reference to Exhibit 10.64 to AAG's Annual Report on Form 10-K for the year ended December 31, 2017 (Commission File No. 1-8400)).†
10.116	Form of American Airlines Group Inc. 2013 Incentive Award Plan Restricted Stock Unit (Cash-Settled) Award Grant Notice and Award Agreement (incorporated by reference to Exhibit 10.125 to AAG's Annual Report on Form 10-K for the year ended December 31, 2013 (Commission File No. 1-8400)).†
10.117	Form of American Airlines Group Inc. 2013 Incentive Award Plan Restricted Stock Unit (Stock-Settled) Award Grant Notice and Award Agreement (incorporated by reference to Exhibit 10.127 to AAG's Annual Report on Form 10-K for the year ended December 31, 2013 (Commission File No. 1-8400)).†
10.118	Form of American Airlines Group Inc. 2013 Incentive Award Plan Restricted Stock Unit (Stock-Settled) Award Grant Notice and Award Agreement for Director Grants (incorporated by reference to Exhibit 10.129 to AAG's Annual Report on Form 10-K for the year ended December 31, 2013 (Commission File No. 1-8400)).†
10.119	Form of Indemnification Agreement (incorporated by reference to Exhibit 10.9 to AAG's Current Report on Form 8-K filed on December 9, 2013 (Commission File No. 1-8400)).†
10.120	2014 Short-Term Incentive Program Under 2013 Incentive Award Plan (incorporated by reference to Exhibit 10.8 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2014 (Commission File No. 1-8400)).†
10.121	American Airlines Group Inc. 2023 Incentive Award Plan (incorporated by reference to Exhibit 99.1 of AAG's Form S-8 Registration Statement, filed on May 10, 2023 (Registration No. 333-271802)).†

Exhibit Number	Description
10.122	Form of American Airlines Group Inc. 2023 Incentive Award Plan Restricted Stock Unit (Stock-Settled) Award Grant Notice and Award Agreement.†
10.123	Form of Letter Agreement for Directors Travel Program (incorporated by reference to Exhibit 10.106 to US Airways Group's Annual Report on Form 10-K for the year ended December 31, 2007 (Commission File No. 1-8444)).†
10.124	Amended and Restated Employment Agreement, dated as of November 28, 2007, among US Airways Group, US Airways, Inc. and W. Douglas Parker (incorporated by reference to Exhibit 10.1 to US Airways Group's Current Report on Form 8-K filed on November 29, 2007 (Commission File No. 1-8444)).†
10.125	Form of Letter Agreement, dated April 25, 2017, by and between American Airlines Group Inc. and each of Robert D. Isom, Jr., Stephen L. Johnson and Derek J. Kerr (incorporated by reference to Exhibit 10.1 to AAG's Current Report on Form 8-K filed on May 1, 2017 (Commission File No. 1-8400)).†
10.126	Letter Agreement, dated as of April 28, 2016, between American Airlines Group Inc. and W. Douglas Parker (incorporated by reference to Exhibit 10.1 to AAG's Current Report on Form 8-K filed on April 29, 2016 (Commission File No. 1-8400)).†
10.127	Payroll Support Program 3 Agreement, dated as of April 23, 2021, between American Airlines, Inc. and the United States Department of the Treasury (incorporated by reference to Exhibit 10.1 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2021 (Commission File No. 1-8400)).
10.128	Promissory Note, dated as of April 23, 2021, issued by American Airlines Group Inc. in the name of the United States Department of the Treasury and guaranteed by American Airlines, Inc., Envoy Air Inc., Piedmont Airlines, Inc. and PSA Airlines, Inc. (incorporated by reference to Exhibit 10.2 to AAG's Quarterly Report on Form 10-Q for the quarter ended June 30, 2021 (Commission File No. 1-8400)).
10.129	Term Loan Credit and Guaranty Agreement, dated as of March 24, 2021, among American Airlines, Inc., AAdvantage Loyalty IP Ltd., American Airlines Group Inc., AAdvantage Holdings 1, Ltd., AAdvantage Holdings 2, Ltd., Barclays Bank PLC, as administrative agent, Wilmington Trust, National Association, as collateral administrator, and the lenders party thereto (incorporated by reference as Exhibit 10.4 to AAG's Quarterly Report on Form 10-Q for the quarter ended March 31, 2021 (Commission File No. 1-8400)).#
10.130	Severance Agreement and Restrictive Covenants Agreement, dated as of September 20, 2023, among American Airlines Group, Inc., American Airlines, Inc. and Robert D. Isom.†
10.131	Severance Agreement and Restrictive Covenants Agreement, dated as of September 20, 2023, among American Airlines Group, Inc., American Airlines, Inc. and Stephen L. Johnson (incorporated by reference to Exhibit 10.4 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2023 (Commission File No. 1-8400)).†
10.132	Severance Agreement and Restrictive Covenants Agreement, dated as of September 20, 2023, among American Airlines Group, Inc., American Airlines, Inc. and David G. Seymour (incorporated by reference to Exhibit 10.5 to AAG's Quarterly Report on Form 10-Q for the quarter ended September 30, 2023 (Commission File No. 1-8400)).†
10.133	Credit and Guaranty Agreement, dated as of December 4, 2023, among American Airlines Inc., as the borrower, American Airlines Group Inc., as parent and guarantor, the lenders from time to time party thereto, Citibank, N.A., as administrative agent, and certain other parties from time to time party thereto (incorporated by reference to Exhibit 4.3 to AAG's Current Report on Form 8-K filed on December 4, 2023 (Commission File No. 1-8400)).
14.1	Code of Ethics (incorporated by reference to Exhibit 14.1 to AAG's Current Report on Form 8-K filed on December 9, 2013 (Commission File No. 1-8400)).
21.1	Significant subsidiaries of AAG and American as of December 31, 202 3.
23.1	Consent of Independent Registered Public Accounting Firm – KPMG LLP.
24.1	Powers of Attorney (included in signature page of this Annual Report on Form 10-K).
31.1	Certification of AAG Chief Executive Officer pursuant to Rule 13a-14(a).
31.2	Certification of AAG Chief Financial Officer pursuant to Rule 13a-14(a).
31.3	Certification of American Chief Executive Officer pursuant to Rule 13a-14(a).
31.4	Certification of American Chief Financial Officer pursuant to Rule 13a-14(a).
32.1	Certification pursuant to Rule 13a-14(b) and section 906 of the Sarbanes-Oxley Act of 2002 (subsections (a) and (b) of section 1350, chapter 63 of title 18, United States Code).
32.2	Certification pursuant to Rule 13a-14(b) and section 906 of the Sarbanes-Oxley Act of 2002 (subsections (a) and (b) of section 1350, chapter 63 of title 18, United States Code).
97.1	American Airlines Group Inc. Policy for Recovery of Erroneously Awarded Compensation.

Exhibit Number	Description
101.1	Interactive data files pursuant to Rule 405 of Regulation S-T, formatted in Inline XBRL (eXtensible Business Reporting Language).
104.1	Cover page interactive data file (formatted in Inline XBRL and contained in Exhibit 101.1).

Pursuant to Item 601(a)(5) of Regulation S-K promulgated by the Securities and Exchange Commission, certain exhibits and schedules to this agreement have been omitted. Such exhibits and schedules are described in the referenced agreement. AAG and American hereby agree to furnish to the Securities and Exchange Commission, upon its request, any or all of such omitted exhibits or schedules.

* Confidential treatment has been granted with respect to certain portions of this agreement.

** Portions of this exhibit have been omitted in accordance with Item 601(b)(10) of Regulation S-K.

† Management contract or compensatory plan or arrangement.

ITEM 16. FORM 10-K SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

American Airlines Group Inc.

Date: February 21, 2024

By: /s/ Robert D. Isom
Robert D. Isom
Chief Executive Officer and President
(Principal Executive Officer)

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

American Airlines, Inc.

Date: February 21, 2024

By: /s/ Robert D. Isom
Robert D. Isom
Chief Executive Officer and President
(Principal Executive Officer)

KNOW ALL PERSONS BY THESE PRESENTS, that each individual whose signature appears below constitutes and appoints Robert D. Isom and Devon E. May and each or any of them, his or her true and lawful attorneys and agents, with full power of substitution and resubstitution, for him or her and in his or her name, place and stead, in any and all capacities, to sign any and all amendments to the registrants' Annual Report on Form 10-K for the fiscal year ended December 31, 2023, and to file the same with all exhibits thereto, and all other documents in connection therewith, with the Securities and Exchange Commission, granting unto said attorneys and agents, and each or any of them, full power and authority to do and perform each and every act and thing requisite and necessary to be done, as fully to all intents and purposes as he or she might or could do in person, hereby ratifying and confirming all that said attorneys and agents, and each of them, or his substitute or substitutes, may lawfully do or cause to be done by virtue hereof.

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of American Airlines Group Inc. and in the capacities and on the dates noted:

Date: February 21, 2024	/s/ Robert D. Isom Robert D. Isom Chief Executive Officer and President (Principal Executive Officer)
Date: February 21, 2024	/s/ Devon E. May Devon E. May Executive Vice President and Chief Financial Officer (Principal Financial Officer)
Date: February 21, 2024	/s/ Angela K. Owens Angela K. Owens Senior Vice President and Corporate Controller (Principal Accounting Officer)
Date: February 21, 2024	/s/ Jeffrey D. Benjamin Jeffrey D. Benjamin, Director
Date: February 21, 2024	/s/ Adriane M. Brown Adriane M. Brown, Director
Date: February 21, 2024	/s/ John T. Cahill John T. Cahill, Director
Date: February 21, 2024	/s/ Michael J. Embler Michael J. Embler, Director
Date: February 21, 2024	/s/ Matthew J. Hart Matthew J. Hart, Director

Date: February 21, 2024	/s/ Susan D. Kronick Susan D. Kronick, Director
Date: February 21, 2024	/s/ Martin H. Nesbitt Martin H. Nesbitt, Director
Date: February 21, 2024	/s/ Denise M. O'Leary Denise M. O'Leary, Director
Date: February 21, 2024	/s/ Vicente Reynal Vicente Reynal, Director
Date: February 21, 2024	/s/ Gregory D. Smith Gregory D. Smith, Chairman
Date: February 21, 2024	/s/ Douglas M. Steenland Douglas M. Steenland, Director

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of American Airlines, Inc. and in the capacities and on the dates noted:

Date: February 21, 2024	/s/ Robert D. Isom Robert D. Isom Chief Executive Officer and President (Principal Executive Officer)
Date: February 21, 2024	/s/ Devon E. May Devon E. May Executive Vice President and Chief Financial Officer (Principal Financial Officer)
Date: February 21, 2024	/s/ Angela K. Owens Angela K. Owens Senior Vice President and Corporate Controller (Principal Accounting Officer)
Date: February 21, 2024	/s/ Stephen L. Johnson Stephen L. Johnson, Director